Geodesic Dome Wedge Method

The 40 Hour Cabin

Built 2026-10-03. Every figure in this book was computed at build time from the geometry in the DomeSim project.

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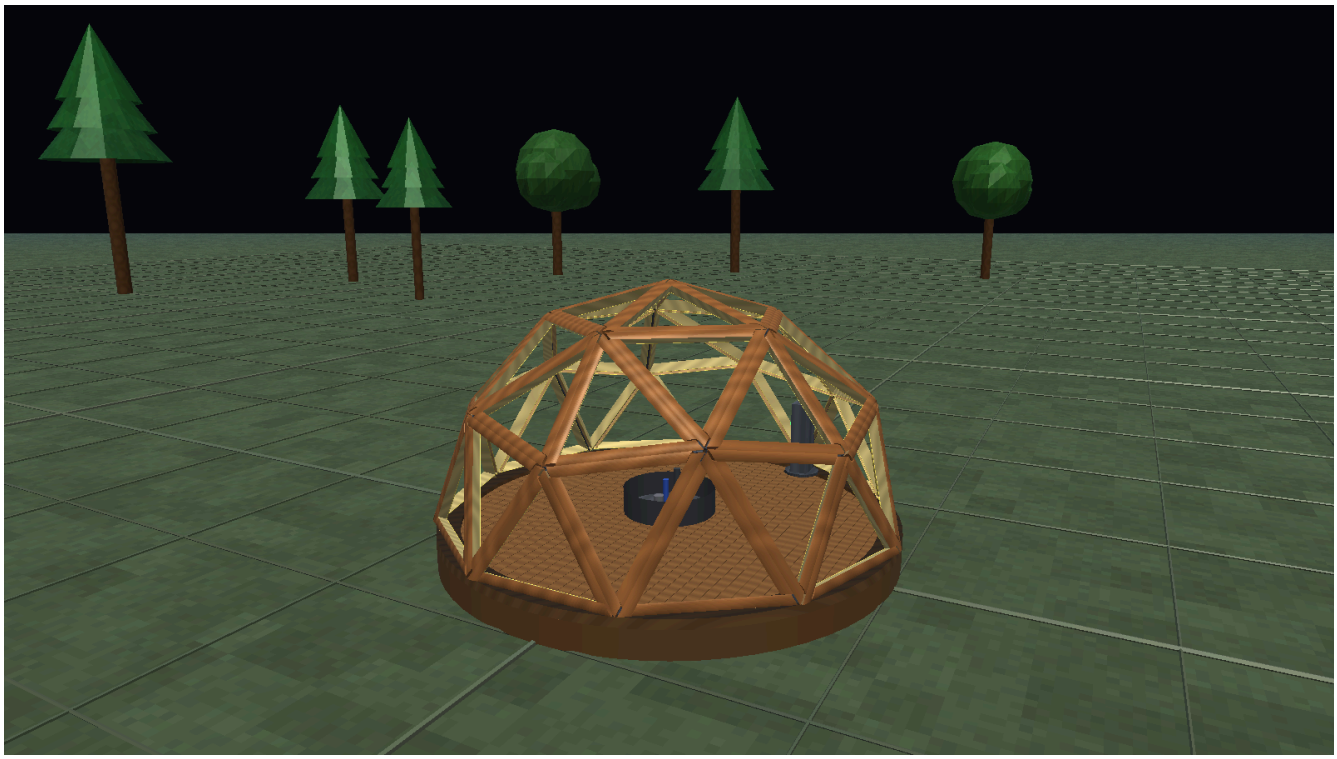
The 40 Hour Cabin



The frame, on its own land.

A dwelling of 120 split-log members, 40 panels and 277 square feet of floor — cut from standing timber with a chainsaw, with a frame that stands up in a working week.

Built from the geometry in the DomeSim project. Every figure in this book was computed at build time by the same software that draws the models.

The Engineer’s Note

Read this before anything else in the book, and before you pick up a saw.

This book is not engineering. Nothing in it is a structural rating. Snow load, wind load, the ground under the pad and the way a frame is fastened are specific to your site, and they are a licensed engineer's numbers, not this book's. Where a chapter reaches one of those questions it says so and stops. Before anybody sleeps in a building made this way, have an engineer look at the drawings and at the frame.

This book is not your building code. Permits, setbacks, fire, egress and occupancy rules are local. Ask your building department what they need before you fell the first tree, not after you raise the last panel.

Electricity and water are trades. The utility column carries a full-size electrical feed and real plumbing. Have a licensed electrician connect and inspect the electrical work and a licensed plumber the water and drain, and follow what your inspector asks for.

Felling is the most dangerous hour in this book. Chapter 22 lists it as the first process and, on purpose, does not teach it. Learn it from a qualified instructor, in person. Wear a helmet with a face screen, hearing protection, cut-resistant chaps, gloves and boots; never fell alone; plan your escape route before the first cut.

The rest of the work is heavy. Ripping logs throws chips and sawdust, so wear eye protection as well as the rest of your saw gear. Wedges and panels are lifted above your head, so work in pairs from a stable platform. The shell comes apart in pieces that are a job for several people or a crane, and the book gives their weights so nobody has to guess.

Some of this book is a design possibility, not a tested product. The book keeps three kinds of claim apart – known geometry, tested construction, and design possibility – and says which is which. The floating dome in Chapter 55 is the clearest example: it is priced, not rated.

You build at your own risk. The author and publisher accept no liability for injury, loss or damage arising from the use of this book.

Forty Hours

The goal is a house built so efficiently that one person can go from standing timber to a standing shell inside a single working week.

Not a kit. Not a shell delivered on a lorry. A tree, a chainsaw, and 40 hours.

That is not an aspiration bolted onto the method afterwards. It is where the number came from, and here is the arithmetic.

The clock

Splitting. One session at the log — 4 hours with a meagre chainsaw — produces 30 wedges. The frame needs 120, so every member in the building is **16 hours** at the log.

Everything else. Taken end to end, one member costs about 15 minutes from tree to standing: split, crosscut, drilled, joined, and up in the shell. That is an average over the fast work and the slow, not a stopwatch on one operation.

```
120 members x 15 minutes  =  30 hours
```

Splitting is 53 per cent of that, which is why Chapter 22 says the rip is the process worth improving first.

Getting it wrong. Add 25 per cent for the stick that splits off the radius, the panel laid down in the wrong order, and the hour spent looking for the drill.

```
30 hours + 25%  =  37.5 hours
```

Which is a 40 hour week, with a little in hand.

The honest part

You will not do it in five days.

This is hard physical work. Splitting a log is not typing; you get perhaps four good hours at it before the quality of your swing goes, and the quality of your swing is what keeps a split on the radius. Nobody does forty hours of this linearly, and anybody who tells you they did was counting the hours and not the days.

It comes in **spurts**. A weekend at the log. An evening on the bench. A day when three people turn up and the shell goes from nothing to closed.

That does not weaken the number — it is the reason the number is worth having. Forty hours of work is a thing you can *schedule* against a life: five Saturdays, or two weeks of evenings, or one very determined week off. A building measured in months is a building you never start, because months cannot be fitted into a year that already has things in it.

So: the forty hours is a **budget of effort**, not a calendar. The point of counting it is that it is small enough to count.

And the rest of the book

The shell is a week. The skin and the floor — Chapters 2 and 36, the two parts of this building that decide whether it is still standing in thirty years — are not, and this book spends more pages on them than on the frame for exactly that reason.

The frame is the part you can do in a week. It is also the part you are allowed to get wrong.

The whole building, counted the same way

So that nobody has to find it on the last page: the forty hours is the frame and its shell – 37.5 hours with error allowed for. The whole standard building, counted the same way – frame, panels, shell, utility column and the rest – is **67 hours** of labour, and Chapter 53 shows every one of them. The title is the part one person can do in a week. The rest is the part worth doing slowly.

PART 1

What Actually Has to Be Right

“A dome is a frame, a skin and a floor -- and the frame is the part you are allowed to get wrong.”

Three chapters that invert the usual advice, and one practical consequence: build the frame loose and the two layers that matter stay changeable for the life of the building.

CHAPTER 1 · STORY

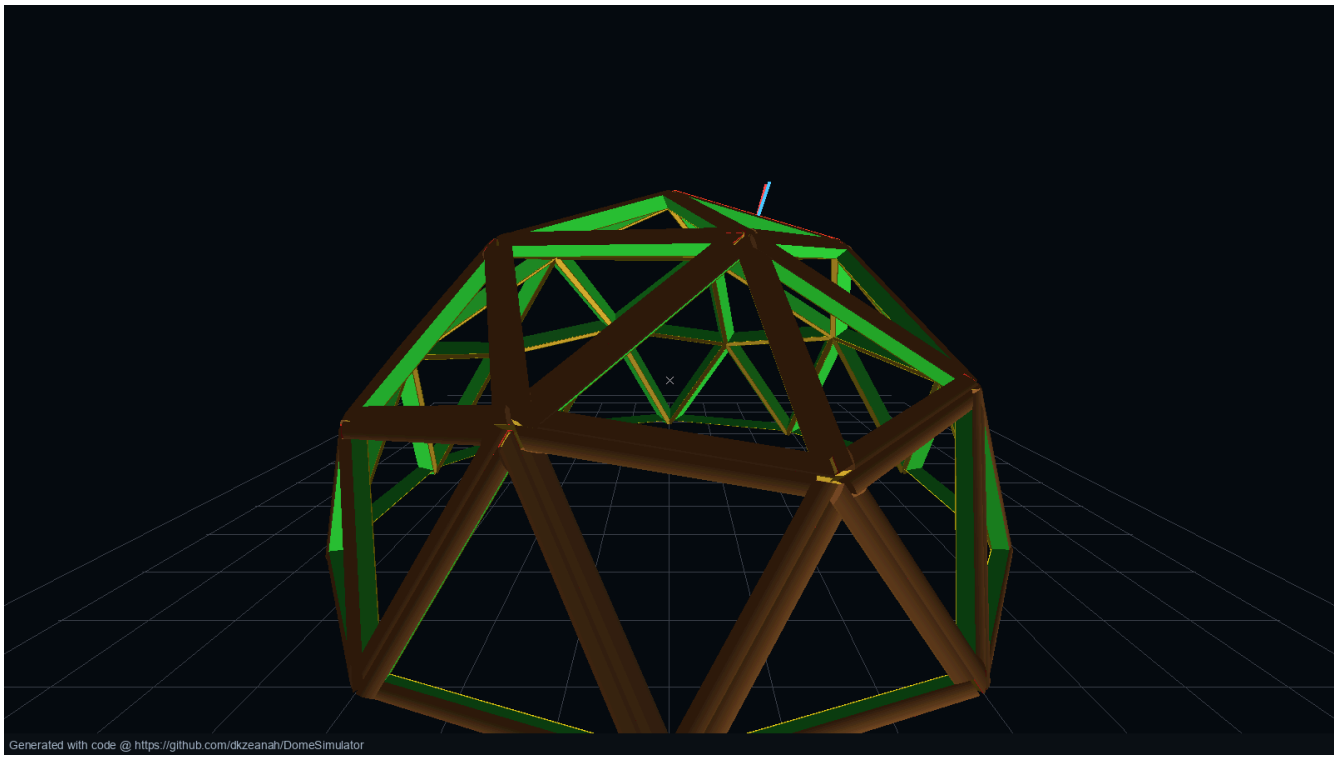
Your Skull Is Not Symmetrical Either

The frame is the part you are allowed to get wrong, and almost every dome book has this backwards

Every dome book I have read opens by telling you to be precise.

Measure twice. Hold your angles. A geodesic dome is a mathematical object and you are about to approximate one out of wood, so the closer you get to the maths the better your building will be.

I believed that for about a year, and it cost me most of a year.



The reference build. Nothing about it is as accurate as it looks.

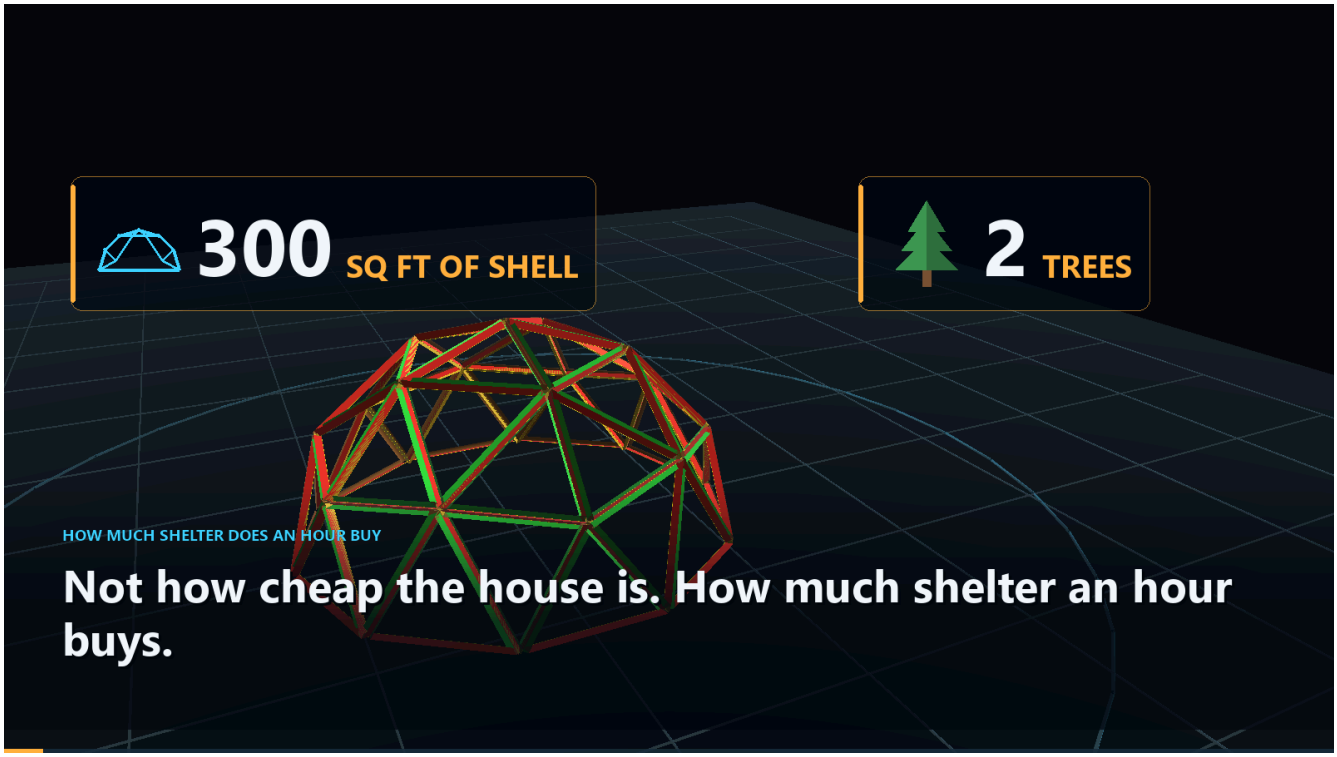
The afternoon I stopped believing it

I had a frame up. Not this one — an earlier one, made of whatever the chainsaw had produced that week, with no two sticks alike. I had built it to prove to myself that the method could not work, so that I could stop thinking about it and go and buy some lumber like a normal person.

It stood. It stood with a stick in it that was out by most of an inch, and a vertex where three members met that I could put my whole thumb into.

I pushed it. I hung off it. I put a ladder against it and climbed on it, which I do not recommend and did anyway.

It did not care.



How much shelter an hour buys, which is the question this book is an answer to.

Why it did not care

A dome carries its loads *axially*. The members are in compression along their own length — pushing end to end, the way a stack of bricks pushes — and a triangulated shell passes a load from one triangle to the next all the way round, so no single member is ever being asked to do very much on its own.

Put a stick in slightly out of position and two things happen. The triangle it is in is very slightly a different triangle. And the load path shifts by that same very slight amount.

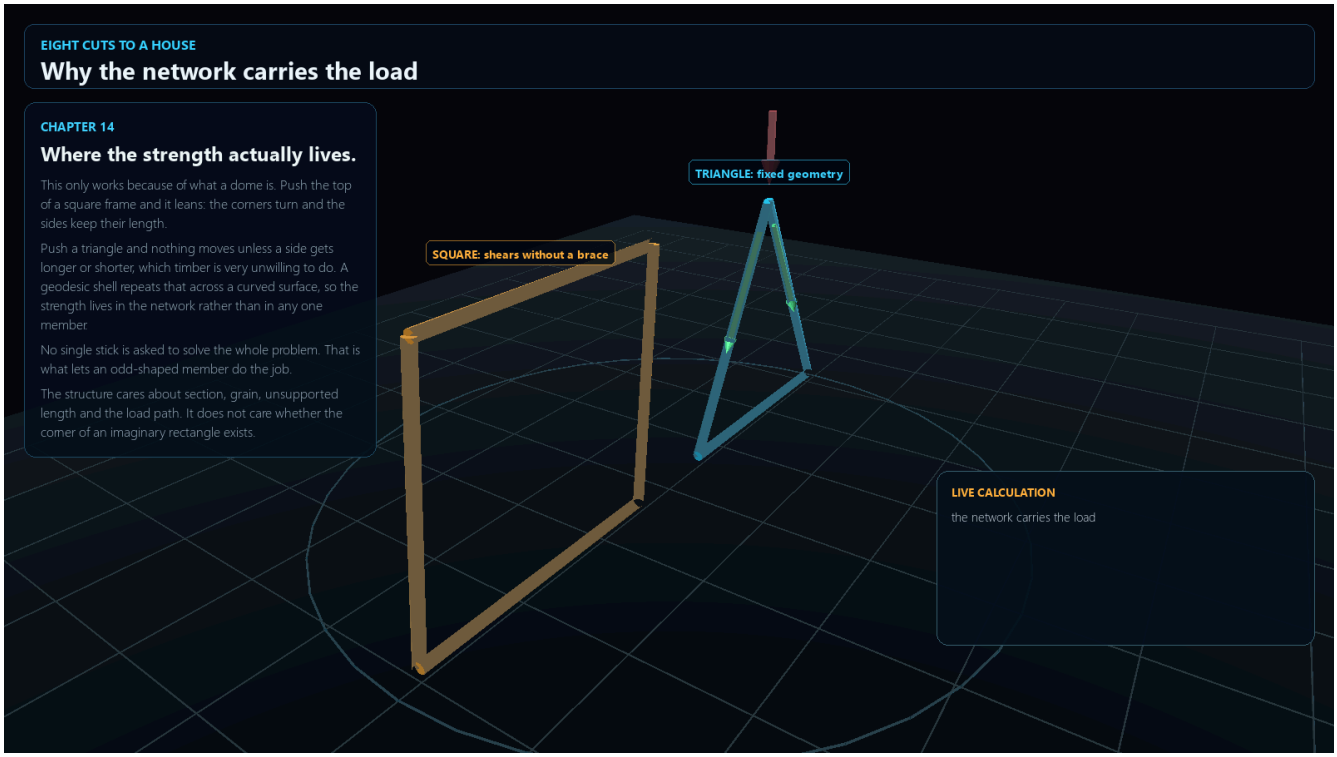
Neither of those is an error that grows. A frame is not a tower of tolerances stacking up toward a failure; it is a shell that absorbs the difference and carries on. The error goes into the seam — and there is already a key in the seam, because there was always going to be one.

Why a triangle, and not a square

Push the top corner of a square frame sideways and it leans. Nothing breaks: the four sides keep their lengths and the corners simply turn. A square is a mechanism, and it needs stiff corners – nailed, braced, glued – to stop being one.

Push the corner of a triangle and nothing moves unless one of its sides gets longer or shorter. Timber is very unwilling to do either along its length. So a triangle needs nothing from its corners at all.

A geodesic shell repeats that triangle across a curved surface, 40 times on this dome. The strength lives in the **network**, not in any one member. No single stick is asked to solve the whole problem; each one is asked to stay its own length, and that is all.



Where the strength actually lives.

That is what lets an odd-shaped member do the job. The structure cares about the member's section, its grain, its unsupported length and the path the load takes through it. It does not care whether the corner of an imaginary rectangle exists – which is the whole reason a split log, with no rectangle anywhere in it, can be a stick in a building.

The lumpy skull

You have an asymmetric skull.

Genuinely: measure it. One side of your head is further from the midline than the other, one eye socket is fractionally lower, your jaw is not centred. This is true of essentially everybody and it is why nobody's face looks right when you mirror it.

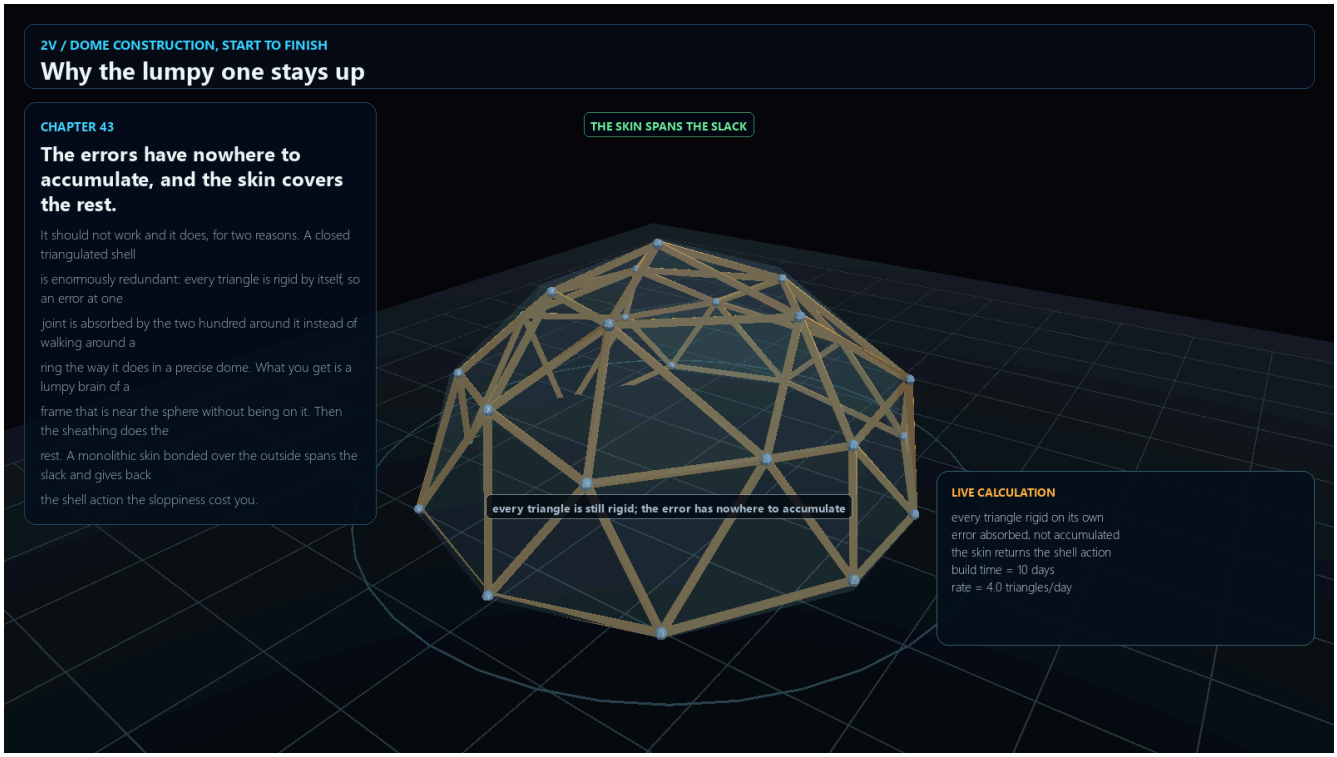
Nobody likes you any less for it, and your skull opposes a baseball bat exactly as well as a perfectly symmetrical one would — because the thing that stops the bat is the *shape*, a closed curved shell in compression, and not the precision.

That is a dome. The geometry does the work. The accuracy is decoration.

There are two reasons the lumpy one stays up, and both are worth knowing before you trust it.

The errors have nowhere to accumulate. A closed, triangulated shell is enormously redundant. Every triangle is rigid on its own, so a joint that is out is absorbed by the triangles around it instead of walking round a ring and adding up the way an error does in a precise, hubbed dome, where every strut has to land on the same point as five others. What you get is a frame that is near the sphere without being on it – a lumpy brain, not a ball.

And the skin covers the rest. One continuous layer bonded over the outside spans the slack between members and gives back the shell action the sloppiness cost. Which is the first hint of the argument the next chapter makes: the frame can be forgiving because the skin is not allowed to be.



The errors have nowhere to accumulate, and the skin covers the rest.

What this book does with that

It spends it.

If the frame does not need precision, then the frame does not need a mill. It does not need a planer, or a moisture-controlled shed, or lumber bought to a tolerance somebody else held for you. It can be made of logs split with a wedge, on the ground, in a field, by one person who is not very good at it yet.

That is the whole method. Everything else in this book — the 45.0-degree sector, the pinwheel joint, the one flat board that makes 40 identical panels — is a consequence of deciding that the frame is the part you are allowed to get wrong.

And what it does not spend

Here is the part that took me another year to learn, and it is the reason this book exists rather than being a pamphlet about splitting logs.

The permission is not general. It applies to the frame and it applies to nothing else.

The skin has to be right, because water does not negotiate and does not announce itself. The floor has to be right, because everything you own sits on it and because the ground sends its water up.

So the advice most dome books give is not merely unhelpful. It is aimed at the wrong layer. It asks for care where care is cheap to skip, and it treats as finishing work the two layers that decide whether the building is still there in thirty years.

The next two chapters are about those two layers. Then we can go and cut something.

CHAPTER 2 · EXPLAIN

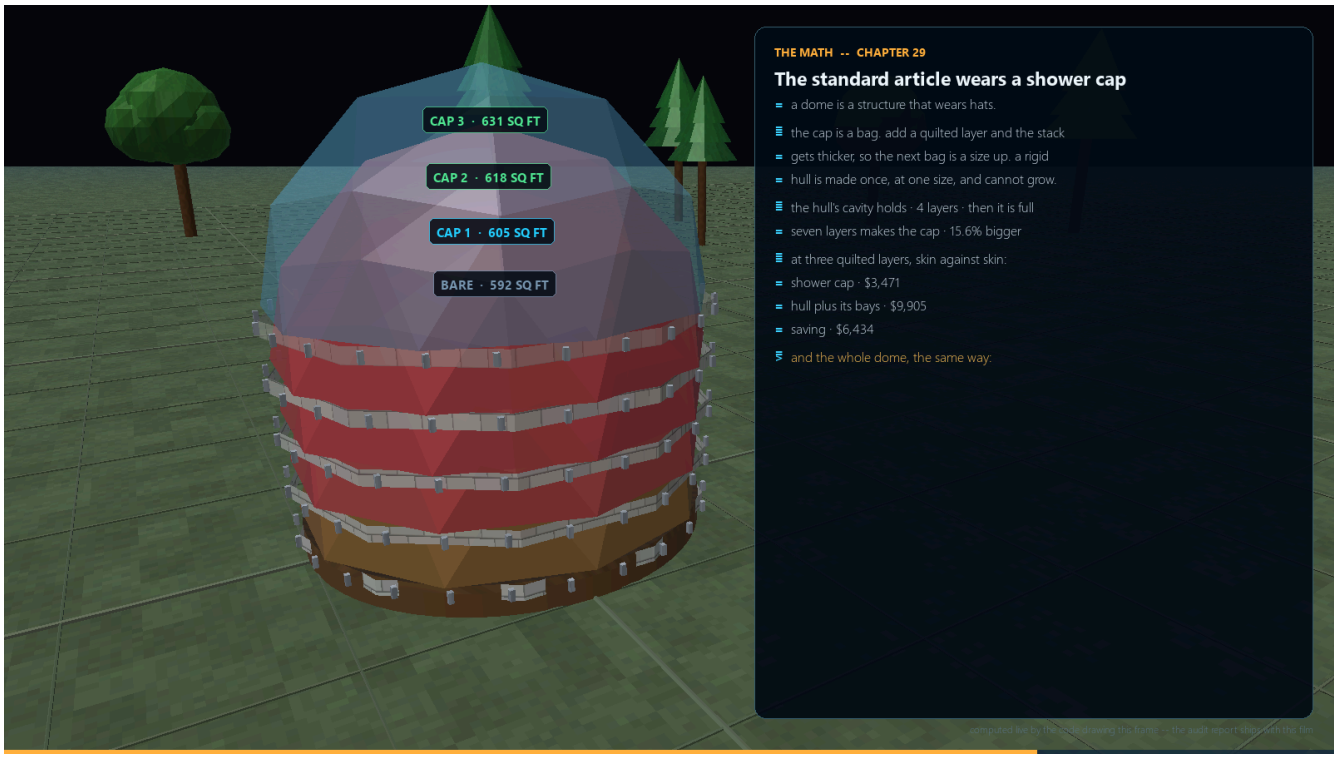
Water Is the Silent Killer

The skin is the part that has to be right, and it is the part every dome book treats as a finish

A frame that is out by an inch stands for fifty years.

A skin that leaks for one winter takes the frame with it.

That asymmetry is the most useful thing in this book, and almost every dome book has it the other way round.



One watertight layer over everything else.

Why water gets to decide

Structural failure is loud. Something cracks, something sags, something you can see from across the room. You get warning, and the warning arrives in a form that makes you do something about it.

Rot is not loud. Rot is a slow chemical argument between a fungus and your building, conducted at about twenty per cent moisture content, entirely out of sight, and it is usually two or three years old by the time anything shows on a surface you look at.

By then you are not repairing a leak. You are replacing members, and the members are the thing this whole method spent its cleverness on making cheap — so you are not even losing much money. You are losing the fortnight.

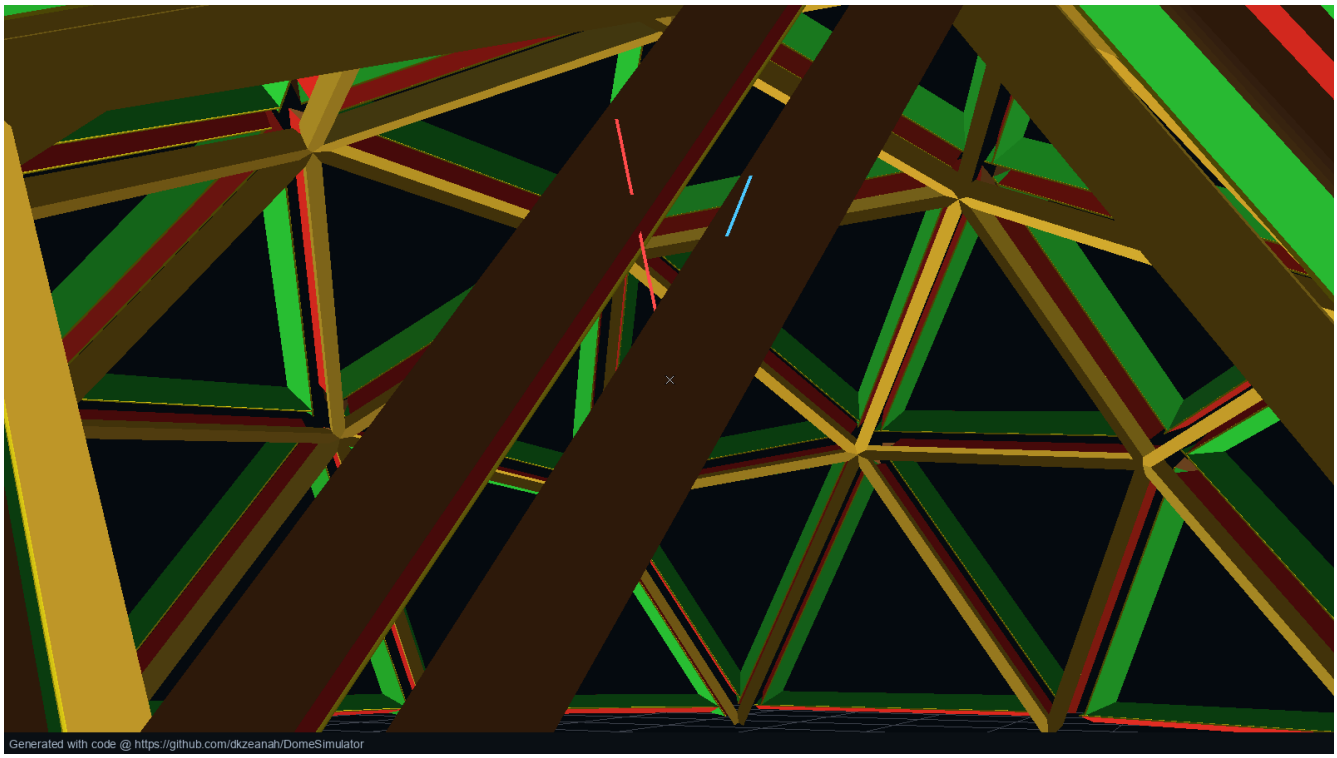
That is the trade nobody states plainly: the frame is cheap to build and expensive to replace, because replacing it means taking the building apart.

Where a dome actually leaks

Not through the panels. A panel is a large flat sheet with no holes in it and it is the easiest part of any building to keep water out of.

Domes leak at seams, at penetrations, and at the apex, in that order — and a 2V dome has a great many seams.

This one has 55 interior seams, which is 309 feet of joint. Every foot of that is a place where two panels meet and where somebody has to have done something deliberate.



Every seam in the dome, and the key in each one.

Three hundred feet of joint is not a detail. It is the single largest risk surface the building has, and it is spread out so evenly that no one part of it looks important enough to worry about.

The move that removes the category

You do not solve three hundred feet of joint by making each joint better.

You solve it by putting one continuous layer over all of it, so that the joints stop being the outside of the building.

That is what the shower cap is. One watertight membrane over the whole shell, strapped down, with nothing structural fastened through it and no seam in it that is not a welded one. The panels underneath do not have to be watertight. The seams underneath do not have to be watertight. They have to be dry, and they are dry because something above them is doing that job and only that job.

The chapter on the cap (28) has the physics and the arithmetic. What matters here is the ordering: **one layer whose only job is water, and it goes on last and comes off first.**

And the floor

The floor is the second thing, and for a different reason.

It is the surface you are in contact with all day. It carries everything you own. It is the only part of the building that is in permanent contact with the ground, which means it is where ground moisture arrives, and unlike a wall it cannot dry to the outside because there is no outside — there is dirt.

A floor that is cold, or damp, or that flexes, is a floor you notice every hour you are awake. A frame that is out of true is a frame you never think about again after the first week.

Get the floor right. Chapter 36 says what right means, and what the five ways of doing it cost on this footprint.

The hierarchy, stated once

1. **The skin.** Water. One continuous layer, no compromises, no penetrations you did not design.
2. **The floor.** Comfort, dryness, and everything you own.
3. **The frame.** Standing up, which it will do whether or not you were careful.

If you read nothing else in this book, invert your instincts in that order. It is worth more than every angle in the appendix.

The catch, and the next chapter

There is an obvious objection, and it is a good one.

The skin and the floor are the expensive parts. They are the parts you cannot cut out of a tree with a chainsaw. Telling somebody that the two layers that matter are also the two layers that cost money is not obviously helpful.

The answer is that they are also the two layers you can *defer* — and the next chapter is about the four things you do while building the frame that keep that option open for the life of the building.

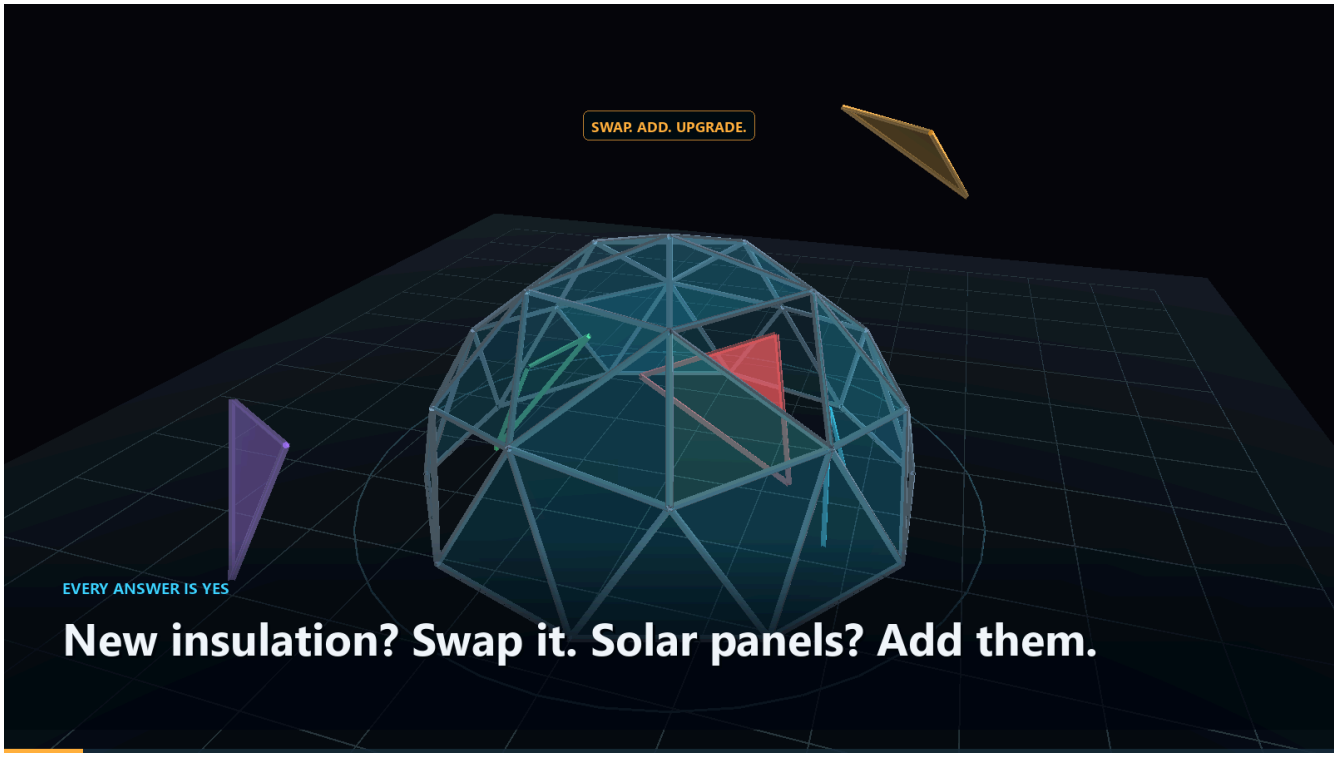
CHAPTER 3 · EXPLAIN

Build It So You Can Change Your Mind

The skin and the floor are the two things that matter and the two things you can defer -- if the frame is built for it

The two layers that matter most are the two layers you can put off.

That sounds like a consolation and it is actually the design. A frame is a commitment: 120 members, cut to two lengths, and once the first panel is closed the shape of the building is decided. A skin is not. A floor is not. Those are things the building *wears*, and a building can change what it wears.



New insulation? Swap it. Solar panels? Add them.

The condition

It only works if you build for it, and the decisions are all made in the first week, before there is a building to look at.

None of them costs anything meaningful at the time. All of them are effectively impossible to add later.

The four things

Threaded inserts in the wedge faces, not screws into end grain.

A panel that will be taken off and put back needs somewhere to bolt to, and end grain is not it — the second time you drive a screw into the same hole it is holding about half of what it held the first time, and the fourth time it is holding nothing. An insert set into the flat sawn face of a wedge is a machine thread in a piece of metal. It does not care how many times you use it.

This is the single highest-leverage decision in the build, and it happens while the panel is still flat on the jig.

A cap sized a layer larger than it needs.

The outer skin goes over whatever is under it, so its size depends on how much is under it. A cap cut to fit a bare shell will not go over a shell wearing two quilted layers, and you find that out on the day you want to add the second layer, in November.

Chapter 30 has the arithmetic — each hat is a size up, and the sequence is not one number. Buy one size up from what you need on day one.

A floor that clamps rather than is built in.

A floor framed into the base ring is part of the structure. A floor that is a hub, radial spokes and decking, clamped to something in the middle, is a component — it can come out, go back, be replaced with a better one, or be left behind.

It also turns out to be the thing that makes a floating dome possible at all, which was not the reason for doing it.

Services in the seam channel, not in the wall.

The seam leaves a gap that has to be filled with something. Fill it with a hose and the building's plumbing is in its structure, reachable from outside by taking off one panel, and no pipe has ever been buried in anything.

An A seam's channel takes a line up to 2.878 inches across. That is not a tight fit for the things a small building actually needs.

What deferring costs

Honestly: some money, later, in exchange for less money now.

The standard article in this book's cost model lists at $12,036, built to $10,030. The shell inside that is $3,265. The bare version of the same cap — no quilted layers at all — is $3,265, and each layer you add afterwards is a few tens of dollars of material and an evening.

So the deferral is real and it is cheap. What you are buying with the four decisions above is the right to spend that money in the order your life actually allows, rather than in the order a building would prefer.

The failure this prevents

I have watched somebody build a beautiful frame, run out of money, put a tarpaulin over it for a winter, and lose the frame.

Not because the tarpaulin was a bad idea. A tarpaulin is a perfectly good temporary skin. They lost it because there was nowhere for a real skin to *attach* when the money came back — no inserts, no strap points, nothing but a frame that had been designed on the assumption that the skin would be figured out later by somebody with more time.

Later is not a plan. Later is a thing you design for.

Every answer is yes

Built that way, the questions people ask about a dome all get the same answer. New insulation? Swap it. Solar? Add it to a panel. A greenhouse wall? Sure. A workshop extension? Bolt it to the rim. Some technology nobody has invented yet? There should be a panel for that – and if the frame is a set of sockets (Chapter 21), there will be.

Where this leaves us

Three chapters in, the argument of this book is complete:

- the frame is allowed to be messy, which is what lets a split log be a structural member;
- the skin and the floor are not, because water and comfort do not forgive;
- and the two layers that are not allowed to be messy are the two you can postpone, upgrade and swap — if you spend four decisions on it in week one.

The rest of the book is how. It starts where the material does, with a tree.

CHAPTER 4 · EXPLAIN

A House With a Parts List

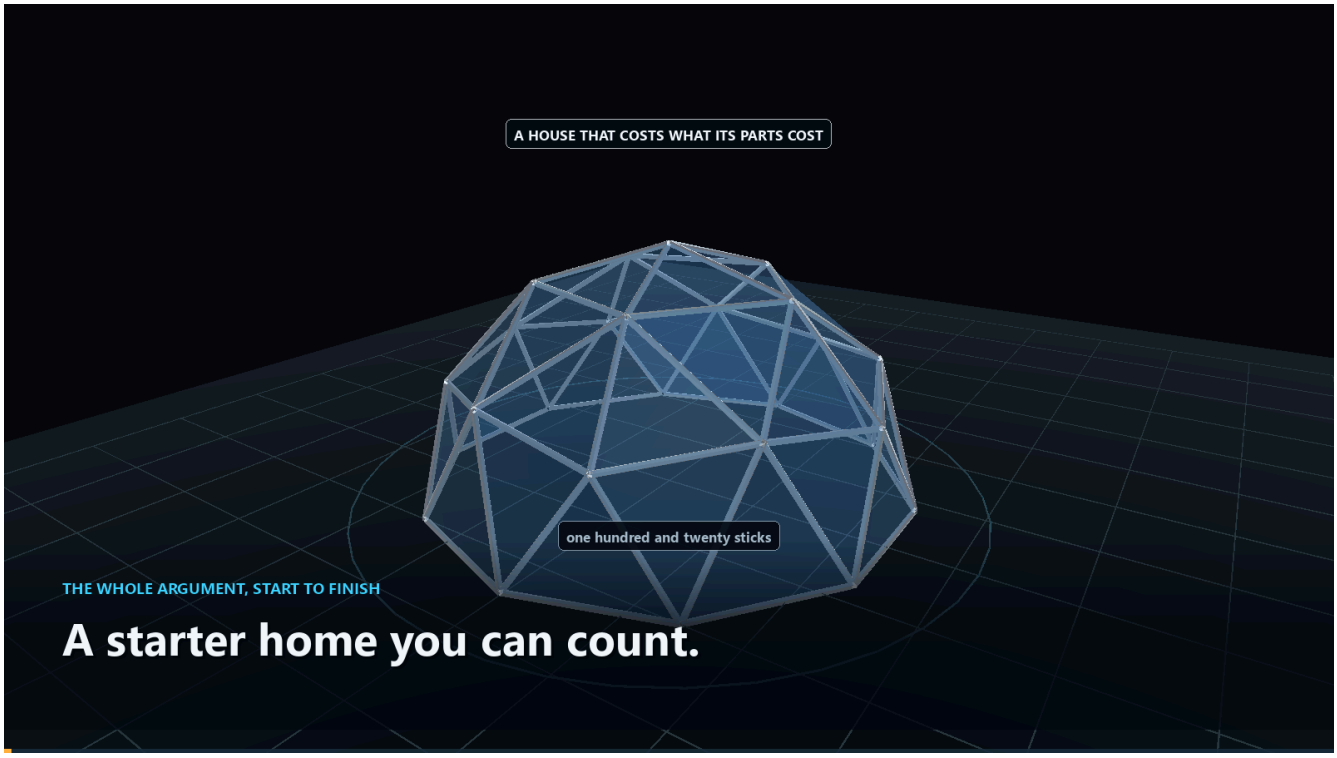
Count the model, and a building stops being a price opinion: the cut list makes itself, and the labour is a flat rate

The whole argument, start to finish

Here is everything this book is going to argue, said once, so you can watch it being proved.

A dome made this way is a shape that spends less on skin and gives that back every winter. It is made of two lengths of wood, cut on one flat board. It is built by a method that forgave a chainsaw, so it will forgive you. It has a price list with a receipt behind every line and an energy bill you can work out for yourself. And none of it is an opinion, because every number in it is counted off the same model of the building that draws its pictures.

That last part is the one this chapter is about.



Everything this project knows, in one film.

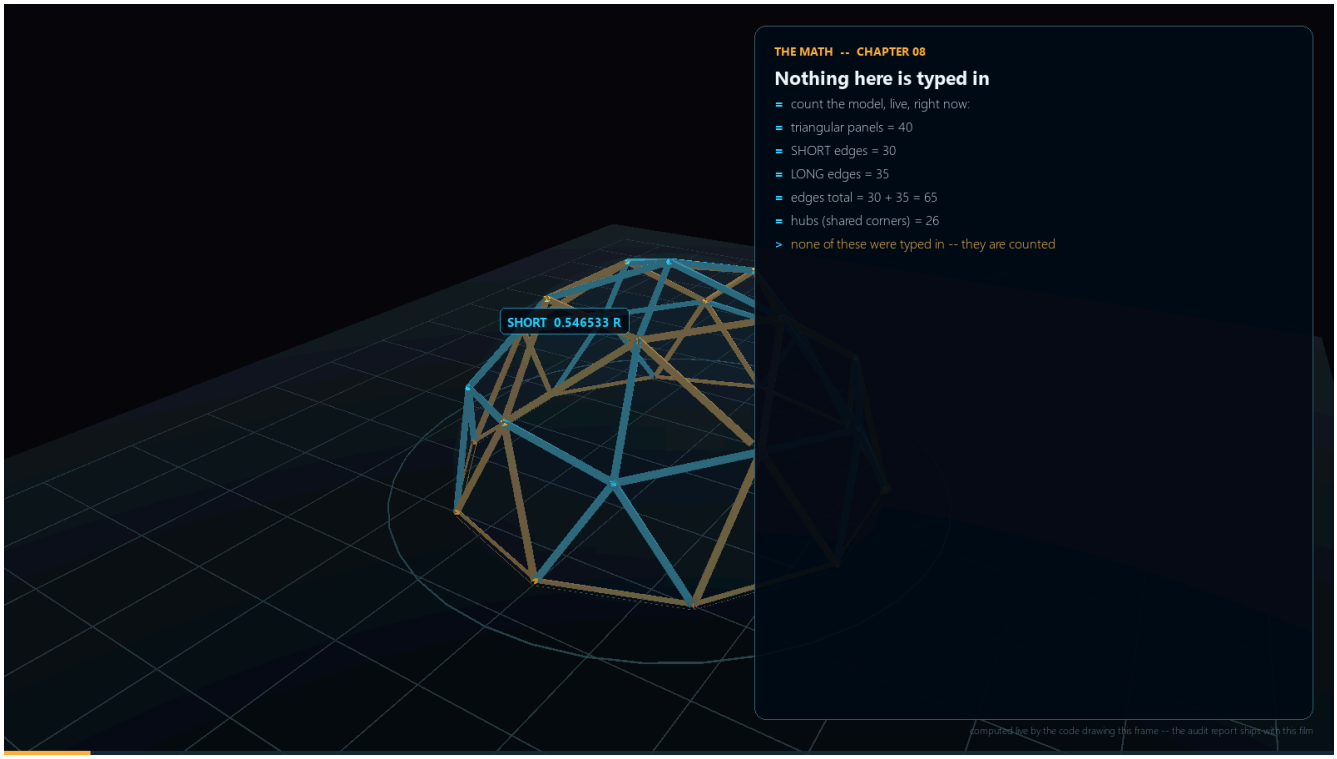
Nothing here is typed in

Ask a builder what a house will cost and you get an opinion, informed by experience and padded for the things that go wrong. Ask this dome, and it counts itself:

triangular panels	40
edges of the short length	30
edges of the long length	35
edges in all	65
corners where edges meet	26

Nobody wrote those numbers into this paragraph. The software that solves the dome and draws its pictures is the software that counts it, and the book fills those numbers in at the moment it is printed. Every figure in the chapters that follow – every angle, every dollar, every gallon – reaches the page the same way. Where something cannot be computed, a price or a bead size or how much oil rough wood drinks, the book says where it came from and marks it as an assumption.

That is not pedantry. It is the difference between a parts list and a price opinion. A parts list can be checked, and when one input changes, every figure that depends on it changes with it.



The panel on the right is counting the dome behind it.

The cut list makes itself

Here is the chain, as arithmetic.

Pick the size: one number, the radius of the sphere the dome is a slice of. This book's reference build has a radius of 116.4984 inches.

Two fixed factors turn that radius into the two chord lengths every geodesic reference publishes: 72.000 inches and 63.670. Chapter 16 explains why you do not cut either of them: the pinwheel joint bites 1.818 inches off a long member and 0.990 off a short one, and that bite is measured, never guessed. So the saw is set to 70.182 and 62.681.

Then the order falls out:

```
long members           60, each 70.182 in
short members          60, each 62.681 in
members in all         120
timber in the frame    664 ft
log sections           15, split 8 ways each
```

Change the radius and every line of that table recomputes. That is what it means for a house to have a parts list: the size is a decision, and everything after it is consequence.

Pick the radius; the cut list and the order fall out.

The flat rate, proved

This is the number that turns a weekend project into a product, so prove it rather than repeat it.

Scale the dome from 10 feet across to 30 and count again:

```
10 ft dome       79 sq ft of floor
30 ft dome      707 sq ft of floor
members, either way            120
panels, either way              40
joints, either way              65
```

The floor grows 9 times over, because area grows with the square of the size. The parts list does not grow at all. The sticks get longer; the list of things you do to them stays exactly the same length – the same splits, the same two saw settings, the same panels on the same board, the same raising.

Labour is priced by operations, not by square feet. That is why the labour in a dome is close to a flat rate, and why the bigger house is the better deal. The honest limit is the stick: a member is a split log, and a log only comes so long, which is why Chapter 11 picks the size from the stick rather than from the floor plan.

Scale the dome and count what changes.

One skeleton, four buildings

The same frame is already four products in the software that wrote this book: a dome home, a storage shed, a greenhouse and a storm shelter. Four buildings at four price points, and the same triangles under every one of them.

That is the pattern the rest of the book keeps finding. The skeleton is the constant and everything that makes it a particular building – the skin, the floor, the openings, the services – is a choice made on top of it. Chapter 54 takes that all the way, to 15 structures from one body.

Four buildings, one skeleton.

CHAPTER 5 · STORY

Why I Built the Ugly One

The beautiful maths needed an ugly test: salvage, a folded bracket, and nothing landing where it should

Everything in the last chapter assumes care. Measured deductions, a jig proved before it is trusted, a cut list worked out to the thousandth of an inch.

The first dome that taught me anything was the opposite experiment, and I built it on purpose.

I took whatever the chainsaw had made that week. I skipped the compound cuts entirely. Where two members met at a seam I folded a flat piece of steel into a bracket instead of mitring anything. I let nothing land where it was supposed to, and then I went to see what the geometry would forgive.

Take whatever the chainsaw made, and see what the geometry forgives.

Here is why that was the right test.

A careful hubless dome replaces the hub with **precision**. Every strut is cut exactly so that it lands exactly where five or six others land, and the hub that would have hidden the difference is no longer there to hide it.

The ugly one replaced the hub with **tolerance**. Nothing landed exactly, and the triangles absorbed it, the way Chapter 1 describes.

That difference is the whole experiment. If triangulation only works for careful people with good timber and a workshop, it is a luxury: a hobby for people who already own the tools. If it works for salvage and a folded bracket, it is a way of housing people. The ugly one stood. So the rest of this book is about how to do on purpose, and well, what that frame did by accident.

Who is behind this

For anyone wondering who is behind this unreasonable quantity of triangles: a Navy veteran, a programmer and an avionics bench technician, now using the G.I. Bill to study toward aerospace engineering. The longer version is at the back of the book. The short version is that a dome, to me, is a home written the way I would write software – modular, reusable, built from parts with clean interfaces so one can change without disturbing the rest – and this book is that idea in timber.

Build one. Then build the next one faster.

So that is the argument, before any of it is proved.

A shape that encloses the most room for the least material, and so spends less on skin and gives it back every winter. Two lengths of wood and one flat board. A method that forgave a chainsaw. A price with a receipt on every line, and a parts list that does not grow when the house does.

Every number in this book came out of code you can run, and the parts that are borrowed are named out loud. A sphere encloses the most room with the least material whether you are a developer or a widow. The geometry does not check your credit.

Build one. Then build the next one faster.

The geometry does not check your credit.

PART 2

The Geometry, From Scratch

“One irrational number places twelve points in perfectly even space. Everything else in this book is consequences.”

The whole derivation, on screen: why triangles, why an icosahedron, where the golden ratio really sits, and the two lengths that come out of it.

CHAPTER 6 · EXPLAIN

Why It Is Triangles

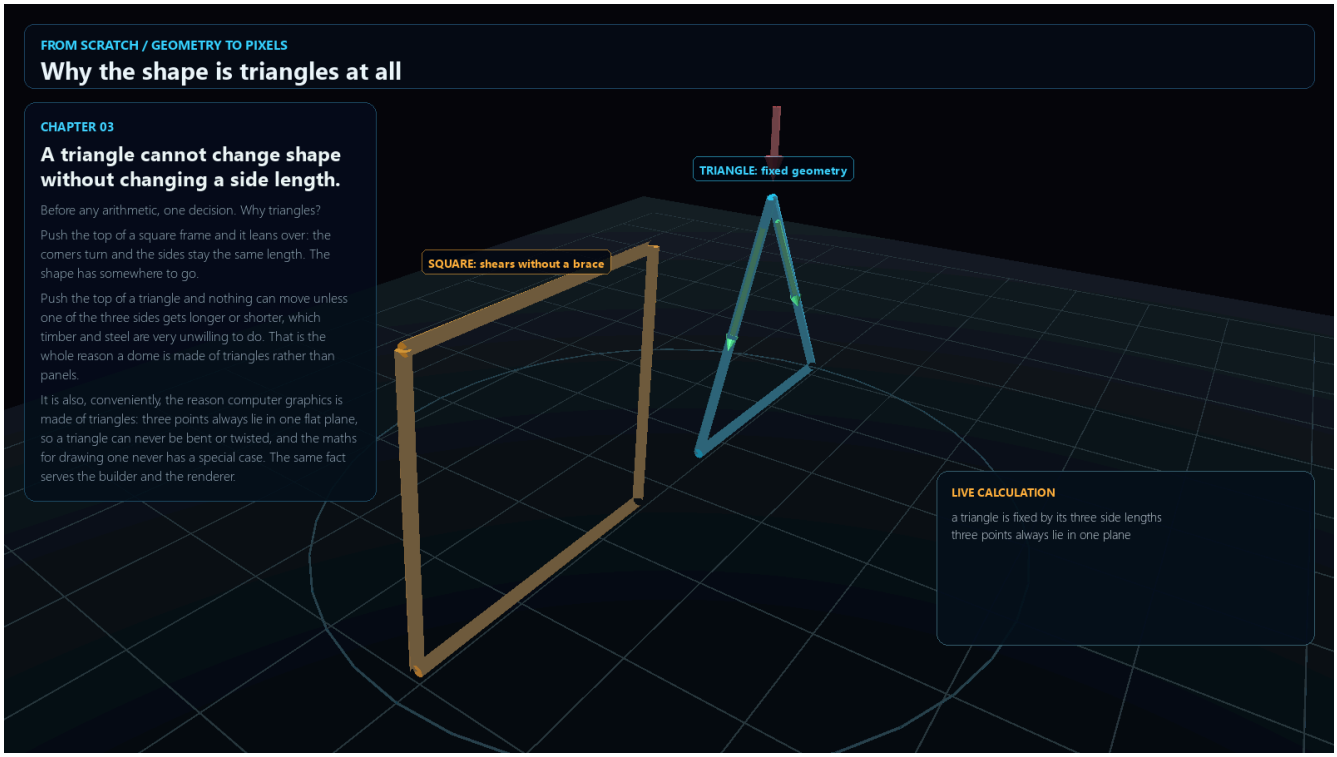
Every other polygon is a mechanism. A triangle is the only shape that cannot change without something breaking

Take four sticks and pin them at the corners into a square. Push one corner.

It folds. You now have a rhombus, and you can keep pushing until it is flat, and at no point did anything break or bend or come loose. A four-bar linkage is a **mechanism**. It is a machine for changing shape, and you have built one by accident.

Take three sticks and pin them at the corners. Push one corner.

Nothing happens. Nothing *can* happen — the only way that triangle changes shape is if a stick changes length or a pin fails. There is no motion available to it.



Why the shape is triangles at all.

That is the entire structural argument

A square frame needs a rigid corner, because the corner is what stops it folding. Rigid corners are welds, gussets, plates, moment connections — and they are where every framed building spends its engineering.

A triangulated frame does not. The geometry is holding the shape, so the corner only has to stop the sticks sliding apart. A pin will do. Two flat faces pressed together will do, which is Chapter 20 and is the whole reason this method can use a joint you could make with a handsaw.

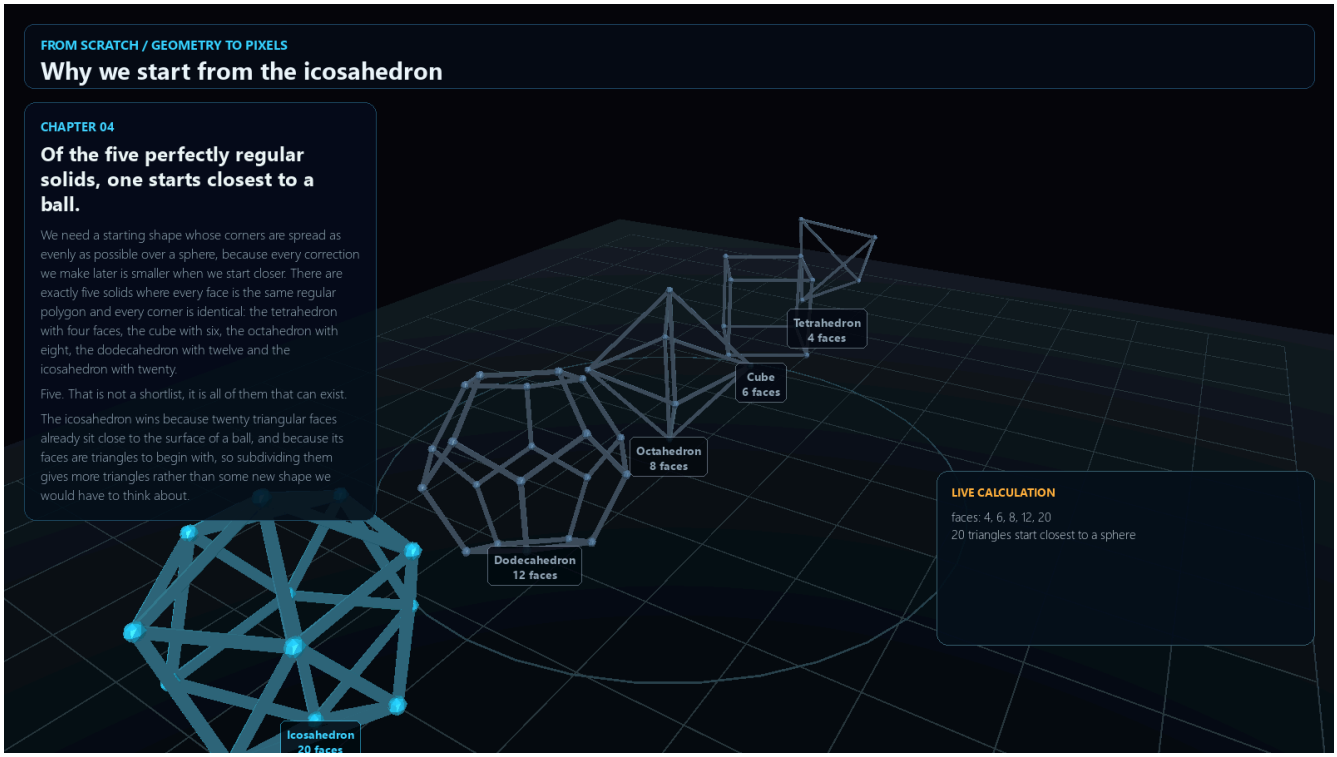
And it is why the frame can be lumpy

Chapter 1 claimed a dome forgives error. This is why.

A mechanism amplifies error — push a linkage slightly out and it moves a lot. A triangle cannot move at all, so the error has nowhere to go and nothing to grow into. A triangle whose sides are each a shade off is simply a slightly different triangle, sitting a shade differently in the shell, and its neighbours absorb the difference at the seam.

The shape is doing the work. That is the sentence this whole book keeps returning to.

And why an icosahedron



The solid everything starts from.

If triangles are the unit, the question becomes: how do you wrap a ball in them?

The answer has been known since Plato. There are exactly five solids whose faces are all the same regular polygon meeting the same way at every corner, and of those, three are made of triangles: the tetrahedron with 4 faces, the octahedron with 8, and the icosahedron with **20**.

Twenty is the most triangles you can get with every one identical and every corner the same. So it is the roundest starting point available, and starting rounder means less distortion when you subdivide.

Everything after this is subdivision. Cut each of those twenty faces into smaller triangles, push the new points out onto the sphere, and you have a geodesic dome. Cut each edge once and you have the 2V of this book. Cut it twice and you have a 3V.

But before any of that, you have to *place* the twenty faces — which means placing twelve points in perfectly even space, which is the next chapter and is the one piece of this derivation that looks like magic.

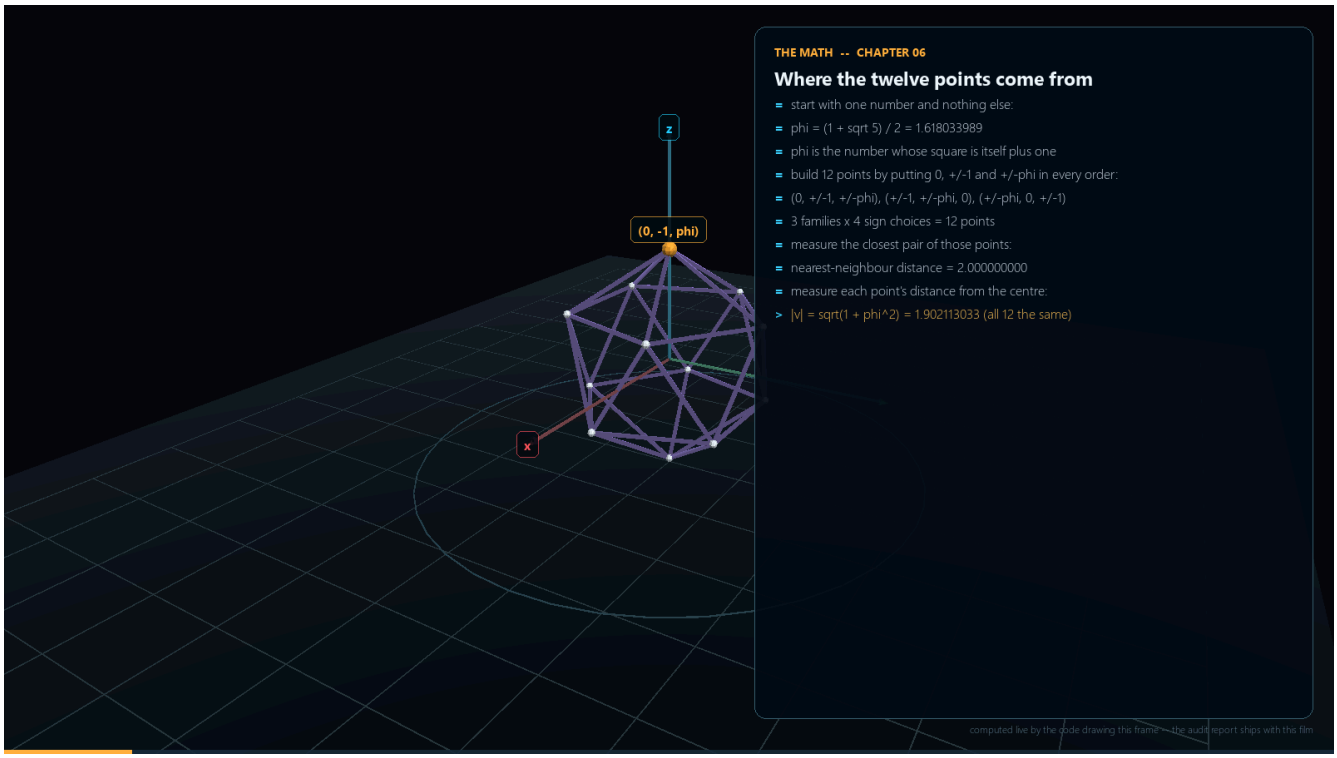
CHAPTER 7 · EXPLAIN

Twelve Points From One Number

Where the golden ratio actually sits in this method -- and the place everybody expects it and it is not

The golden ratio has a real job in this method and it is not the one people expect.

Here is the job.



Twelve points placed by one irrational number.

The number

φ — phi — is **1.618033989**.

It is defined by one property: it is the number whose square is itself plus one. $\varphi^2 = \varphi + 1$. Solve that and you get $(1 + \sqrt{5}) / 2$, which is where the decimal comes from.

That is the whole definition. No rectangles, no seashells, no Parthenon.

What it does here

Write down three numbers: 0, 1 and φ.

Now make points by putting them in every order with every combination of signs:

`(0, ±1, ±φ)`
`(±1, ±φ, 0)`
`(±φ, 0, ±1)`

Three families, four sign choices each. **Twelve points.**

Measure the distance from each point to its nearest neighbours: exactly 2.000, every time. Measure each point's distance from the origin: **1.902113**, every time, because it is $\sqrt{(1 + \varphi^2)}$ and every point has the same coordinates in a different order.

Twelve points, all the same distance from the centre, all the same distance from their neighbours. That is an icosahedron — the most even way to place twelve points on a sphere — and it fell out of writing one irrational number into a list of coordinates.

Nobody measured anything. Nobody iterated toward a solution. The evenness is a consequence of φ's one property, and this is the only place in the whole derivation where something arrives for free.

Why that property does it

Briefly, because it is worth knowing rather than accepting.

The icosahedron's twelve vertices can be seen as three golden rectangles standing in three perpendicular planes, interlocked. A golden rectangle is one whose sides are in the ratio $1 : \varphi$ — and the reason *that* ratio works is $\varphi^2 = \varphi + 1$, which is exactly the relationship needed for the three rectangles' corners to land the same distance apart.

So the defining property of φ is not decorative here. It is the thing being used.

Where it survives to

Now subdivide. Cut every edge of those twenty faces in half, push the new points out to the sphere, and measure what you get.

Two lengths, and only two — Chapter 9 proves there are no others. Expressed as multiples of the radius:

```
short   0.546533 R
long    0.618034 R
```

Look at the long one. **0.618034 is exactly 1/φ.**

0.618033989, to nine places, and it is not an approximation — it is $\varphi - 1$, which is the same number by φ's own defining property.

So the golden ratio places the twelve starting points, and then it comes all the way through the subdivision and the projection and lands in your cut list. When you crosscut a member at 72.000 inches, you are cutting the radius divided by the golden ratio.

And the place it is not

Here is the part that surprises everybody, including me.

The **ratio between the two struts is not golden.**

```
long / short = 1.130826
```

Not 1.618. Not 0.618. 1.130826, which is not φ, not 1/φ, not φ^2, and not any simple expression of φ at all.

People assume it is. It feels like it should be — the whole thing came out of φ, the long factor *is* 1/φ, so surely the two lengths are in a golden relationship. They are not, and a dome cut on that assumption is a dome with sixty wrong sticks in it.

The short factor 0.546533 is its own number. It comes out of the subdivision arithmetic and it does not simplify.

The lesson, which is about more than phi

A pattern that holds four times is not a rule.

φ places the points. φ gives the long chord. Both are exact and both are checkable. The third thing — the ratio — looks like it must follow and it does not, and the only reason this book knows that is that something computed it rather than assuming it.

That is why every number in these pages is derived rather than typed, and it is why the next chapter computes the same two factors four separate ways and prints the difference between them.

CHAPTER 8 · EXPLAIN

Halve Every Edge, Then Push It Out

One division puts the icosahedron on a unit sphere; halving its edges and pushing the midpoints back out is the whole of 2V

Chapter 7 placed twelve points in perfectly even space, and twenty triangles strung between them make an icosahedron. That is a ball, of a kind: a very faceted one, with flat faces the size of a door.

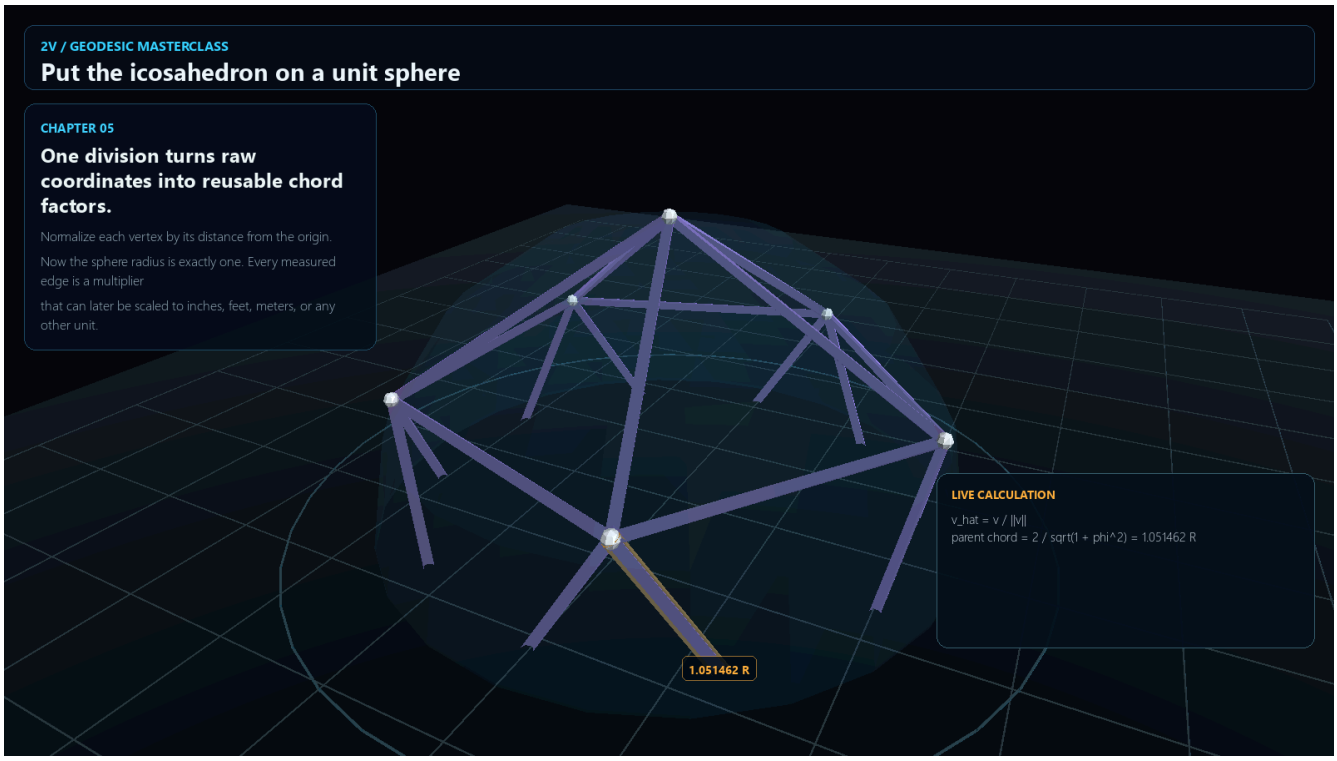
Two moves turn it into a geodesic dome. Halve every edge. Then push the new points out to the sphere. This chapter is those two moves, and the reason the second one is the whole idea.

Put it on a sphere of radius one

The twelve points come out of the golden ratio at an awkward size: every one of them sits 1.902113 units from the centre. Not one, and not anything you would choose.

So divide every point by its own distance from the centre. Each point keeps its direction and lands at exactly one. The icosahedron now sits on a sphere of radius one, and its edge, which measured two units, measures 1.051462.

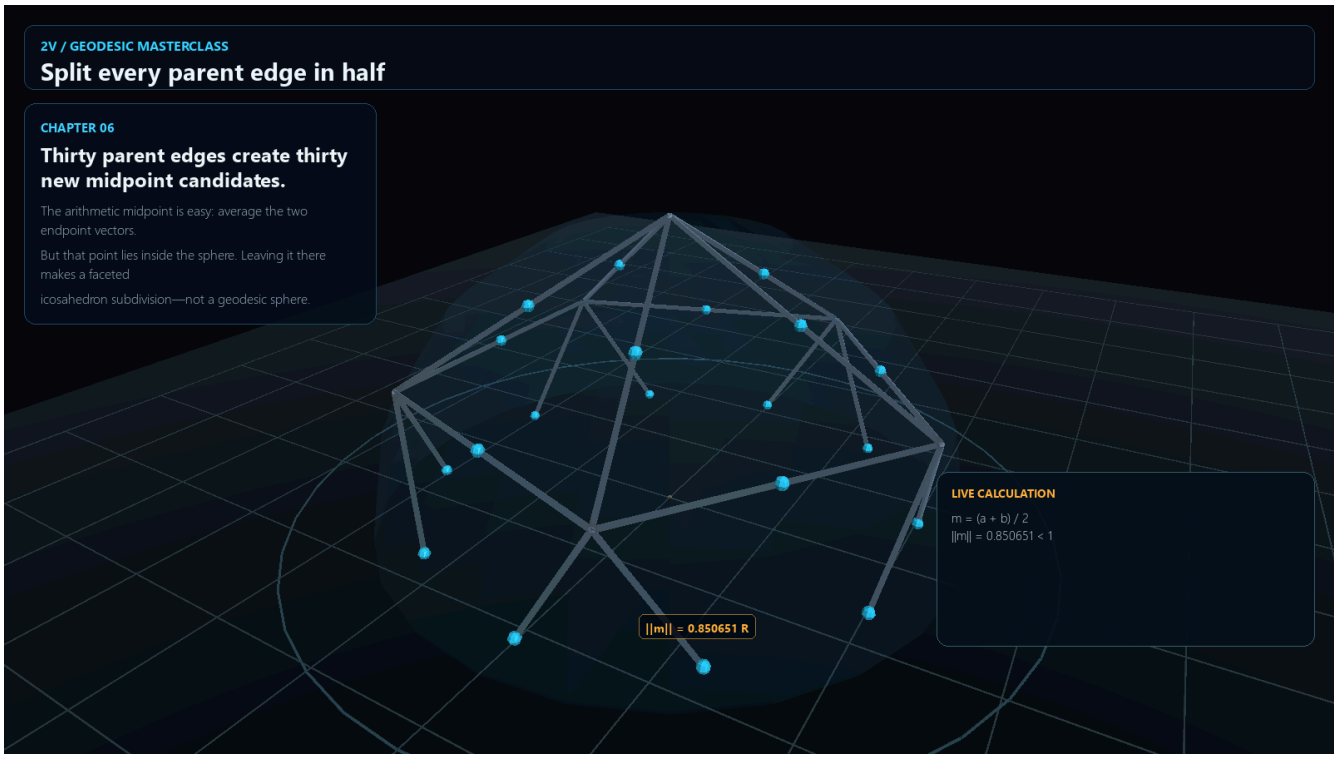
That number is not a length. It is a **chord factor**: a multiplier. Every distance in the model is now a multiple of a radius you have not chosen yet. Choose a radius in inches and multiply, and you have inches; choose one in metres and you have metres. The same numbers serve a playhouse and a hangar, which is why the size of this book's dome is decided last (Chapter 11) rather than first.



The icosahedron on a sphere of radius one.

Halve every edge

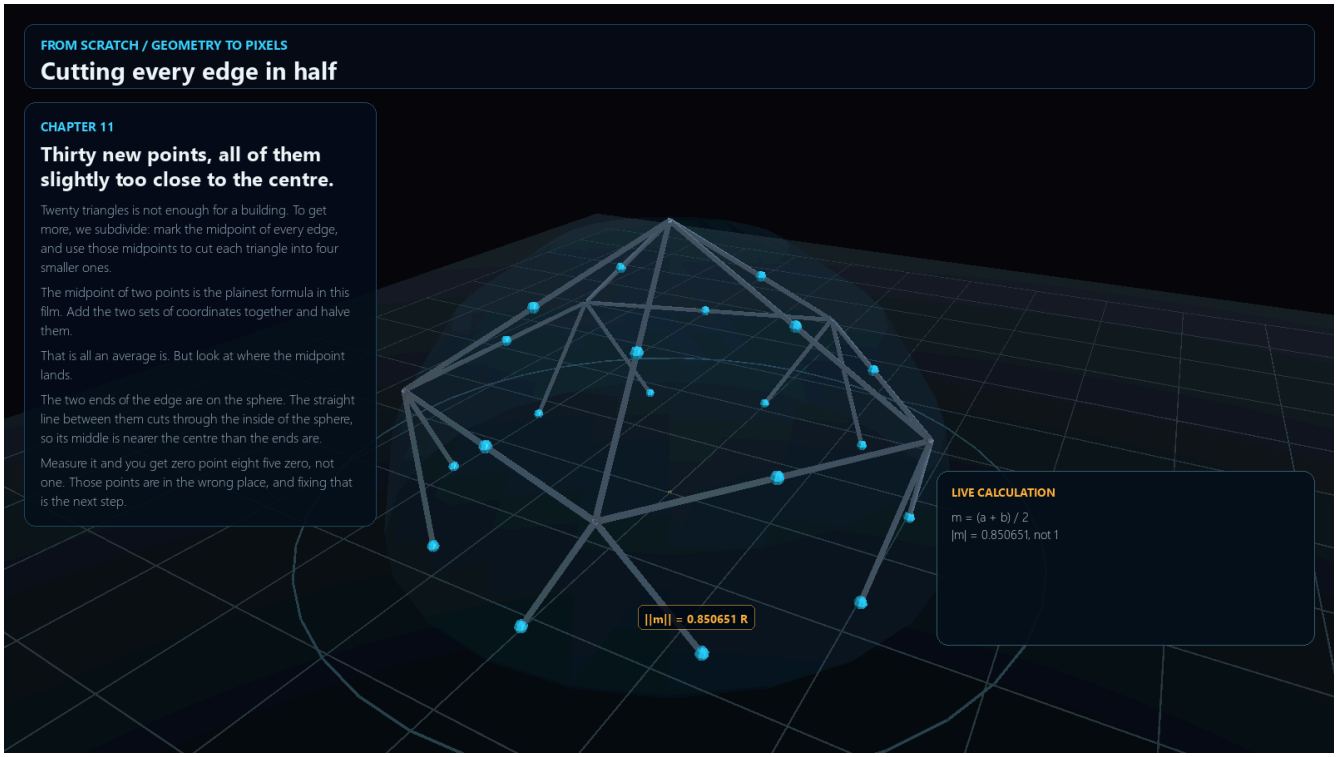
The icosahedron has 30 edges. Find the halfway point of each one – add the two end points and halve them – and you have 30 new points. Join them up and every one of the 20 faces is cut into four smaller triangles.



Thirty parent edges, thirty midpoints.

But look where the new points are. Both ends of an edge are on the sphere; the straight line between them cuts *inside* the ball, the way a chord cuts inside a circle. Every midpoint sits only 0.850651 of the way out: 14.9 percent of the radius short of the sphere. On this book's dome that is 17.4 inches.

Leave the midpoints there and you have an icosahedron with more triangles. Flat faces, just smaller ones. Not a dome.

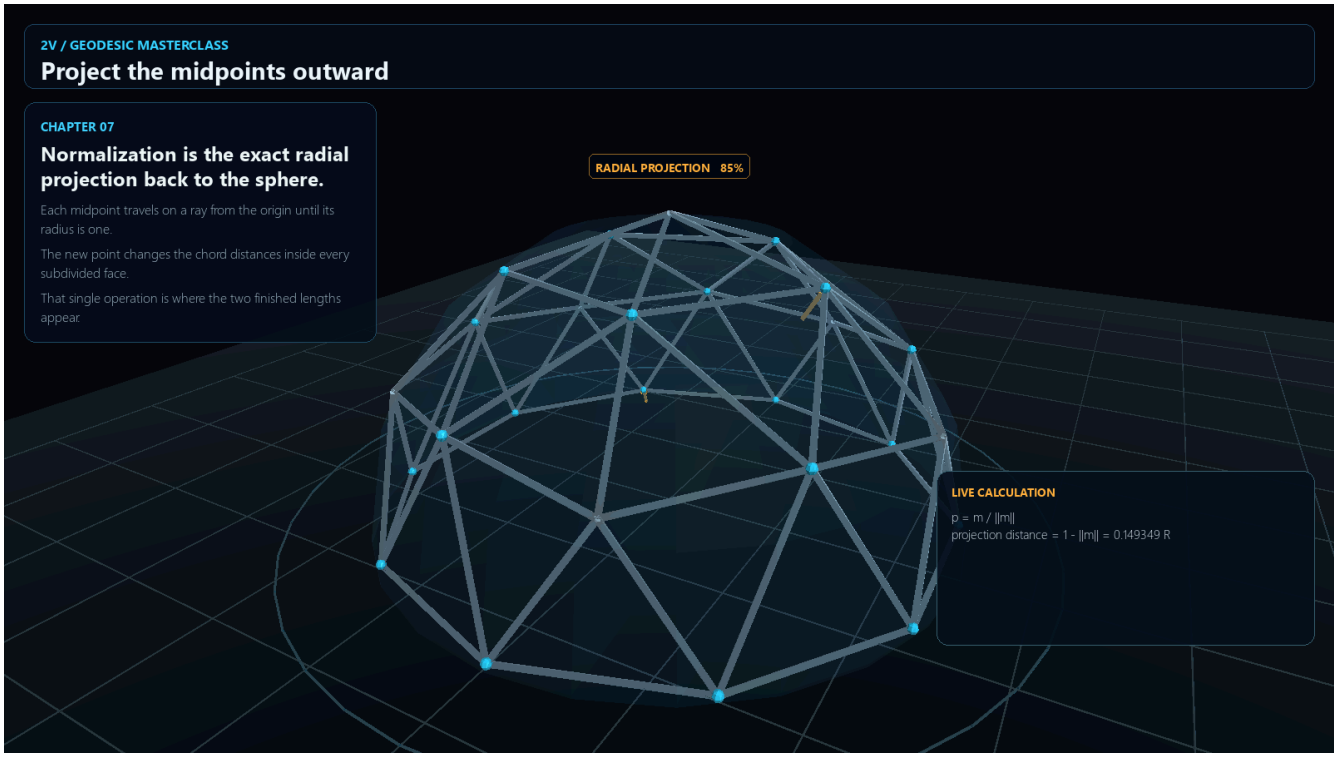


The straight line cuts the corner.

Push the midpoints out

So push every midpoint straight out from the centre until it reaches the sphere. And here is the elegant part: that is the same division as before. Divide the point by its own distance from the centre. Its direction does not change – it slides out along its own ray – and its distance becomes one.

Each midpoint travels 0.149349 of a radius outward. Each face keeps its layout and simply bulges.



Same direction, distance set to the radius.

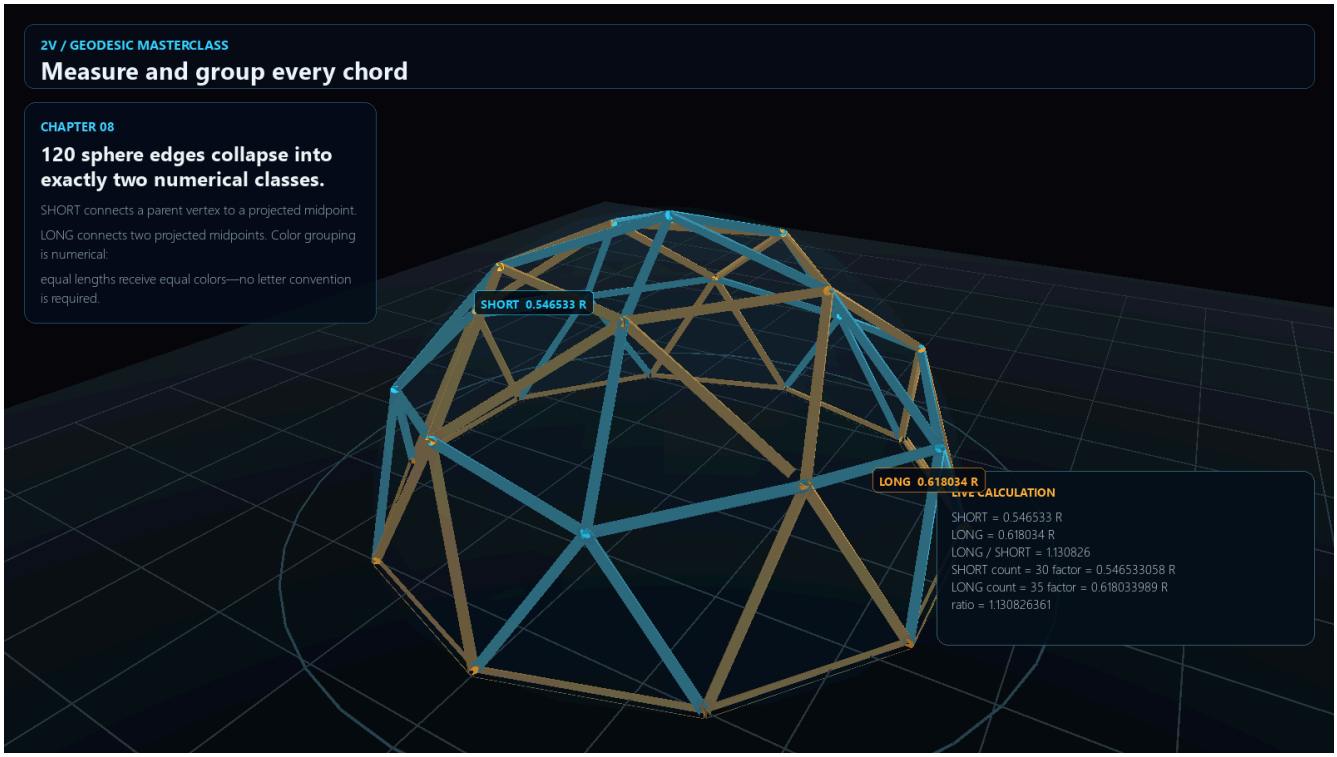
This one division is the entire difference between a faceted ball and a geodesic dome. And it does something the halving did not: it breaks the equal edges. Before the push, the four small triangles in each face were identical. After it, all three corners of the middle triangle have moved out, so it grows more than the three at the corners of the face, each of which has only two of its three points moved. The middle triangle comes out equilateral in the long length; each corner triangle keeps two short sides from its parent corner and gains one long one. The dome now has edges of two lengths.

Two lengths come out

Measure every edge of the whole sphere again: 120 of them. They collapse into exactly two numbers:

```
short edge    0.546533 x the radius
long edge     0.618034 x the radius
```

Those are the two chord factors of a 2V dome, and they are why this book cuts two lengths of member and only two. Chapter 9 derives them four independent ways and prints the difference between the answers. The long one is exactly one over the golden ratio; the short one is not golden at all (Chapter 7).



A hundred and twenty edges, exactly two lengths.

Euler's check

How do you know the shape in the computer really closed up, with no missing face and no doubled corner? One addition.

For any closed surface without holes, corners minus edges plus faces is two.

```
the icosahedron      12 - 30 + 20 = 2
after 2V             42 - 120 + 80 = 2
```

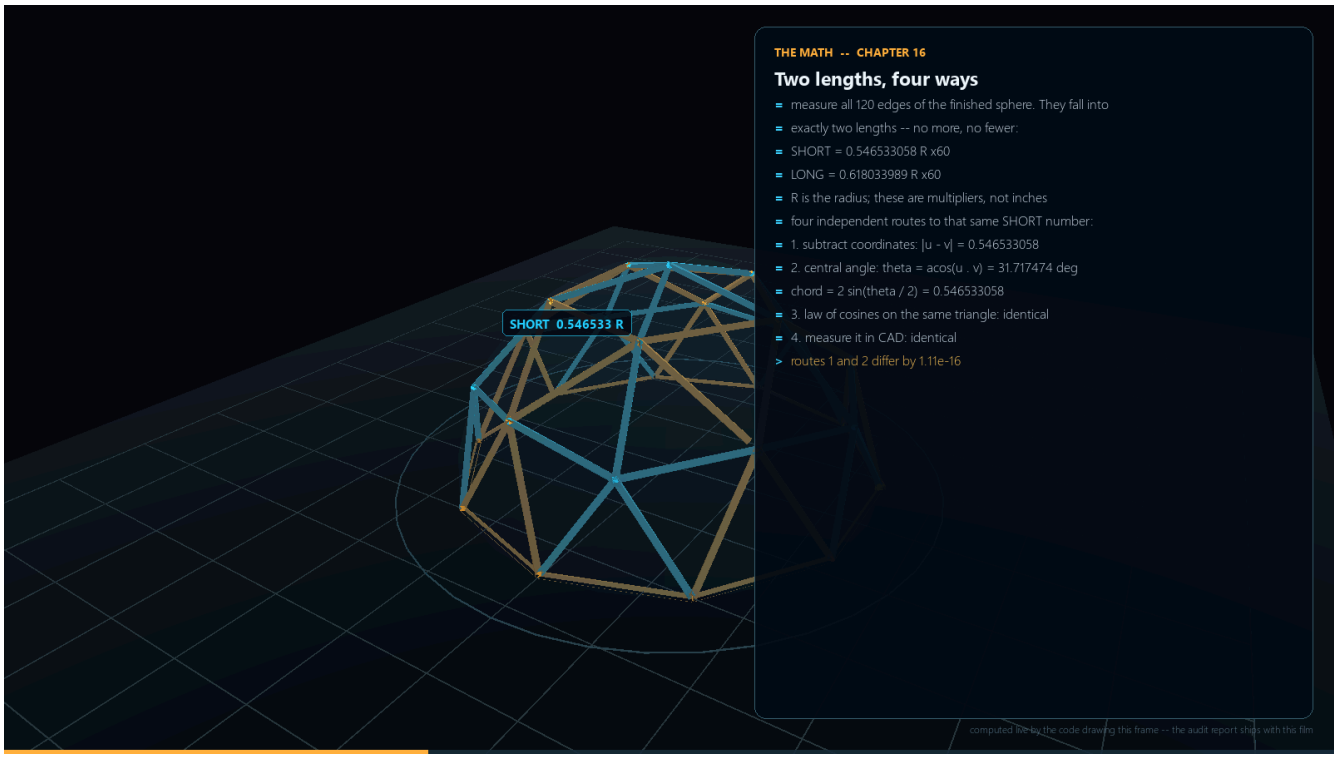
Anything other than two is a bug: a hole, a missing triangle, a corner counted twice. It is the cheapest proof in geometry, and the software that drew every picture in this book runs it.

CHAPTER 9 · EXPLAIN

Two Lengths, Four Ways

Four independent routes to the same chord factor, and the residual between them printed rather than promised

Never trust one calculation.



The two chord factors, derived four ways.

What is being checked

After subdivision and projection, the finished sphere has 120 edges. Measure all of them and they fall into exactly two lengths — no more, no fewer:

```
short   0.546533 R
long    0.618034 R
```

Those two numbers decide every stick you will ever cut. If either is wrong, every member is wrong, and the frame will not close — but it will not fail *obviously*. It will be a bit out, everywhere, and you will spend the assembly blaming your sawing.

So it is worth more than one derivation.

Route one: subtract the coordinates

The two points are known — they came out of Chapter 7's twelve, then a midpoint, then a projection. Subtract one from the other and take the length.

```
|u − v| = 0.546533
```

Direct, and it depends on nothing but arithmetic.

Route two: the central angle

Two points on a unit sphere subtend an angle at the centre. Take the dot product of the two unit vectors, take the arc cosine, and you have that angle. Then the chord across it is $2 \cdot \sin(\theta/2)$.

Completely different arithmetic — trigonometry rather than subtraction — and it uses the fact that the points are on a sphere, which route one never assumed.

Same answer.

Route three: the law of cosines

The same triangle, solved the way a surveyor would: two sides of length one and a known included angle.

Same answer again, by a third road.

Route four: measure it

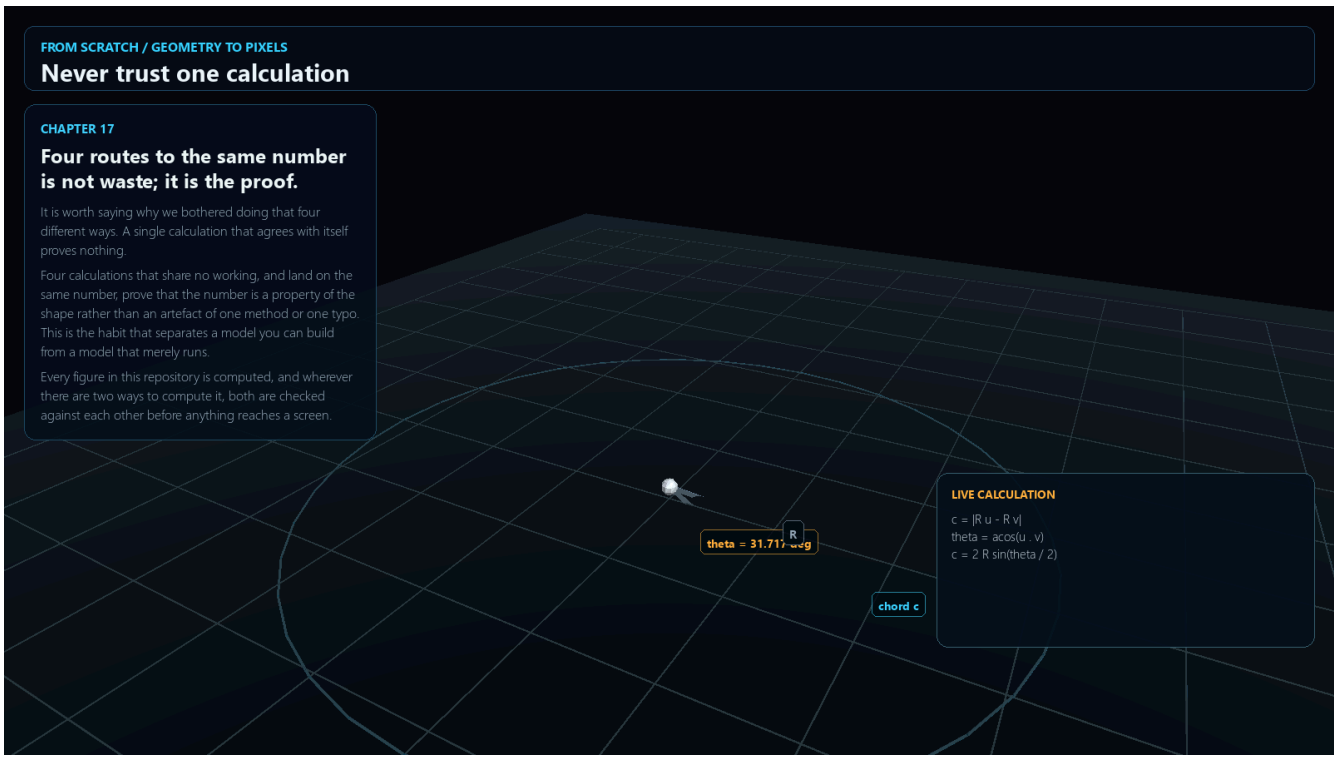
Build the thing in CAD and put a dimension on it.

This one matters more than it looks. The first three are all the same person's reasoning in three costumes; if the underlying model is wrong they will agree with each other and all be wrong together. Measuring the drawn object catches an error in the *model* rather than in the algebra.

What the residual is

Routes one and two differ by about $\mathbf{1.1 \times 10^{-16}}$.

That is not "close". That is the gap between two different sequences of floating-point operations on numbers that are mathematically identical — the smallest disagreement a computer can express at this magnitude. It is the answer you get when two methods agree completely.



Two ways of computing the same number, and the residual between them.

Why the book does this everywhere

This is the pattern, not a one-off.

Where two ways of computing something exist, this project computes both and prints the difference. The strut table in Chapter 16 does it for the pinwheel's bite. The seam angles do it for the fold. And the audit described in Chapter 56 does it for the whole book, checking every printed figure against what the code can produce.

The reason is not rigour for its own sake. It is that **a wrong number in a building method does not announce itself.** It becomes a pile of sticks that nearly fit, two weeks in, with the tree already down.

Four routes and a residual of 10^{-16} costs an afternoon once. Finding out the other way costs the tree.

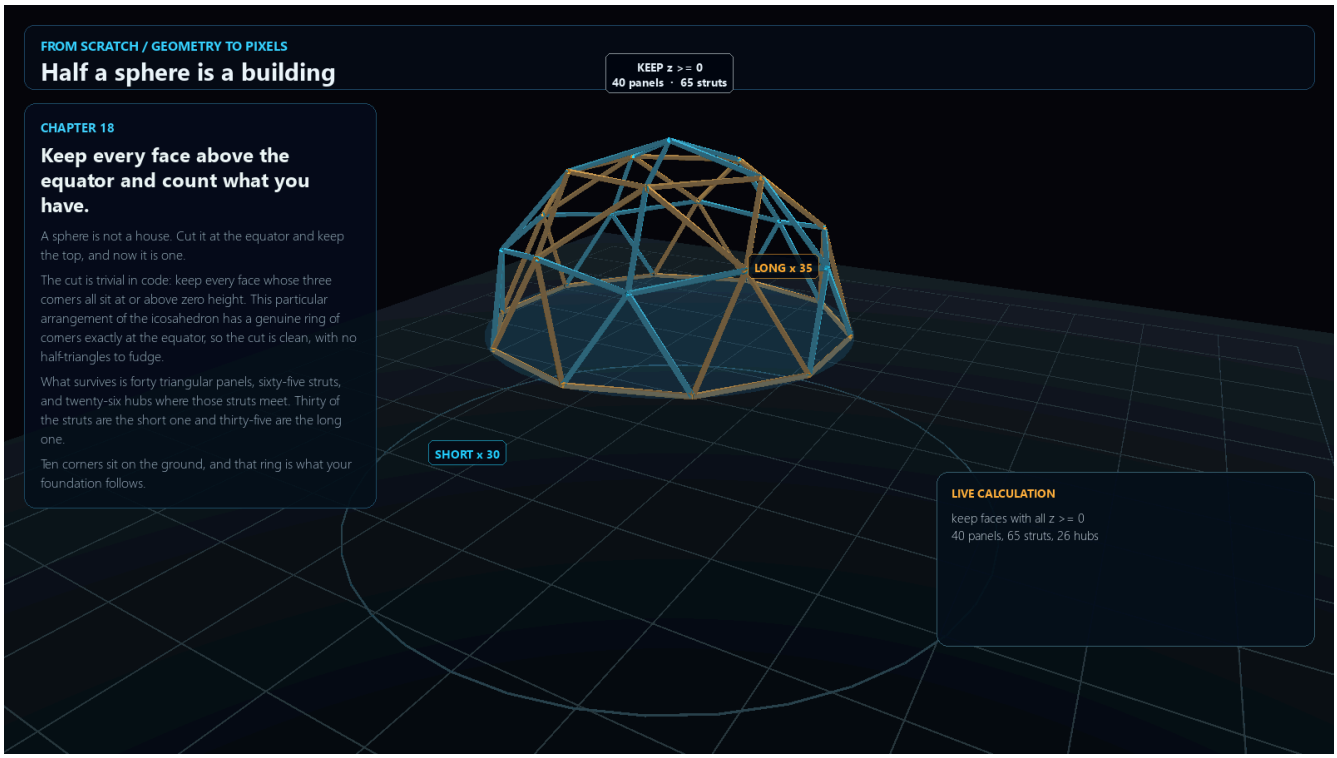
CHAPTER 10 · EXPLAIN

Half a Sphere Is a Building

Where the sphere gets cut, what survives the cut, and how every count in this book falls out of it

A sphere is not a building. It has no floor and no door and half of it is underground.

So you cut it, and where you cut it is a design decision with consequences.



Where a sphere gets cut to become a building.

Cutting at the equator

This book cuts at the equator: a hemisphere, exactly half.

The 2V geometry cooperates. There is a ring of vertices exactly on the equator, so the cut lands on existing points and the base comes out as a clean regular polygon — 10 sides, each one a long chord, sitting flat on the ground with nothing to trim.

That is not true of every frequency and it is not true of every cut. Take a 3V dome and the equator falls between vertex rings, so you either move the cut or accept a base that is not level. Take a 2V and cut it above or below the equator and you get a shallower or deeper shell, a smaller or larger floor, and a base ring that is no longer made of whole chords.

The equator is where the geometry is kindest, and this book takes the kindness.

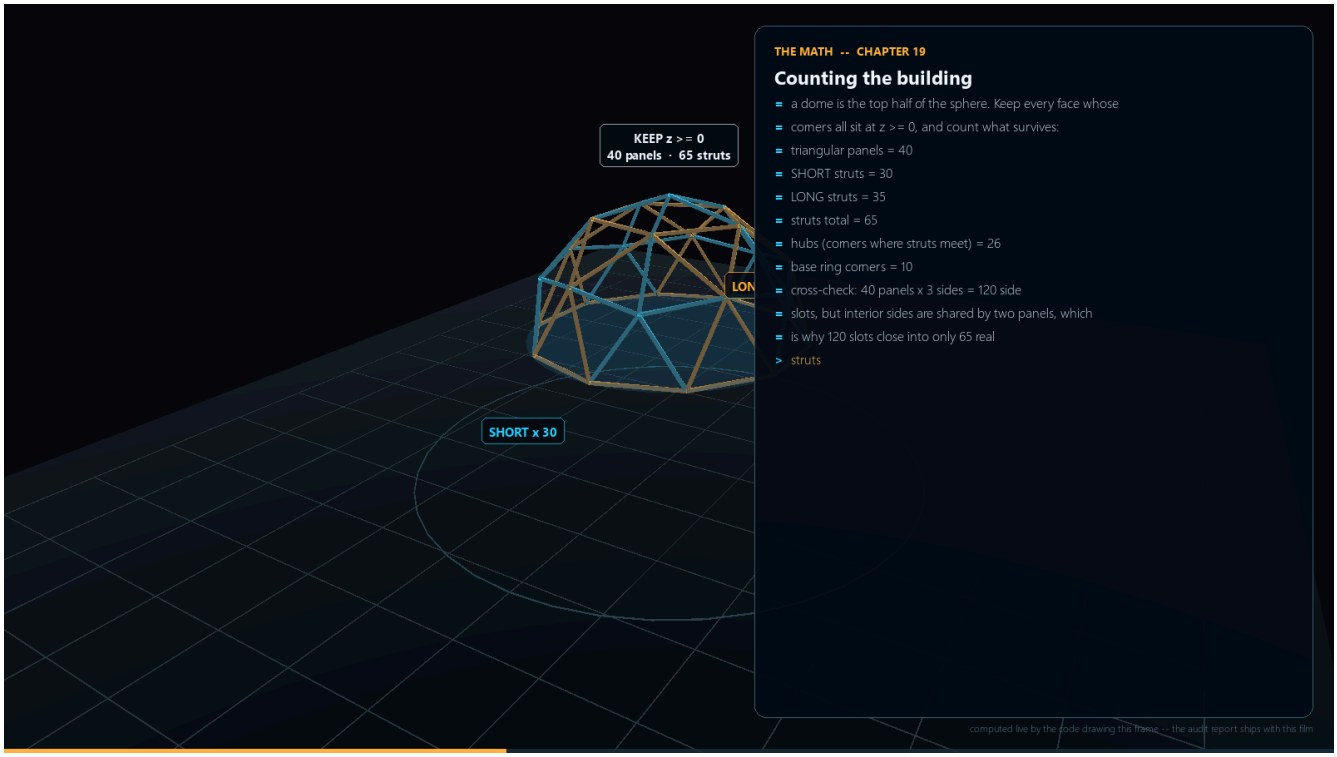
What the cut gives you

across the base	19.42 ft
to the apex	9.71 ft

```
floor inside the ring 277 sq ft
shell surface        550 sq ft
```

Apex height equals the radius, because a hemisphere's apex *is* the radius. That one identity saves more arithmetic than any other fact in this book.

Counting the building



Every part of the building, counted.

Everything else falls out of the topology rather than being counted by hand:

```
vertices             26
geodesic edges       65
triangular panels    40
  equilateral (AAA)  10
  isosceles   (BAB)  30
structural members   120
interior seams       55
base edges           10
```

Two of those are worth a second look.

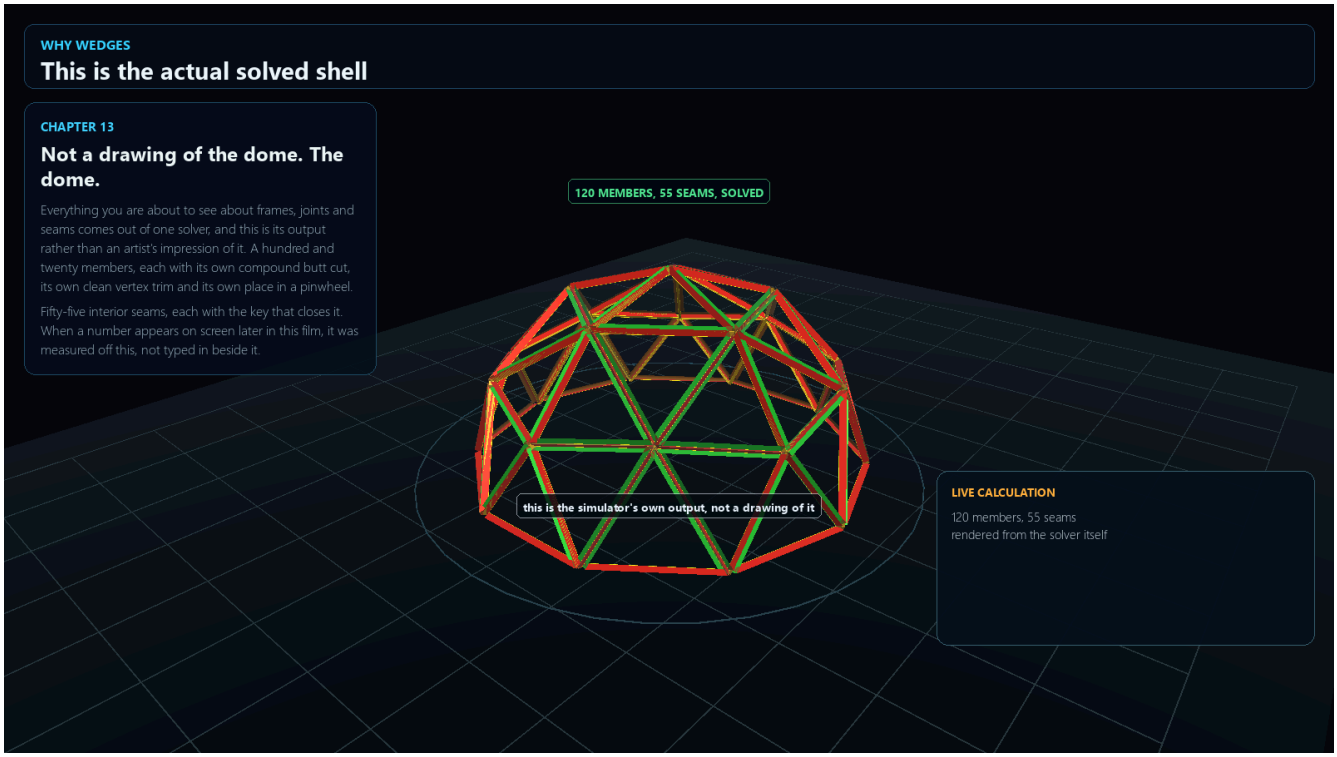
120 members for 65 edges. The duplication of Chapter 20: each panel brings its own three sticks, so every interior edge has two.

55 interior seams, not 65. The 10 base edges are the rim and have nothing on the other side of them, so they are not seams. That distinction matters the moment you start counting keys, or hose, or feet of gasket — and it is the kind of thing that is obvious once stated and silently wrong in a bill of materials otherwise.

This is the actual solved shell

Every picture of the frame in this book is the output of one solver, not an artist's impression of it. It takes the hemisphere above and builds each of the 120 members as a real split-log wedge: its own compound butt cut, its own trim at the vertex, its own place in a pinwheel. It closes the 55 interior seams, each

with the key that fills it. When a number about a joint or a seam appears later in this book, it was measured off that model, not typed in beside it.



Not a drawing of the dome. The dome.

What actually moves when the log changes

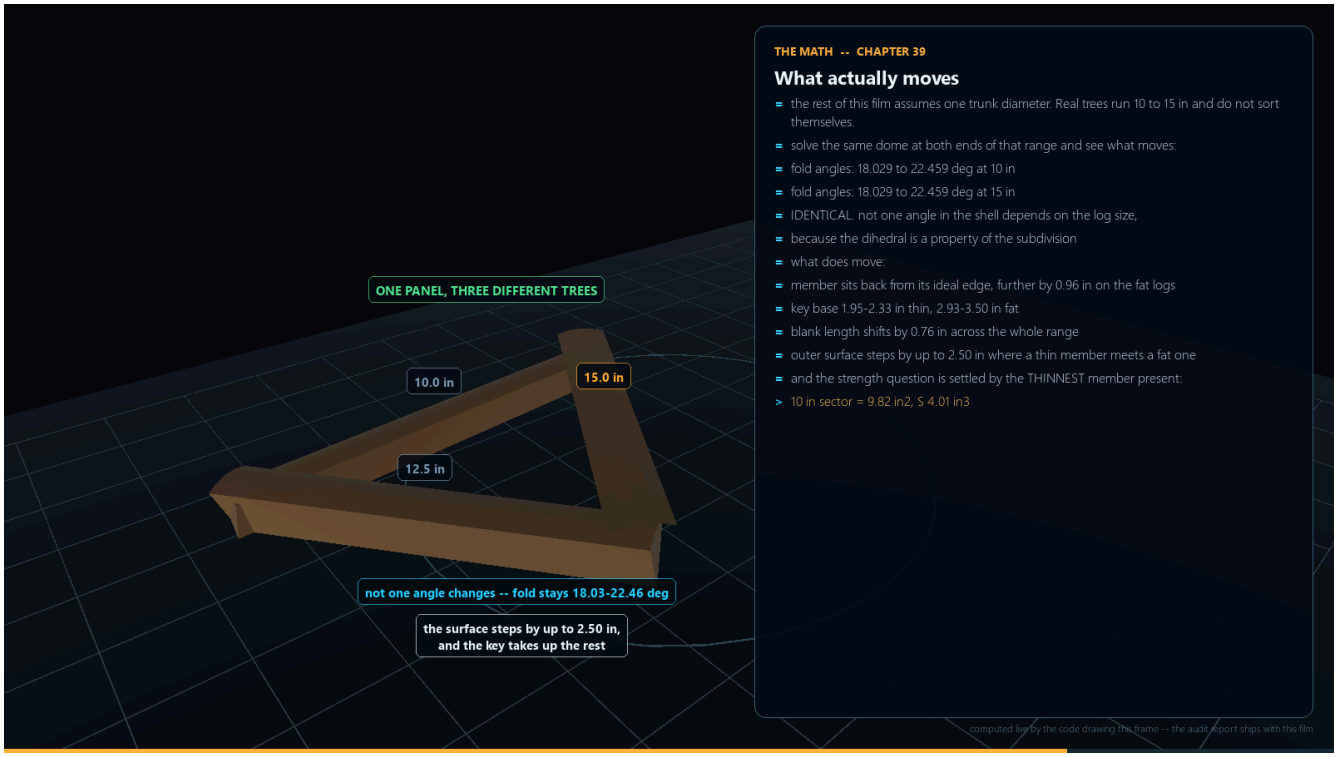
The rest of this book assumes one log diameter. Real trees do not sort themselves; a stand of pine gives you a spread of trunks. So solve the same dome for a thin log and a fat one, 10 inches and 15, and see what changes.

```
fold angles, 10 in log     18.029 and 22.459 deg
fold angles, 15 in log     18.029 and 22.459 deg
key base, 10 in log        1.95 to 2.33 in
key base, 15 in log        2.93 to 3.50 in
```

The fold angles are identical. Not one angle in the shell depends on the size of the log, because the angle between two panels is a property of the subdivision, not of the wood. The geodesic solution does not know what a log is.

What does move is the key, which gets wider as the log gets fatter, and the member's offset from its ideal edge. The key was already absorbing the spread between the dome's two fold angles, so this is one more job it was already shaped for.

And the strength question is settled by the thinnest member on the building, not the average one. Even at 10 inches, a wedge's weakest section modulus is 3.20 cubic inches, against 3.06 for a two-by-four on edge. Every log in that range makes a member stiffer than the stud it replaces.



The same shell solved at both ends of the range.

Why it is a decagon and not a circle

The floor figure above is the **decagon** inside the base ring, not the circle it sits in.

The circle is bigger — about seven per cent bigger — and the dome does not reach it. Quoting the circle is the single most common way a dome's floor area gets overstated, and it is overstated by exactly the amount of the ten little segments between the chords and the arc.

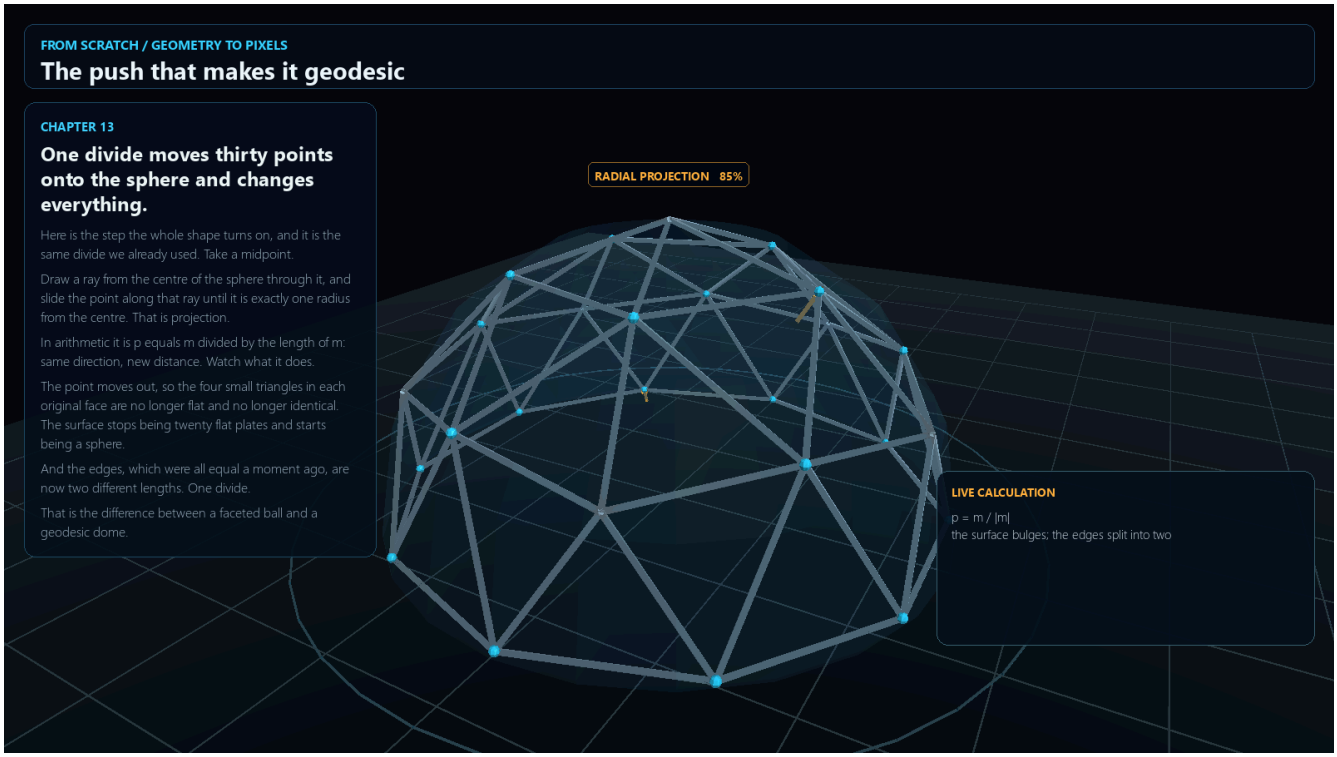
This book quotes the decagon everywhere. It is the honest number and it is the one you can put furniture on.

Choosing a Size, At Last

One number decides the whole building, and this book picks it from the stick rather than from the floor plan

Everything up to here has been ratios. Nothing has had a unit on it.

That is deliberate: the geometry is scale-free, so it is worth solving once and then choosing a size — rather than choosing a size and re-solving the geometry inside it, which is how most people do it and is why most people only ever build one dome.



The push that makes it geodesic.

One number does it

Pick any one of these and the rest follow:

- the radius
- the long member's chord
- the diameter
- the apex height (which is the radius)
- the floor area

They are all the same number wearing different units. Multiply the radius by 0.618034 and you have the long chord; by 0.546533 and you have the short one; everything else is those two and the counts from Chapter 10.

This book picks the stick

The reference build is sized from the **member**, not from the floor plan.

72.000 inches — six feet — because:

- it is what comes comfortably out of a twelve-foot log section, with the crosscut in the middle;
- it is what one person carries without thinking about it;
- it fits in a pickup bed, diagonally, with the tailgate up.

The dome that falls out of that is 19.42 feet across with 277 square feet of floor. **The diameter is the answer to the member, not the other way round.**

That is backwards from how buildings are normally sized and it is the right way round for a method whose whole argument is about what one person can make and move.

What it costs to size from the member

One real inconvenience, stated plainly.

No panel fits a four-foot sheet. The narrowest face is 52.516 inches across its shortest altitude and a sheet is 48. Every panel of the reference build needs a seam in its sheathing or a wider sheet.

That is a direct consequence of sizing from the stick, and anybody who would rather have the sheets than the six-foot member should build a different diameter. Chapter 16's table will give them one.

What a bigger radius buys

Pick the size for a reason, because the radius does not scale everything the same way. Double it and you get four times the floor, four times the skin you have to buy and keep watertight, and eight times the air you have to heat. Floor and skin grow with the square of the radius; volume with its cube.

And headroom at the wall is the thing people underestimate. At the very edge of a hemisphere there is none at all: the shell meets the floor at the rim. The first ring of a 19.42-foot dome is the band you sit against, not the band you stand in, and a bigger dome only pushes that band outward.

Audit the boards you have

A chord factor turns a radius into a length. It also runs backwards: a length you have already cut implies the radius it was cut for. That makes it a way to check boards you already own.

Say you have two members measured at 72.0 and 63.5 inches. Each one, on its own, implies a radius:

```
from the long board     72.0 / 0.618034 = 116.498 in
from the short board    63.5 / 0.546533 = 116.187 in
```

They disagree, because tapes and saws are not exact. The fair answer is the radius that misses both by the least – a least-squares fit:

```
best radius             116.362 in
long board misses by    +0.084 in
short board misses by   -0.096 in
```

Two lessons come with it. Keep the centre-to-centre geometry separate from the physical cut length: what you measure on a board is the cut, which is the chord less what the joint takes (Chapter 16). And the ratio of the two boards, 1.1339, should be close to the geometry's own 1.1308; if it is far off, one of the two lengths is wrong, and the fit will not tell you which.

If you would rather size from the floor

Work backwards.

Floor area for a decagon inscribed in a circle of radius R is $5 \cdot R^2 \cdot \sin(36°)$, so a target floor of A square feet wants

```
R  =  sqrt( A / (5 · sin 36°) )
```

and then the long chord is 0.618034 R and the cut is that minus 1.818 inches. A 400-square-foot floor wants a radius of about 11.7 feet and a long member a shade over seven feet — which is past comfortable one-person handling, which is the trade.

Five sizes worth knowing

The reference designs, from a demonstration you build to get it wrong on, up to the point where a 2V starts to strain. Each one is a full cut list in the tables at the back, and each one is the same 120 members and the same nine processes.

The only thing that changes between them is two numbers.

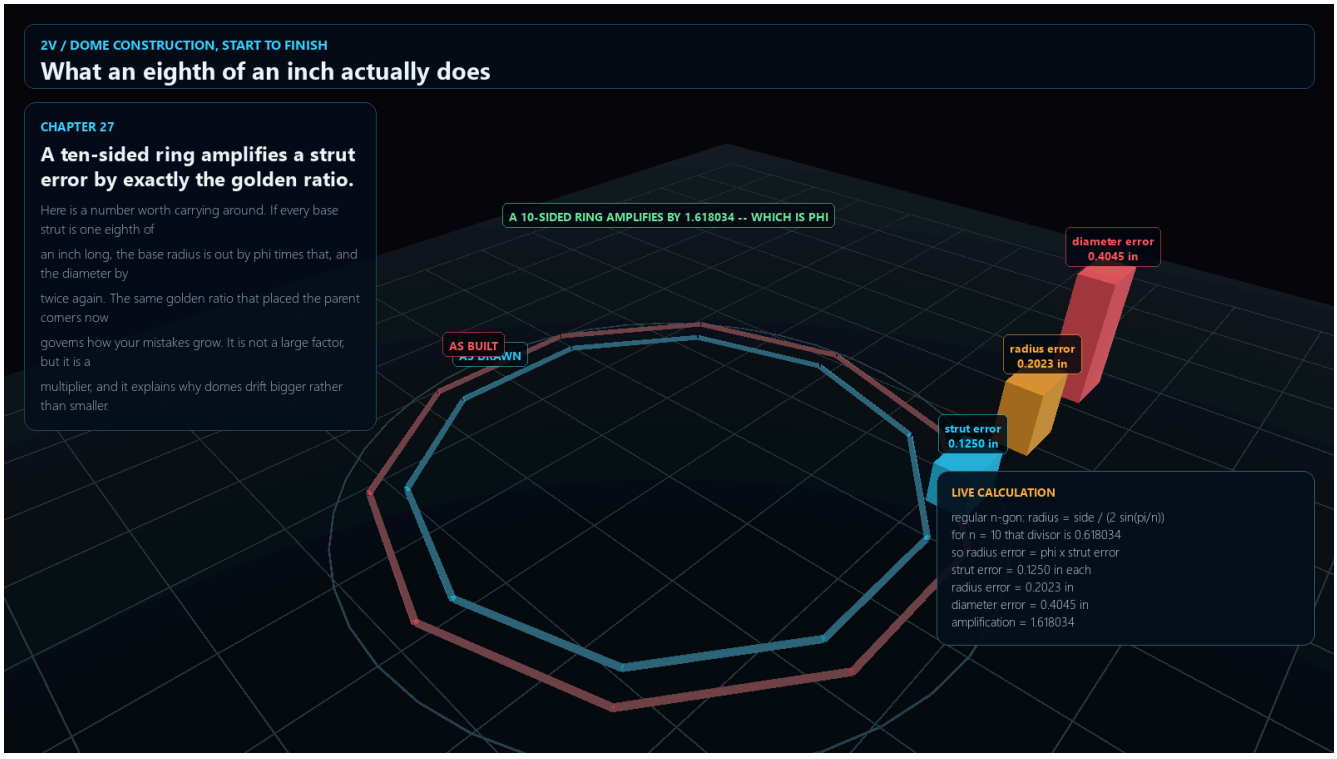
What an eighth of an inch does

Here is a number worth carrying to the site.

The base ring is a regular ten-sided polygon, and the radius of a regular polygon is its side divided by a fixed constant. For ten sides, that constant is exactly one over the golden ratio. So every error in a base member reaches the foundation multiplied by 1.618034:

```
each base member long by        0.125 in
the base radius grows by        0.202 in
the diameter by                 0.405 in
```

The same golden ratio that placed the twelve parent corners now governs how your mistakes grow. It is not a large factor, but it is a multiplier, and it is why a hand-built dome drifts bigger rather than smaller.



The base ring multiplies your mistakes by phi.

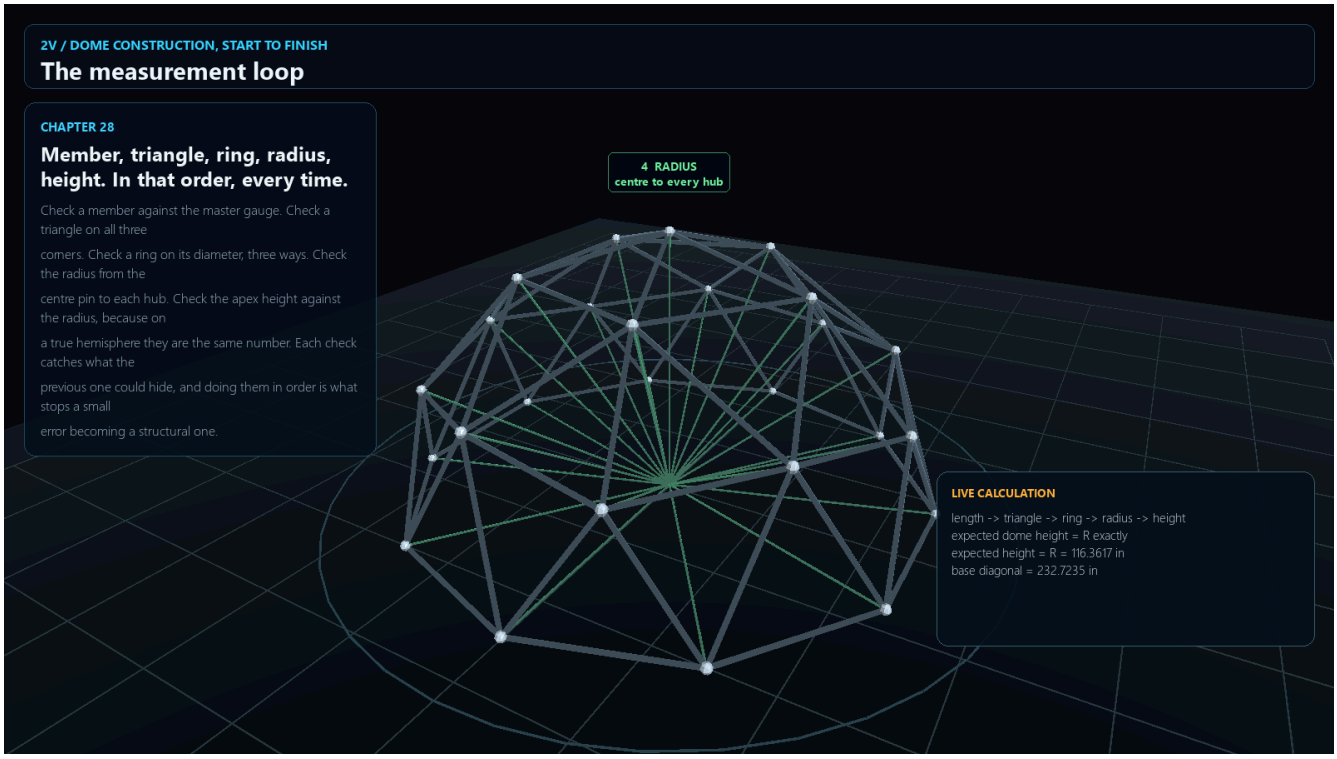
Close the measurement loop

So measure in a fixed order, every time, and do not go on until each check passes:

1. **The member**, against a master gauge stick.

2. **The triangle**, on all three corners, on the jig.
3. **The ring**, on its diameter, three ways across.
4. **The radius**, from a centre pin to every corner.
5. **The height**, from the floor to the apex: on a true hemisphere it is the radius, so the two numbers have to match.

Each check catches what the one before it could hide. A member can be right and a triangle wrong; a ring can be round and the wrong size. Doing them in order is what stops a small error becoming a structural one, and it is why the ring is checked before the next ring goes up. For anything you will live in or load, your local code and an engineer have the last word.



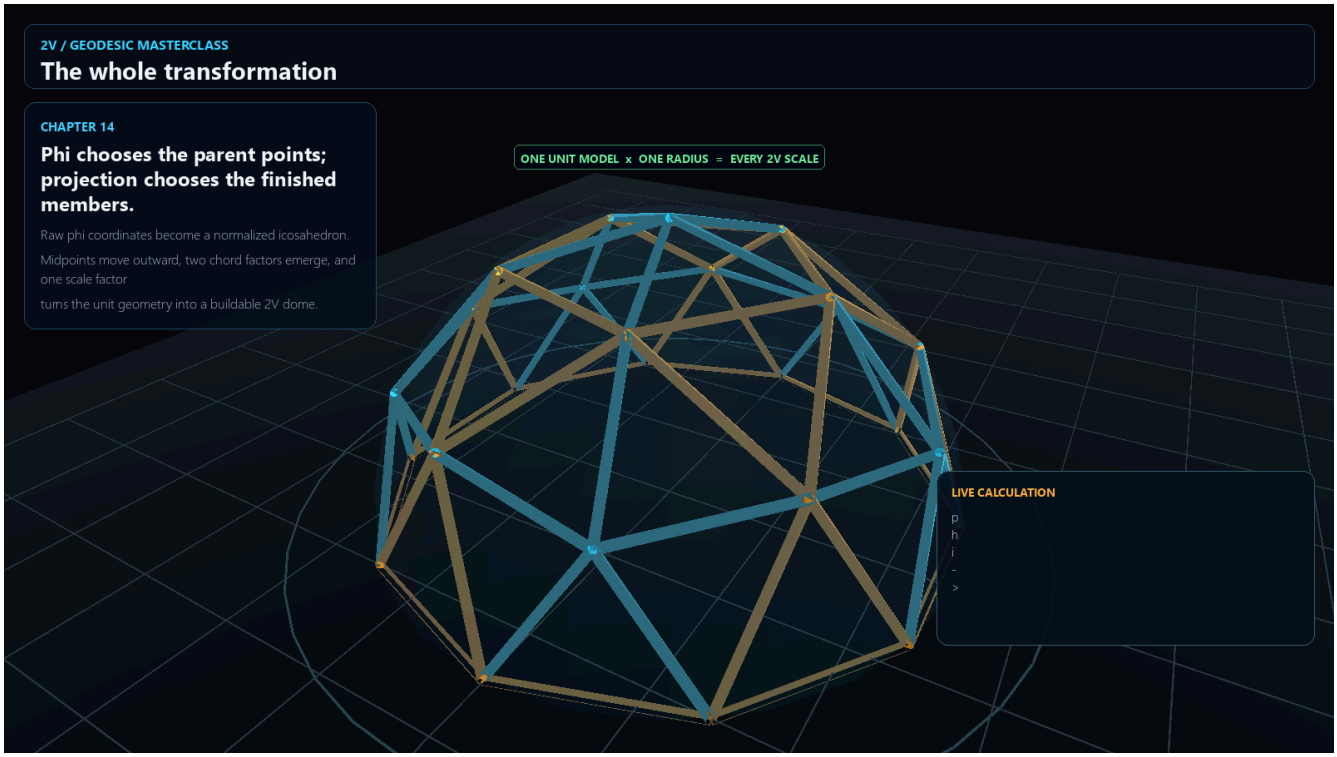
Member, triangle, ring, radius, height.

The whole transformation

That is the whole derivation, and it is short enough to say in one breath.

The golden ratio places twelve corners. One division puts them on a sphere of radius one. Halving every edge and pushing the midpoints back out leaves exactly two edge lengths, as multipliers. One radius turns those into two chords; the joint's bite turns the chords into two saw settings; one flat board turns the settings into 40 identical panels; the rings turn panels into a shell; and a skin turns the shell into a room.

Every number on the way came from the geometry, and every one of them can be recomputed rather than trusted.



From the golden ratio to a building you can stand inside.

PART 3

The Stick

“Every radius of a round log is the same as every other. That is the whole trick, and the tree held the tolerance.”

How a standing tree becomes 120 structural members, what a 45-degree sector is, the two lengths you actually cut, what drying does to them, and the one face that gets a finish.

CHAPTER 12 · STORY

The Woodpile Is Standing Up

What one tree holds, how it comes down, and why the trees decide the size of the building rather than the other way round

This is a house frame before it is cut.

Not a pile of lumber from a yard, and not a drawing. A tree, standing on the land you are going to build on, already the right shape for something. The question this Part asks is what happens if the tree gets a vote in the design: if instead of forcing it into rectangles, you ask what it is already good at and build around that.

The answer, it turns out, is a frame. Split a round log through its heart and every piece is a structural member, and nothing but the width of the saw blade leaves the log. The frame is already on your land. It is standing up.

A whole house frame, before it is cut.

What is standing there

Before anything is cut, measure what you have. A trunk is a cone with its tip cut off, so its wood is a formula in its two end diameters and its length. This is the tree the films that come with this book work through – one pine, 15 inches at the butt, 5.5 where it stops being useful, 60 feet between:

```
one pine, measured before anything is cut:
  15.0 in across at the bottom of the usable length
  5.5 in across where it stops being useful
  60 ft of usable trunk between them
a trunk is a cone with its tip cut off, so its volume is
  V = pi L (r1^2 + r1 r2 + r2^2) / 3
  r1 and r2 are the two end radii, L the length
```

```
  one board foot is 144 cubic inches, so divide by 144:
  solid wood = 442.1 board feet
buck it into 5 lengths of 12 ft:
  section 1: 15.0 -> 13.1 in, 155.3 bf
  section 2: 13.1 -> 11.2 in, 116.2 bf
  section 3: 11.2 -> 9.3 in, 82.8 bf
  section 4: 9.3 -> 7.4 in, 55.0 bf
  section 5: 7.4 -> 5.5 in, 32.9 bf
```

That is the starting balance. Every way of turning the tree into a building, in this chapter and the next, is measured against it.

The fell line

Felling gets its own section because nothing after it is safe until it is right, and because the part of it that does the work is not the saw.

1. **Cut the notch** on the side you want the tree to fall toward: a wedge of wood taken out of the trunk, about a quarter of the way in.
2. **Cut the back cut** from the other side, a little higher than the floor of the notch.
3. **Stop before the cuts meet.** The strip of wood left between them is the **hinge**. It steers the tree all the way to the ground; cut through it and the tree goes wherever it likes.

The hinge steers the tree. The saw only sets it free.

That is the whole technique in a paragraph, and it is not enough to learn it from. Fell your first trees with somebody who has done it, wear the gear, and clear an escape path at an angle behind you before the first cut. This book's saw is an ordinary hardware-store chainsaw; the skill is not in the machine.

The hinge steers the tree.

All at once

Then everything happens at once. The crown comes off. The trunk is bucked into sections the length of the longest member you need. And every section splits the same way: in half, in half again, and once more, into 8 sectors. One tree's whole contribution to the frame comes off the saw in the same few operations, repeated, and the arithmetic of it is short:

```
one trunk: 15.0 in at the butt, 5.5 in at the top, 60 ft usable
bucked into 5 sections of 12 ft
solid wood in those sections = 442.1 board feet
SPLIT RADIALLY:
  8 sectors per section, 45 deg each = 40 full-length sectors from one tree
  the saw passes remove 51.2 bf of kerf
  wood kept = 442.1 - 51.2 = 390.9 bf = 88.4 %
  measured across whole trees on site: 88 %
SAWN INTO TRUE TWO-BY-FOURS, same sections:
  rectangles that grid-pack inside the round = 25 pieces
  25 x 8 bf = 200.0 bf = 45.2 %
  square milling in practice recovers 50-65 %, losing wood to slabs, edgings, trim, drying degrade and planer shavings
crosscut once, one tree gives 80 members; the dome needs 120 and 2 trees give 160
```

The honest cost is the stick itself, not the wood: from a small log, a single wedge is the weaker member in bending, and it is the triangulated shell, not the wedge, that makes it enough. Chapter 15 puts numbers on exactly where that stops being true.

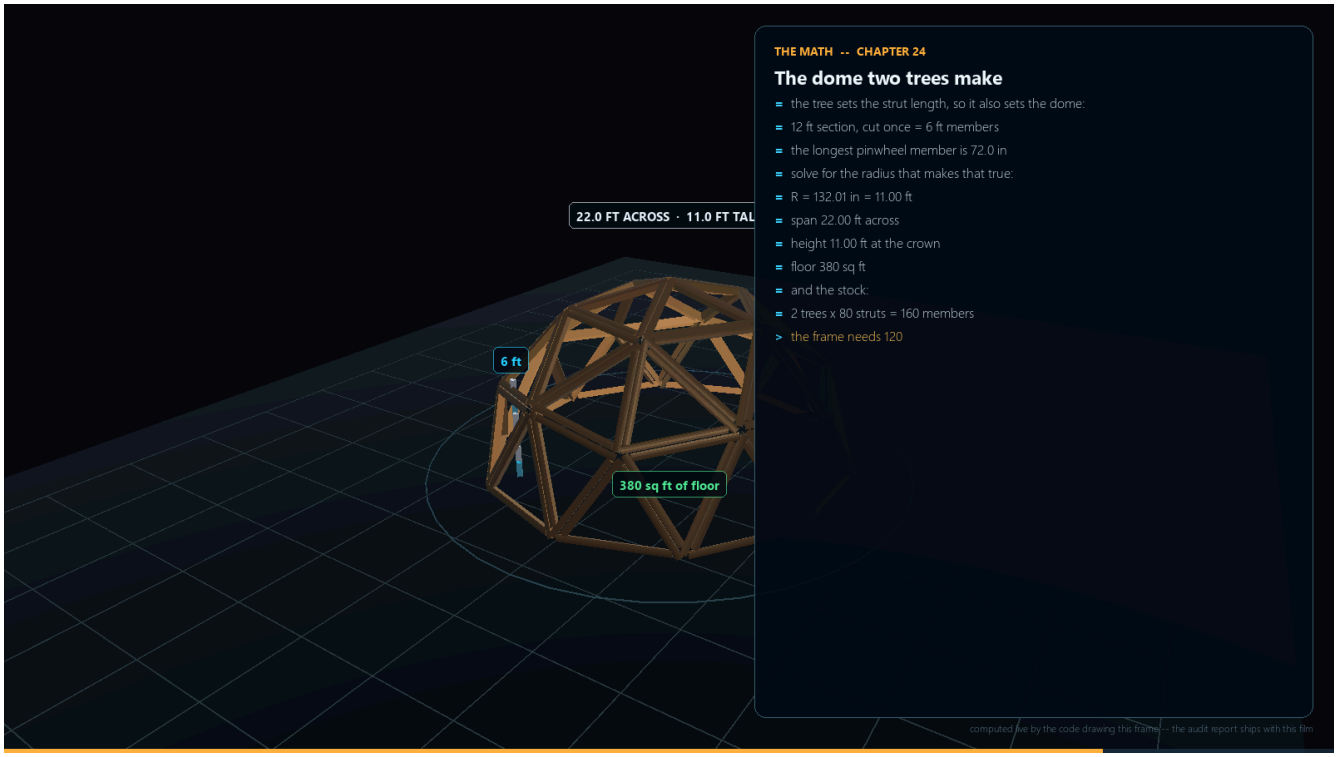
The trees size the building

Here is the inversion that runs through this book. A normal building is designed first and the timber is bought to suit. Here the timber comes first, and the building is solved **backwards from the longest member the wood can give**.

Buck a section, cut it once across the middle, and that half-length is your longest member. The dome that member makes is then fixed by geometry – a size nobody chose:

```
the tree sets the strut length, so it also sets the dome:
  12 ft section, cut once = 6 ft members
  the longest pinwheel member is 72.0 in
  solve for the radius that makes that true:
  R = 132.01 in = 11.00 ft
  span   22.00 ft across
  height 11.00 ft at the crown
  floor  380 sq ft
and the stock:
  2 trees x 80 struts = 160 members
  the frame needs 120
  160 - 120 = 40 left for the floor and the fit-out
  timber standing in the frame = 672 ft of member
```

That is the worked tree of these films: a different, slightly larger dome than this book's reference build, because those two trees gave a slightly longer member. The method is the same; only the tree changed. And that is the point: **the number that depends on the tree is the trunk's diameter**, because it sets every member's section and so its strength. Everything else – the angles, the two lengths, the panel count – belongs to the geometry and does not care what tree you started with (Chapter 10).



The dome two trees make.

One log at a time, 8 rip cuts to a section, and the frame of a house comes off the land it will stand on. The next chapter is why splitting beats squaring, and what it gives up.

CHAPTER 13 · STORY

Stop Squaring. Start Splitting.

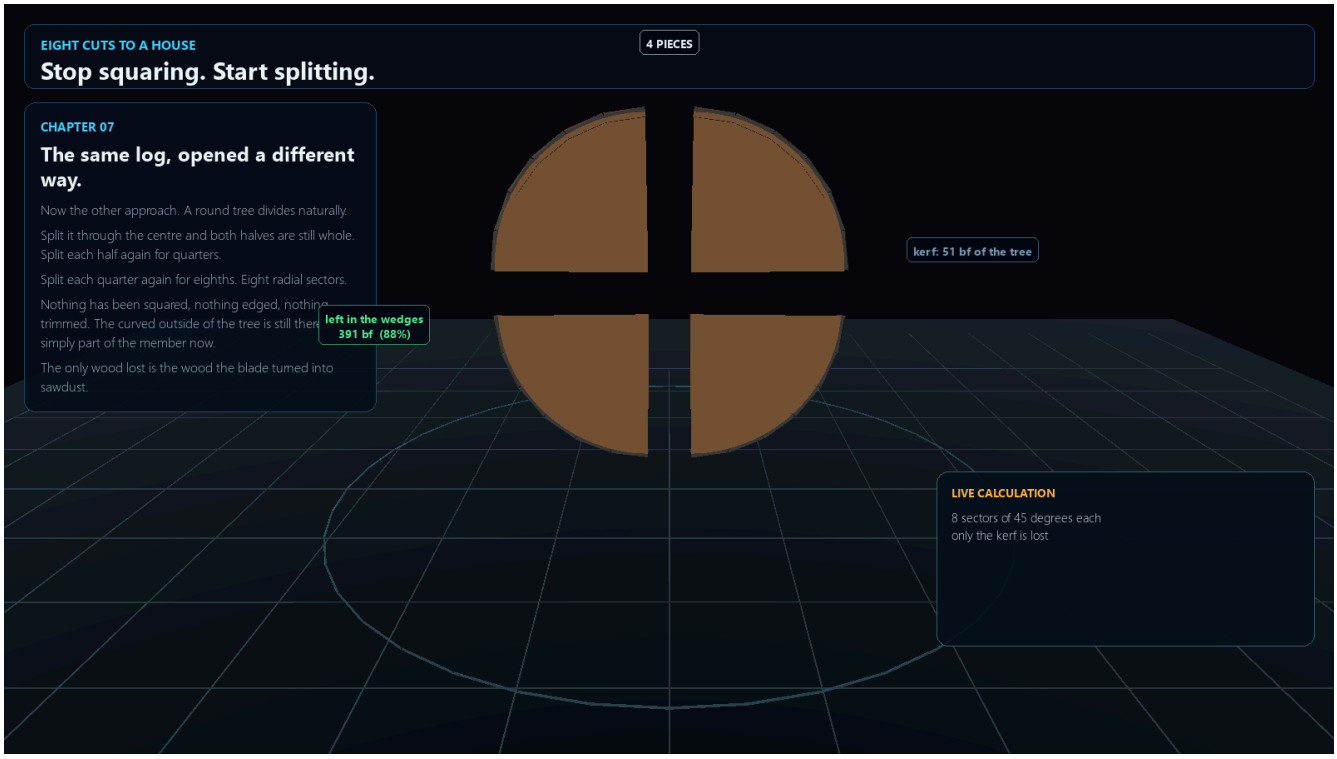
A round log is already rotationally symmetric, and three splits give you eight identical sticks nobody had to measure

I was sitting on a log when this stopped being difficult.

Not thinking about it. Sitting on it, having spent the morning on connectors.

The connector question does have an answer, and it is a flat V bracket: two flaps meeting at the panel's angle, four holes in each flap, and each flap run out to as much as half the member's length so the screws are spread along the stick instead of clustered at its end. Chapter 20 has the detail. It works, it is cheap, and you can make forty of them on a bench brake in an afternoon.

What it does not do is make the sticks alike. It joins dissimilar sticks by absorbing the difference, which means the connector is now doing that work, in every joint, forever.



The same log, opened a different way.

The wrong question

The question I had been asking was: *how do I join sticks that are all different?*

Everybody's answer to that is the same. Mill them. Square them up, run them through a planer, and now they are all the same because you made them the same.

That answer costs a mill, a planer, a shed to keep the mill dry, the fuel to run it, and — this is the part nobody mentions — about half the tree. A round log does not contain a rectangle. Squaring one is the act of cutting a rectangle out of a circle and calling the rest waste.

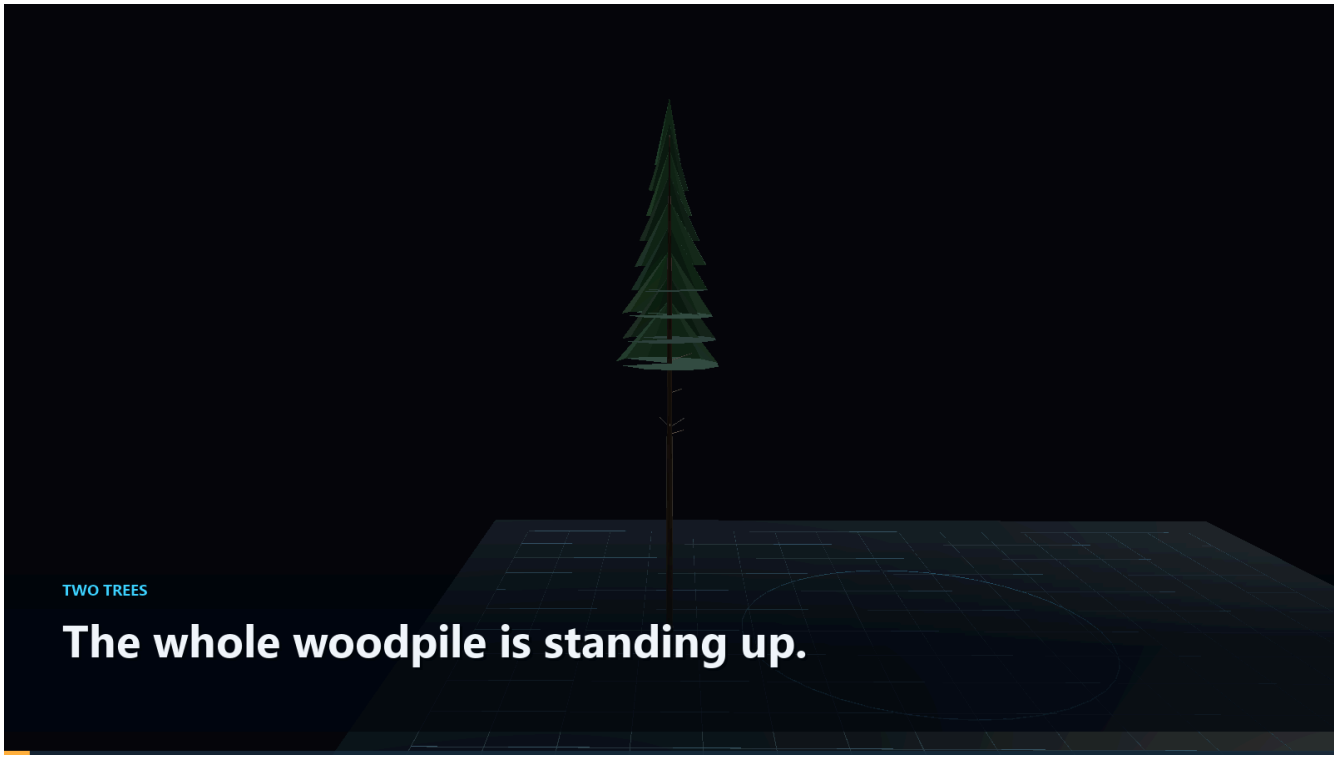
The better question

Sitting on the log, I noticed the log.

A round trunk is rotationally symmetric. Every radius is the same as every other radius. That is not a thing you achieve; it is a thing the tree did while you were not there.

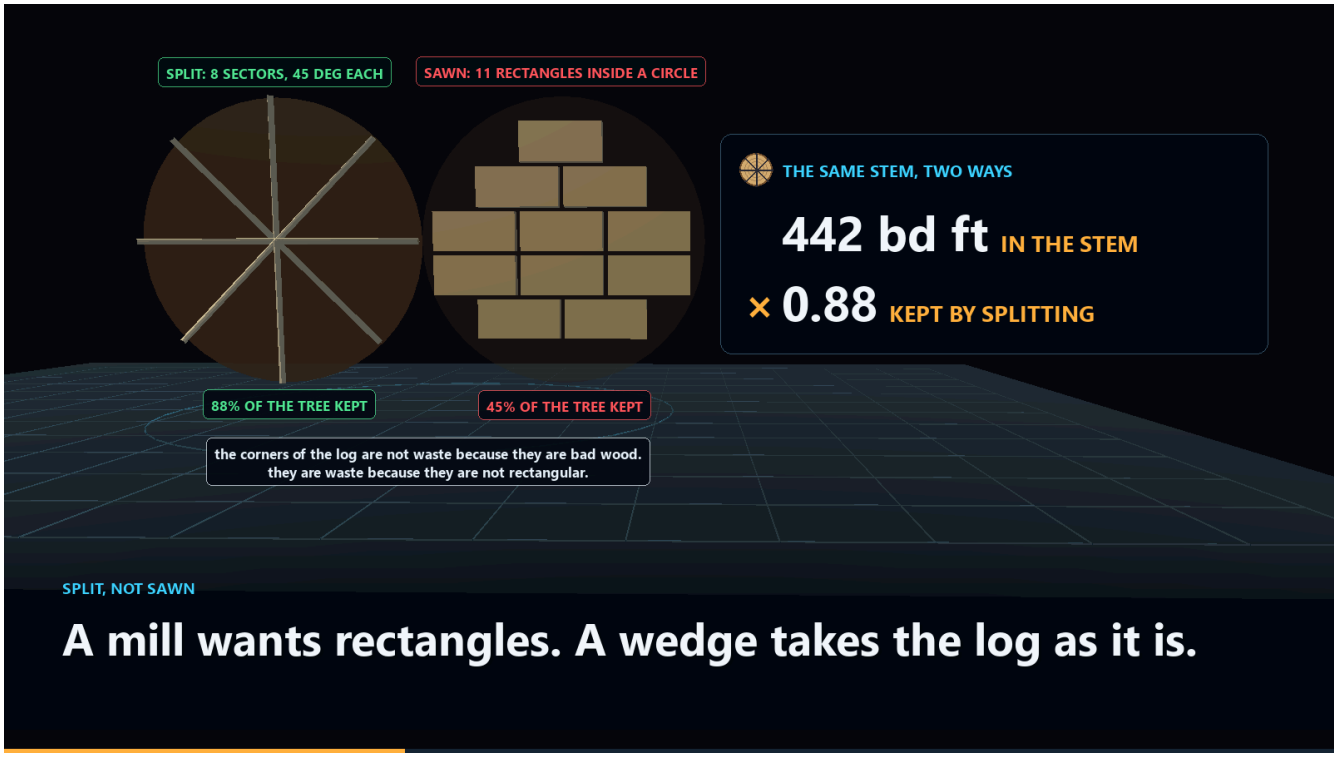
So split it. Split it through the middle and you get two identical halves — identical not because you were careful but because the log was round before you touched it. Split those and you get four. Split those and you get 8.

Three passes. Eight sticks. Every one of them a 45.0-degree sector of the same circle, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.



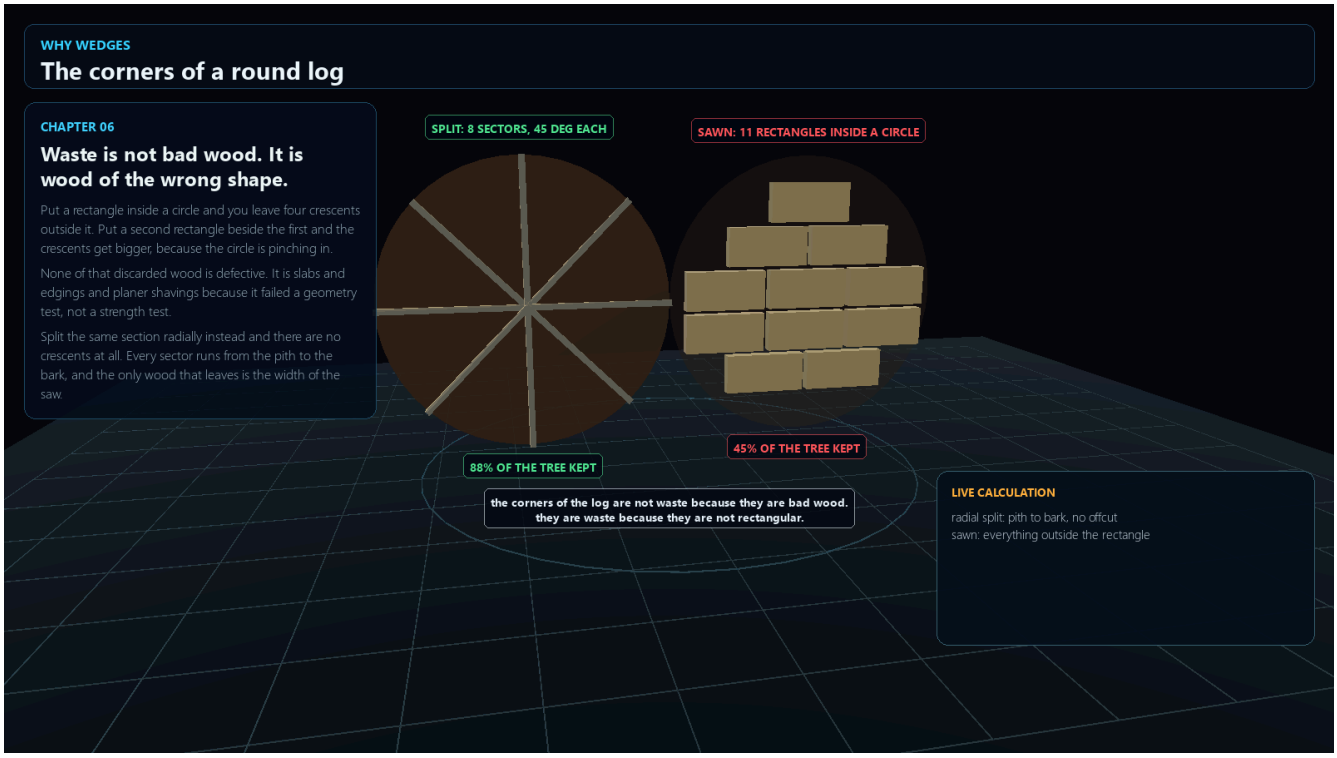
The whole woodpile, standing up.

That was the afternoon. It is not a clever idea and it took me a year to have it, because I was busy asking a question that had no answer.



How much of the stem survives splitting, against sawing.

What the mill is actually for



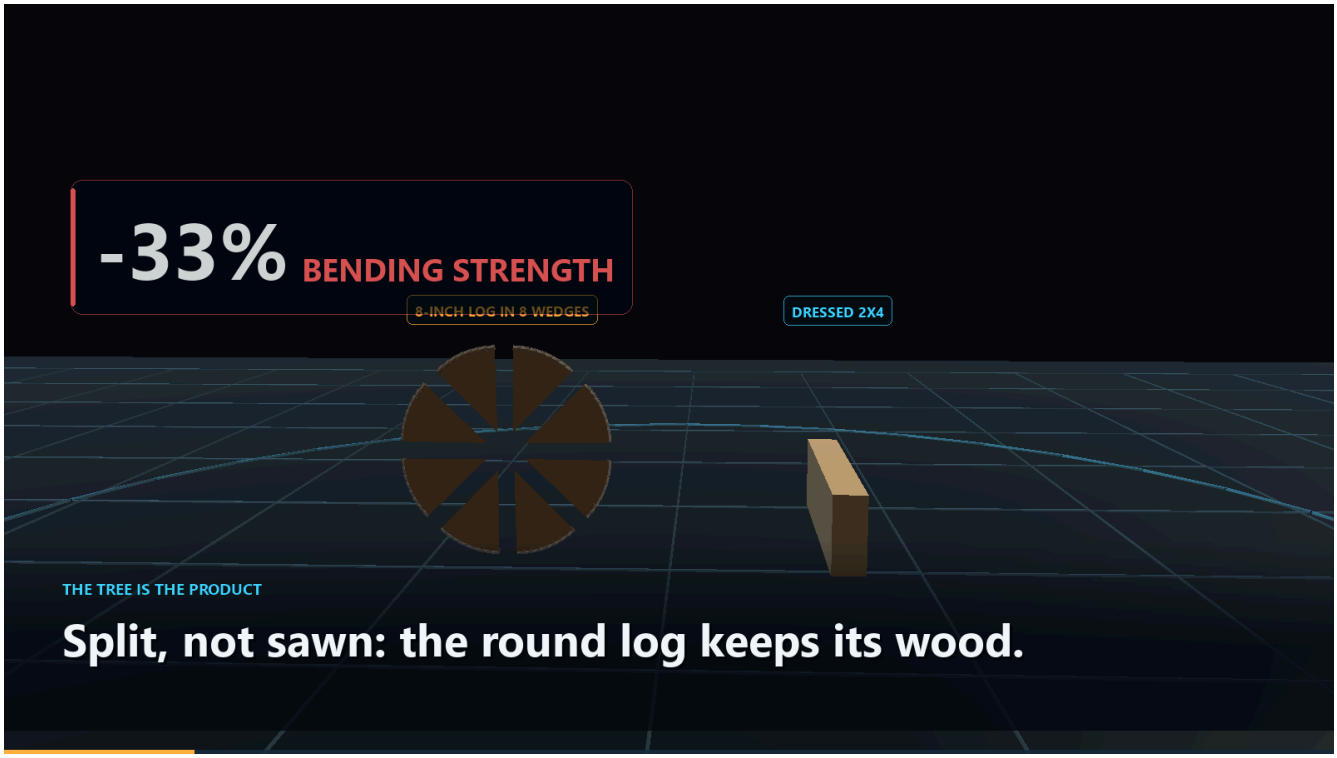
The corners of a round log are not bad wood. They are wood of the wrong shape.

One thing to be careful about, because the recovery argument is easy to overstate.

The corners a mill throws away are not bad wood. They are perfectly good wood of the wrong *shape*. A sawmill exists to turn trees into a standard rectangular product that can be stacked, graded, priced, shipped and sold by people who will never see the tree. It is extremely good at that, and the waste is the price of the standardisation, not of incompetence.

Splitting does not beat sawing at making lumber. It beats sawing at making *this* — a structural member for a frame that does not care about rectangles.

That distinction matters because it tells you when the method stops working. Build something with right angles in it and you want a mill. Build a triangulated shell and you do not.



The tree is the product.

How many rectangles fit in a circle?

Conventional milling squares the log first: four slabs come off the outside to make a rectangular cant. Then the cant is sawn into boards, each board edged, trimmed, dried and graded. Every step removes wood, and every one costs a machine, a setup and a handling. What survives is beautifully standard. What is on the floor is the curved outside of the tree – which is also the densest wood it grew.

How much survives is a packing problem: true two-by-fours, laid out in rows inside each section's small end, where a row can only be as wide as its narrowest edge. For the worked tree of the last chapter:

```
to get true 2 in x 4 in sticks, rectangles have to be packed
into a circle. A mill saws parallel rows, and a row is only as
wide as its narrowest edge, because a stick has to be full
width along its whole length:
  section 1 (13.1 in small end): 0 + 2 + 3 + 3 + 2 + 1 = 11
  section 2 (11.2 in small end): 0 + 2 + 2 + 2 + 1 = 7
  section 3 (9.3 in small end): 0 + 1 + 2 + 1 = 4
  section 4 (7.4 in small end): 0 + 1 + 1 = 2
  section 5 (5.5 in small end): 0 + 1 = 1
  total = 25 true 2x4x12 per tree
  each one holds 8 bf, so 25 x 8 = 200 bf
  out of 442.1 bf standing, that is 45.2% recovery
the brief this lesson works from estimates 32 pieces
```

```
  (256 bf, 57.9%),
  which is kinder to the mill than this packing. Both numbers
  are carried through to the comparison.
```

Note what the working does with its own brief. The brief it started from estimated a kinder figure for the mill than honest packing gives, and both are carried forward rather than the convenient one.

The same two trees, both ways

Put the two conversions side by side, for the same two trees:

```
the same two trees, converted both ways:
  radial wedges      782 bf   160 struts
  packed 2x4s        400 bf   100 struts
  ratio = 782 / 400 = x1.95
against the brief's kinder estimate for the mill:
  brief's 2x4s       512 bf   128 struts
  ratio = 782 / 512 = x1.53
  = 53% more wood, which is the figure the brief quotes
either way the direction is the same, and the member count
  moves with it:
  160 wedge struts against 128 sawn ones = 32 more
```

Somewhere between those two ratios more structural wood comes out of the same two trees split than sawn – and one fewer machine was needed to get it.

What this does not claim

One honest paragraph, because the argument does not need overstating.

This is not the claim that any eighth of any tree equals a graded two-by-four. It does not. Species, moisture, knots, checks, decay, grain running off the axis, taper and the connections all still matter, and a building people live in has to be designed and checked around the members actually being used. The figures in this chapter are geometry and volume: what the tree contains and what the frame needs. They are not a structural calculation, and they are not a substitute for one. Chapter 15 is where the structural numbers start.

What comes off the log

At the reference build's 12.0-inch trunk, each sector is 4.592 inches across its bark face and 6.0 inches deep from point to bark. The frame needs 120 of them, which is 664 feet of stick.

Nothing in that sentence required a tape measure on the tree. It required a chainsaw, a steady rip along the radius, and a willingness to let the log decide the section.

The next chapter is what that section actually is, and which way round to put it — which is the first decision in this build you cannot take back.

The round tree is not defective lumber

So, plainly: a wedge from a small tree is not a stronger stick than a stud. It is a cheaper stick, a faster one, one that leaves far more of the tree standing as structure instead of shavings, and one you can make without owning a factory. From a big enough tree it is also simply the stronger stick, and Chapter 15 gives the diameter where that flips.

What makes it enough, either way, is the building. A triangulated shell puts its members mostly in compression along their length, doubles every interior edge, and asks only that the ends can be made to meet. Efficiency is not always a better machine for making a conventional material. Sometimes it is a structure that no longer needs the conventional material at all.

The round tree is not defective lumber. The wedge is not an unfinished two-by-four. It is the member, and the dome is the geometry that lets you use it.

What the Saw Actually Does

Two timed sittings, the fuel written down, a prediction the geometry makes without being told -- and the cuts counted both ways

Everything in the last two chapters could be argued about. This chapter was measured.

Measured, and guessed

Before any rate, here is what the figures rest on, sorted into the two piles this book always keeps apart. Geometry, board feet and angles are computed. The rest was either timed, weighed and read off a rack on site, or it is an estimate, and it says which:

```
 geometry, board feet and angles in this film are computed. The rest
splits into two piles, and the difference matters:
MEASURED ON SITE -- timed, weighed or read off a price rack:
  session_a_wedges                12.00 wedges
  session_a_hours                  2.00 h
  session_a_consumables_usd       10.00 USD
  session_b_rip_feet              42.00 ft
  session_b_hours                  1.00 h
  session_b_depth_in               6.00 in
  session_b_fuel_tanks             2.00 tanks
  session_b_oil_tanks              2.00 tanks
  planning_wedges                 30.00 wedges
  planning_hours                   6.00 h
  stud_shelf_usd                   8.88 USD
  stud_with_tax_usd               10.00 USD
  measured_log_diameter_in        12.00 in
  builder_volume_multiplier        2.50 x
  used_volume_multiplier           2.00 x
  wedge_length_ft                  6.00 ft
  wedge_recovery_share             88.0 %
  traditional_recovery_low         50.0 %
  traditional_recovery_high        65.0 %
ASSUMED -- overheads nobody times, so they are estimates:
  store_round_trip_miles          40.00 mi
  truck_usd_per_mile               0.55 USD/mi
  store_run_hours                  1.50 h
  studs_per_load                  40.00 studs
  cull_share                       10.0 %
  transit_damage_share              3.0 %
  labour_usd_per_hour             25.00 USD/h
```

Every estimate sits on the store-bought side of the comparisons that come later, which is the side it flatters least to guess about.

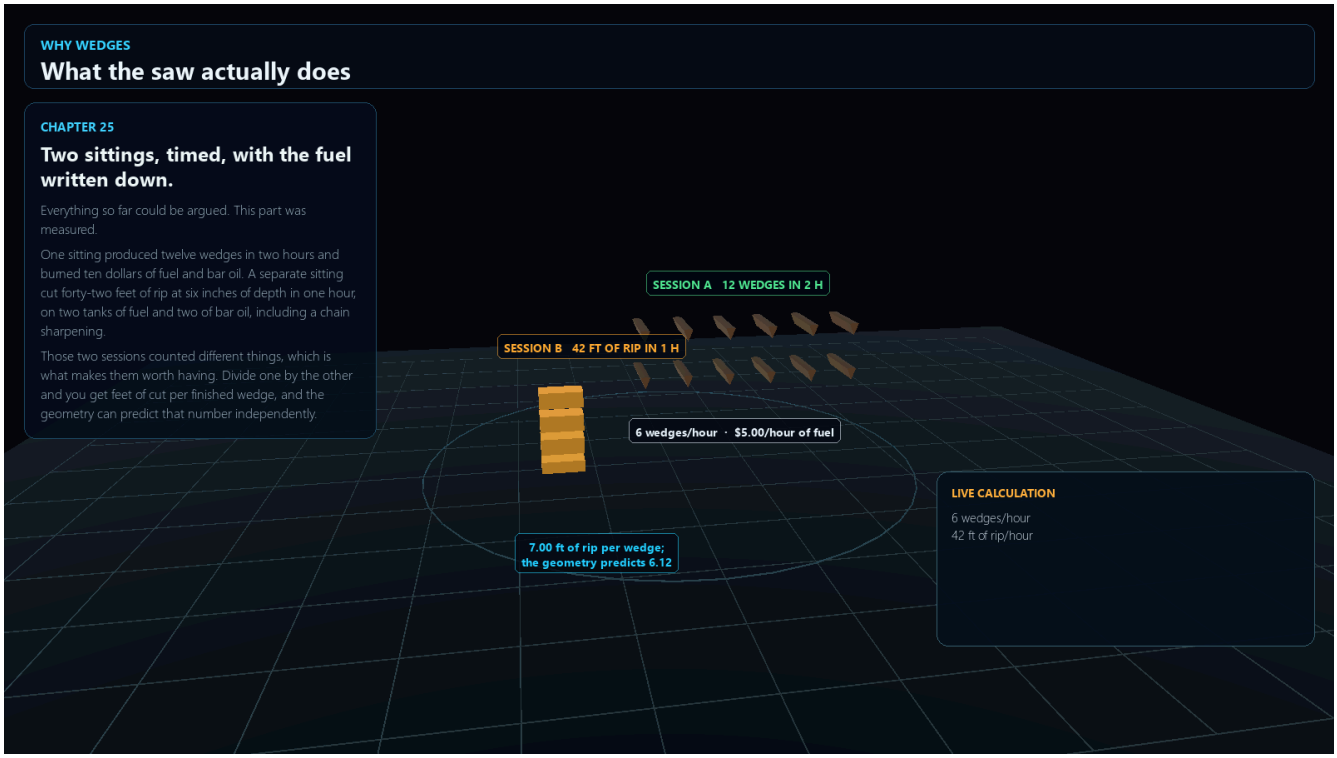
Two sessions, and do they agree?

Two separate sittings with the saw, each measuring something different. One counted finished wedges against the clock. The other measured feet of rip cut in an hour at full depth, with the fuel and bar-oil tanks counted and one chain sharpening included.

Because they counted different things, they can be divided – feet of cut per finished wedge – and the geometry can predict that same number without being told it:

```
SESSION A, timed:
  12 wedges in 2.0 h = 6.0 wedges/hour
  $10.00 of fuel and bar oil = $5.00/hour, the only cash spent
SESSION B, timed separately:
  6 passes at 7 ft = 42 ft of rip in 1 h, at 6 in+ of depth
  2 tanks of fuel and 2 of bar oil, one sharpening
the two sessions measured different things, so divide them:
  42 ft/h / 6.0 wedges/h = 7.00 ft of rip per wedge
and the geometry predicts the same number independently:
  8 sectors takes 7 passes down a 7 ft section
  7 x 7 / 8 = 6.12 ft per wedge
the pure cutting is 88 % of the measured time; the other 12 % is rolling the log, setting up and sharpening
planning uses 5.0 wedges/hour, 17 % below the measured rate
```

Two independent measurements and a geometric prediction agree closely enough for the rate to be worth planning on, and the planning figure is set deliberately below the measured one.



Two sittings, timed.

What the harvest took

At the planning rate, ripping the frame's 120 members takes 24 hours of saw time. The whole harvest is laid out as 8 days: 2 of felling, 2 of bucking and 4 of ripping.

Ripping is the part that was timed with the fuel written down: 2 tanks an hour for 24 hours, 48 tanks. The saw's tank is an estimate – the listings disagree, between 0.28 and 0.37 litres – so the fuel is a range: somewhere between **3.6 and 4.7 US gallons** for all the ripping. Felling and bucking were never metered, and they are left out rather than guessed.

A few gallons of fuel for the frame of a house. That is the figure the whole method is built on, and it is a measured one.

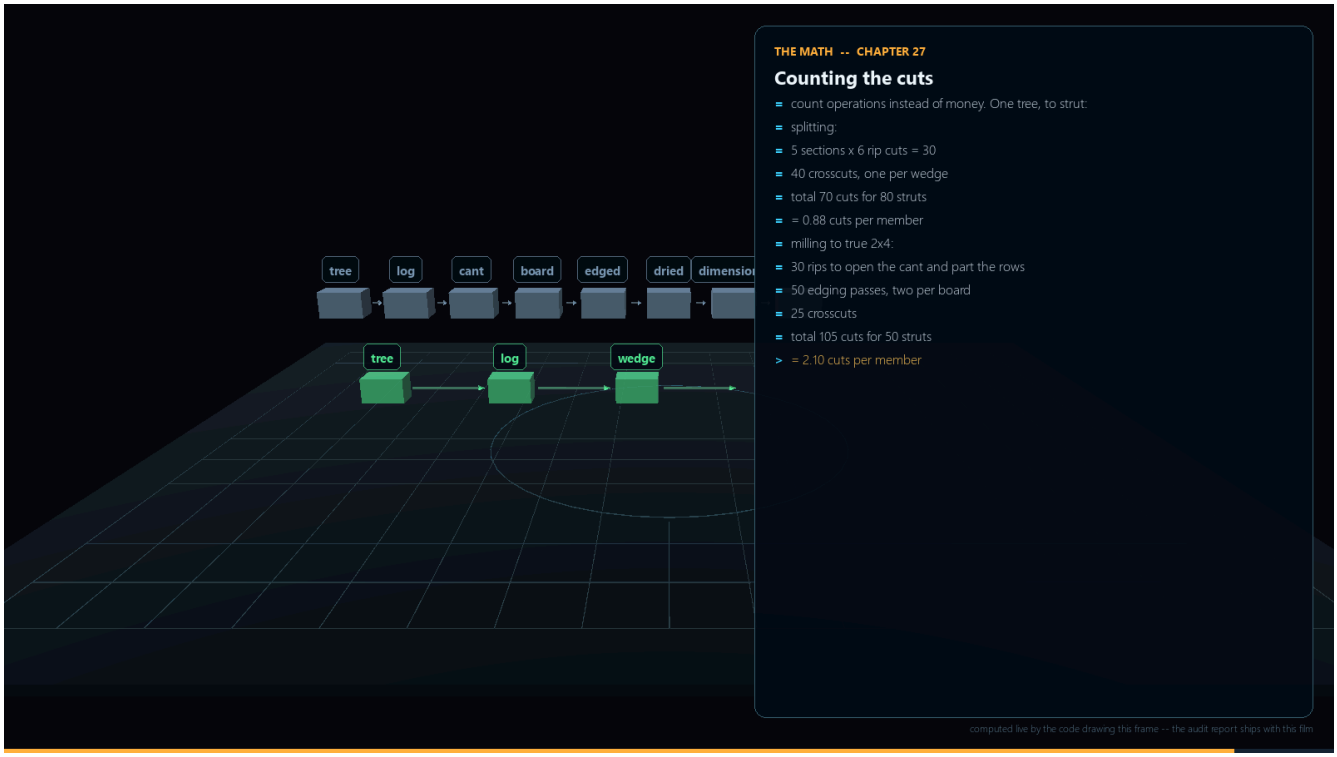
Counting the cuts

The real measure of a system like this is not money. It is the number of actions per piece of finished building: measuring, marking, setting up, cutting, edging, planing, sorting, stacking, hauling and handling the waste. Standard lumber is only convenient because somebody else already paid for all of those. When the tree is standing next to the site, the question becomes: what is the shortest honest path from that tree to a finished member?

Count it both ways, for one tree:

```
count operations instead of money. One tree, to strut:
  splitting:
    5 sections x 6 rip cuts = 30
    40 crosscuts, one per wedge
    total 70 cuts for 80 struts
    = 0.88 cuts per member
  milling to true 2x4:
    30 rips to open the cant and part the rows
    50 edging passes, two per board
    25 crosscuts
    total 105 cuts for 50 struts
    = 2.10 cuts per member
and that is only the cuts: not the setting out, the handling,
  the sorting, the stacking or the hauling between machines
```

And that is only the cuts. Every other action on the milling side – the setting out, the sorting, the hauling between machines – comes on top. The shorter chain is also the one that keeps more of the tree.



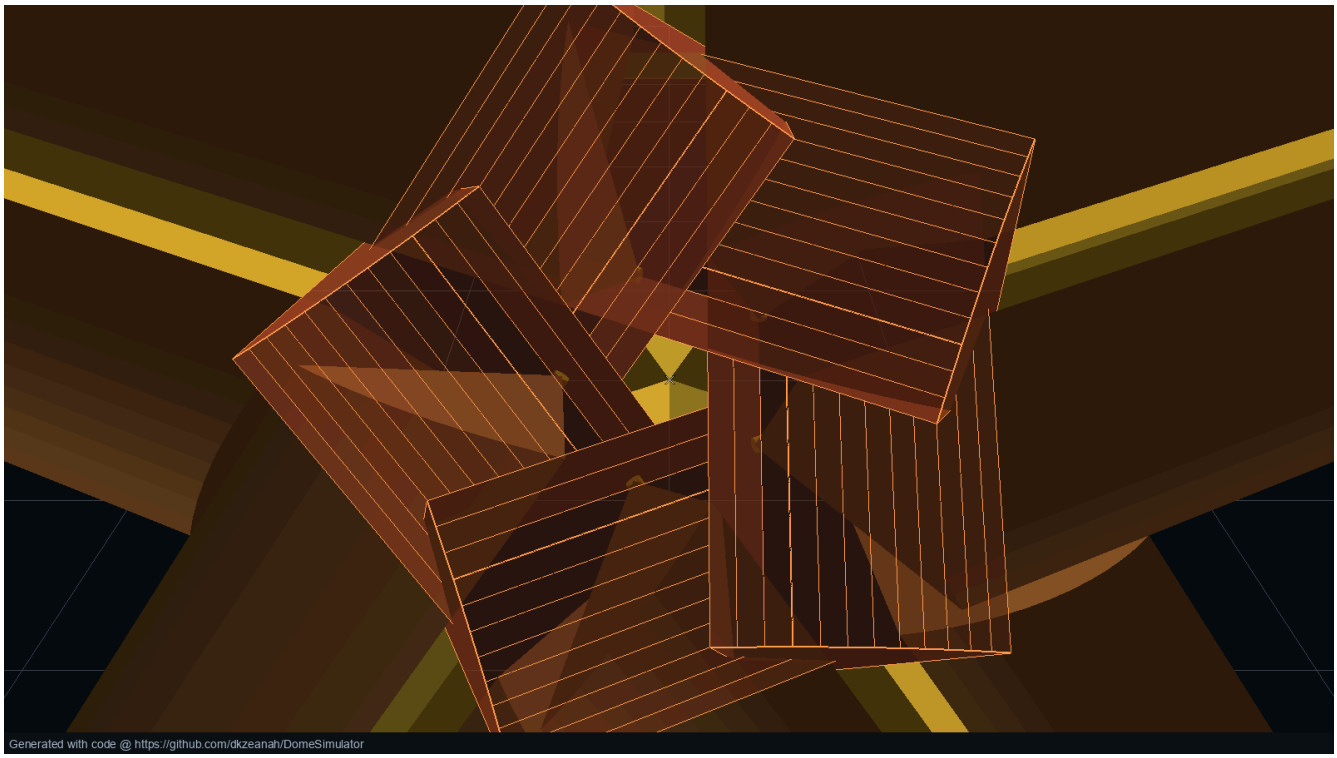
One tree to a finished member, counted both ways.

One Eighth of a Tree, Used as a Stick

What a 45-degree sector actually is, and why its shape is an advantage rather than something to apologise for

A member in this frame is a 45.0-degree slice of a 12.0-inch log. It has a point, two flat sawn faces, and a curved bark face.

That is the whole part. Nothing is machined.



The end of a member: two sawn faces and a bark face, and nothing machined.

What the shape gives you for free

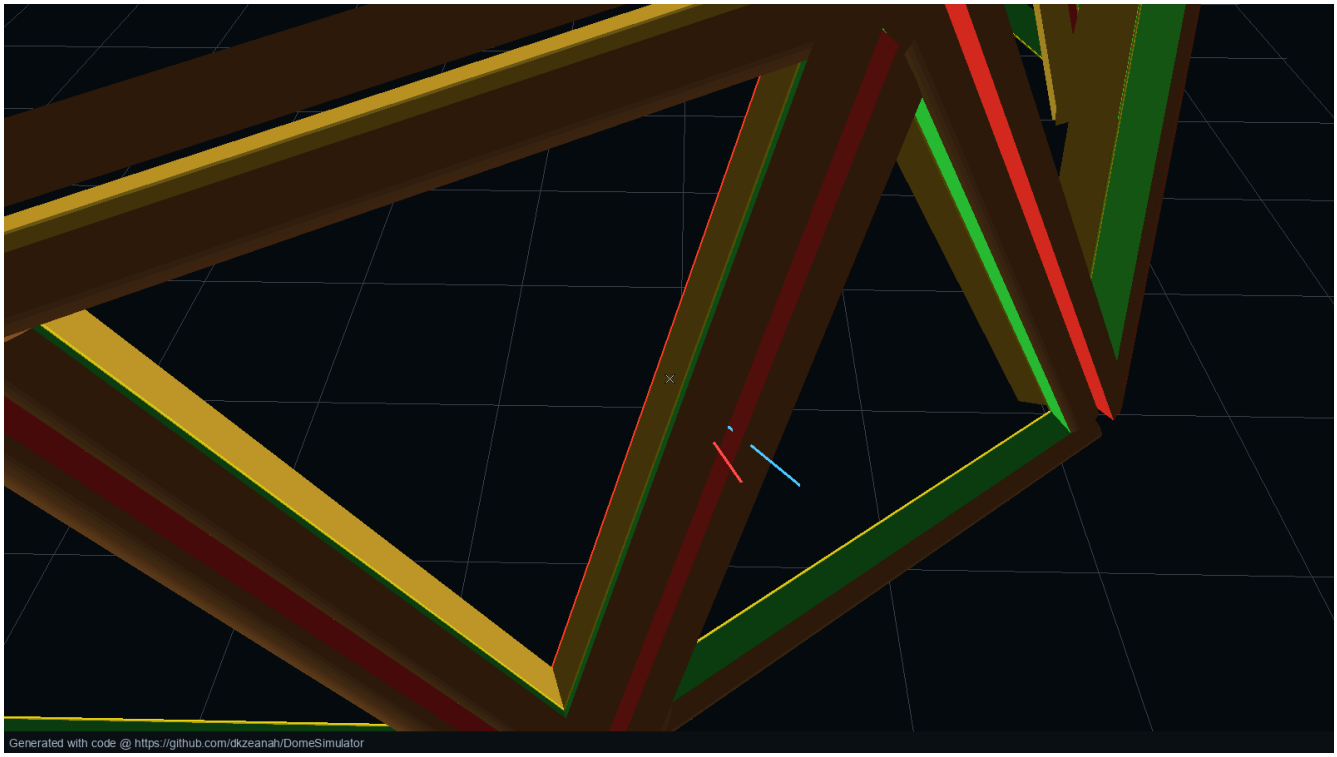
Two reference surfaces. The split faces are flat and they are at a known angle to each other — 45.0 degrees, because that is what a 8-way split means. Every jig in this book locates off those two faces. You never reference the bark, which is lumpy, and you never reference a measurement, which is a thing that can be wrong.

A point that tells you which way round it is. A rectangular stick has a four-fold ambiguity: drop it in rotated ninety degrees and it still looks fine. A sector has one obvious orientation, and — as Chapter 24 shows — the fixture is built so a stick dropped in the wrong way round physically does not seat.

Depth where you want it. The section is deep from point to bark (6.0 inches here) and narrower across the face (4.592 inches). Stand it point-inward and the depth is in the direction the shell wants stiffness.

Which way the point faces

This is the first decision in the build you cannot take back, because it changes the seam, the key, and the shape of every panel.



Point inward: the reference build.

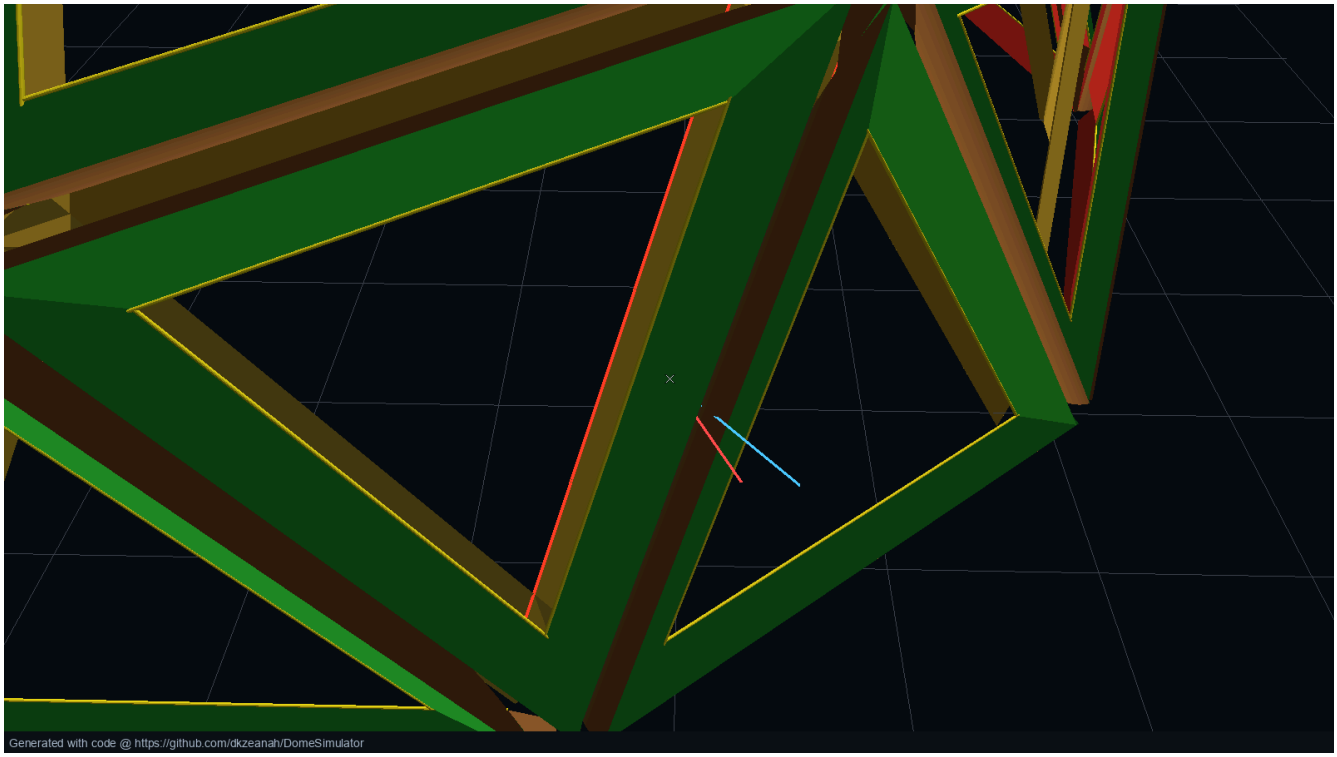
The reference build faces the point **into the dome**. The bark faces are outside, the flat sawn faces look at each other across every seam, and the leftover sector angle opens toward the inside.

The consequences, from the solve: an A seam folds 18.029 degrees and leaves 26.971 degrees of sector angle over; a B seam folds 22.459 and leaves 22.541.

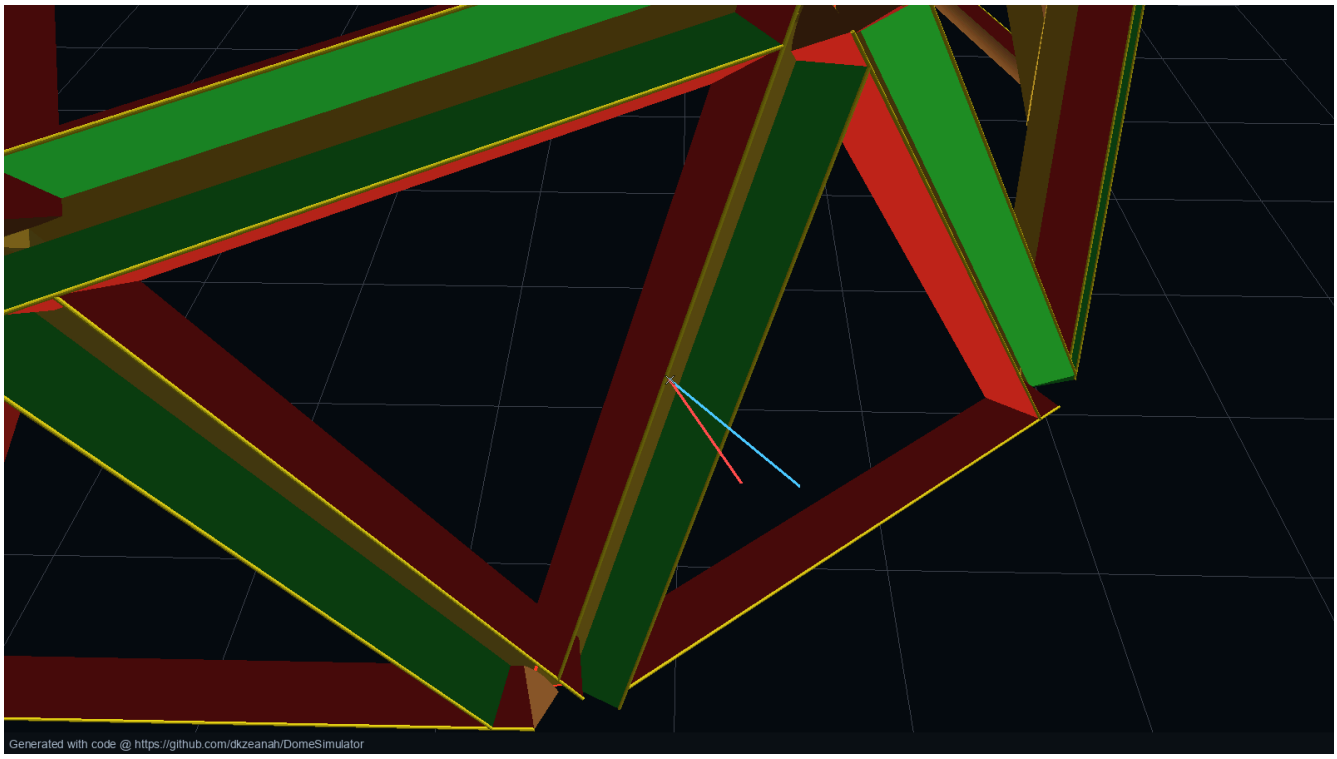
Each pair sums to 45.0. That is not a coincidence and it is the closest thing this method has to a theorem: **the sector you split out of the tree is exactly the seam budget the dome needs.**

The other three

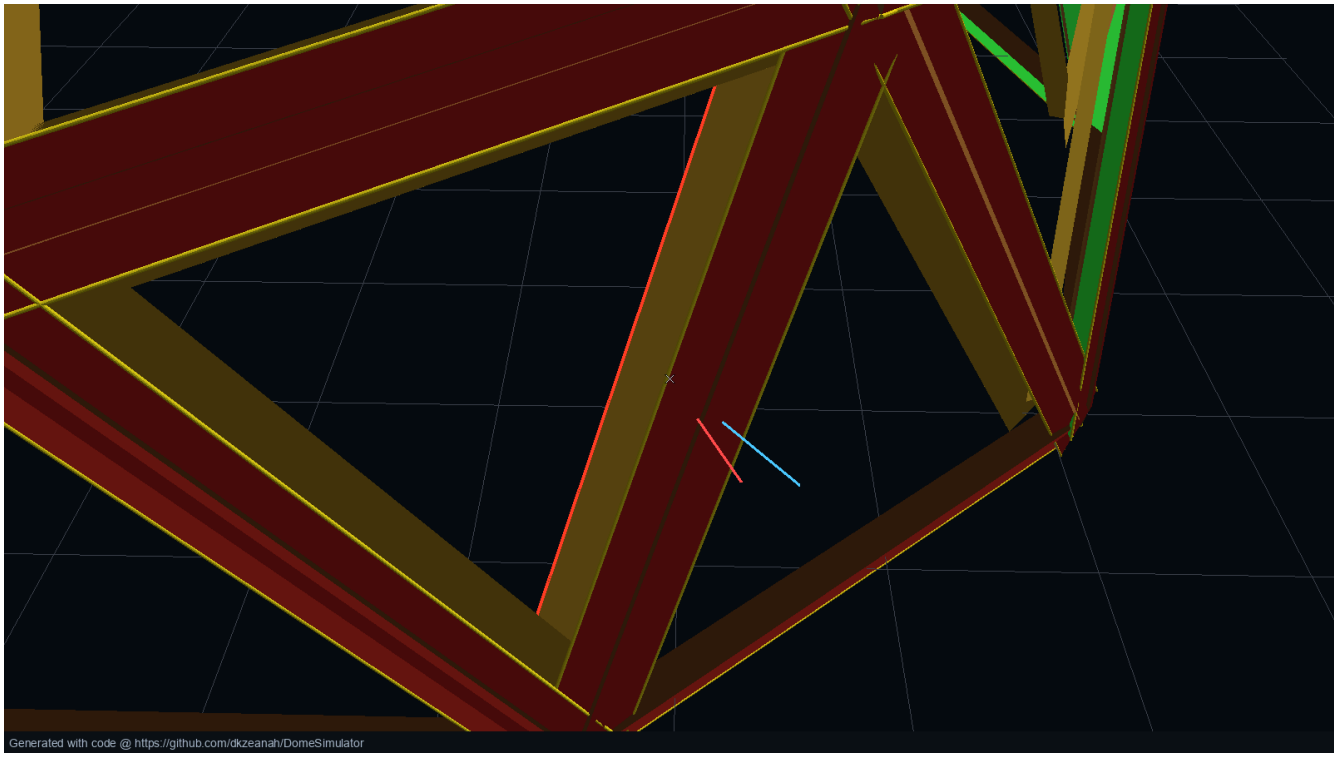
The solver will build all four. They are not equally good and they are not equally bad, and the book shows them because a reader who wants something different should be able to see what they are choosing.



Points apart, each into its own triangle.



Both points outward, at the sky.



Points toward each other, across the seam.

Point outward puts the flat faces where the weather is, which is a real argument for it: a flat outer face is easier to seal a panel against than a curved one. It costs you a key more than twice the size, and it puts the narrow point where the shell wants depth.

Points apart, and points across, both produce a seam that wants a key around eleven and a half inches across its base. That is not a key. That is a beam you are now also making.

The member, measured

A sector is exactly an eighth of the round, so its size is a formula in the log's diameter: its area is the circle's over eight, its depth is the radius, its width across the bark face is a chord. At the films' design diameter:

```
one sector is 45 degrees of the round:
  area  = pi r^2 / 8
  width = 2 r sin(180 deg / 8)   across the bark face
  depth = r                      pith to bark
at the design diameter of 12.5 in:
  area  = 15.34 sq in
  width = 4.78 in
  depth = 6.25 in
a true 2x4 is 2 x 4 = 8 sq in, so this
  sector is worth 1.92 of them
and the depth is the useful part: bending stiffness grows with
  the cube of depth, and this section is deep and narrow
the break-even diameter, where one eighth equals one 2x4:
  pi r^2 / 8 = 8  ->  diameter = 9.03 in
  check: sector at 9.0 in = 8.00 sq in
```

The depth is the useful part. Bending stiffness grows with the cube of depth, and this section is deep and narrow – the right shape for a member whose job in the shell is mostly to stay its own length.

Section properties, and the crossover

Area is not strength. What decides how a member bends is its second moment and its section modulus, and those grow faster than area as the log gets bigger. So there is a diameter where the sector overtakes the stud, and it is worth knowing exactly:

```
a raw sector is literally 1/8 of the tree, and how strong that is depends entirely on how big the tree was.
AT THE SIMULATOR'S 8 IN TRUNK:
  area 6.28 in2, S 2.05 in3
  = 120 % of a dressed 2x4's area but only 67 % of its bending
AT THE 12 IN TRUNKS ACTUALLY BEING CUT:
  area 14.14 in2, I 26.98 in4, S 6.92 in3
  = 269 % of a dressed 2x4's area and 226 % of its bending
  and 177 % / 130 % against a FULL-dimension 2x4
section grows with the square of the radius and bending with the
  cube, so there is a diameter where the sector overtakes the stud:
  more area than a dressed 2x4 above 7.31 in
  stiffer than a dressed 2x4 above 9.14 in
  stiffer than a FULL 2x4 above 11.00 in
and neighbouring panels never share a stick, so all 55 interior edges carry TWO sectors:
  539 % of a stud's area, 452 % of its bending
```

Under that crossover, the wedge is the weaker member and this argument fails. That is the number that does not help, and it is the reason this book names a minimum trunk rather than pretending any tree will do. Above it, every member is stiffer than the stud it replaces – and every interior edge carries two of them.

What the rotation is worth

The wedge is not symmetric, so turning it about its own long axis keeps the same wood but changes how deep it is in the direction the shell bends. Measured for all four ways round, at the simulator's smaller trunk:

```
the wedge is not symmetric, so which way it faces is a decision.
turn it about its own long axis and the SHAPE stays the same
  while the STIFFNESS does not, because bending in a shell is
  radial: in and out along the dome's own radius.
  DOME-IN    I  5.32 in4   S  2.05 in3
             a pair at one seam: both points inward, toward the centre of the dome
  PANEL-IN   I  2.50 in4   S  1.63 in3
             a pair at one seam: points apart, each into the middle of its own triangle
  DOME-OUT   I  5.32 in4   S  2.05 in3
             a pair at one seam: both points outward, away from the centre, at the sky
  PANEL-OUT  I  2.50 in4   S  1.63 in3
             a pair at one seam: points toward each other, across the seam
area is 6.28 in2 in all four -- rotation moves no wood
but S runs 1.63 to 2.05 in3, a factor of 1.25
```

The same stick, turned before it goes down: the reference build takes the free stiffness and points every member at the centre of the dome.

Why the bark stays on

It is free, it is already there, and taking it off is an operation on 120 sticks.

The honest version: bark is not a finish and it will come off in its own time on the outside faces, in patches, over years. On a frame that is inside a weatherproof skin — which is the argument of Chapter 2 — that is a cosmetic event rather than a structural one.

If you want it off, take it off green, with a drawknife, in the first week when it is still wet and comes away in strips. Do not plan to do it later.

Why six feet is the useful length

The members are short on purpose. A knot or a bend does not ruin wood; it ruins whatever length has to be thrown away around it, and that length is set by the piece you were trying to make:

```
a knot or a bend does not ruin wood. It ruins whatever length has
  to be thrown away around it -- and that length is the piece you
  were trying to make.
this dome's members run 5.13 to 5.90 ft, cut from 6 ft stock
a 16 ft stick from the same trunk is 2.7 times longer
one defect costs, out of 60 usable ft:
  short members  6 ft = 10.0 % of the tree
  long sticks   16 ft = 26.7 % of the tree
and the trunk gives 10 short pieces against 3 long ones, so there are more places for a
  defect to fall harmlessly between
a real bend is cut out at its middle: go six feet clear either side
  and the sections above and below it are both still good
in square milling the same bend follows the board down its whole
  length, because the rectangle has to be straight before it exists
```

Short members are also easier to carry, season, sort, store and replace, and a modest tree can produce them – nobody waits a century for old growth. The geodesic frame turns all of that into an advantage, because it is built from many short members by design.

The one number to hold on to

A member's section is set by the log, and the log is set by what you have. Everything downstream — the seam key's size, the depth of the wall, how many trees you need — follows from the trunk diameter and the split count, and Chapter 49 prices what happens when you change either.

What does *not* change is the two lengths you cut them to, and that is the next chapter.

Two Lengths, and Neither Is the Chord

72.000 and 63.670 inches are what the geometry says; what you cut is those minus a bite that does not scale

A 2V dome has two member lengths. This is the page you come back to.

The two chords

A chord is the straight-line distance between two vertices of the sphere. It is the number every geodesic reference publishes, and it is a property of the shape rather than of your building:

```
A chord   =  0.618034 x radius
B chord   =  0.546533 x radius
```

At this book's radius of 116.4984 inches, that is 72.000 and 63.670 inches.

The A chord being a round 72 is not luck. The dome was sized from the stick: a six-foot member is what comes comfortably out of a twelve-foot log section, what one person carries, and what fits in a pickup. The diameter is the answer to the member, not the other way round.

The chord is not what you cut

Neither end of a member reaches the vertex it is named after.

One end butts into the *side* of its neighbour, short of the corner. The other runs past the mathematical vertex so that the previous member can butt into its side. Chapter 20 is why. The consequence is that the joint takes a bite out of each end, and the cut length is:

```
cut  =  chord  -  bite
```

For this frame:

```
A members   72.000  -  1.818  =  70.182 in
B members   63.670  -  0.990  =  62.681 in
```

The bite does not scale

This is the part that catches people.

The bite is set by **how wide the member is**. A member does not get wider because the dome gets bigger — you are still splitting the same size of log — so the bite is the same number at every diameter.

Solve the same pinwheel at a four-foot radius and at a fifteen-foot one with the same stick and the bite comes out identical to four decimal places.

So: **scale the chord, subtract the bite.** Not chord times a ratio.

If you take one thing from this chapter, take that.

The bite used here

```
A members   1.818 in
B members   0.990 in
```

Both at this book's 4.592-inch member. A wider member takes a bigger bite, roughly in proportion, so a frame cut from heavier stock wants its own solve rather than this number.

Why the distinction is worth a page

Scaling the bite with the dome is the natural mistake, because everything else about a dome does scale.

It is also invisible at the reference build, where the two arithmetics agree exactly. It only shows up at other sizes: at eight feet across, scaling makes every stick more than an inch too long; at twenty-four feet, half an inch short. Every stick, all 120 of them, in the two sizes a first-time builder is most likely to pick.

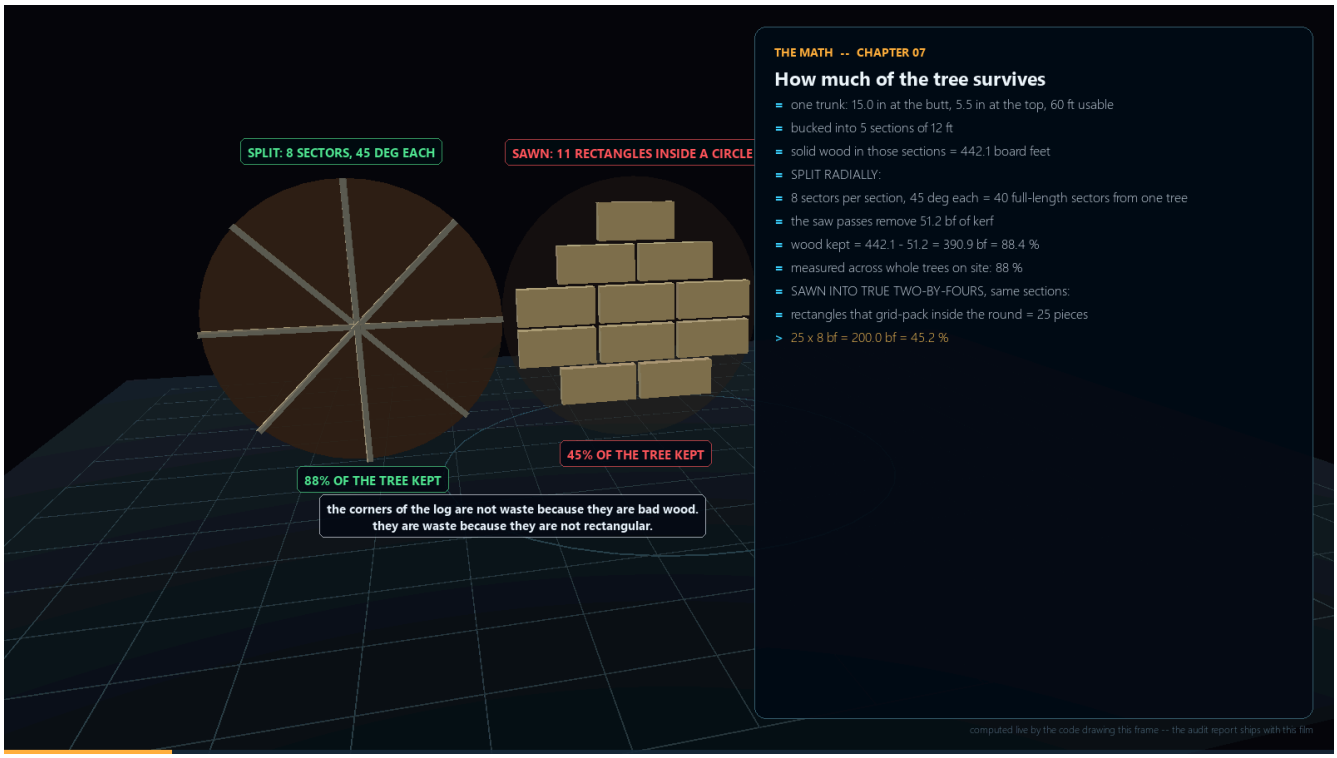
That is why every figure in this book is generated rather than typed, and checked against the solver on each build — Chapter 56. An error that is exactly right in the one case you test is the kind that survives proofreading.

Two models, and they disagree

This repository solves the pinwheel twice.

The raw-wedge solver treats a member as a real 45.0-degree log sector. That is where the bites above come from, it is what the simulator draws, and it is what the reference build is cut to.

A second, simpler model treats a member as a rectangular band lying with its face on the edge line. It gives a bite of 9.253 inches for the same stick — several times larger.



How much of the tree survives, measured.

Neither is a mistake. They are answers about differently shaped sticks.

The films that come with this book carry a third answer, and it should be named rather than quietly dropped. An earlier pinwheel model in them gives **four** stick lengths rather than two – it lets where an end stops depend on which corner it stops in, and so splits each edge length in two:

```
a 2V hemisphere has two edge lengths, SHORT and LONG.
the pinwheel gives four stick lengths, because where an end
stops depends on the corner it stops in:
  SHORT x30   62.523 in   (edge 72.146 in)
  SHORT x30   63.194 in   (edge 72.146 in)
  LONG  x30   71.082 in   (edge 81.584 in)
```

```
  LONG  x30   72.000 in   (edge 81.584 in)
  120 sticks in 4 lengths, three per panel
cut them in that order and the saw stop moves four times for
  the whole building
  longest  72.000 in
  shortest 62.523 in
  spread   9.477 in
```

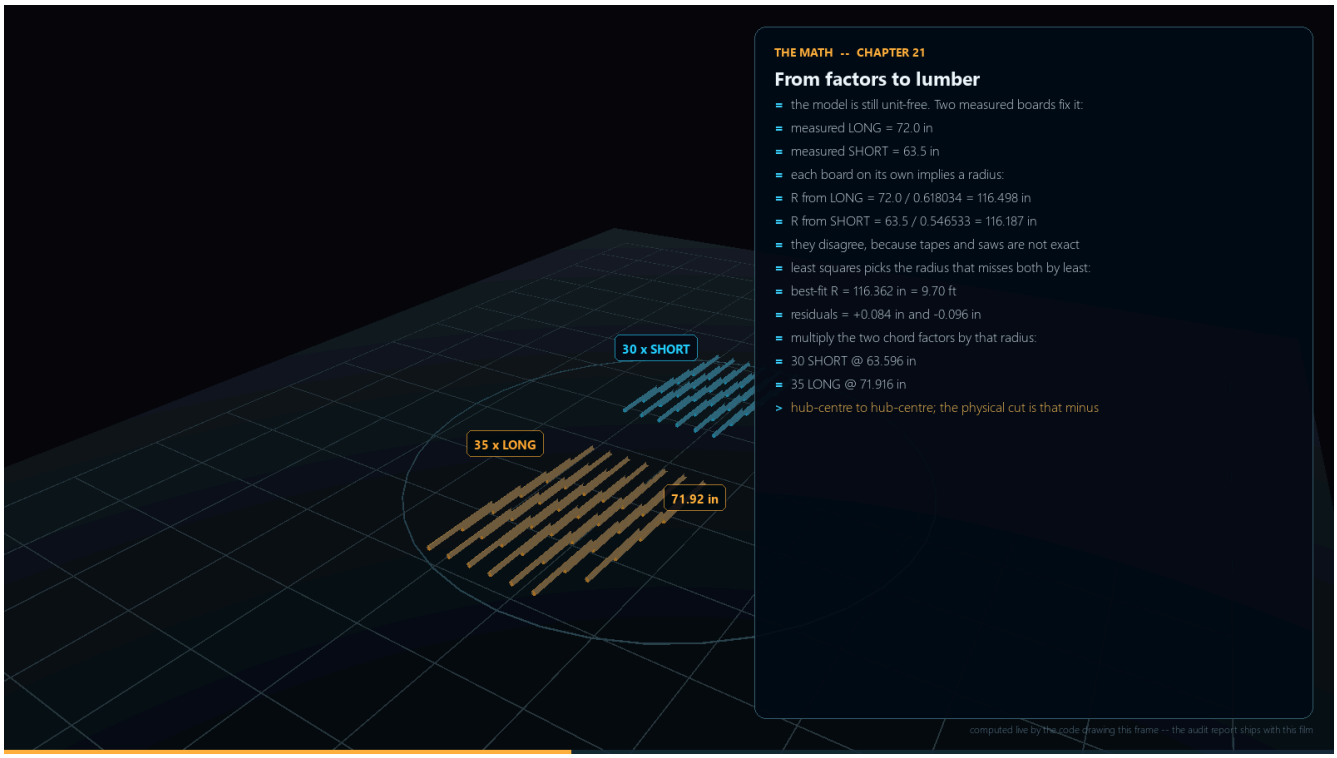
That model also sized a larger dome. The raw-wedge solver, which this book is cut to, gives two blank lengths, 60 of each, because this method never asks the saw to land the head of a member exactly: each head is left long and cut flush in place on the jig (Chapter 24), so the corner-to-corner difference is taken up there, not by a third and fourth saw setting. Where the two disagree, this book follows the solver, and the four-length table is printed here only so that a reader who has seen it in a film knows which one to cut to.

The numbers in this book are the first. **If you are cutting rectangular stock rather than log sectors, this table does not describe your frame**, and the difference is inches on every member.

The row to check yourself against

19.42 feet across: A chord 72.000, A cut 70.182.

If your own arithmetic gives that, the rest will behave.



Chord factors turning into a cut list.

The order

So the whole order for the frame of a 19.42-foot dome with 277 square feet of floor fits on a card:

```
long members      60 at 70.182 in
short members     60 at 62.681 in
in all            120, from 15 log sections
```

Add spares. Two or three of each length costs very little and saves an entire afternoon the first time a piece splits or a cut goes wrong.

CHAPTER 17 · EXPLAIN

Green Wood Moves, and a Split Moves Least

A wedge closes its own angle as it dries -- which is why it checks less than a squared timber, and why the key is cut last

A tree is mostly water. The frame of this dome comes off the saw weighing 2,478 pounds in green pine, and a good part of that weight is going to leave through the surface of the wood over the next few months, whether you plan for it or not.

When water leaves wood, the wood gets smaller. Not evenly: wood shrinks about twice as much going round the growth rings as it does going across them. For the pine this book prices, green to bone dry, that is 7.4 percent round the rings and 4.8 percent across them.

That one fact explains two things about the method. It explains why a split wedge checks less than a sawn, squared timber. And it explains why the key is the last thing you make.

Two sawn faces meeting at the ridge and opening to the room: the V the key fills.

Why a boxed heart tears and a wedge does not

Picture the end of a squared beam cut with the pith in the middle: a boxed heart. As it dries, every ring wants to get shorter round its circumference, and it cannot, because the ring is a closed loop held at full size by the wood inside and outside it. The stress builds until the wood gives, and it gives along a radius. That split running in from the surface is a check.

Now picture one of this book's wedges. It was split along a radius to begin with. Its rings are not closed loops any more; they are arcs, each one free at both ends. When an arc wants to get shorter it simply does, and the two sawn faces swing toward each other a little. The wedge **closes its own angle** instead of tearing.

The tree held the tolerance when you split it. It keeps holding it while the wood dries.

What the key sees

Here is the consequence you have to plan for.

A wedge dried from green down to 12 percent moisture closes from 45.0 degrees to 44.28: 0.72 degrees. The V in every seam is the sector angle less the fold between the two panels (Chapter 20), and each of the two members turns its face away by half of that. So the V opens by the whole of it:

the tighter V, cut green	22.54 deg
the same V, once dry	23.26 deg
width at the room side, green	2.35 in
width at the room side, dry	2.35 in

The width at the room side barely changes, because the faces also get shorter as the wood shrinks across the rings: 6.0 inches deep becomes 5.83. What changes is the **tilt**. Each face swings 0.037 inches at the room side, so a key cut to fit the green V now touches along one edge and rocks.

That is small, and it is enough. A key is only worth its space if it bears on both faces for its whole depth.

The rule: dry the frame, then make the keys. Split and build green if you must, because green wood cuts and splits more easily. But print or cut the keys last, to the V the dried frame actually has, and bolt the seams up after the frame has come down to its working moisture. The channel's own air is what gets it there.

How long, and how much water

Drying time goes roughly with the square of the thickness the water has to travel through. A wedge's thickest point is the biggest circle that fits inside its sector, 3.3 inches across, against the 12.0-inch log it came from. So:

- a wedge dries about **13 times faster** than the round log would;
- and about **5 times slower** than a two-by, which is thinner still.

Splitting is, among other things, a way of drying a log.

The water it gives up, green down to 12 percent, is about 563 pounds: 67 gallons, from the frame alone. All of it has to leave through the surface of the wood and then leave the building. If the frame is already standing when it does, that water is in the seams, and Chapter 32 is about moving it.

Wood below 19 percent is what lumber grading calls dry: under that line, rot cannot get going. Above it, it can. Framing indoors settles to about 12.

Drying the stack

1. **Stack the wedges off the ground**, on bearers, in the shade, with the round face up so the rain that gets in runs off the bark side.
2. **Sticker every layer.** Put thin spacers across the stack at every layer so air reaches all three faces. A wedge laid sawn face on sawn face dries from one side and cups.
3. **Cover the top, not the sides.** A roof over the stack keeps rain off; open sides let the wind through, and the wind does the drying.
4. **Seal the end grain.** Water leaves end grain many times faster than it leaves the side, which is why logs check from the ends. A coat of wax or paint on each end slows it to match the rest.
5. **Buy a moisture meter.** It is the cheapest tool in the book and the only one that tells you when to make the keys.

The figures in this chapter use handbook shrinkage for pine, and they are declared as such:

fibre saturation	0.3 kg/kg	standard
shrink radial	0.048 -	standard
shrink tangential	0.074 -	standard
board thick in	1.5 in	nominal
char mm	3 mm	assumption
oil m2 per l	8 m2/L	estimate
oil coats	3 coats	assumption

CHAPTER 18 · HOWTO

Char the Outside, Oil It, and Leave the Faces Bare

Yakisugi and linseed on the round face only, because the two sawn faces carry the key, the bolts and the seam's air

Every wedge has three faces, and only one of them wants a finish.

The two sawn faces are a joint. One of them butts into the side of the next member in the panel (Chapter 20); both of them meet their neighbours' faces across a seam; and they are the walls of the channel the seam's air runs down (Chapter 29). Anything you put on them changes the fit of the key and slows the one thing the channel is there to do, which is carry water out of the wood.

The third face is the round one: the bark face, turned out toward the weather. That is the face that stands in the sun and rain for the weeks the frame is up before its cap goes on, the face that lives in the vented gap under the cap afterwards, and the face a frame without a cap shows the world. It gets the finish.

Round face out, point in: the only face the weather ever sees.

The whole frame has 24 square metres of bark face and 62 of sawn face. The finish in this chapter goes on the smaller number.

Charring the round face

Charring wood is old. The Japanese name for it, yakisugi, means burnt cedar, and the idea is simple: a layer of char on the surface has nothing left in it for fungus or insects to eat, sheds water, and does not break down in sunlight the way bare wood does. It is also a finish that costs a tank of gas.

Char goes on wood, not on bark. Chapter 15 lets the bark stay on if you want it, and that is a fine choice for a frame inside a skin – but then skip the char. To char the round face, take the bark off first, green, with a drawknife, in the first week while it still comes away in strips. The steps below assume that face is bare wood.

1. **Char before you build**, with the wedges on bearers on bare ground, well away from the stack, the chips and the sawdust. Never char a standing frame.
2. **Use a propane torch with a wide head**, and work along the bark face in one steady pass, holding the flame until the surface cracks into a fine alligator pattern. Stop there; a deep char just crumbles away.
3. **Keep the flame off the sawn faces and the ends.** Stop short of the two sawn edges so they stay crisp. The key, the butt joint and the bolts all bear on those faces.
4. **Brush it.** A stiff brush along the grain takes off the loose soot and leaves the hard char. Brush more for a lighter, textured face; less for a darker one.
5. **Let it cool and blow it off** before any oil goes near it.

A single pass of about 3 millimetres of char takes 3.9 percent off each member's cross-section, all of it from the bark face, which is the part of the section furthest from the seam. That is the number that does not help, and it is small: the frame's strength argument in Chapter 1 does not rest on the last few millimetres of the bark side.

Char is not fireproofing. A charred surface is slower to catch than a bare one, and that is all it is. Do not count it in anything to do with fire.

Oiling it

Linseed oil is pressed from flax seed, and it does not dry by evaporating: it takes up oxygen from the air and sets into a tough film inside the surface of the wood. Over char, it binds the char so it stops dusting off on your hands, and it deepens the colour.

- **Raw or boiled.** Raw linseed soaks in deeper and sets slowly, over days. Boiled linseed has driers added so it sets in a day or so. Over char, either works; raw is the traditional choice and the more forgiving.
- **Thin coats, wiped off.** Brush or rag it on, let it soak for twenty minutes, then wipe off everything that has not gone in. A thick coat that sits on the surface stays tacky for weeks and collects dust.
- **3 coats**, a day or more apart.

How much the frame takes, at a declared 8 square metres a litre on rough or charred wood:

```
bark faces only, 3 coats      9 litres
every face, 3 coats          32 litres
```

Oiling everything would take more than three times the oil to finish surfaces that should not be finished at all.

What oil will not do. Linseed is a food for mildew in wood that stays damp. On a bark face that dries between wettings, that does not matter. On a sawn face inside a seam, where the whole point of the channel is to move moisture, a film of oil slows the drying and feeds the mould. Leave the faces bare until the frame is dry, and then leave them bare anyway.

Oily rags catch fire on their own

A rag soaked in linseed oil can set itself alight. The same reaction that sets the oil gives off heat, and a crumpled rag holds that heat in until it reaches the point where cloth burns. It happens hours later, in a bin, with nobody there.

Lay used rags out flat, one layer thick, outdoors, until they are stiff. Or drop them in a metal can full of water with a lid on. Never put them in a pile, a bag or a bin with anything else, and never leave them in the shop or in the frame.

The figures in this chapter are declared, not measured:

```
fibre saturation      0.3 kg/kg       standard
shrink radial         0.048 -         standard
shrink tangential     0.074 -         standard
board thick in        1.5 in          nominal
char mm               3 mm            assumption
oil m2 per l          8 m2/L          estimate
oil coats             3 coats         assumption
```

Linseed Oil, Where It Belongs

Read the can, set up the rag station, oil the faces people touch, and do not ask a finish to be a roof, a seal or a repair

A video that this project's research pipeline ingested – *20 LOST Linseed Oil Tricks Old-Timers Swore By That Outperform Anything at the Hardware Store, by The Lost Handyman* – offers twenty uses for linseed oil, and some of them are good ones for a dome workshop. This chapter sorts them by what survives contact with the building: exposed interior wood, shelves and trim, and hand tools on one side; seams, weather and windows, where a finish has to work with the rest of the assembly, on the other.

The source is a list of claims, not a comparison test, so this chapter shows the working, the quantities and the limits before it calls anything a saving. Chapter 18 already covers the round face of the frame; this one is everything else.

The rag station comes first

Before the can is open, set up for the used cloths. Linseed oil cures by taking up oxygen, and that reaction gives off heat. A heap of oily cloths traps the heat and can ignite with no spark at all.

Follow the label and your local fire service. The usual practice: cover used rags completely with water and an oil-breaking detergent in a metal container with a tight lid, stored outside and away from anything that burns, and arrange proper disposal. Keep them out of pockets, sawdust and ordinary bins. A wet rag that dries out in a heap is dangerous again.

Read the can

Raw, boiled and polymerised linseed oil are not interchangeable.

- **Raw** oil cures slowly, over days.
- **Boiled** oil, as sold in hardware stores, is usually raw oil with drying additives; it is not boiled, and you should never try to boil oil yourself.
- **Polymerised** oil has been heat-treated by the maker and cures differently again.

Read the ingredients, the safety sheet and the intended uses. A name like *natural*, *boiled* or *Danish* does not identify a formulation. For anything that touches food, choose a product sold for that use and follow its cure instructions – the source's rule that only raw oil is food-safe is too broad in both directions.

Two labels, two schedules

Two products from one maker show why a generic timing cannot be applied to every can. These are the manufacturers' claims, marked as such, not measurements:

```
Danish oil        rub dry after at least 5 min, cure at least 8 h
oil-and-wax       rub dry after at least 60 min, cure at least 24 h
```

Both are minimums. In cold, damp conditions neither guarantees the surface is ready. Use the current label of the product you actually bought.

Prepare the member, then apply thinly

1. **Inspect.** Reject decay; deal with splits and loose joints through the repair the design calls for. Oil cannot restore lost strength.
2. **Dry.** Let the wood reach the moisture the finish needs, and check it with a meter (Chapter 17).
3. **Clean.** Remove dirt and unsuitable old coatings, smooth splinters, clear the dust.
4. **Mask.** Mark the places that must bond to glue, tape, sealant or a gasket, and keep the oil off them.
5. **Test an offcut** for colour and cure before the real piece.

6. **Apply a very thin coat**, evenly, with the applicator the label specifies. After the product's soaking interval, rub away everything that has not gone in until the surface is dry to the touch.
7. **Cure** with the ventilation and temperature the label asks for, and look at the surface before the next coat. Never bury a sticky layer under more oil.

Watch the end grain, which drinks more, but stop drips running into joints. And never close an uncured finish inside a panel or an air passage people breathe.

Apply thinly, then wipe dry.

How much, and how long

Measure wood, not floor area. The solver's stock schedule is 664.3 feet of member, and each flat radial face is 6.0 inches deep. The film works an allowance from declared planning inputs, marked by kind:

coats	2 coats	assumption
faces	2 flat faces/member	assumption
coverage	350 sq ft/US gal/coat	estimate
handling	15 % extra volume	assumption
minutes	2 min/member/coat	estimate
danish_wait	5 min minimum	claimed
danish_cure	8 hours minimum	claimed
wax_wait	60 min minimum	claimed
wax_cure	24 hours minimum	claimed
label_coverage	1000 sq ft/US gal	claimed

That gives a planning envelope of 664 square feet, and:

the declared coats	3.80 US gal
with the handling allowance	4.37 US gal
at the label's best coverage	1.33 US gal

Two cautions about those numbers. First, the envelope counts *both sawn faces* of every member as a reference area. It is not an instruction to oil them – Chapter 18 argues they should stay bare, and its own allowance for the round faces alone is 9 litres. The two figures answer different questions; the

difference between them is the sawn faces. Second, coverage is the input that decides everything, and the label's best figure is a claim made on smooth wood, not a rough-sawn wedge. Measure how much you use on a known offcut through the whole coat schedule, and scale that.

And the number that argues against it: hand application is labour. At an estimated 2 minutes a member a coat, the coats above are **8.0 hours of active work**, before any preparation, access, inspection or waiting for cure. A cheap can can still make an expensive job.

Where it does not belong

Not in the seam's working surfaces. Oil on an exposed interior face is a finish; oil on a sealing land is a contamination. Adhesives and tapes need clean, compatible surfaces. Keep the printed key, the gasket lands, the drainage path and the services clean, check both makers' compatibility before coating wood beside a gasket or sealant, and never count an oil coat as an air seal, a rain seal or a cure for condensation.

Finish decisions follow the assembly.

Not as the weather skin. The source describes linseed oil as outdoor protection and waterproofing. A dome still needs its watertight layer (Chapter 28), flashed openings, drained joints and a way to dry. Plain oil establishes none of those, proves no resistance to sunlight, rot or insects, and damp, linseed-rich wood can grow mildew. Outdoors, use a finish rated for the exposure.

Not as glazing in a dome roof. Linseed-based putty has a real place in traditional sash windows, with mechanical retainers holding the glass – putty alone must never carry it. It is not a design for sloped or overhead dome glazing, which needs a complete system specified for that position, and the video's homemade chalk-and-oil putty does not belong in the dome's weather seams.

The parts people touch, and keeping them

Inside, linseed oil is at its best: shelves, trim and sound furniture. Test for yellowing and for compatibility with any old coating. An oil-and-wax blend gives a low sheen, but wax complicates later gluing and recoating. Floors need a floor-rated system; kitchen surfaces need a product sold for food contact.

Tools. The handle trick transfers well when the handle is sound: inspect the hammer, mallet and jig handles for cracks and loose heads, then finish them thinly and let them cure before gripping. Oil does not repair a broken handle. A drying oil can protect clean, non-moving steel in storage, but keep it out of bearings, hinges, the saw chain and brake, electrical contacts and fastener threads, which need their own products.

Maintenance. Keep a finish record: product, batch, date, preparation and where it went, with a sample piece in the building's notes. Inspect for wear, tackiness, mildew and water marks; find the cause before renewing. Grey wood darkens beautifully under oil, but colour is not strength – oiling does not reverse decay, and it earns no credit for waterproofing a foundation.

So: use linseed oil where a maintainable wood finish suits the job – accessible interior faces, trim and sound tool handles. Keep seams and bonding surfaces clean, give the roof its own weather system, choose glazing for the actual opening, base the quantity on measured wood, budget the hand labour, and end every day with every oily cloth accounted for.

PART 4

The Panel

“Forty panels come off one board, and nothing on the board is adjusted between them. Panel forty is panel one.”

The joint that removes the hub, the nine processes that make the whole shop, the one fixture that makes forty panels the same, and where the door goes.

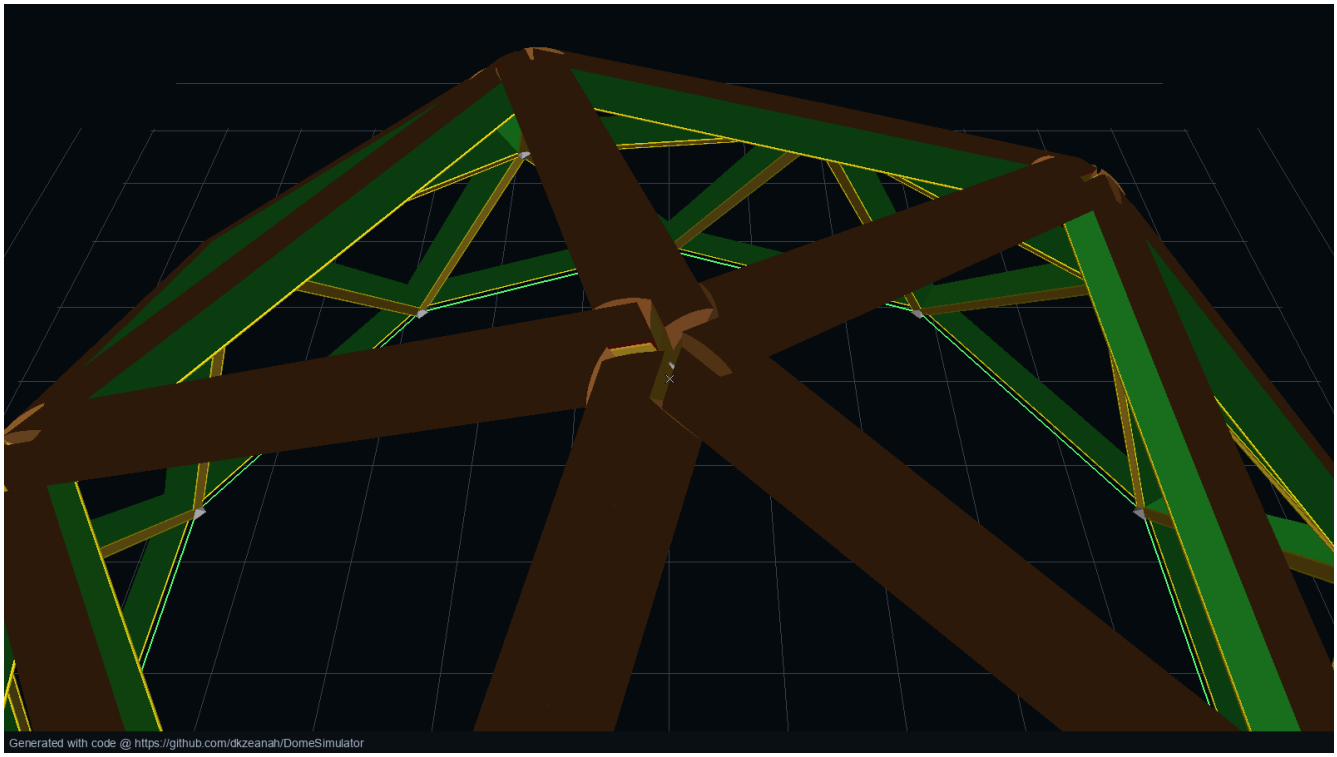
CHAPTER 20 · EXPLAIN

Nothing Meets at the Corner

Three sticks, each butting into the side of the next, and not one of them reaching the vertex it is named after

In a classical geodesic frame, five or six struts arrive at a point and something has to hold them there. That something is a hub, and hubs are why most people give up on domes.

This frame has no hubs, because nothing arrives at the point.



Five triangles meeting, and an empty corner.

The pinwheel

Each panel is built complete before it goes anywhere near its neighbours: three members, closed into their own triangle.

Inside that triangle, each member's end butts into the **side** of the next one, a little way along from the corner. Not into its end. Into its face. The three of them chase each other round the triangle, which is why it is called a pinwheel, and the mathematical vertex in the middle of that arrangement has nothing in it at all.

Three consequences, and they are the argument for the whole method:

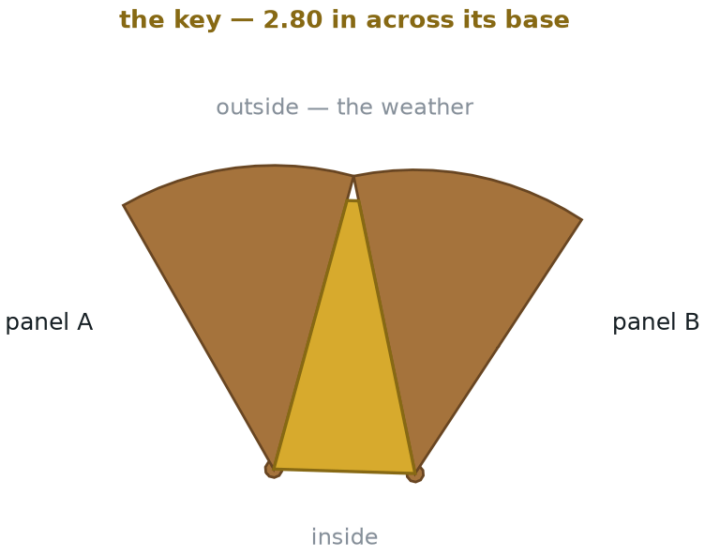
No hub. There is nothing to buy, machine or get wrong.

Flat sawn face on flat sawn face. A split wedge already has two flat faces at a known angle. The joint presses one against another. Nothing is mitred to meet at a point.

The corner is empty, so the corner cannot be crowded. The failure mode of a hubbed dome — five struts whose ends all want the same cubic inch — does not exist here.

What is actually at a seam

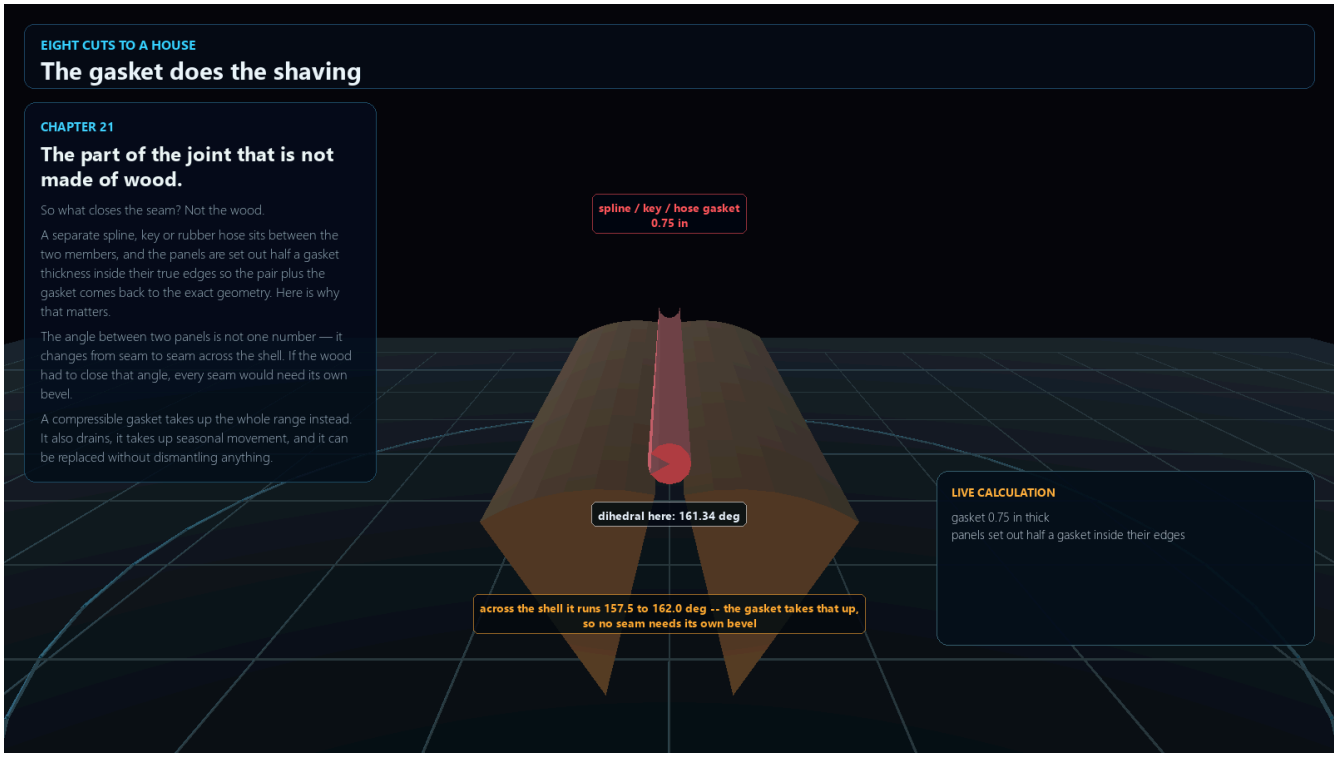
Where two panels meet there are **two** members, one belonging to each, with the key between them.



A seam in cross-section, at the angle the solver puts it: two sectors, the key between them, drawn to the solved dihedral.

That picture is a real cut. A plane through the seam, perpendicular to its own run, intersected with the solver's own geometry — not a diagram of the idea. Two 45.0-degree sectors, points inward, and the 26.971 degrees of sector angle they cannot close between them.

The key fills that. It is 2.798 inches across its base on an A seam and 2.345 on a B, and it is not a manufactured part



The gasket does the shaving: the part of the joint that is not made of wood.

: it is the truncated point of another raw sector, cut from the same log at the same angle.

Why put the variation there? Because the fold between two panels is not one angle. Across this dome it runs from 18.029 to 22.459 degrees. That spread is small in degrees and large in workshop terms: cut into the members, it would be a saw setup per seam and sticks that are no longer interchangeable. Moved into the key, the wood stays generic and the fussy part is small, cheap and replaceable.

The price of a panel being a thing

40 panels times three members is 120. The shell has only 65 edges.

So 55 members exist twice over, once for each panel meeting there. That is nearly twice the timber a shared-strut frame would use.

It is not lumber somebody forgot to remove. It is the price of a panel being an **object** rather than a position in an assembly — and what you buy with it is everything in Chapter 24: a panel you can build flat, check flat, carry, sheathe and lift as a finished thing, on a bench, before anything is standing.

Two trees instead of one and a half, against forty panels you can make indoors in winter. That has been an easy trade every time I have had to make it.

The pinwheel, measured

The pinwheel is not free. Each member lies with its bark face on its own edge line, so its centre sits half a member's width inside that line, and that offset is what opens the gap at the vertex no stick reaches. The bearing area where one member's end lands on the next one's face is the joint's real capacity; the vertex gap is what has to be trimmed clean so that panels do not fight each other at a corner. Both come out of the same solve, and both are what the jig in Chapter 24 has to hold.

Where the error goes

Chapter 1 claimed the frame absorbs error. This is the mechanism.

A member that is a bit long, a bit short, or at a bit of an angle changes the width of the gap at its seam. The gap already has a key in it. A key is a wedge, and a wedge accommodates a range of gaps by sitting a little deeper or a little shallower.

So the error does not accumulate around the triangle and it does not propagate into the neighbours. It lands in a joint that was always going to be filled, and it changes how far in the filling goes.

This is also why the head end of every member is left long and cut in place — Chapter 24 again. There is exactly one end of one stick where error is allowed to leave the building as offcut, and the method spends it deliberately.

What the butt cut actually is

One correction, because this project published the wrong version of it and some of that is still out there.

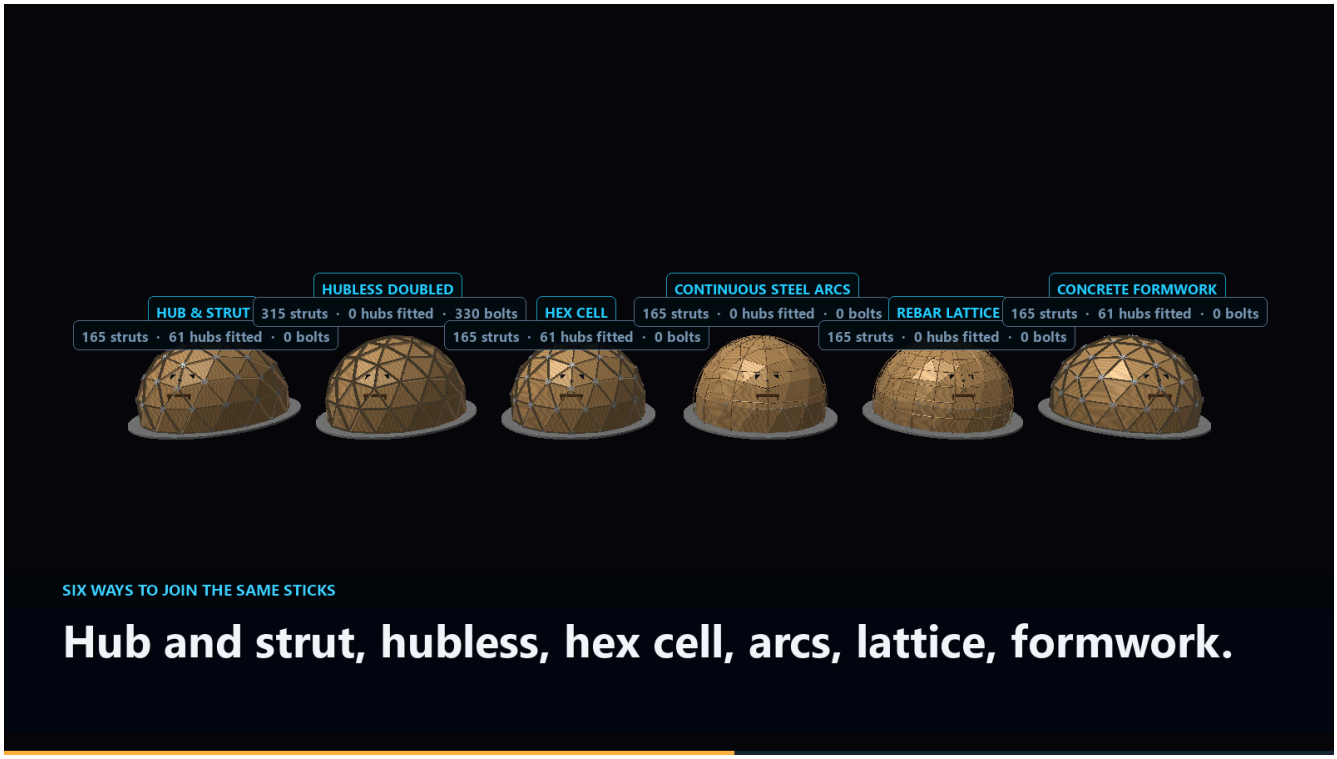
The butt is a **compound** cut. It has a bevel and a mitre.

The bevel is constant at 22.5 degrees — half the sector angle — because it comes from how the log was split rather than from where the member sits in the dome. The mitre takes 3 values across all 120 members.

The wrong version is still on screen in one of the films that come with this book: the *Eight Cuts to a House* film's "pinwheel, measured" screen says every end in the frame is a square crosscut. Its other figures – each member sitting half its width inside its edge line, no wood reaching the vertex – stand. Its claim about the ends does not.

So the honest claim is not "no compound angles". It is **three saw settings for a hundred and twenty members**, which is a better claim anyway, and true.

What the pinwheel removes is the shared vertex. It does not remove the mitre.



Six ways to join the same sticks. This book uses the second.

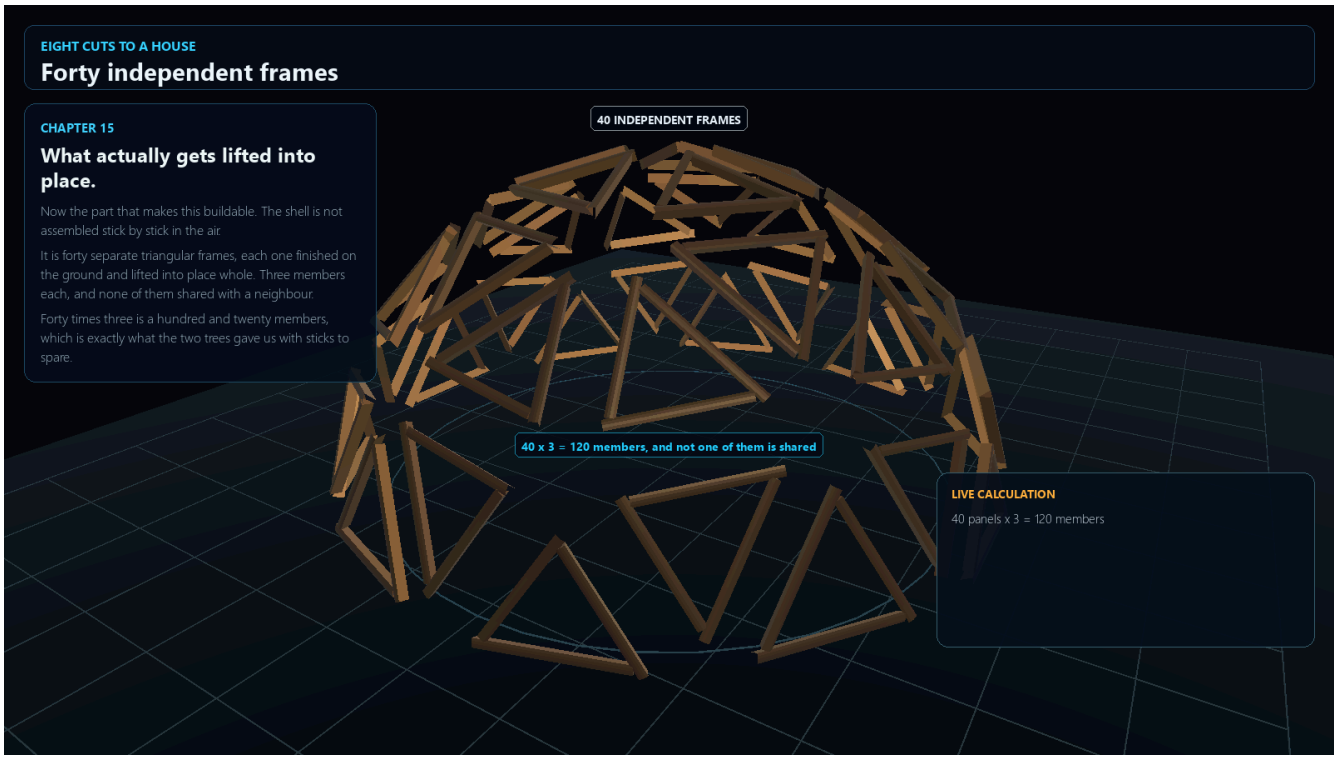
CHAPTER 21 · EXPLAIN

Forty Frames, Lifted Whole

Each triangle is finished before it meets its neighbours -- which makes a panel a module you can swap, and the air a place nobody works

The shell of this dome is not assembled stick by stick in the air. It is 40 separate triangular frames, each one finished on the ground and lifted into place whole: three members each, closed into their own triangle, none of them shared with a neighbour. 40 frames of three is 120 members.

That changes where the precise work happens. Each triangle is built flat, on a bench, at waist height, where a square end and a screw are easy (Chapter 24). What happens up a ladder is bolting finished frames to each other. Nobody is holding three sticks at an angle in the wind. There is not one connector between two triangles that has to be bought, machined or waited for – every joint between panels is a lap between two flat sides.



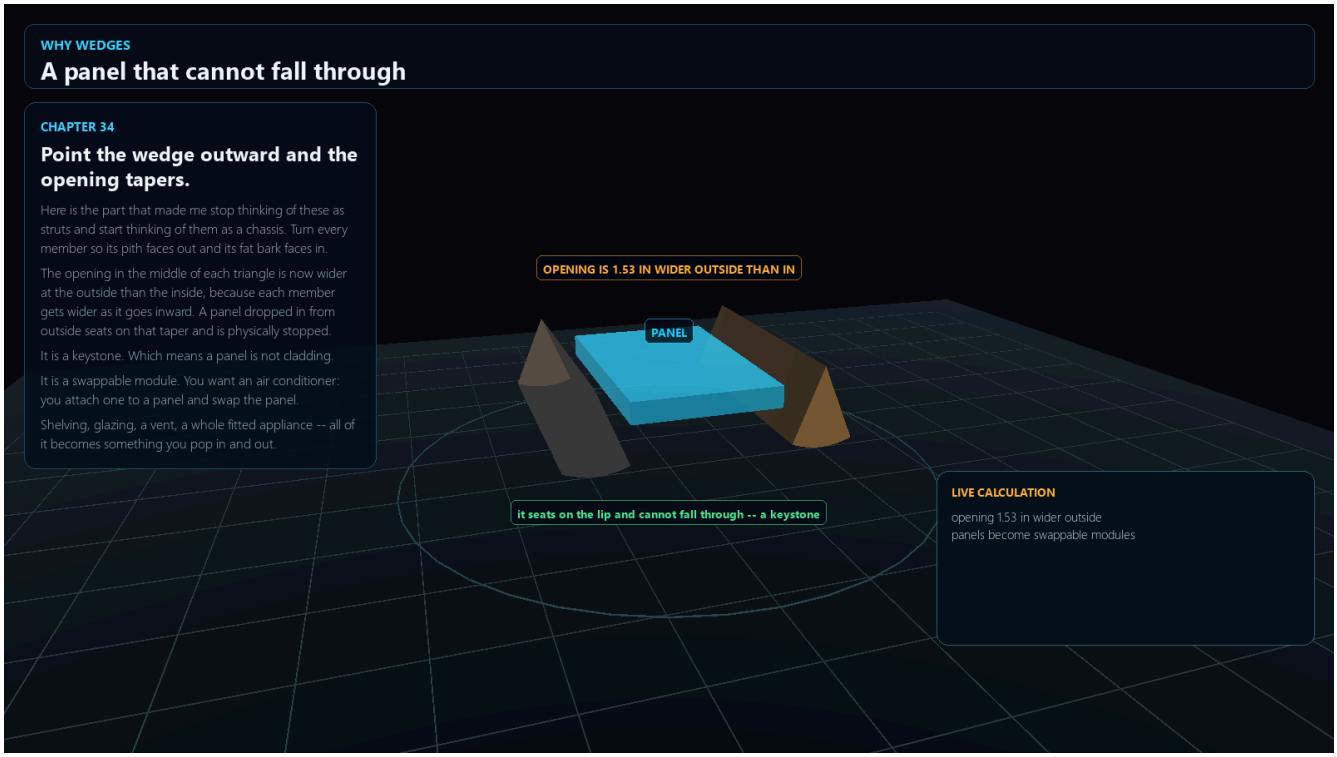
Forty frames, built flat, lifted whole.

A panel that cannot fall through

Here is the property that turns a frame into a chassis.

A wedge gets wider toward its point. So the opening in the middle of a closed triangle is not a plain hole; its sides slope, because each member's faces lean. Which way they lean depends on which way the members point. Turn them so the opening is wider on one side than the other, and a panel dropped in from that side **seats on the taper and is physically stopped**. It is a keystone.

Which means a panel is not cladding nailed onto a frame. It is a module in a socket. You want an air conditioner: it goes in a panel. You want a window, a vent, a solar collector, a hatch: each is a panel that drops into the same seat. This is the frame-level reason for the argument of Chapter 3.



The opening tapers; the panel seats.

Four rotations, four buildings

Stiffness was one reason to choose which way the wedge faces (Chapter 15). Turning it changes the building in other ways too, measured off the same cross-section:

- **Point in** (this book's reference): the panel is caught from the inside, so it goes in from inside and cannot fall out; and the services run on the warm side.
- **Point out**: the panel is caught from outside, and the channel's services sit toward the weather.
- **Turned sideways**, either way: the opening's sides go parallel. No lip, so panels need fixings, but the sawn faces come much closer to flat, which is what a sheet of sheathing wants to land on.

Same stick, same wood. The choice is which building you are making.

What holds it together

The first frame of this project was held together with folded steel V brackets: a strip bent to the corner angle, seated inside the vee between two struts, with long legs screwed down each one. A V bracket does not care what section it is holding, which is exactly why the salvage dome of Chapter 5 could use whatever the chainsaw made.

The wedge frame does not need it, because every member is the same shape as every other one, so the joint gets to be simpler:

- **Inside a triangle**, the members are glued on their mating faces and screwed in from the far side of the neighbour, so the fastening pulls the joint closed rather than hanging off it.
- **Between triangles**, panels bolt to each other edge to edge, and the bolt passes through the seam key that was going in anyway (Chapter 20).

No brackets to order, no hubs to weld, no single part the whole build waits on.

CHAPTER 22 · EXPLAIN

Nine Processes, Whatever Size You Build

Nine operations turn standing timber into a shell. Not nine categories with sub-steps hiding inside them -- nine setups

The last chapters counted parts. This one counts *verbs*, and the verb list is the shorter of the two.

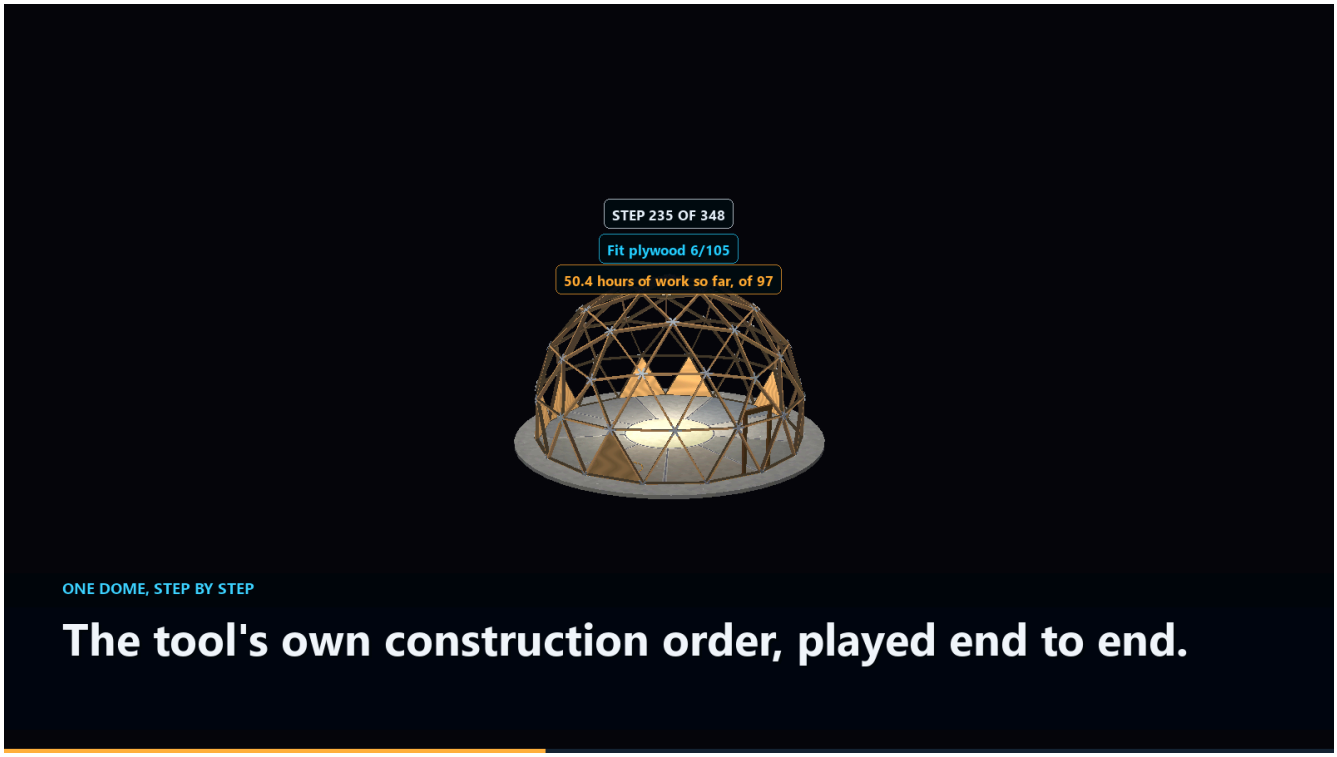
There are 9 things you do to turn standing timber into a shell. Nine. Not nine categories with sub-steps hiding inside them — nine operations, each one a distinct setup with its own tool, its own bench and its own way of going wrong.

You can learn them in an afternoon of reading and be bad at all of them by the end of the first week, which is exactly the right number of things to be bad at.

Here they are, in the order they happen:

> *fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass*

That is the whole shop.



The tool's own construction order, played end to end.

Everything in this book that is not thinking is one of those nine.

Why verbs matter more than parts

Parts are things you buy. Processes are things you **get better at**.

A list this short is a list you can actually attack. Nine places to put your attention is a manageable number of places. A list you cannot hold in your head is not a list, it is a fog — and most building methods are a fog, which is why people who are perfectly capable of building things do not build houses.

Fix one process and every dome you ever build gets cheaper. That is not true of fixing one part.

The nine, one at a time

1. Fell. Drop the tree, buck it into sections the length of your longest member plus waste. One chainsaw. The failure is felling into something you did not intend; the fix is the same as it has always been and is not in this book.

2. Rip. Rip each section radially into 8 sectors with the chainsaw, every cut through the heart of the log. The failure is a rip that wanders off the radius on a section with a knot in it; the fix is to buck around the knot and accept a shorter stick.

3. Crosscut. Cut each stick to length — 70.182 or 62.681 inches — with the compound butt at one end and nothing yet at the other. This is the one cut that can be batched, and Chapter 24 is why. The failure is cutting both ends; the head end is not yours to cut yet.

4. Fold. Assemble three sticks into a panel on the jig, pull the butts up tight, and flush-cut the three heads in place. "Fold" because the triangle closes around itself. Forty times. The failure is doing it in the wrong order and trapping the last stick, which the jig's stage nine is specifically about.

5. Drill. Pilot and set the threaded inserts in the wedge faces — the ones Chapter 3 argues for. Done on the bench while the panel is flat, which is the only time it is easy.

6. Screw. Join panel to panel, key into seam, hardware into insert. The failure is treating this as structural; it is not. The shell holds its shape because it is a shell.

7. Raise. Ten panels on the base ring, thirty above them, five around the apex. Chapter 27 has the sequence. The failure is bracing too little on the first course, which is the only course that needs it.

8. Sheathe. Close the triangles, then the breather, then the cap. The one watertight layer of Chapter 28.

9. Glass. The openings — and the reason it is last is that an opening is a panel you did not close, so it costs nothing to leave the decision until the shell is up and you can see where the light actually goes.

What is not on the list

No milling. No planing. No jointing, thicknessing, moulding or sanding. No welding, no machining, no hub fabrication, no casting.

Every one of those is a process that some other building method needs and this one does not, and every one of them is a machine, a shed to keep the machine in, and a skill.

The list is short because of the decision in Chapter 1. A frame that is allowed to be lumpy does not need the processes whose only purpose is precision.

Why the list does not grow with the house

Here is the part that matters for anybody thinking about more than one building.

Make the dome twice as wide and the parts list grows: more timber, more sheathing, bigger everything. The *process* list does not change at all. It is still fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass.

Build a second dome and it does not change. Build ten and it does not change.

That is what makes this a manufacturing argument rather than a carpentry one. Every hour you spend getting better at one of nine operations is an hour invested in every dome you will ever build, including the ones you have not thought of yet.

7 of the 9 happen on a bench rather than on a ladder, which is the other half of why this is a one-person method.

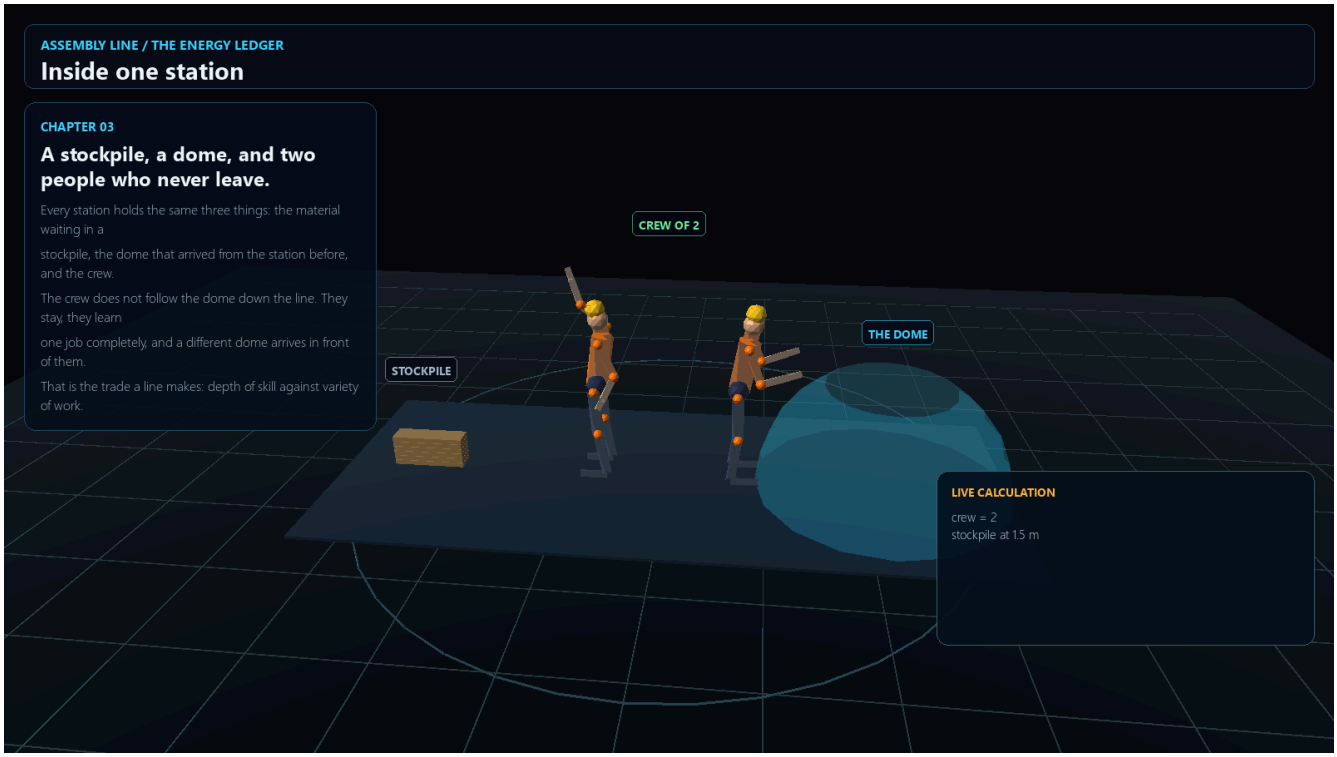
And it is why Chapter 43 can talk about a network of buildings without the argument falling apart: the thing that would normally not scale — the skill — is nine items long.

The honest limit

Two things this does not say.

It does not say the nine are equally hard. Fold is most of the work and rip is most of the danger, and a list that gives them one line each is flattening something real.

And it does not say nine is a natural number. Count differently and you get seven, or eleven. What is not arbitrary is the *order of magnitude*: this is a method with nine-ish operations, and the alternative is a method with fifty.



Inside one station: the manufacturing view of the same nine.

That difference is the argument, and it survives any reasonable recount.

What the Work Costs the Body

Follow two people through every motion of a factory-built dome: the parts that look like effort are cheap, and holding still is the bill

Chapter 22 counted the work as operations. This chapter counts it as food.

The study it comes from is not this book's dome. It is a factory line building a larger product dome, followed part by part through fifteen stations, with two people at each one:

product	Iris-25, home, radius 4.39 m, 4V, Studio
stations	15
crew	2 x 82 kg, 1.75 m, age 34
resting metabolic rate	84.7 W = 1749 kcal/day

A line does not reduce the work; one crew doing every station does exactly the same labour. What a line buys is less **waiting**: split the work across stations and a dome comes off the end every cycle instead of every build. And the line runs at the speed of its slowest station – there, the frame. Speeding up any other station buys nothing. The lessons below hold for any build where people place parts, including one person building this book's dome in a field.

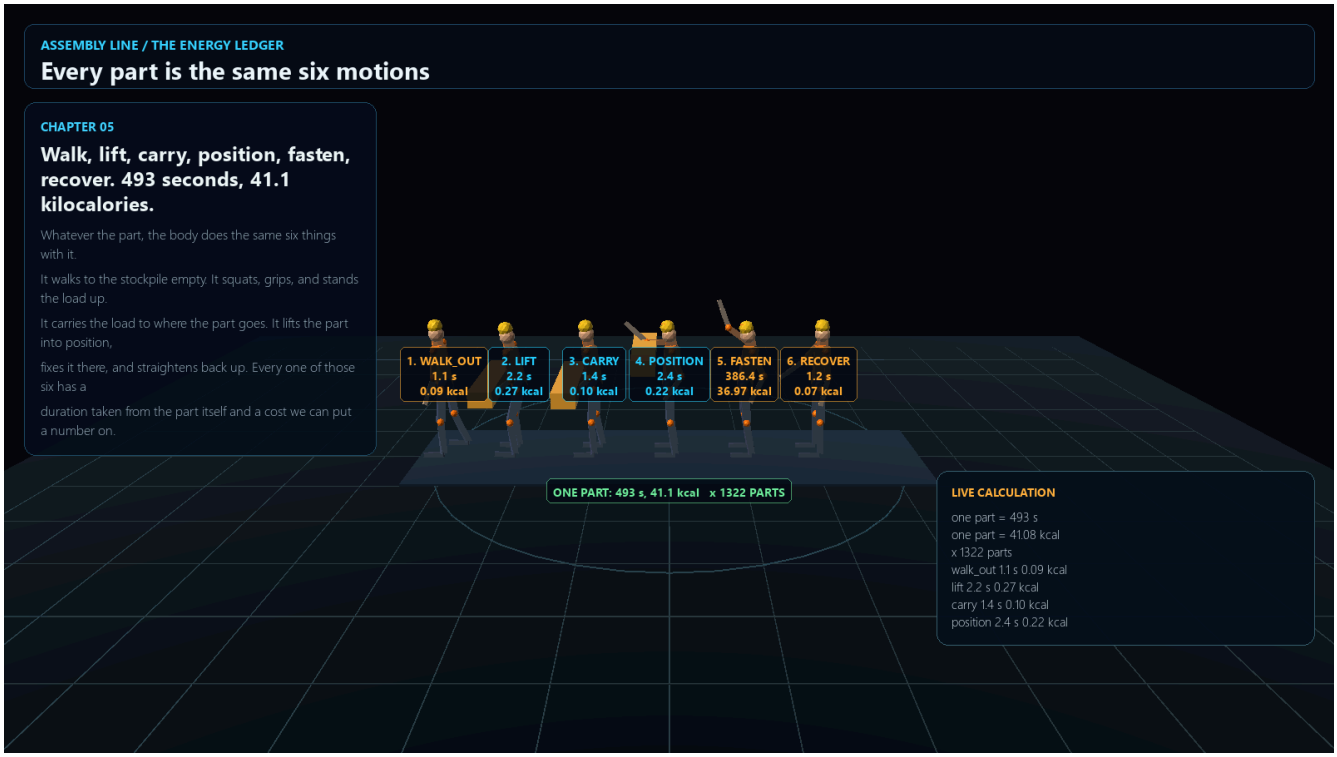
Six motions

Whatever the part, the body does the same six things with it. It **walks** to the stockpile empty. It squats, grips and **lifts**. It **carries** the load to where it goes. It **positions** it. It **fastens** it. It **recovers**, straightening up.

Four results from following those motions through the whole building:

- **The walk is cheap per trip and not cheap in total.** Which is why where the stockpile sits is a decision, not a detail left to whoever unloads the truck.
- **Carrying is not linear in the load.** The published equation for walking with a load has its load term squared: twice the weight costs more than twice the energy. That one fact is the argument for carts, conveyors and a closer stockpile.
- **Height changes the price.** Above 1.75 metres the arms are above the heart, the posture costs more and the crew tires faster. How much of a shell falls into that band is decided by its geometry – a design decision disguised as a shape.
- **Fastening is almost the whole bill.** It raises nothing, does no work against gravity, and still consumes most of what the crew burns, because it is minutes per part of holding a posture, gripping a tool and resisting its torque. The body pays for holding still.

And the rest is not slacking. Industrial engineering has added a recovery allowance to every task time for a century, with more for overhead work, because a schedule written without one is a schedule that will not be met.



Walk, lift, carry, position, fasten, recover.

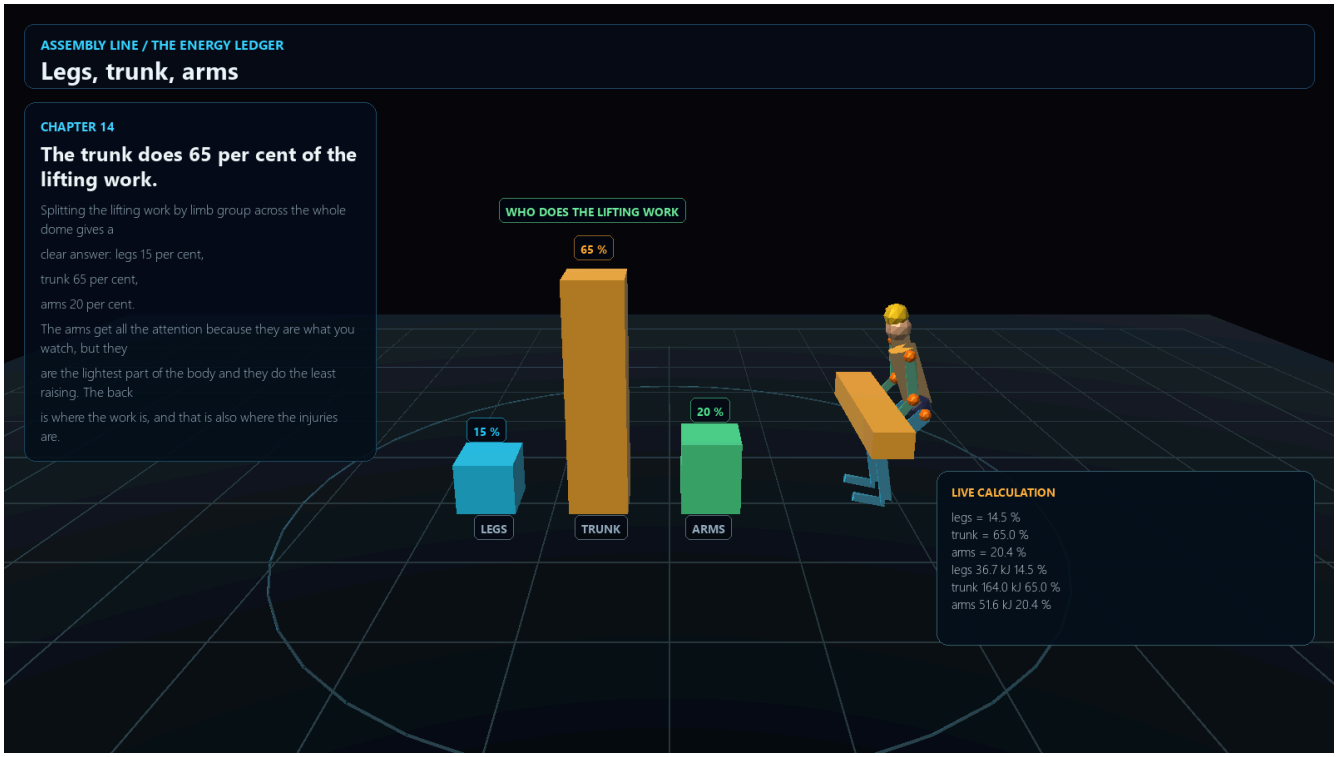
The body is the heaviest thing you lift

To cost a movement you have to know what is being moved, and the biggest thing moved is the worker. The model uses the standard anthropometric tables of biomechanics, built for a 82 kg person: each body segment a fixed share of the total mass. The trunk alone is about half of it. So in every placement the body rises and falls along with the part, and outweighs it several times over. Split by limb, across the whole building:

```
mechanical lifting work by limb group:
  legs            36.7 kJ   14.5 %
  trunk          164.0 kJ   65.0 %
  arms            51.6 kJ   20.4 %
```

The arms get the attention because they are what you watch, but the back does the work – and the back is where the injuries are. That is the honest case for a lower stockpile, a taller bench, and parts presented at waist height.

Above 23 kilograms, manual-handling guidance says a lift takes two people, and the model splits every such part between them. Work above the shoulders costs more per second and needs more recovery for the same output, and whenever the geometry pushes work up there, the cost lands on a body rather than a spreadsheet.



The trunk does most of the lifting.

Exact, and modelled

One part of this is exact: raising a part is its mass, times gravity, times the height it rises. No estimate anywhere in it.

Turning that into food is a model, and it says so. A muscle holding a panel steady does no mechanical work and still burns fuel, so there is no route from joules of lifting to kilocalories without published figures for how hard each task is. The model names every one it takes on authority:

```
external constants this model takes on authority:
  met_definition              1 kcal/kg/h    Metabolic equivalent of task, standard definition
  met_lift                    5.5 METs       Compendium of Physical Activities: lifting items continuously
  met_position                4 METs         Compendium: carrying and placing building materials
  met_fasten                  3.5 METs       Compendium: installing or repairing, general
  met_fasten_overhead         4.2 METs       Compendium: work above shoulder height
  met_recover                 2.5 METs       Compendium: standing, light activity
  pandolf_terrain_factor      1 dimensionless Pandolf, Givoni and Goldman (1977) load-carriage equation
  concentric_efficiency       0.25 fraction  Standard exercise-physiology value for positive muscular work
  rmr_mifflin                 kcal/day       Mifflin-St Jeor (1990) resting metabolic rate
  pfd_allowance               0.15 fraction  Industrial engineering: personal, fatigue and delay allowance
  pfd_overhead_extra          0.1 fraction   Ergonomics practice: added recovery for above-shoulder work
  met_pause                   1.5 METs       Compendium: standing at rest
  sustainable_shift_watts     350 W          Occupational physiology: sustainable 8-hour working rate
```

That is why two very different efficiencies are both true:

```
mechanical work done    0.363 MJ = 87 kcal
  as a share of food    0.2 %
  ceiling at 25 % muscle  25 %  (everything else is posture, grip and stabilising)
```

During a lift itself, a good share of the fuel becomes height, close to what muscle can manage. Across the whole building, mechanical work is a fraction of a percent of the food, because almost none of a working day is spent lifting.

The ledger

Every motion of every part, totalled:

```
elements                 1322
material in the dome    16,751 kg
lifted by one worker    15,100 kg
time per worker         191.05 h = 23.88 shifts
energy per worker       54,870 kcal
energy for the crew     109,741 kcal
energy per shift        2,298 kcal
mean working rate       334 W = 3.50 METs
  sustainable bound     350 W
recovery allowance      17.5 % of the total (33.50 h)

food energy by motion:
  fasten       49,366 kcal   90.0 %
  pause         4,120 kcal    7.5 %
  carry           358 kcal    0.7 %
  walk_out        338 kcal    0.6 %
  lift            326 kcal    0.6 %
  position        272 kcal    0.5 %
  recover          90 kcal    0.2 %
```

Station by station, the energy follows the part count and the time, not the tonnage: a station placing many light pieces slowly costs its crew more than one placing a few heavy ones quickly. This is the table to read when deciding where a jig, a lift assist or an extra pair of hands would change someone's day:

```
by station:
  frame         266 parts     3,925 kg   36.48 h   10,531 kcal
  fiberglass    161 parts     1,691 kg   32.33 h    9,407 kcal
  osb           160 parts     2,806 kg   22.49 h    6,550 kcal
  sheetrock     160 parts     2,347 kg   21.90 h    6,357 kcal
  shingles      160 parts     1,032 kg   18.96 h    5,514 kcal
  insulation    160 parts       637 kg   14.58 h    4,220 kcal
  wrap          160 parts       410 kg   12.55 h    3,638 kcal
  solar          33 parts       404 kg    7.15 h    2,094 kcal
  floor          22 parts     1,258 kg    6.07 h    1,615 kcal
  interior        8 parts       767 kg    4.81 h    1,280 kcal
  fixtures        7 parts       428 kg    4.10 h    1,090 kcal
  water          11 parts       113 kg    3.06 h      813 kcal
  column          3 parts       375 kg    2.42 h      660 kcal
  power           8 parts        72 kg    2.42 h      642 kcal
  hatch           3 parts       487 kg    1.73 h      459 kcal
```

And the total, in food, because until now it was on no drawing of the building:

```
the crew's total, in food:
      1,372  slices of bread at 80 kcal
      1,045  bananas at 105 kcal
         44  days of a 2,500 kcal diet
```

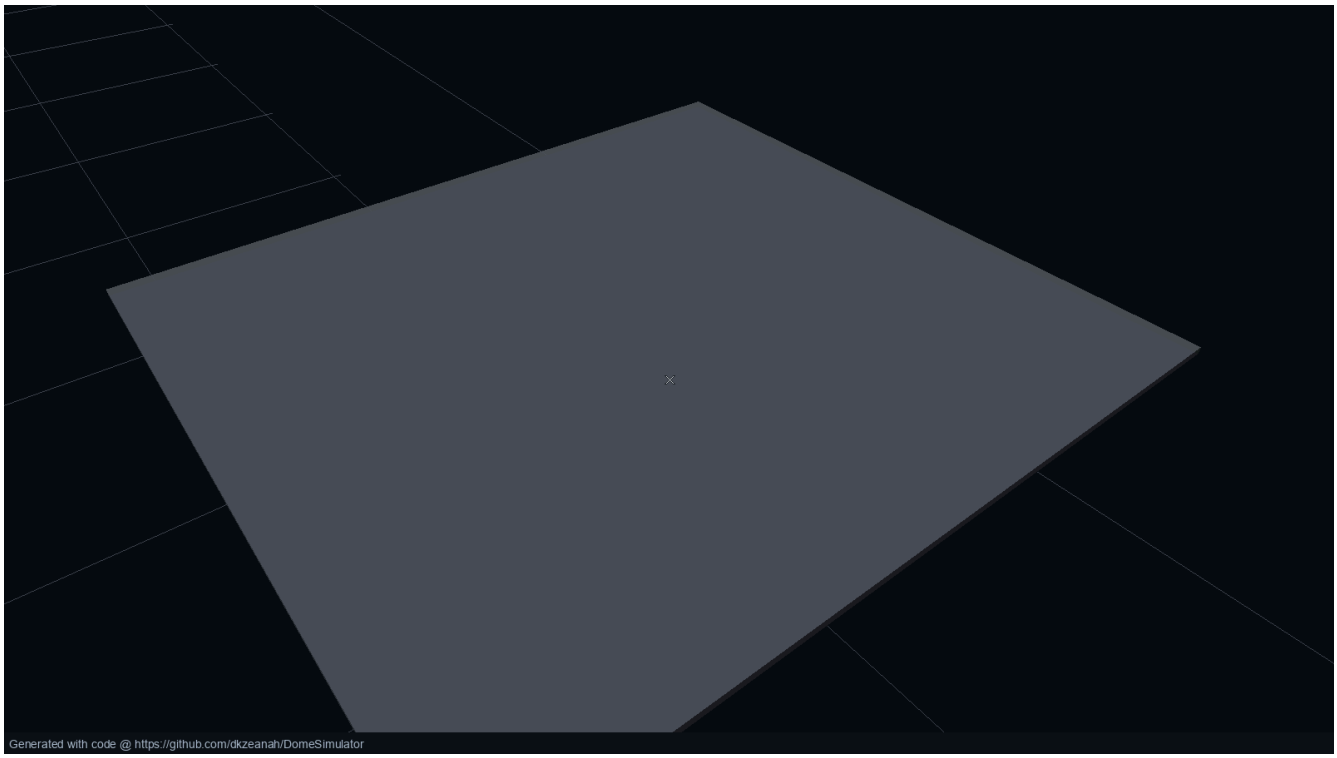
The parts that look like effort turned out to be cheap. The part that looks like nothing – holding a position while fastening – is almost the whole bill. So design the posture, not just the part.

One Flat Board, Forty Identical Panels

The only thing in the whole build that has to be true is the bench, and everything else is measured from it

The only thing in this entire build that has to be true is a sheet of plywood on two sawhorses.

Everything else is measured from it, and nothing on it is adjusted between panels — so panel forty is panel one.



One flat sheet on sawhorses, and nothing else.

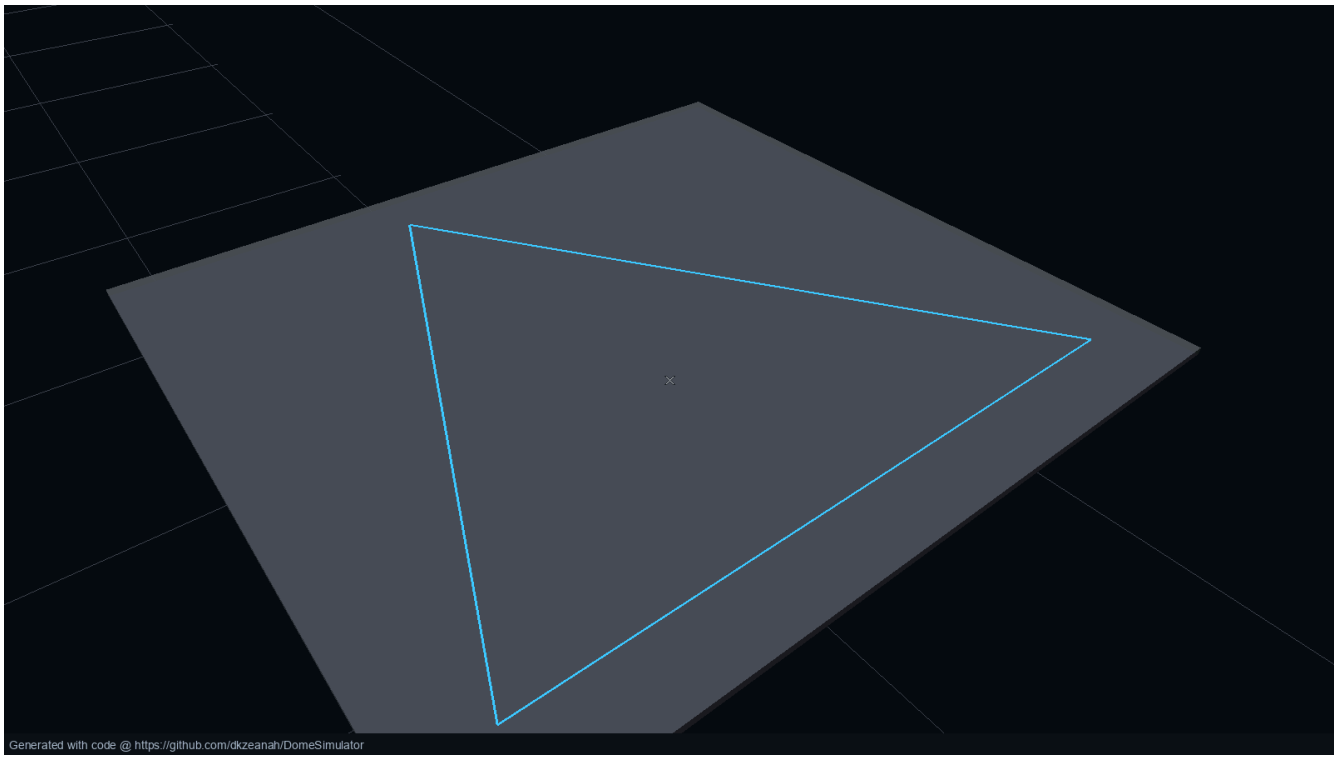
Building the jig

1. The bench

One sheet, flat, level enough that a marble does not run. Flatness is the whole specification: every angle below is measured from this surface, so the panel cannot wind or twist as long as the board does not.

Check it with a straightedge on both diagonals. If it rocks, shim the horses, not the sheet.

2. The triangle, struck once



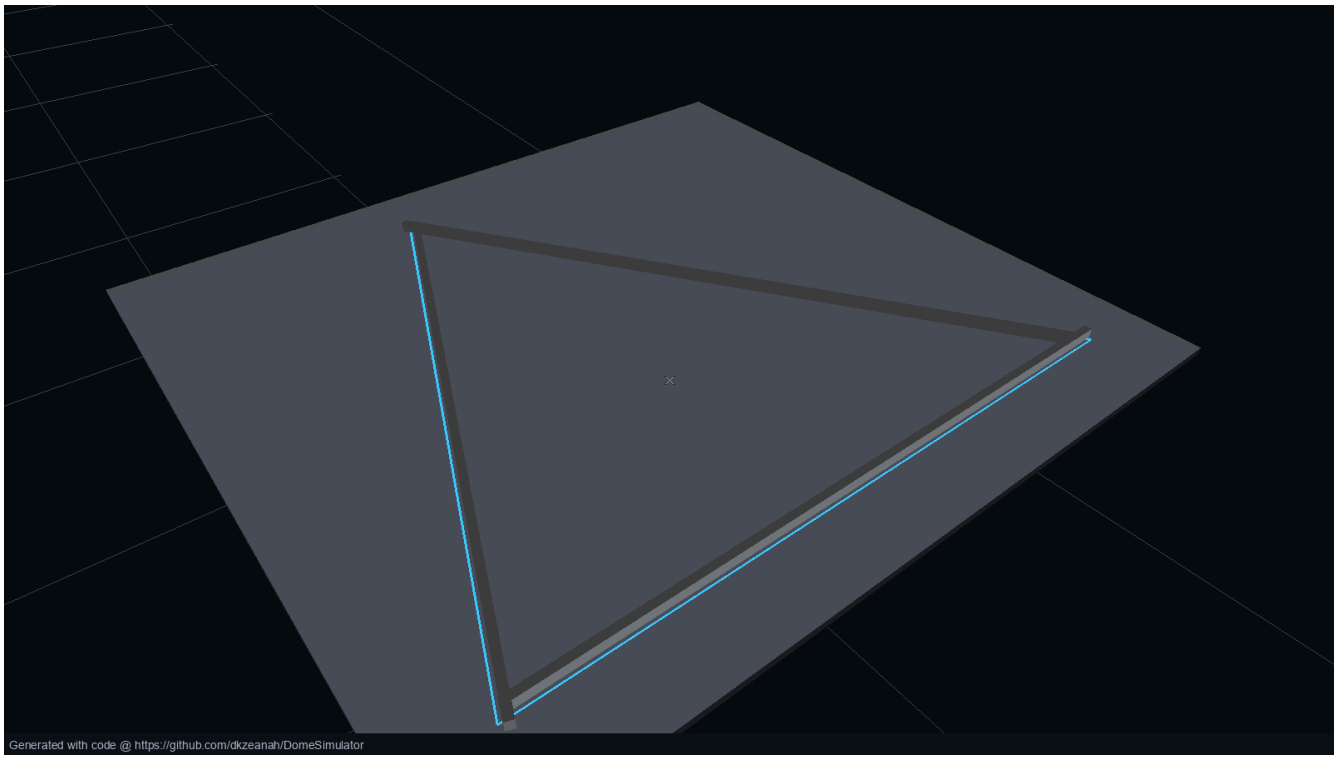
The panel's triangle, struck once from the solution and never measured again.

Snap the panel's mathematical triangle onto the board — the 72.000 and 63.670 inch sides of Chapter 16.

Strike it **one time**, from the geodesic solution, and never measure it again. Forty panels laid out from three lines struck once is a different proposition from forty panels laid out from forty tape measurements, and the difference is the entire reason this works.

The tool prints this triangle full size as a drawing you can lay on the sheet. Chapter 56 says how.

3. Axis rails



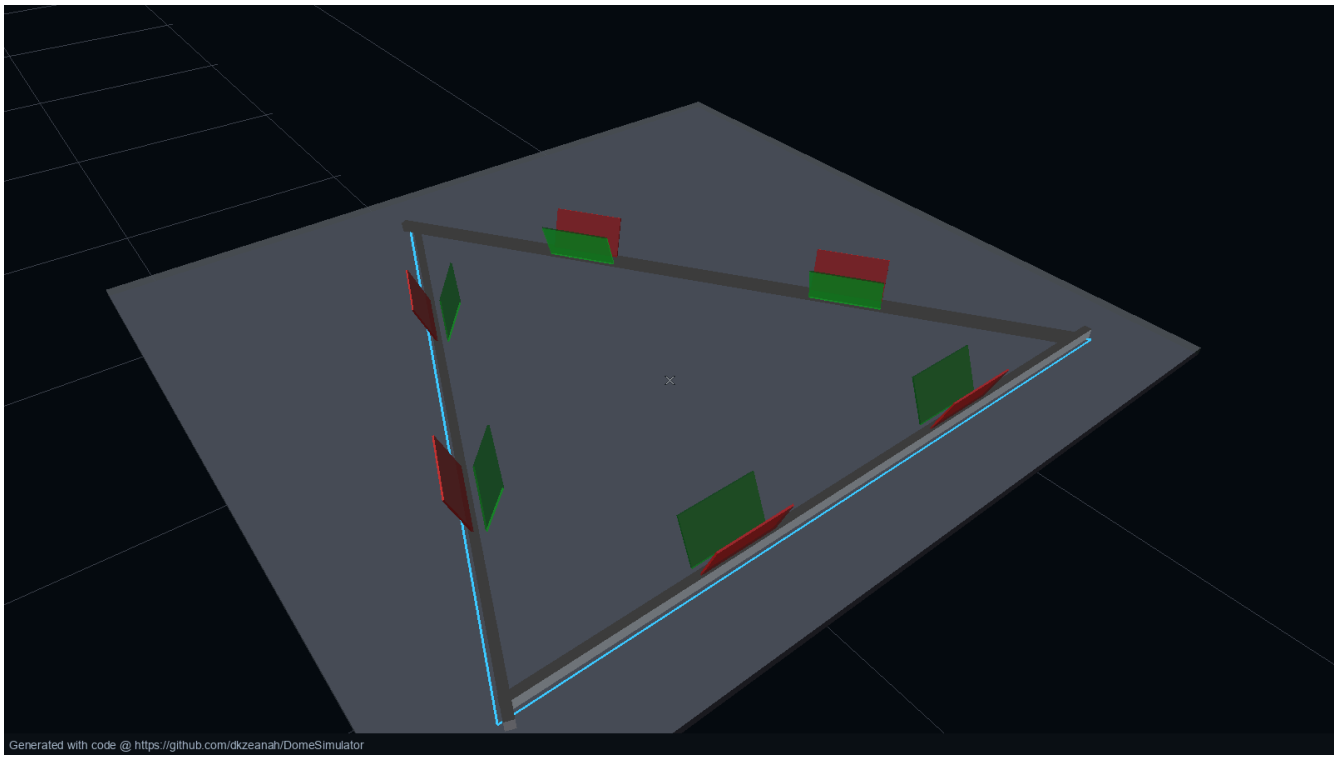
Three cleats, one per member, each parallel to that member's point-axis.

Three low cleats, one per member, each set parallel to that member’s point axis and just to the seam side of it.

These fix **position across the panel**. Push the stick against its rail and the sector’s point line sits exactly where the solution says.

The rail references the sawn point, never the bark. A lumpy log does not move the joint.

4. Red and green fences



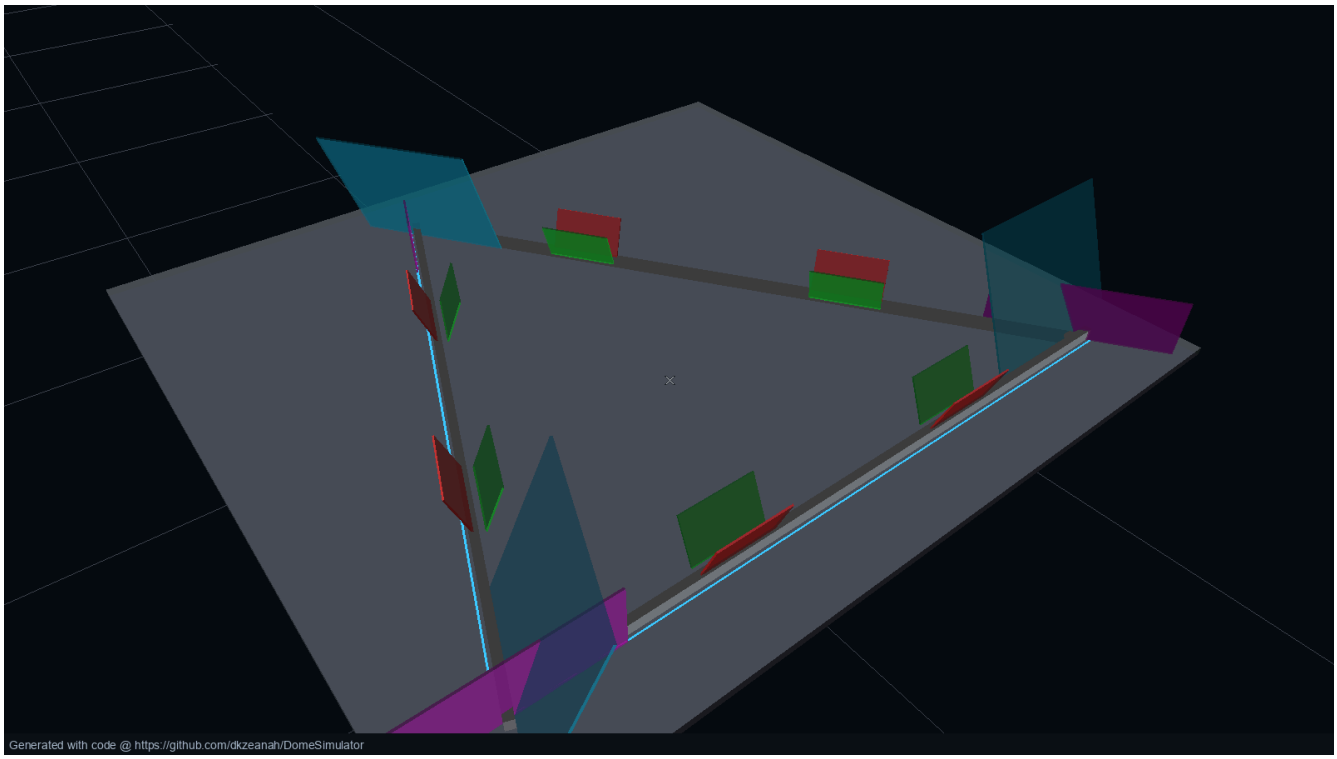
Red to red, green to green, or the stick does not seat.

Two short cradle plates per member, standing against the two sawn radial faces of the wedge.

These fix **rotation about the stick's own axis**, and they are the reason you cannot get a member in the wrong way round. A sector dropped in rotated ninety or a hundred and eighty degrees still *looks* fine. These plates make it not fit.

Mark them. Red side to red plate, green side to green plate. It feels childish for about two panels and then it saves you one.

5. The two cut planes



The magenta plate is the flush-cut fence; the cyan is the face the next butt lands on.

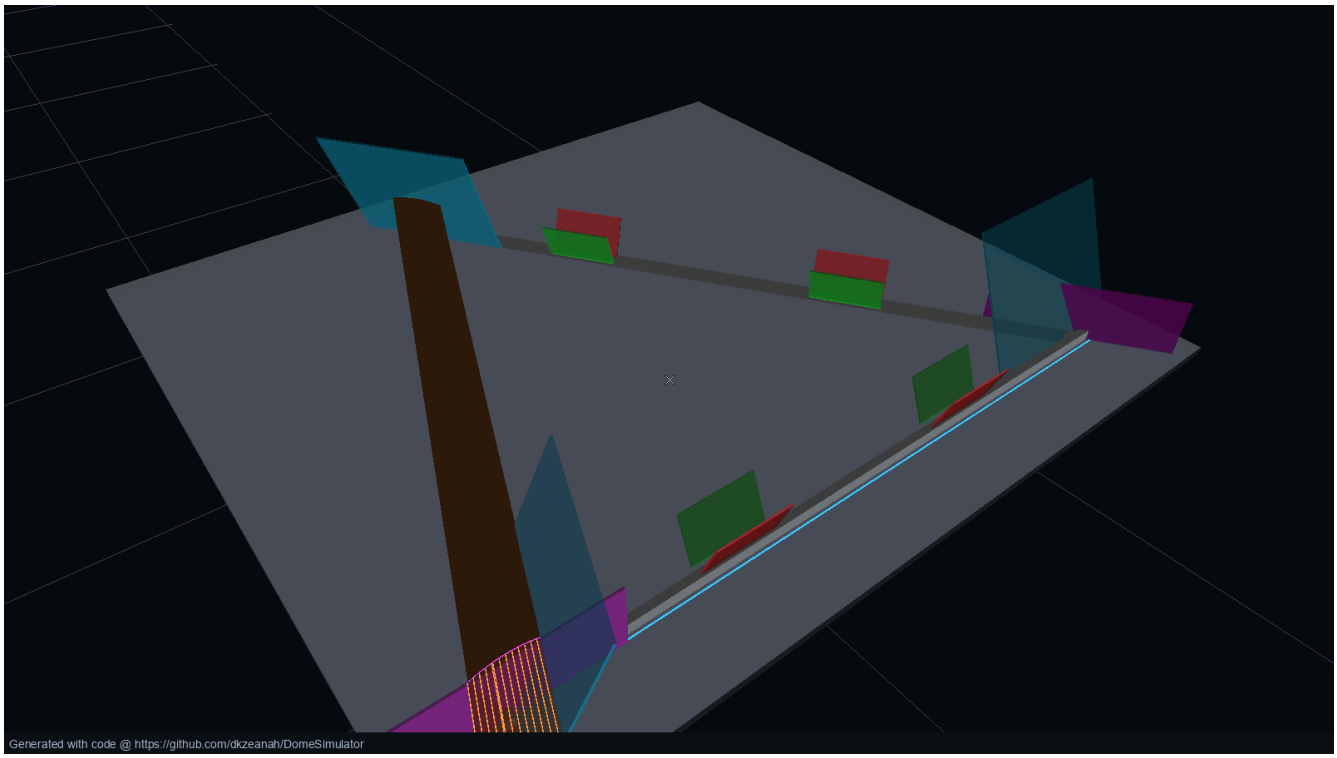
At each head end, a plate standing in the plane of the triangle's edge. That is the flush-cut fence: a saw run flat against it cuts the head to exactly the right plane without anybody setting a bevel or a mitre.

At each butt end, a plate in the plane of the receiving face the next stick's butt will land on.

That is the fixture. A sheet, three cleats, six cradle plates, six guide plates. Nothing on it moves again.

Making one panel

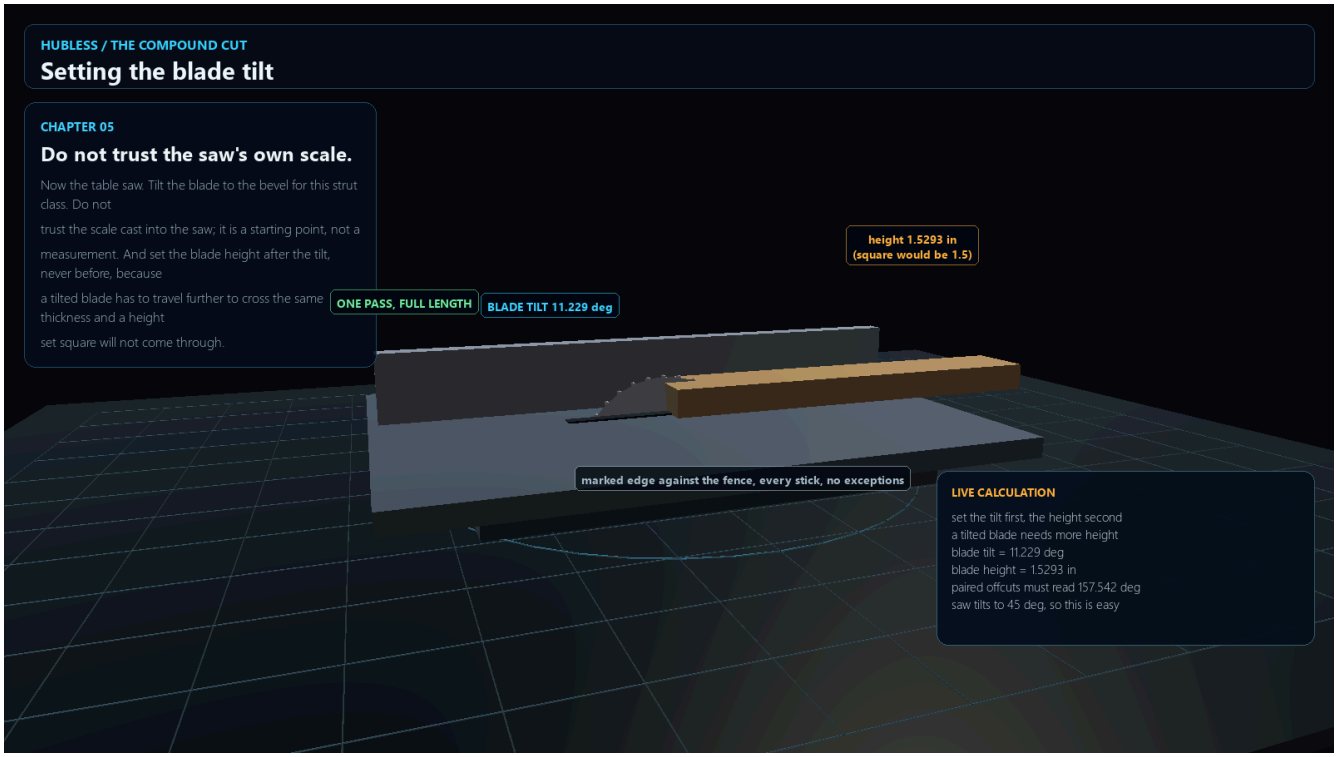
6. The butt cut, off the jig



The one cut made before assembly, and the only one that can be batched.

One stick, still full length at both ends, laid up to the cyan plate. Cut the butt.

This is the **only** cut made before assembly, and every stick of the same family takes the same one — so it runs through a saw bench in a batch rather than being fitted one at a time. Sixty A butts, then sixty B butts.



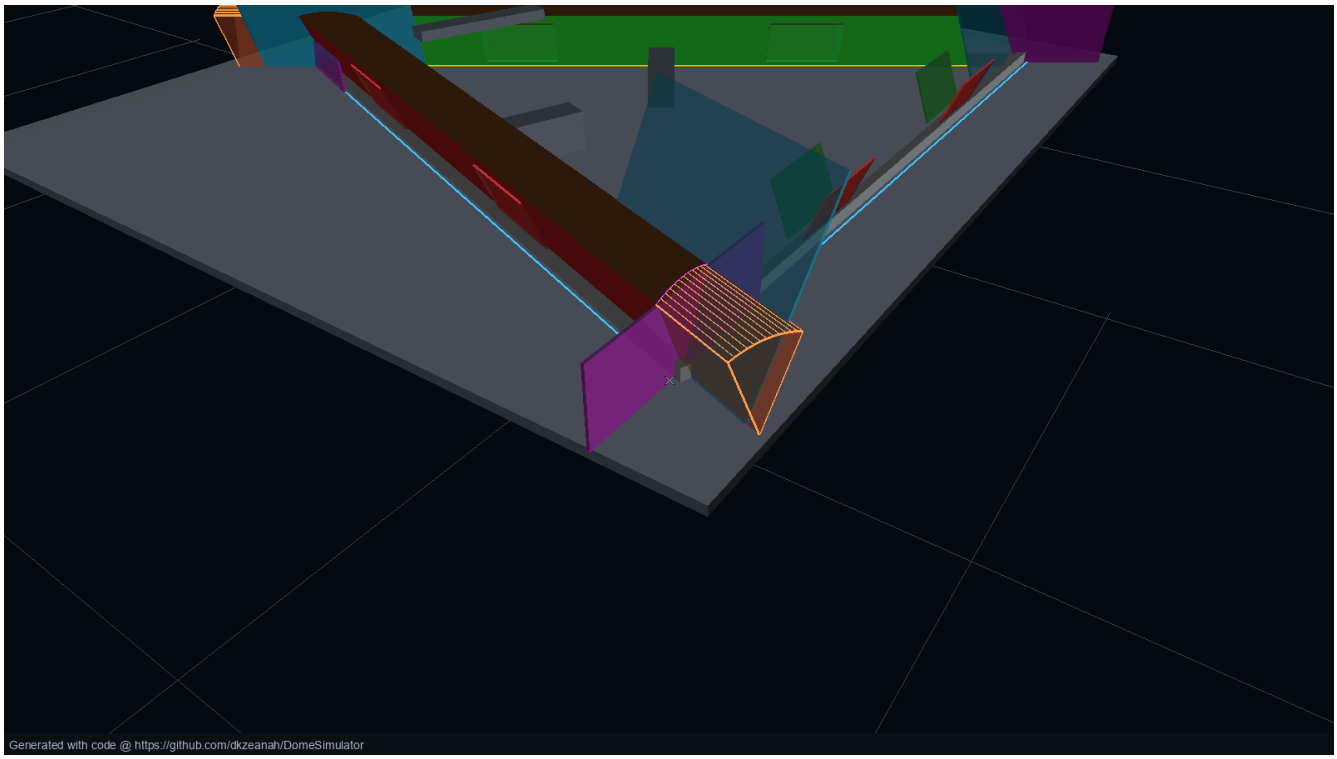
Setting the blade tilt.

7. First stick down

Butt now cut, dropped against its rail and its two fences. The head end is still long and hangs off the board.

Two ends, two completely different operations. The butt arrives finished. The head arrives long on purpose and you do not touch it yet.

8. Second stick, butt onto side

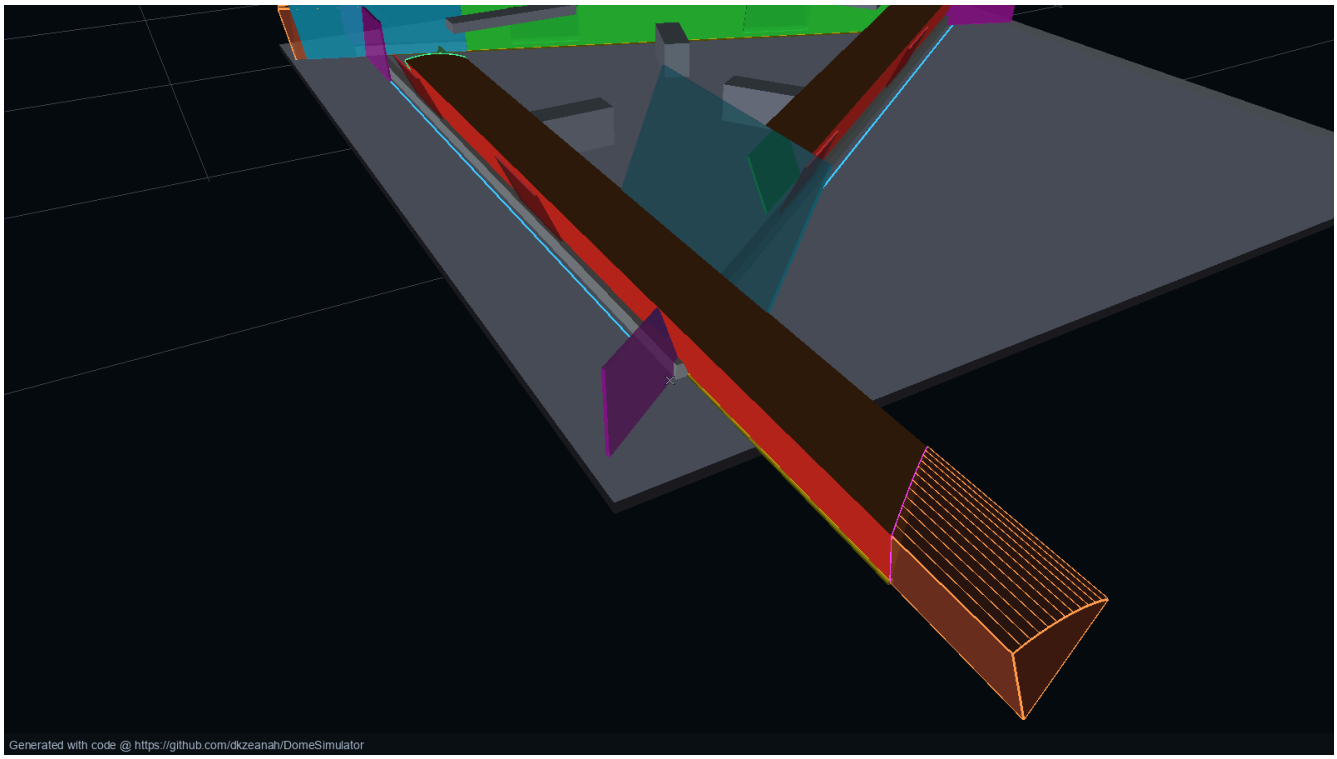


End to side: a flat sawn face on a flat sawn face.

The second stick lands so the first stick's butt bears on its **radial face** — on the side of it, not on its end.

Nothing is mitred to meet at a point. A flat sawn face is pressed onto a flat sawn face, which is what a split wedge already has without any further machining.

9. The last stick slides in



The last stick is captured at both ends, which is why the head is left long.

The third stick is captured at **both** ends once the other two are down, so it cannot drop in from above. It comes in along its own axis, sliding home into the pocket.

This is the one awkward move in the whole method, and it is why the head end is left long: the extra length is what gives the last stick somewhere to come from.

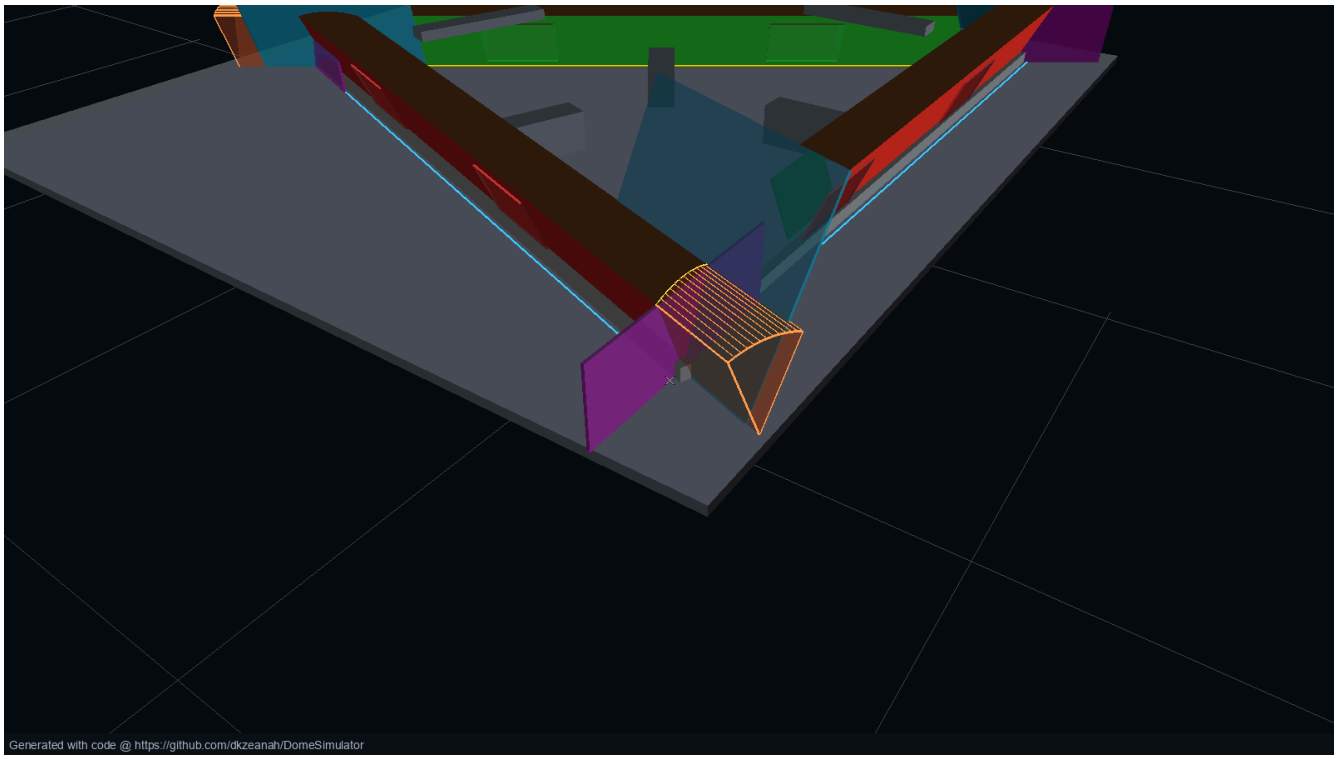
If you have trapped it, you put the sticks down in the wrong order. Take one out and start again; you will do this once.

10. Pull it up tight

Three butts bearing on three sides. The loop is closed, the frame holds its own shape, and the jig is now only keeping it flat while you finish.

All three heads still hang off long.

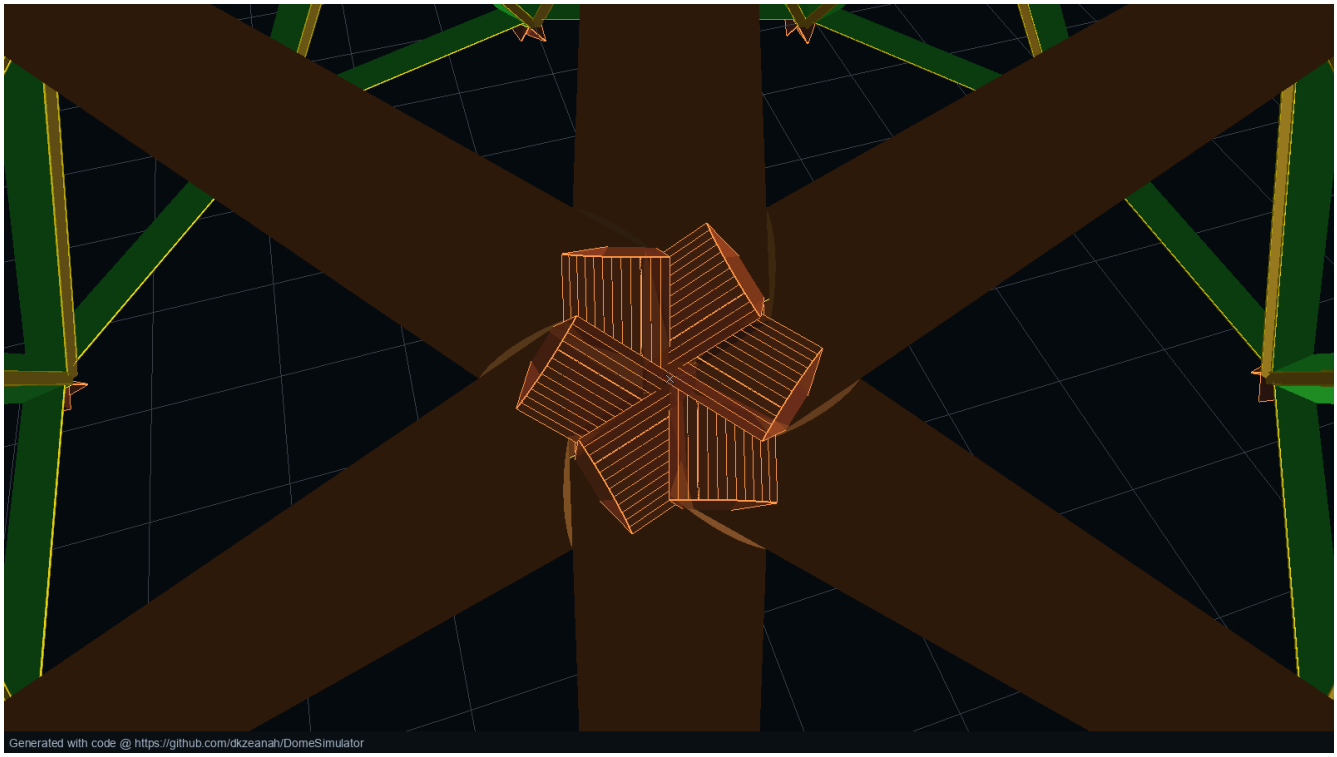
11. Flush-cut the three heads



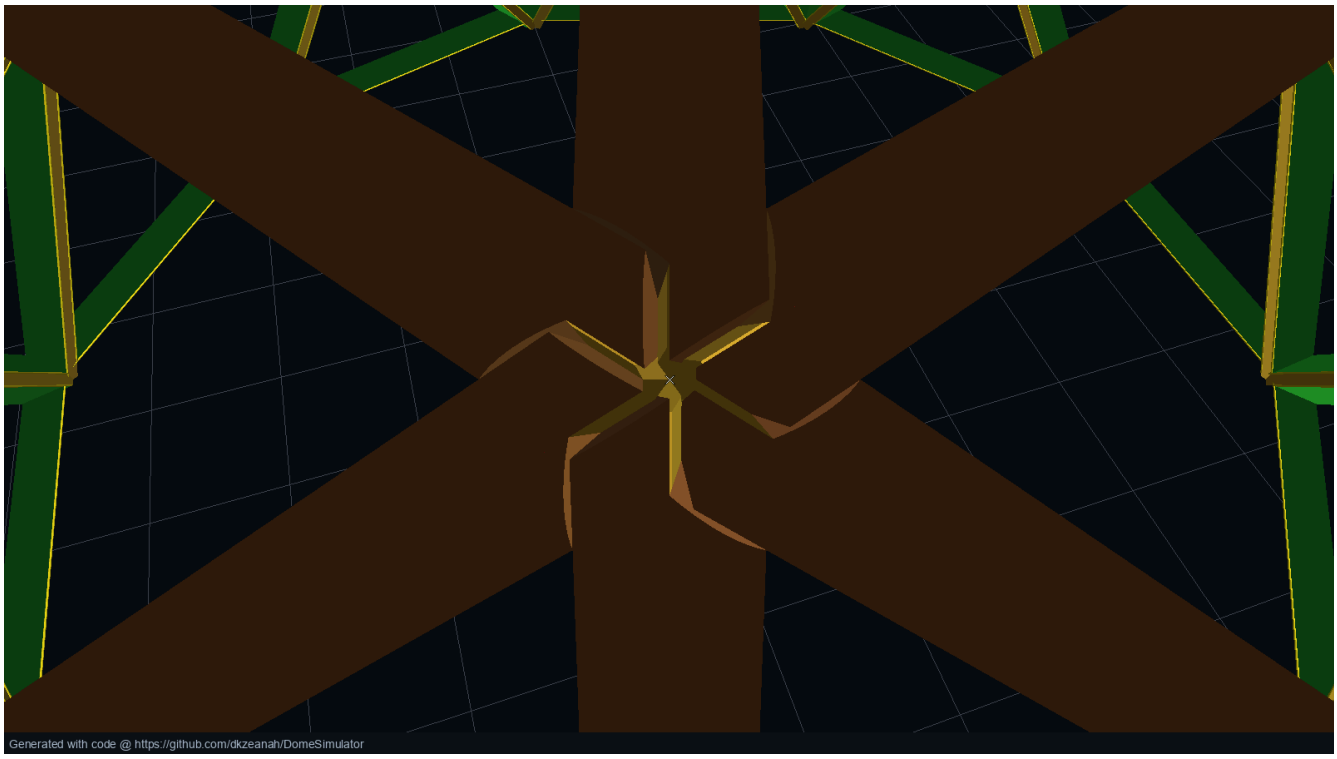
The head is never measured. It is cut in place.

Saw each head back to its magenta fence. The stock hanging past the board falls off.

This is the whole trick. The head is never measured. It is cut in place, to the plane the neighbouring member has already established — so error in stock length, or in the butt angle, or in how round the log was, leaves as offcut instead of accumulating around the triangle.



Uncut, every head trespasses across the vertex.



And after.

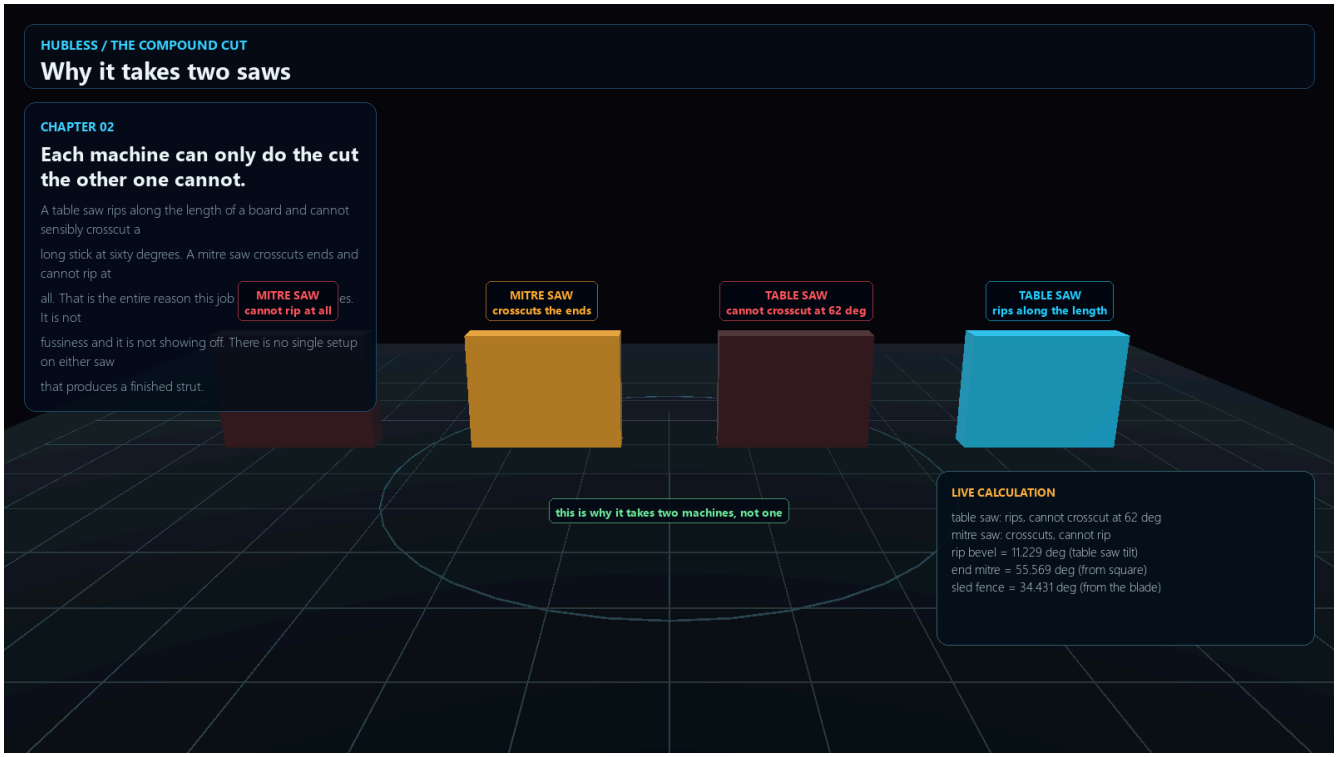
Those two pictures are the same corner of the same dome. In the first, every head still carries its stock and they trespass straight across the geodesic vertex into the neighbouring panel. In the second they end inside their own triangle and nothing crosses.

12. Panel off the jig

A finished triangular frame, its three corners clean inside the mathematical triangle. The jig is empty and unchanged.

Drill the inserts now, while it is flat — Chapter 3 — and sheathe it now if you are sheathing on the bench, which you should be.

Then do it 40 times, in two families: 10 equilateral and 30 isosceles.



Why the compound cut takes two saws.

What the jig is really buying

Repeatability, obviously. But more than that: it moves the hard work indoors and off the ladder.

Every operation above happens at waist height, on a bench, with both feet on the ground, in whatever weather. The part of dome building that everybody pictures — the awkward bit up a ladder holding something heavy at an angle — happens once, in Chapter 27, and takes three days out of a fortnight.

And it is worth being exact about what the fixture enforces, because it is only three things: where each member sits across the panel, which way up it is turned, and the plane its head is cut to. Everything else is free. Counted, the way the films count it:

```
40 triangles, 120 members, 120 corner joints -- all off one flat board
the fixture comes in 12 steps and holds three things:
  a RAIL per member, set to the sawn point-axis, not the bark
  a RED and a GREEN plate that make a wrongly-turned wedge physically refuse to seat
  a FENCE at each head end, which is the plane the saw runs on
each member arrives with ONE cut already made:
  the butt, cut to the angle its neighbour presents, batched
and one deliberate error: the head is left 6 in too long
three butts are pulled up, the loop closes, and only then is
  each head sawn against its fence -- unmeasured, in place
the offcut is 60 ft per dome = 8.3 % of raw length
and 40 panels need only 4 physically different fixtures
```

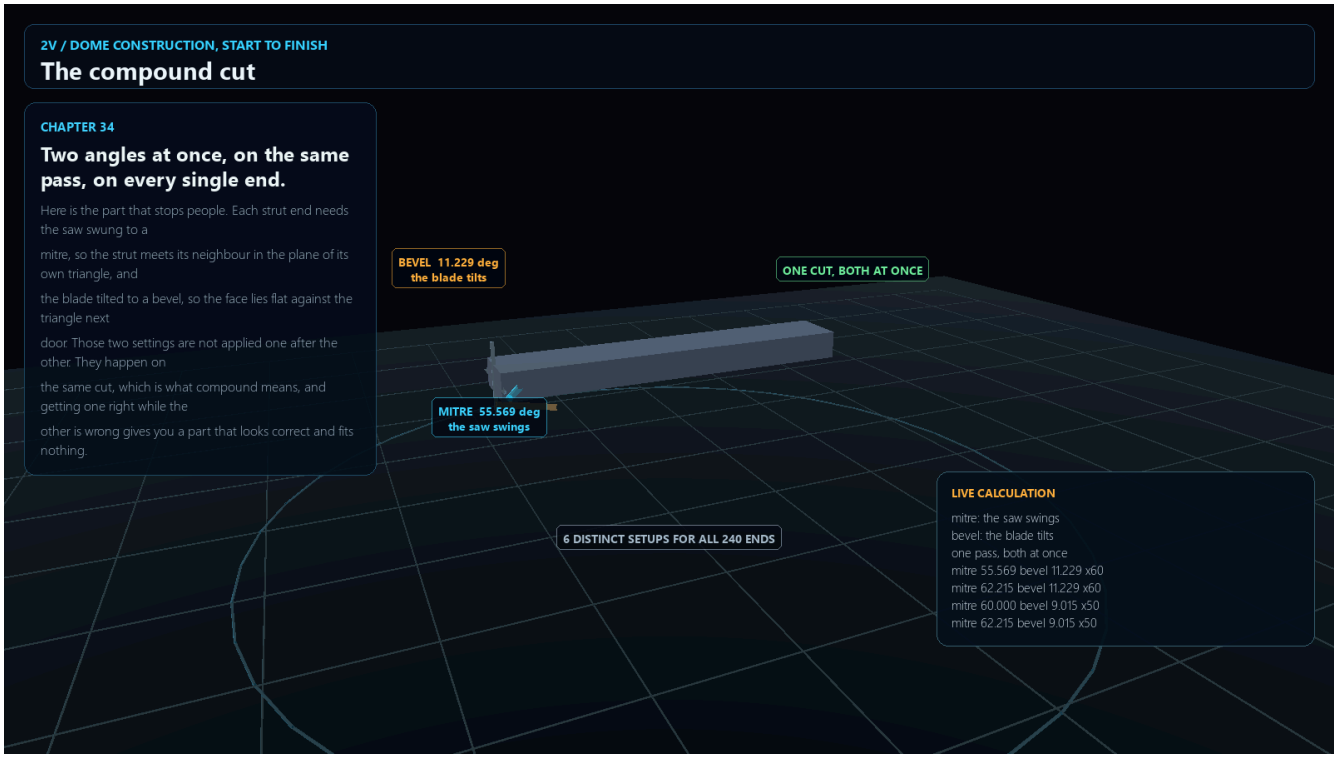
Every stick has some error in it. Cut both ends to a measurement and those errors add up around the triangle until the last corner will not close. Cut one end in place, against the neighbour, and the error has nowhere to accumulate: it leaves in the piece that falls on the floor. The offcut is what the accuracy costs, and it is cheap.

If Your Sticks Are Rectangles

The same hubless dome from milled stock: two angles on every end, two machines, six saw settings, and a sled for the angles no saw reaches

Everything up to here cuts split log sectors. But the hubless frame – forty closed triangles bolted edge to edge – works with ordinary milled stock too, and plenty of people will start with a stack of two-by-fours rather than a stand of pine. This chapter is that version. Its joint is harder than the wedge's, and seeing why is the best argument for the wedge.

Every end of a rectangular strut needs two angles at once. The saw is **swung** to a mitre, so the strut meets its neighbour in the plane of its own triangle, and the blade is **tilted** to a bevel, so its face lies flat against the triangle next door. They happen on the same pass – that is what *compound* means – and getting one right while the other is wrong gives you a part that looks correct and fits nothing.



Two angles, one pass, every end.

Every setting the dome needs

The audit assumes:

```
stock 3.5 x 1.5 in, 10 in blade
mitre saw reaches 50 deg, table saw tilts to 45 deg
```

and the whole dome reduces to **6 distinct setups**:

```
6 distinct setups covering 240 strut ends:
  SET-1  mitre  55.569  bevel 11.229  x60
      table saw      blade tilt      11.229 deg
      mitre saw      table swing     55.569 deg   <- NOT REACHABLE
```

```
     crosscut sled  fence to blade  34.431 deg
     blade height for the rip: 1.5293 in (square would be 1.5)
     sled travel across the strut: 4.243 in
  SET-2  mitre  62.215  bevel 11.229  x60
     table saw      blade tilt      11.229 deg
     mitre saw      table swing     62.215 deg   <- NOT REACHABLE
     crosscut sled  fence to blade  27.785 deg
     blade height for the rip: 1.5293 in (square would be 1.5)
     sled travel across the strut: 3.956 in
  SET-3  mitre  60.000  bevel  9.015  x50
     table saw      blade tilt       9.015 deg
     mitre saw      table swing     60.000 deg   <- NOT REACHABLE
     crosscut sled  fence to blade  30.000 deg
     blade height for the rip: 1.5188 in (square would be 1.5)
     sled travel across the strut: 4.041 in
  SET-4  mitre  62.215  bevel  9.015  x50
     table saw      blade tilt       9.015 deg
     mitre saw      table swing     62.215 deg   <- NOT REACHABLE
     crosscut sled  fence to blade  27.785 deg
     blade height for the rip: 1.5188 in (square would be 1.5)
     sled travel across the strut: 3.956 in
  SET-5  mitre  60.000  bevel  0.000  x10
     table saw      blade tilt       0.000 deg
     mitre saw      table swing     60.000 deg   <- NOT REACHABLE
     crosscut sled  fence to blade  30.000 deg
     blade height for the rip: 1.5000 in (square would be 1.5)
     sled travel across the strut: 4.041 in
  SET-6  mitre  62.215  bevel  0.000  x10
     table saw      blade tilt       0.000 deg
     mitre saw      table swing     62.215 deg   <- NOT REACHABLE
     crosscut sled  fence to blade  27.785 deg
     blade height for the rip: 1.5000 in (square would be 1.5)
     sled travel across the strut: 3.956 in
```

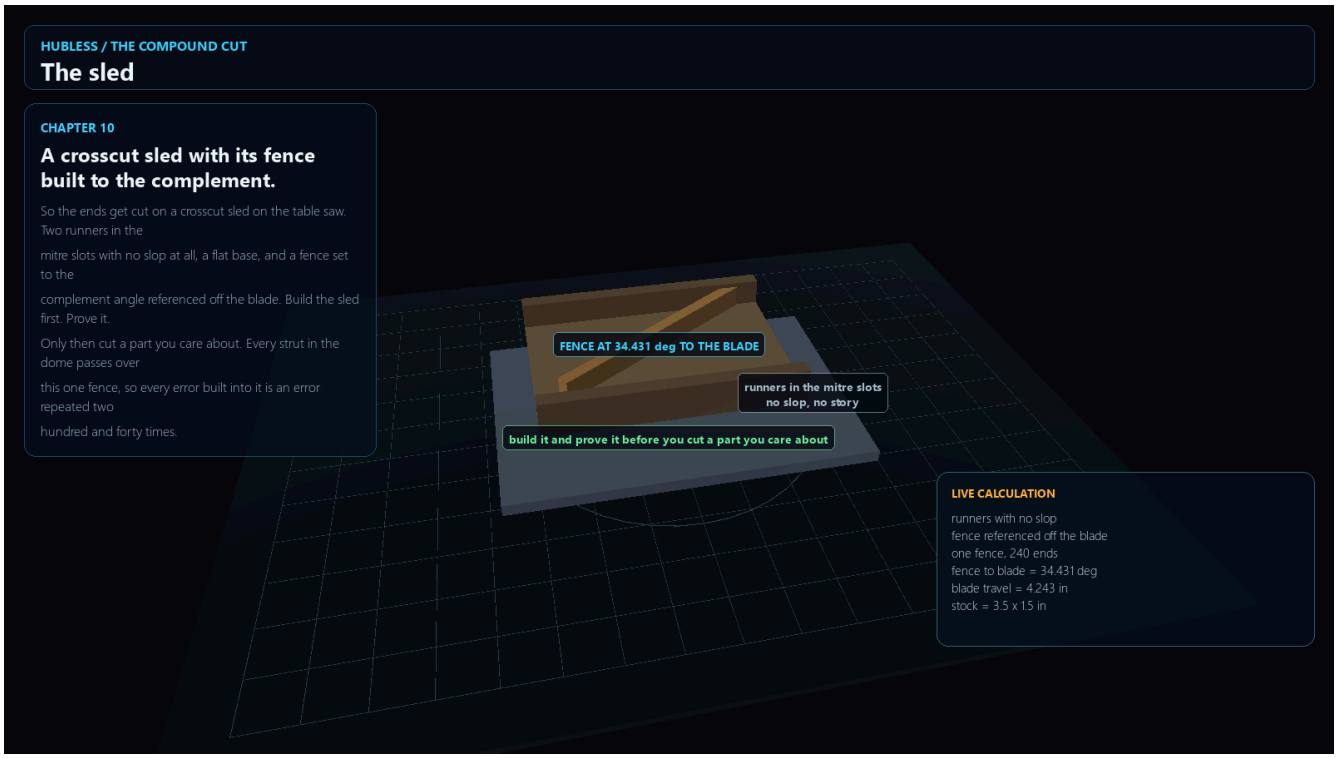
Read the right-hand column. **Not one of the mitres this dome asks for is on a common mitre saw's scale** – they all lie past its 50-degree stop. The geometry is easy; the tool is the problem.

The way through is that a mitre and its complement are the same cut approached from the other face. Swing to the complement instead – the sled column above – and turn the workpiece a quarter turn in a sled. Same joint, reachable setting. And each machine can only make the cut the other one cannot. The table saw rips the bevel down the strut's whole length, tilted to at most 45 degrees, which no mitre saw can do; the ends are crosscut on a mitre saw where its swing reaches, and on a sled fenced to the complement where it does not. Two triangle shapes means two assembly jigs, and the six settings cover every end of all 120 struts.

The sequence

1. Rip every stick to one width Every angle after this is measured from the edge of the board. If the boards are not all the same width, nothing downstream can be right, and no amount of care at the mitre saw will rescue it. Rip the whole pile in one session, with the fence untouched. **2. Mark the mating edge** Only one long edge of each strut gets the bevel: the one that will lie against the neighbouring triangle. Mark it on every stick before any of them go near the blade. This is the single most common way to produce a pile of mirror-image scrap. **3. Set the blade tilt** Set the tilt to the bevel for that strut class. Do not trust the saw's scale: rip two offcuts, put the cut faces together, and measure the pair. The pair reads twice the error, so anything you cannot see in the single is obvious in the double. **4. Rip the bevel, full length, one pass** Featherboard before the blade, push stick past it, and keep the marked edge against the fence for every stick without exception. This cut decides how the triangle meets its neighbour. **5. Build the sled before you touch the ends** The mitres this dome wants are past the stop on a common mitre saw, so the ends are cut on a crosscut sled whose fence is built to the complement and referenced off the blade instead of off square. Build it, prove it with the five-cut method, and only then cut a part you care about. **6. Cut the relief lap into the sled fence** The strut sits against the sled fence with one end projecting past it. The blade has to finish that cut flush, so the fence needs a lap cut away at the blade path -- deep enough for the blade to pass the full width of the strut, no deeper. Cut it once, with the sled in the position it

will always be used in. **7. Set the stop block, not the tape** Length comes from a stop block clamped to the sled, never from a pencil line. Two hundred and forty ends measured individually will drift; two hundred and forty ends against one block will not. **8. Cut the first end** Bevelled edge down and against the fence, marked face up, strut butted to the stop, end projecting over the lap. One pass, all the way through, and let the blade stop before lifting. **9. Turn the strut, do not flip it** The second end is not a mirror of the first. Rotate the strut end for end about its long axis in the sled -- a quarter turn, the way the sled was built for -- so the bevel stays on the same side of the dome. Flipping it face over face puts the bevel on the wrong side and the strut will not close its triangle. **10. Batch by setting, never by triangle** Cut every end that shares a setting before you change anything. Changing a setting is where error enters, so change it as few times as the job allows -- six times, not two hundred and forty. **11. Dry-assemble one triangle before cutting the rest** Three struts, no fasteners, laid on a flat floor. If it closes with no gap at any corner and lies flat, the settings are right. If it does not, you have lost three sticks instead of the whole pile.



The crosscut sled, fenced to the complement.

Proving it

Two checks, because a saw's own scale is not to be trusted.

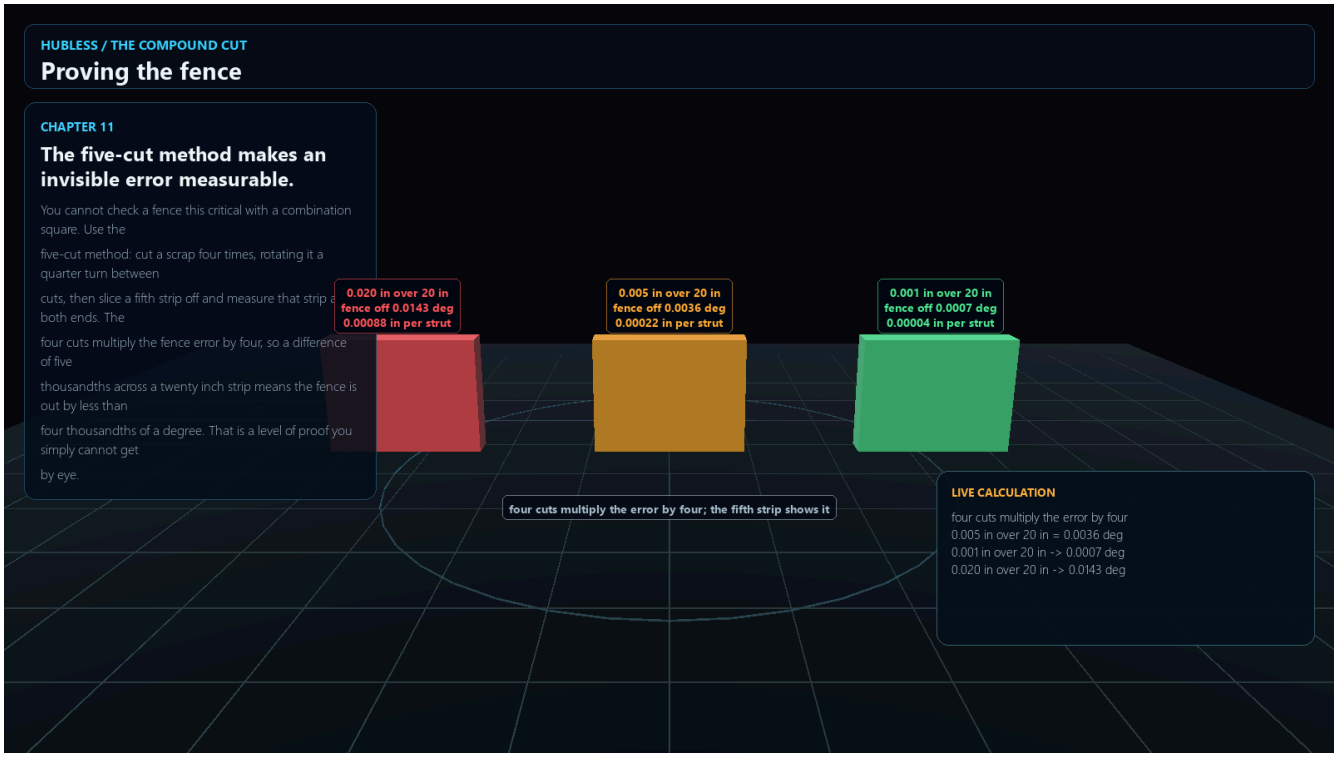
The tilt. Rip two offcuts at the setting, put the cut faces together and measure the pair. The pair doubles the error, so anything invisible on one is obvious on two:

```
bevel verification, by pairing two offcuts:
  bevel 11.229 -> the pair must read 157.542 deg
  bevel 11.229 -> the pair must read 157.542 deg
  bevel  9.015 -> the pair must read 161.970 deg
  bevel  9.015 -> the pair must read 161.970 deg
  bevel  0.000 -> the pair must read 180.000 deg
  bevel  0.000 -> the pair must read 180.000 deg
```

The sled fence. The five-cut method trims one board on all four sides and then once more, and the last strip's taper is four times the fence's error – so an error far too small to see becomes one a caliper reads:

```
five-cut check on the sled fence:
  0.001 in across a 20 in strip -> fence off by 0.0007 deg = 0.00004 in across a strut
```

```
0.005 in across a 20 in strip -> fence off by 0.0036 deg = 0.00022 in across a strut
0.020 in across a 20 in strip -> fence off by 0.0143 deg = 0.00088 in across a strut
```



An invisible error, made measurable.

The five ways it goes wrong

Every one of them looks correct until assembly:

- **Bevel on the wrong edge.** The strut looks correct and mates the wrong way. Marking the edge before ripping is the only reliable prevention. - **Complement confusion.** Setting the sled to the mitre instead of its complement gives a cut that is wrong by exactly the amount that looks plausible. - **Flipped instead of turned.** Flipping face over face between ends mirrors the part. It will look right on the bench and refuse to close a triangle. - **Stop block creep.** A clamped block walks under repeated impact. Check it against a reference stick every twenty cuts. - **Blade not through.** A tilted blade needs more height to clear the same thickness. Set the height after the tilt, never before.

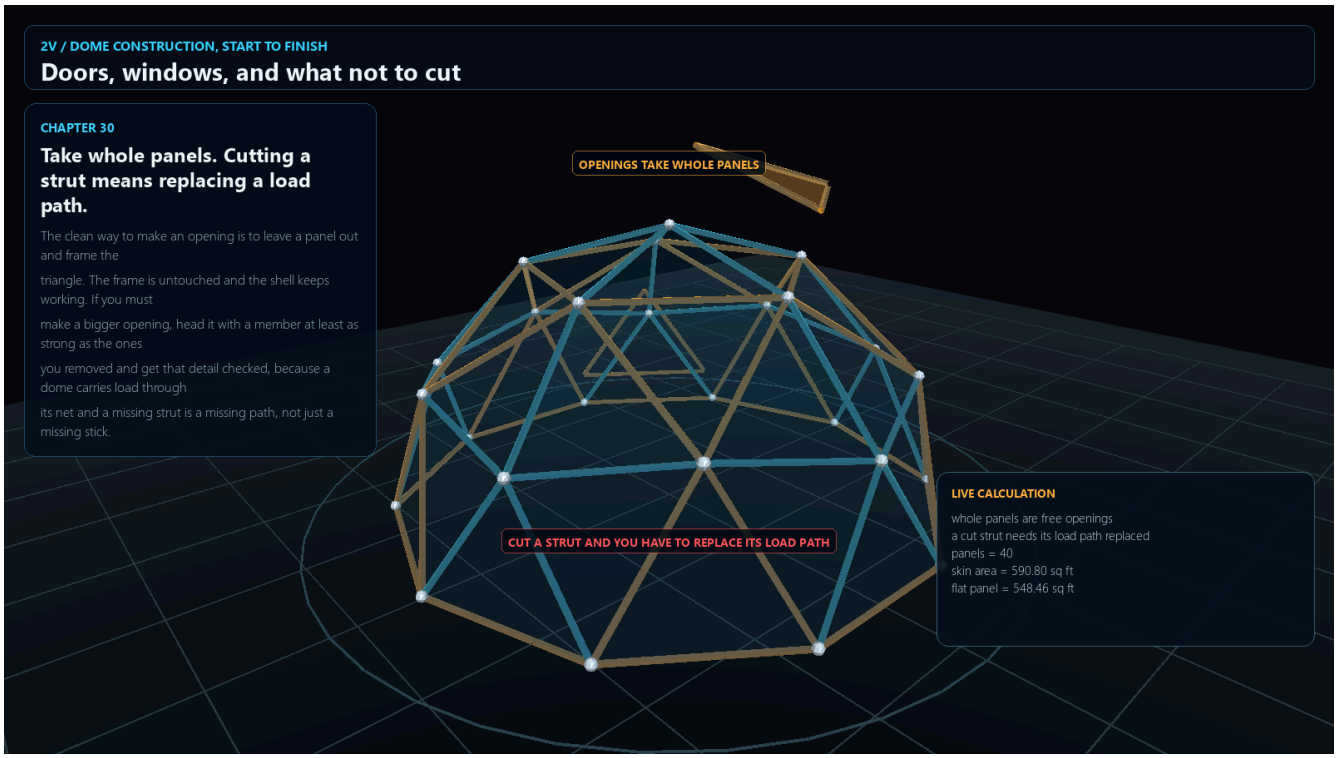
So the whole sequence is: two machines, one sled, 6 settings, and every end of every strut batched by setting rather than by triangle. Six setting changes instead of a change for every end – which is the same lesson as the wedge, arrived at the hard way. The wedge simply never asks for most of these cuts: its bevel is the split, and its head is cut in place (Chapter 24).

Doors, Windows, and What Not to Cut

An opening is a panel you did not close -- which is why it costs nothing, and why cutting one afterwards costs everything

An opening is a panel you did not close.

That is the whole rule, and every expensive mistake people make putting holes in domes comes from not believing it.



Take whole panels. Cutting a strut means replacing a load path.

Why a whole panel

Chapter 6 said a triangulated shell carries load because the triangles cannot change shape. Take one triangle out and the ones around it close the gap between themselves — the load goes round, the way it goes round a missing tile rather than through it.

Cut a **strut**, and you have not made a hole. You have removed a member from three triangles at once and given the shell nowhere for that load to go. Now you need a header, and posts either side, and something to carry what the header carries, and you are framing a wall in a building that did not have walls.

So:

Take whole panels. Never cut a member.

A panel left out is free. A panel cut into is a structural alteration.

Where the door goes

Ten panels stand on the base ring, so the ground floor has ten places a doorway can be, and they are all the same.

That is the first surprise for anybody used to rectangular buildings: you do not choose where the door goes on a plan, you choose *which panel*, and there are ten.

Take two adjacent first-course panels for a comfortable door. One is enough for a doorway you duck through; two gives you a proper opening with room either side.

Frame the perimeter, do not frame the hole. The three members bounding each missing panel are already there and already doing their job. The door frame hangs *inside* them, on the same threaded inserts of Chapter 3, and carries nothing but the door.

The threshold is the base ring. It is already level, already continuous, and already the thing the floor meets. Do not build a sill on top of it.

The sequence

1. **Decide after the shell is up.** This is the one decision in the whole build that gets better for being left late. Stand inside, see where the light actually comes from and where you actually walk, then pick the panel.
2. **Leave the panel out, or take it out.** If you are still raising, simply do not fit it. If the shell is closed, undo its keys and lift it out — the whole argument for panels being objects, collected.
3. **Fit the jamb inside the triangle.** Three sides, following the members, bolted to inserts rather than screwed into end grain.
4. **Hang a door in a rectangle inside that triangle.** Doors are rectangular; triangles are not. The leftover triangle above the head is either a fixed light, a vent, or infill — and a fixed light is the best answer because it puts daylight above head height where it does the most good.
5. **Flash it under the cap.** The cap of Chapter 28 runs over the whole shell and the door is a hole in the shell, so the cap has to be returned into the opening and terminated on the frame. This is the one place in the building where the single watertight layer is interrupted, and it is where a dome leaks if it leaks.

Windows, and why they go high

The wall leans in. That changes everything about glazing.

A window low in a dome looks at the ground a few feet away and lets in very little light, because the aperture is tilted toward the floor. A window high up looks at the sky, which is the bright thing, and it is why every dome photograph you have ever liked has the glass up near the crown.

So: **put glazing above head height** unless you need a view of something specific.

The other consequence of the lean is that every opening in a dome is partly a roof opening. Water arriving at a window in a wall runs down the wall; water arriving at a window in a leaning surface runs *onto the glass and sits at the bottom edge*. Detail every window as you would a rooflight — head flashing over, sill drained outward, and the cap returned properly — not as you would a window in a house.

The panel you do not take

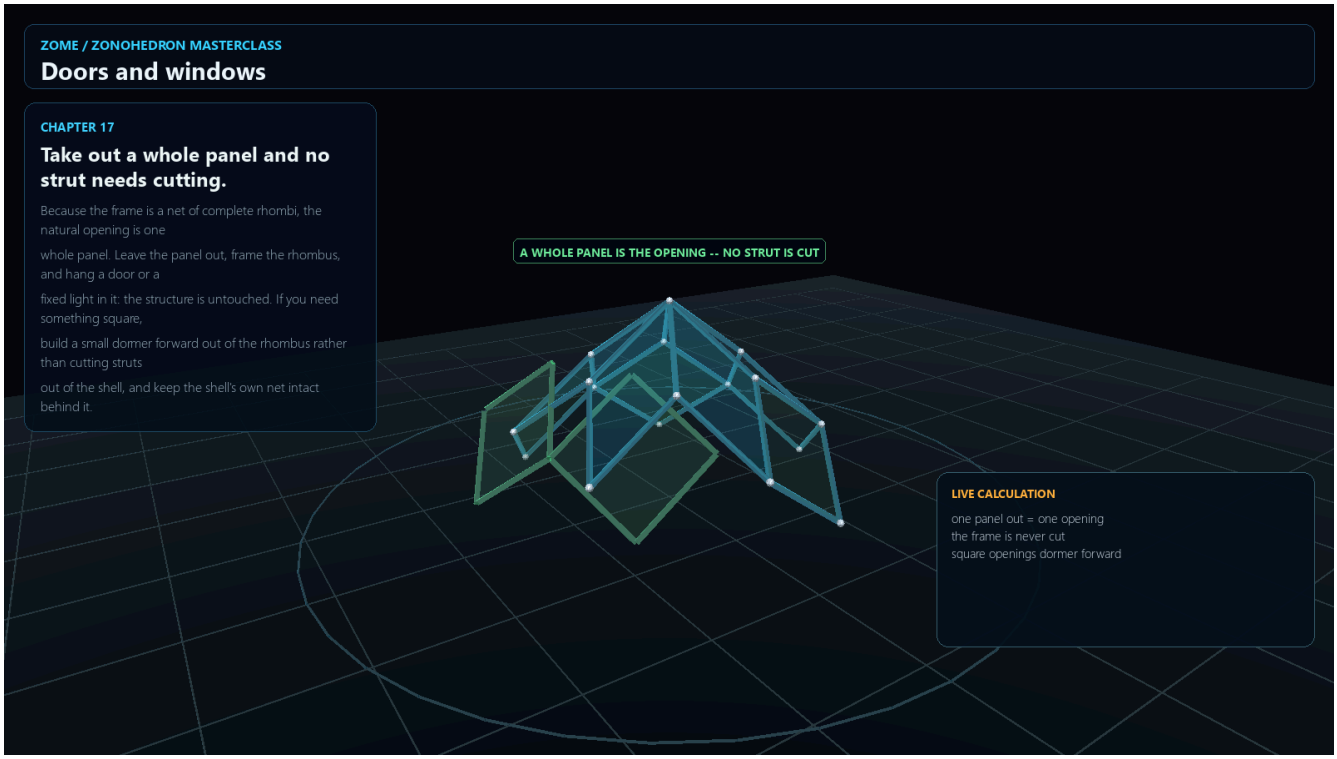
One warning, from the geometry rather than from experience.

The first course is ten panels and they are the ones standing on the ring. Taking two adjacent ones is comfortable. Taking three in a row is the point where the ring between them has nothing above it for a third of its length, and the base ring is the one part of this building that is genuinely carrying in bending rather than in compression.

Three is probably fine. This book does not know that it is fine, because nothing here has been load-tested, and a building with a third of its ground floor open is a different structure from the one that was solved.

If you want a wide opening, take two panels and put the third somewhere else.

Doing it on a zome



Take out a whole panel and no strut needs cutting.

The same answer, on a shape built the opposite way.

A zome's panels are bigger — one parallelogram is a lot of opening — and the rule is identical: take the whole panel, frame inside the struts, cut nothing. That the rule survives a completely different geometry is a good sign that it is about triangulated shells in general rather than about this dome.

What this chapter is not

It is not a schedule of door and window sizes, because those come from what you can buy where you are and from a panel whose dimensions are 52.516 inches across at the narrowest — Chapter 11.

And it is not permission. An opening in a load-carrying shell is exactly the kind of thing an engineer exists for, and the rule at the top of this chapter is what keeps you out of their office rather than a substitute for it.

PART 5

The Skin

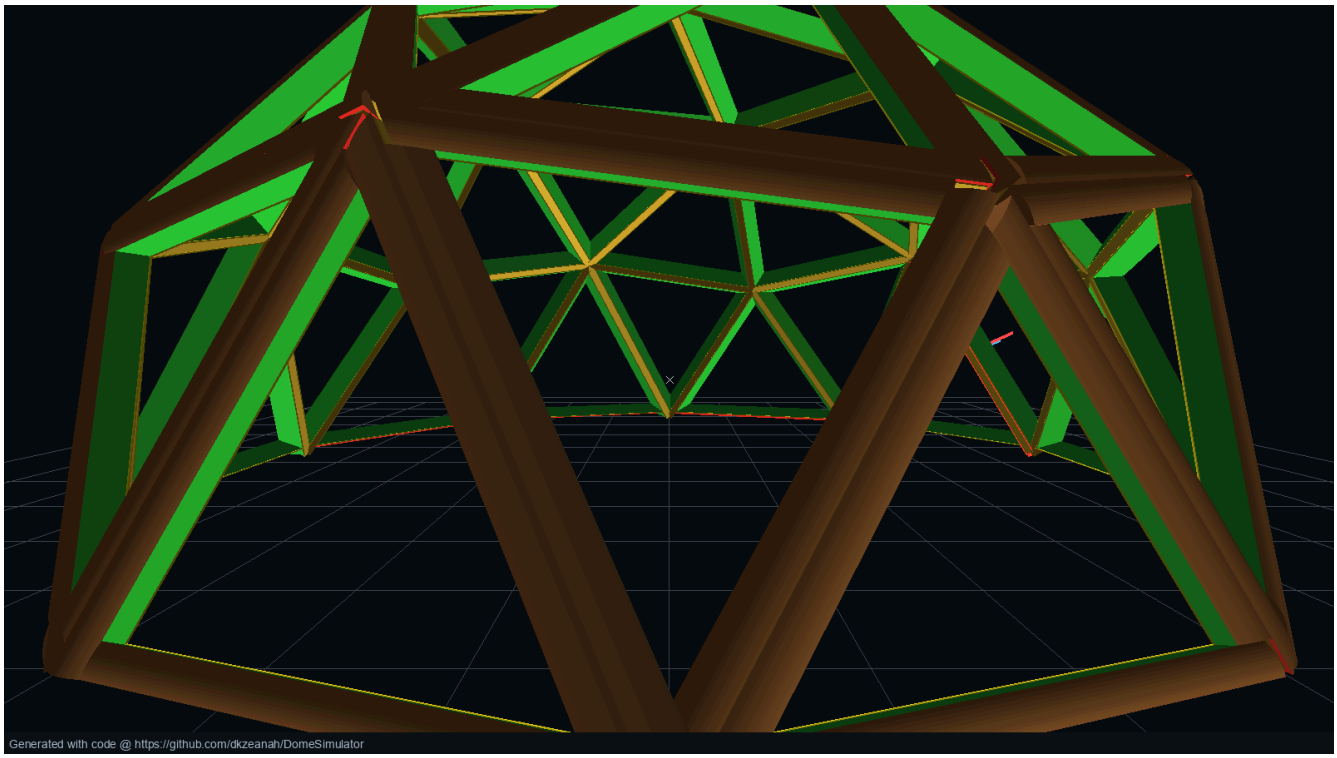
“A frame that is out by an inch stands for fifty years. A skin that leaks for one winter takes the frame with it.”

Raising the shell, the single watertight layer over it, and the hats that go on afterwards -- the part of the building this book says matters most.

Raising It

Twenty panels round the ring, fifteen above them, and five around the apex -- in that order, and braced until the second course closes

Forty panels are stacked against a fence. This part takes three days and it is the only part that looks like building a dome.



The ten panels that stand on the ring.

Before anything goes up

Set the base ring out first. A regular decagon, 10 sides, each one an A chord — so each side is 72.000 inches and the perimeter is sixty feet. Chapter 36 is what it sits on.

Set it out from the centre, every time. Drive a pin where the middle of the floor will be and swing the base radius from it to mark all 10 corners. Do not step round the ring measuring side to side: every small error then adds to the next one.

Get this right and the rest is assembly. Get it wrong and every course above it inherits the error, because the base is the one place in this building where error does accumulate.

Check it by diagonal, not by angle. On a regular decagon the long diagonals, corner to opposite corner, are all the same length – so measure every one of them, and any difference tells you exactly where the setting out drifted.

Sort the panels into two piles. 10 equilateral, 30 isosceles. Mixing them up costs an hour on the second course.

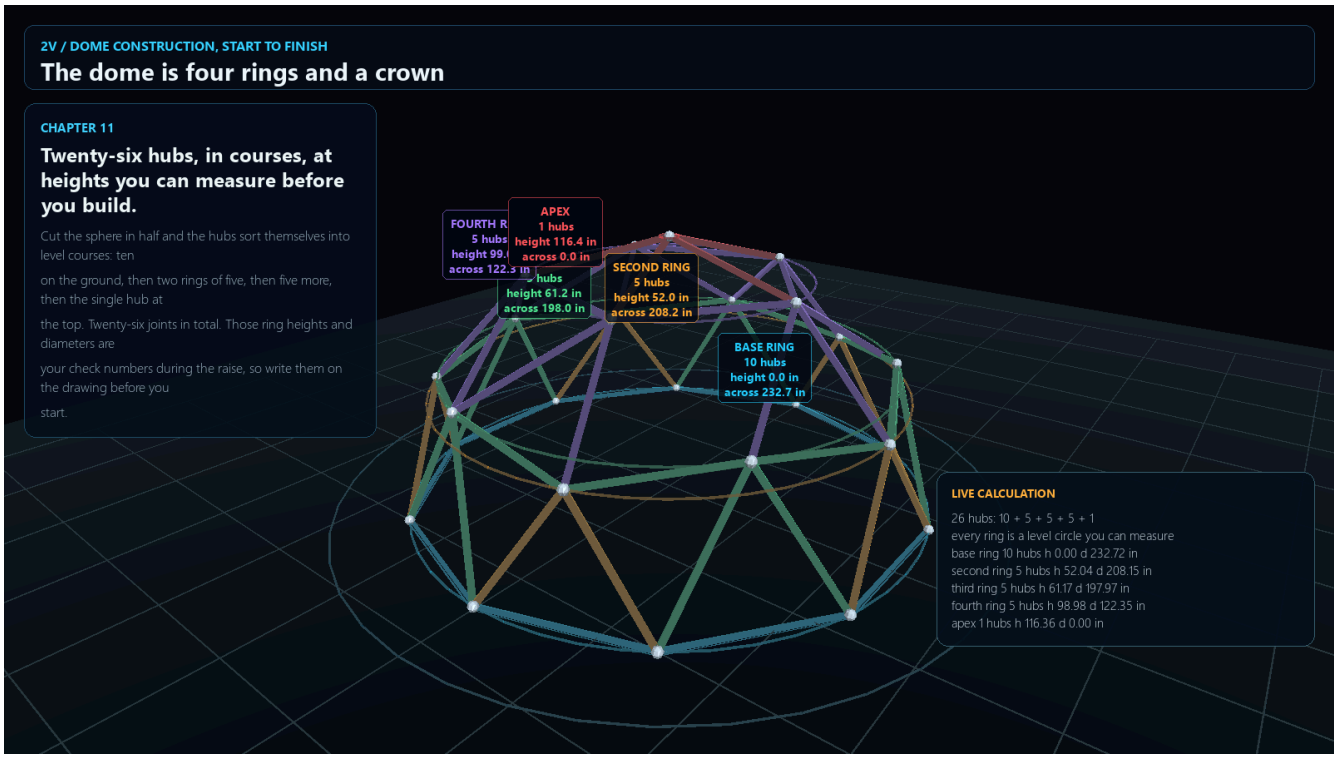
Have more bracing than you think. You will need it for about six hours total and you will need all of it at once.

Four rings and a crown

The corners of the shell sit at fixed heights, and those heights are your check numbers during the raise. Write them on the drawing before you start:

ring	corners	height	across
floor ring	10	0.00 ft	19.42 ft
belt, low corners	5	4.34 ft	17.37 ft
belt, high corners	5	5.10 ft	16.52 ft
pentagon ring	5	8.26 ft	10.21 ft
apex	1	9.71 ft	--

The two belt rows are one zigzag ring that rises and falls between two heights (Chapter 33). Every ring is a level circle you can measure across.



Twenty-six corners, in courses, at heights you can measure.

The first course

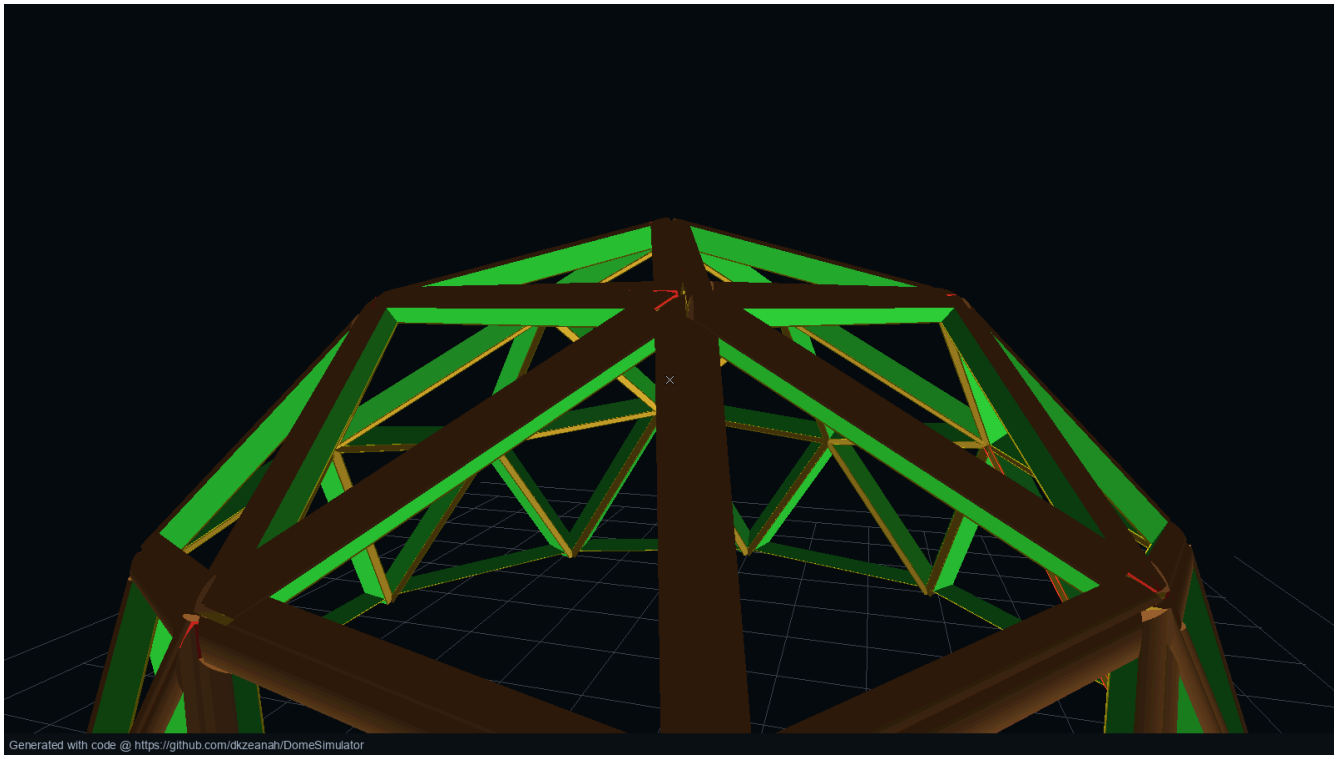
20 panels: 10 standing on the ring with two corners on it, and 10 inverted between them with one corner on it. They lean inward from the moment they go up, which is the first surprise: a dome is not a wall that gradually curves, it is leaning in from the ground.

Stand two adjacent panels, key the seam between them, and brace the pair externally. Then work round. Each new panel keys to the one before, and the run of them wants to fall inward until the ring closes.

The ring closing is the event. Until the tenth panel keys to the first, you have a curved wall held up by props. After it, you have a structure.

Do not take the bracing out yet.

The second course



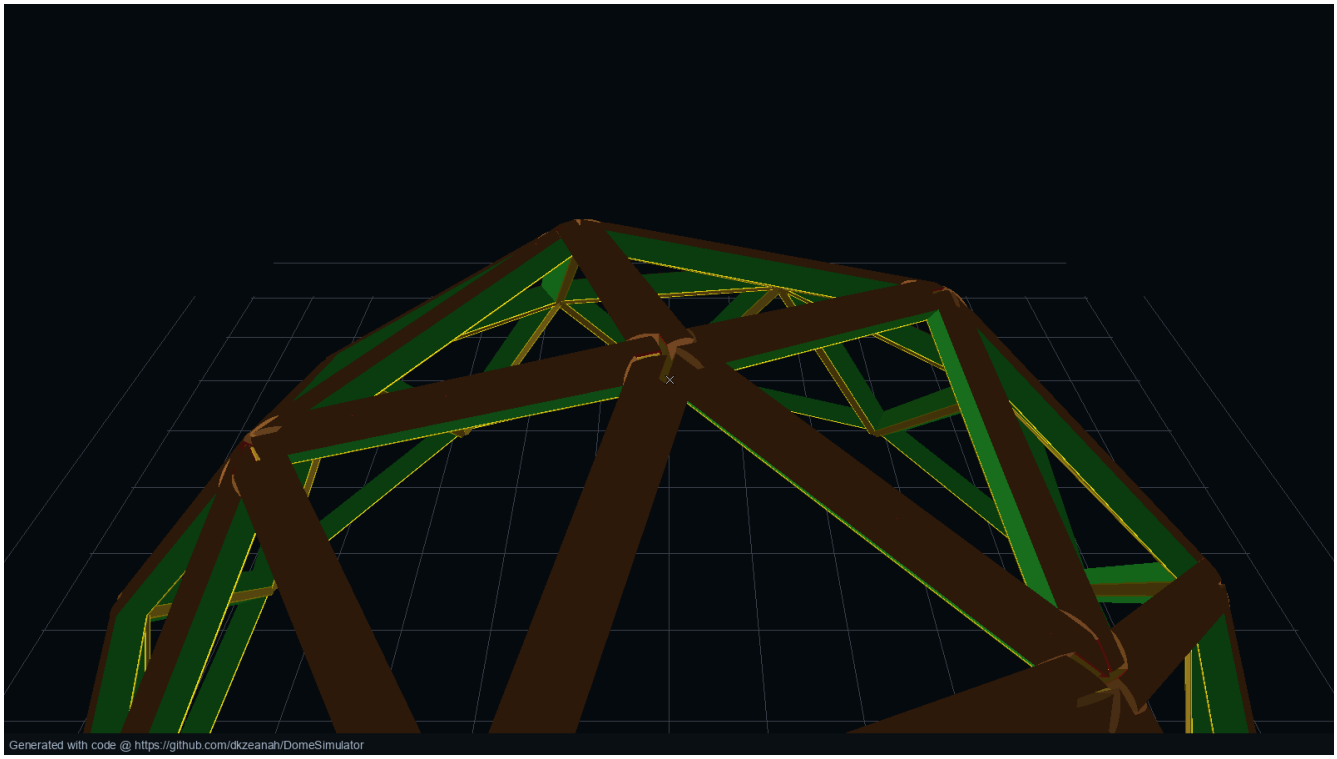
The band above: where the dome starts holding its own shape.

15 panels, leaning further in, each one keying to the course below and to its neighbours either side, up to the pentagon ring.

This is where the dome starts holding its own shape — and it is where the bracing can come out, once the course is closed all the way round. Not before. A half-closed second course is the least stable the building will ever be, because you have added weight and height without adding the ring that resolves it.

Work from opposite sides toward a closure point rather than round in one direction. It keeps the load symmetric and it means the last panel of the course is a panel rather than a wedge-shaped gap you have to persuade.

The apex



Closing the top.

5 panels around one vertex.

This is the last thing to go in and the only part of the assembly that has to be *held* rather than propped. Two people, or one person and a strap from below.

If you have built the mast of Chapter 55, the apex ring is already up there and the last five panels close around it. If not, the apex is five keys and a cap.

What goes wrong

The last panel of a course does not fit. Almost always the ring is out, not the panel. Check the base diagonals before you start cutting anything. The crown is where every accumulated millimetre finally arrives, so it reports the error rather than causing it: if the cap will not close, go back down and measure the ring beneath, which is nearly always slightly out of round or out of level.

Counting the courses wrong. An earlier draft of this chapter said ten panels in the first course, thirty in the second and five round the apex: forty-five panels, on a dome that has 40. The courses above are counted from the solved shell.

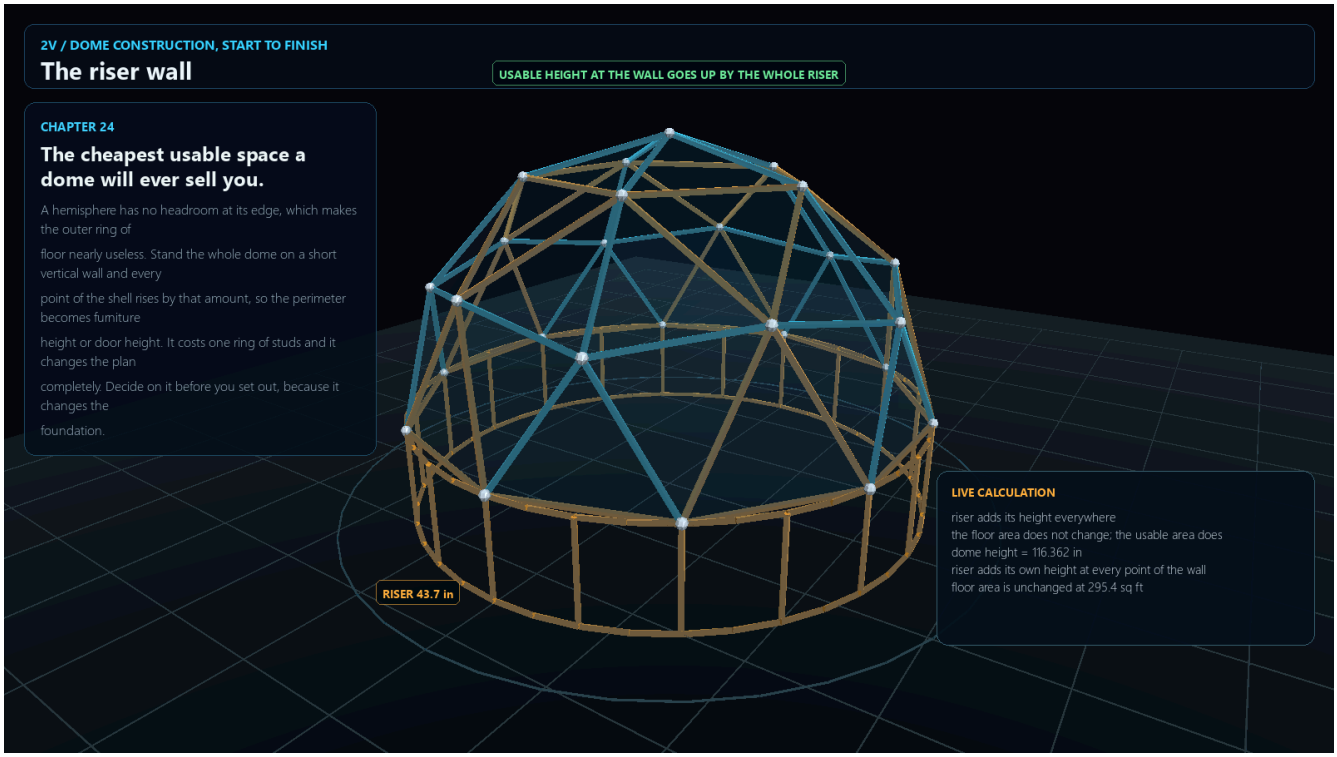
A seam gaps at the outside. The key is not driven far enough. It is a wedge; drive it.

A panel will not key to its neighbour. One of them is in upside down. This happens with the isosceles ones, whose two short sides look alike. Mark the long edge of every BAB panel on the jig, in the shop, before it ever comes outside.

The riser wall

A hemisphere has no headroom at its edge: the shell meets the floor at the rim, which makes the outer ring of floor nearly useless. Stand the whole dome on a short vertical wall and every point of the shell rises by that amount, so the perimeter becomes furniture height, or door height.

The floor area does not change. The usable area does, a great deal, and it costs one ring of studs. Decide on it before you set out, because it changes the foundation (Chapter 36).



The riser lifts every point of the shell by its own height.

Then stop

The frame is up. It is weather-tight in the sense that a bird cage is weather-tight, and it will stand like this indefinitely.

Do not stop here for a winter. That is the failure Chapter 3 describes, and this is the moment it happens: the frame is standing, it looks finished, it is intoxicating, and the skin is the expensive part.

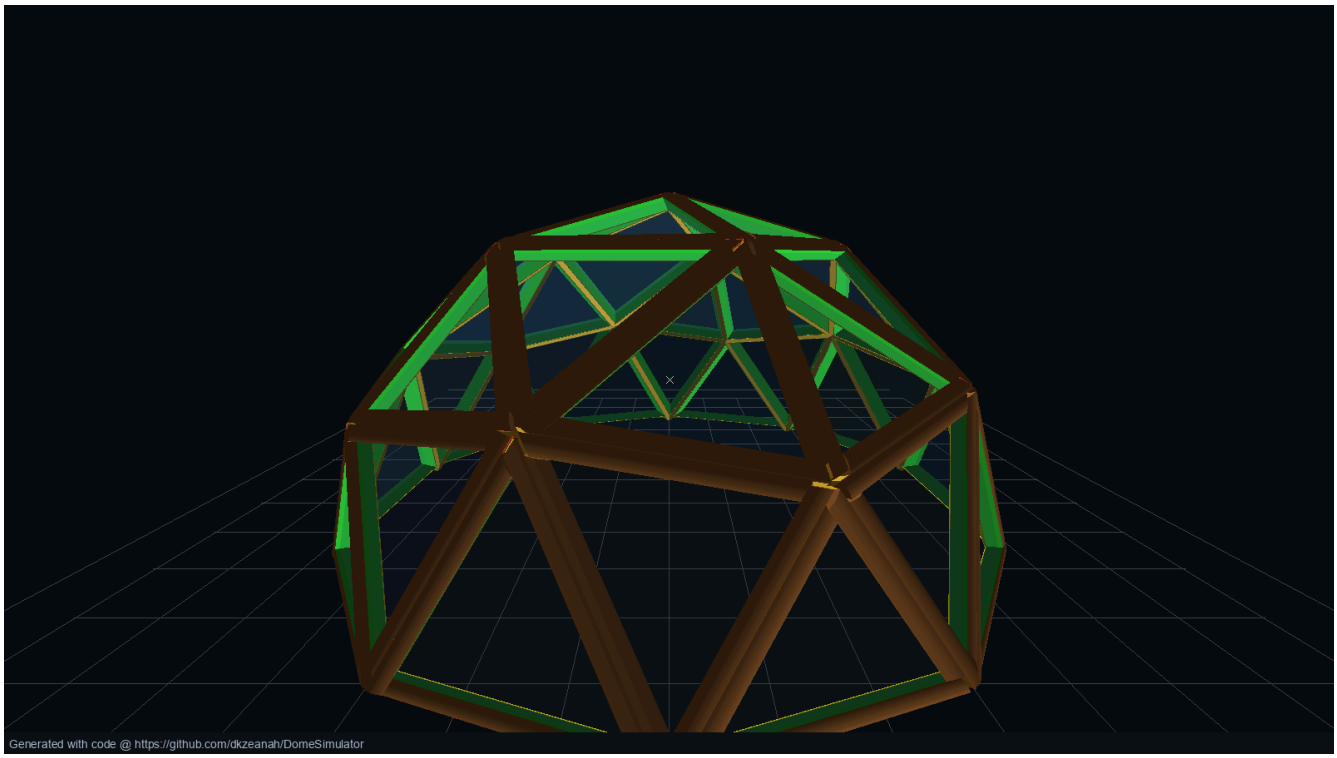
The skin is the part that matters. It is the next chapter, and it is the reason all of this stands up in thirty years' time instead of in three.

One Watertight Layer, and Only One

Two impermeable skins with insulation between them is a trap, and this project shipped that design before it noticed

Put a waterproof layer on the outside of a building and you have kept the rain out.

Put a second one on the inside of the same wall and you have built a bag.



The frame with its skin on.

The design this replaces

The obvious shell is a waterproof membrane *under* the panels and a rain-slick cap *over* them, with the insulation between the two. Two skins, belt and braces.

It is a vapour trap. Water gets into every wall — from the weather, from the air inside, from somebody's breath — and the only question a wall design answers is where it goes next. A wall with an impermeable layer on both sides has no answer. Moisture arrives, cannot leave in either direction, and condenses on whichever surface is colder that night.

So: **one watertight layer, on the outside, like a shower cap** — and under it, a route for vapour to leave. The rest of this chapter is that route.

The stack, outside in

The cap. One continuous rain-slick layer over the entire shell. Welded seams, no fasteners through it, strapped down to ground anchors at the base vertices. Its only job is liquid water and it is very good at that job because it is doing nothing else.

A vented gap. Three quarters of an inch between the cap and what is under it. This is the part people skip and it is the part that makes the rest work: any moisture that gets past the cap, or arrives from inside, has somewhere to go and something moving through it to take it away.

The breather. A vapour-permeable sheet at around ten US perms. Not a poly sheet. It stops bulk water and lets vapour through, which is the exact opposite of what a waterproof membrane does and is why the distinction matters more than the price difference.

The quilted layers. As many as you have. Chapter 30.

The panels. Wood. Structural, not weatherproof, and not asked to be.

The frame. Dry, because everything above it is doing its job.

Which way the vapour goes

Warm air holds more water than cold air. Inside your building the air is warm and full of you, your cooking and your breathing. Outside it is cold.

So vapour moves outward through the wall, and it will condense at whatever point in the wall is below its dew point. The design question is: **can it get out before it gets cold?**

With a breather and a vented gap, yes — vapour passes through the permeable layer into moving air and leaves. With a second impermeable layer, no. It stops at the membrane, at a temperature partway through the insulation, and condenses there, inside the insulation, where you will not see it for years.

That is the whole argument, and it is why the cap is the only waterproof thing in the building.

Four ways to skin the same frame, and what each one costs.

What it costs

A bare cap — no quilted layers at all — is $3,265 on this dome. With layers it climbs slowly; the shell in the standard article is $3,265.

The alternative is a laminated hull: glass fibre in resin over the whole envelope. Genuinely waterproof, sixty years, and $21,513 for the same dome against $12,036 — a real premium for a real difference.

Chapter 49 prices four laminates and the cap side by side. The short version: the hull is better at being waterproof and worse at being changed.

It is not exotic. It is a wood-cored composite skin – a sheet core, chopped strand mat, stitched biaxial cloth, laminating resin, and gelcoat on the weather face – which is how small boats have been built for sixty years, from products you can order this afternoon. In the stem-cell product of Chapter 52 it latches to the pad with over-centre catches, the kind that seal a preserving jar, and lifts off whole from a ring laminated into its rim. So it can come off. What it cannot do is change: a hull is one stiff, finished shape, so the quilted layers of Chapter 30 have no room to grow under it, and when it fails it fails as a repair on a bonded laminate rather than as a component you unstrap and replace.

The shell is a boat hull.

The three things a cap has to get right

It has to be one piece, or welded. A cap made of overlapping panels is just a roof with more seams, which is the problem it exists to remove.

It has to be strapped, not fastened. Every fastener through a waterproof layer is a hole in a waterproof layer. Straps over the outside to anchors at the base put the load into the ground without putting anything through the membrane.

It has to be bigger than it needs to be. Chapter 3: a cap cut to fit a bare shell will not go over a shell wearing two quilted layers.

A thing we found out the hard way: no panel fits a four-foot sheet.

If you skin it in panels anyway

Some builders will sheathe the frame panel by panel and waterproof the panels, and the rules for that are old ones. Start at the rim and work up, so each panel laps over the one below it like a shingle and sheds water downhill. Every corner where several panels meet is a leak point: flash or tape each one before the next course covers it. And give the base a real drip edge that throws water clear of whatever the dome stands on. A dome almost never fails structurally; it fails at its seams.

For a small dome the panels can be the whole building. Cut the 40 triangles as solid sheet, pair the congruent ones into parallelograms and strip-pack them: on one 10 ft x 5 ft sheet the largest dome that fits is 62 inches across, with 21 square feet of floor. A shelter, a kennel, a seed store. Not a house, and it does not pretend to be.

One shell, any skin

The bones do not care what is painted on them. The same panels and the same 120 sticks wear a baseball, a basketball or a disco ball with a facet on every face. And be honest about what a “wall module” is: plywood, insulation and sheet metal, in a frame that holds them. The value is not in the materials, which anybody can buy. It is in the socket they drop into (Chapter 21).

The part that is not solved

Ultraviolet.

A rain-slick outer layer sits in the sun for its whole life, and that is what eventually finishes it — not water. The cap is a consumable on a decade-ish timescale, and the design’s answer is that it is a consumable **by design**: unstrap it, pull it off, put the next one on, in a day, with the building occupied throughout.

That is a genuinely different answer from a roof, and it is the right one for a component whose failure mode is known and gradual. It is still a cost, it is still a thing you will do four or five times in the life of the frame, and this book would rather say so than let you find out in year twelve.

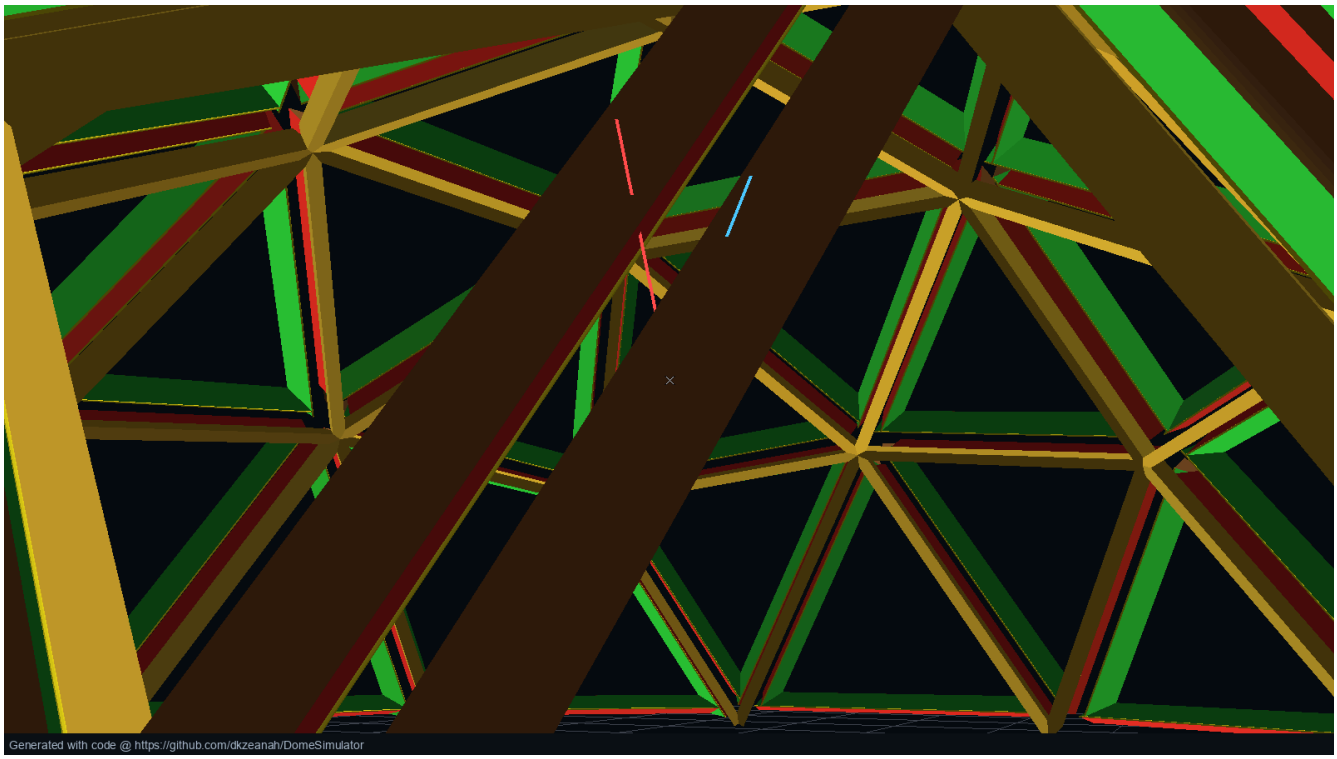
The Seam Does Four Jobs

Gutter, vent, dehumidifier and air barrier -- all of it in the space two sawn faces were always going to leave

The channel is already there.

Two sawn faces meeting at a dihedral angle cannot close flush — Chapter 20 — so every one of this dome's 55 seams carries a void running its whole length. 309 feet of it, threaded through the structure, touching every panel, arriving at the apex and at the base ring.

You are going to fill it with something. The question is what.



Every seam in the dome, and the channel in each one.

Four jobs, one space

Rain in at the ridge. A slot along the outer cap takes water where it already runs — the seam is the low line between two panels, which is where a roof sends water anyway. Under the slot is a gutter with suction inlets, and under that a drain to the tank. The building's guttering is inside its structure rather than bolted onto its edge. *(Chapter 33 corrects this: rain belongs in a gutter at the rim, outside the frame.)*

Air along the wood faces. A perforated liner washes air down both sawn faces of every member. That is the surface most at risk in a timber frame and the hardest to reach, and it is now the surface with moving air on it permanently.

A desiccant leg. A three-way gate routes the airflow either along the faces, or through a silica cartridge, or isolates the water path from the air path entirely. In wet weather when moving air will not dry anything, you pass it through the desiccant instead.

And a plate that condenses on purpose — the part that needs its own section, below.

The reason all four fit is that they want the same thing: a continuous, serviceable void that runs everywhere and reaches the ground. Nobody would build that void for any one of these jobs. It was free.

The air barrier

Here is the part that changes how you think about leaks.

Run the fan so the interior sits slightly **above** outside pressure, and the net flow at every gap in the envelope is outward.

A gap in an inward-leaking building admits weather. The same gap in an outward-flowing one does not — not because it is sealed, but because the air is going the wrong way for anything to ride in on. Wind-driven rain at a seam meets air coming out of it.

The numbers are undramatic, which is the good news:

```
interior volume        1,916 cu ft
at 0.55 air changes an hour   18 cu ft a minute
held above outside     4 Pa
spread over            309 ft of seam
```

4 pascals is far below what anybody feels on a door, and 18 cubic feet a minute is a bathroom extractor. Two fans — 2 — so the barrier can be reversed, because there are days when you want to draw rather than push.

What it costs honestly: it runs continuously, and it only works while it runs. A positive-pressure envelope with the fan off is an ordinary envelope with holes in it. That is an argument for sizing the array in Chapter 58 so this load never has to be shed, and it is an argument against treating the barrier as a substitute for the cap of Chapter 28. It is a second line, not the first.

Condensing on purpose

Thermoelectric plates — Peltier modules — on the cold side of the channel, dropping moisture out of the air as liquid straight into the gutter the rain already uses.

The appeal is obvious. The plate sits exactly where the moisture is highest, it is doing dehumidification whether or not you want the water, and the water it makes needs no new plumbing because the drain is right there.

Now the number.

```
one inch of rain on this roof     185 gal
the plates, running a day         0.20 gal   (360 W)
the plates, running a year        73 gal
which is the same as              0.4 inches of rain
energy per litre                  2.86 kWh
```

A whole year of condensing is worth less than half an inch of rain.

At 2.86 kilowatt hours a litre, thermoelectric condensing is several times worse per watt than a compressor, and a compressor is already not how anybody sensible makes drinking water on a roof.

So the honest position, and it is the one this book takes:

The plates are a dehumidifier that happens to yield liquid. They are not a water supply. The water supply is the roof, and the roof is 185 gallons an inch.

Which leaves them worth fitting for one reason only: there are conditions — warm, humid, still, no rain for weeks — where the channel will not dry itself and the desiccant is saturated, and in those conditions a plate is the only thing in the design that can actively remove water from the timber. Run them on daylight surplus, treat the drips as a bonus, and never size a tank on them.

What the module costs

```
  outer cap with intake slot          309.2 ft              $3.40    $1,051
  vented liner, perforated            309.2 ft              $2.60      $804
```

water gutter with suction inlets	309.2 ft	$2.10	$649
drain line to the tank	309.2 ft	$1.15	$356
three-way gate, one per segment	55.0 each	$46.00	$2,530
silica cartridge drawer	55.0 each	$22.00	$1,210
service cover	55.0 each	$9.00	$495
thermoelectric condensing plate	6.0 each	$14.00	$84
ram fan, continuous duty	2.0 each	$185.00	$370

Seam module, whole dome			$7,549

$7,549 against a dome at $12,036 — a little over half the price of the building it is fitted to.

That is a lot, and it is why this is a chapter about a system rather than a line in the standard article. The parts of it scale differently, too: the cap, liner, gutter and drain are per foot of seam and grow with the dome, while the gates, cartridges and plates are per seam and grow more slowly.

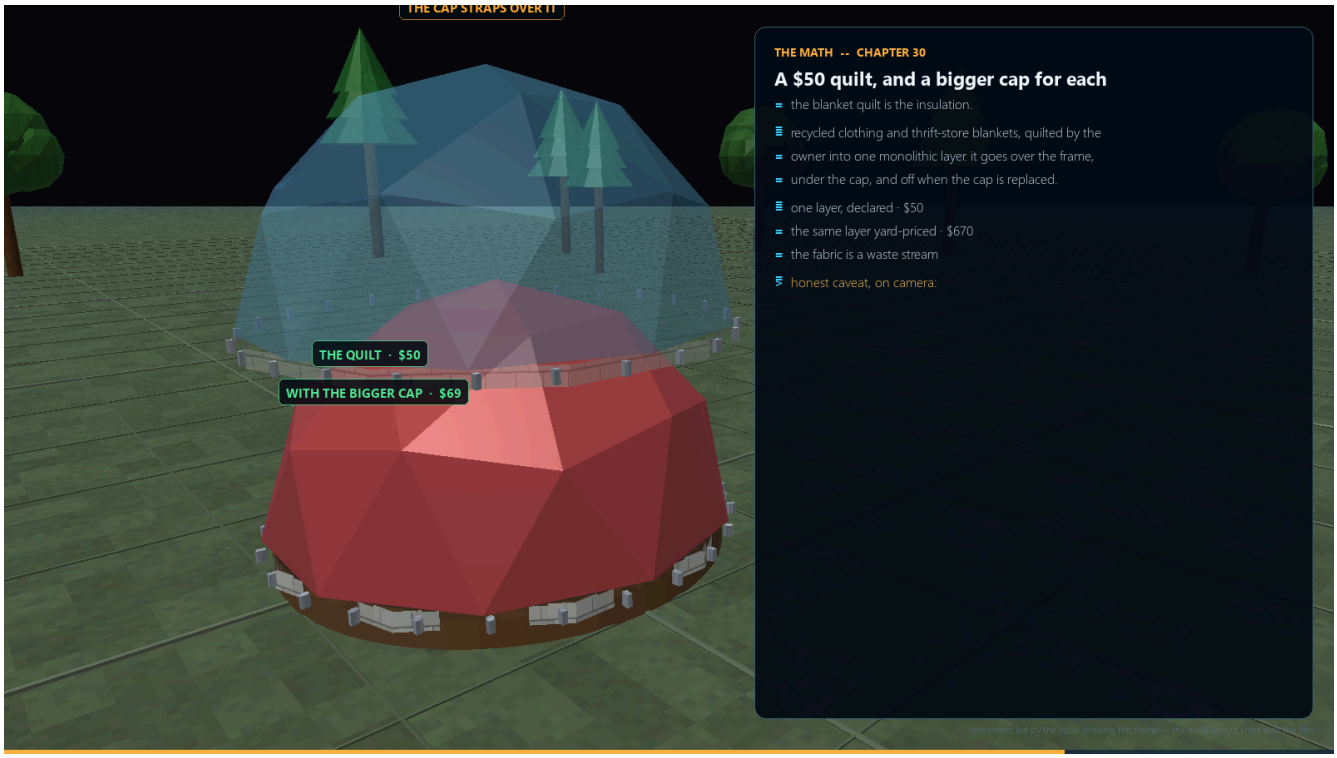
If you fit one part of it, fit the gutter. Water off the roof is the single best return in this chapter, it needs no power, and it is the only piece that pays for itself in a season.

CHAPTER 30 · EXPLAIN

A Dome Is a Head, and It Wears Hats

Bare skin, then layers, then one rain-slick shell over the lot -- the way you dress for the Arctic, and for the same reasons

I learned how to dress a building standing on the outside of a ship in the Arctic Circle, on forward lookout, at night.



A quilt, and a bigger cap for each one.

What you wear out there

Nobody wears one warm coat. You wear layers — several of them, each thin, each doing one job — and on top of all of it goes a pumpkin suit: an immersion suit, bright orange, entirely waterproof, and utterly useless as insulation on its own.

The suit does not keep you warm. The layers keep you warm. The suit keeps the layers *dry*, and a dry layer is worth several wet ones.

The other thing about that arrangement is that it is adjustable in the middle of a watch. Too cold, add a layer. Too warm, lose one. You never take off the suit.

That is the whole shell design in this book, and I did not arrive at it by reading about vapour barriers.

The dome as a head

A dome is a head. It is round, it is the top of something, and it wears hats.

It can wear a bare skin — nothing at all, a frame with panels on it.

It can wear a toboggan: a quilted layer, then another, stacked, each one adding warmth.

And over all of it, always, one shower cap. Waterproof, thin, useless as insulation, and the reason every layer under it is dry.

People understand this instantly in a way they do not understand “vented rain screen with a vapour-permeable weather-resistive barrier”, which is the same idea in a language that makes people stop listening.

Each hat is a size up

This is the part that is arithmetic rather than analogy.

A layer goes *over* the layers already there, so it is bigger than the one under it. Not a lot bigger — the shell is only a few inches thicker each time — but the areas compound, and by the seventh layer the outermost is meaningfully larger than the innermost.

So “one quilt costs fifty dollars” is a sentence about the first quilt. The model works out each layer’s own area, and the cap that goes over all of them is sized for the stack you intend to end up with rather than the stack you have.

The envelope gaining a layer a winter.

It gets warmer every winter

Here is the model’s ladder, with its one assumption stated:

```
REMOVABLE SHELL / upgrade the insulation over time
Model assumption: +R-1.6 per quilted layer
0 added layers -> R-1.5
1 added layers -> R-3.1
2 added layers -> R-4.7
3 added layers -> R-6.3
4 added layers -> R-7.9
5 added layers -> R-9.5
6 added layers -> R-11.1
7 added layers -> R-12.7
Actual assemblies need tested moisture and fire performance.
```

```
The outer shell supplies the weather and wind barrier.
Insulation improves when you add suitable layers; age alone adds none.
```

The improvement comes from adding a layer, not from the house getting older; the outer shell keeps its separate job of keeping weather and wind out.

Here is what the layering model buys that a conventional wall does not.

A conventional wall is insulated once, when it is built, to whatever you could afford that year. Changing it later means opening the wall.

This one is insulated from the outside, in increments, forever. Each autumn you can add a layer — an evening's work with a sewing machine and a box of old clothes — and the building is warmer than it was last winter.

What each quilted layer is actually worth.

The cap goes back over the top and nothing else about the building changes.

A house that improves as you live in it is a different relationship from a house that is as good as it will ever be on the day you move in.

And the quilts are made of clothes

The insulating layers in this design are quilted from recycled textiles — clothing that would otherwise be waste.

That is partly an economic argument and it is mostly a participation one. It is the only part of this entire method that somebody can do at a kitchen table with no land, no saw and no site. A layer is a real object with a number on it that goes on a named building, and it takes an evening.

The arithmetic is in the model: how many garments, how many pounds, what R-value the stack reaches, and what a layer is worth to the person who sews it for somebody else's dome. That last one has to be a real number or this is not a network, it is a request for free labour.

What actually closes a triangle

A wedge is not a rectangle, and the wall uses that. The member's inner face stands proud of where a panel sits, so every bay in the frame already has a lip in it – not machined, just left there by the shape of a split log (Chapter 21). So the wall goes together from the inside out:

1. **An inner panel** drops onto the lip from inside.

2. **The cavity** behind it is as deep as the member, 6.0 inches.
3. **An outer panel** drops into the same bay from outside and compression-fits against the frame.
4. **The shell** lands over everything, and nothing goes through it.

Less skin to start with

Before any hat, the shape has already paid. Compare a dome with a box of the same floor – worked here for the films' 20-foot dome, slightly larger than this book's reference:

```
same floor, both shapes: 314 sq ft
box: side = sqrt(314) = 17.7 ft, walls 8 ft, 0.50 pitch roof
box skin  = walls + roof + gables = 997 sq ft
dome skin = 40 flat triangles     = 583 sq ft
difference = 414 sq ft = 42% less to build, seal, paint, heat
volume enclosed: dome 2,093 cu ft vs box 3,208 cu ft
wind drag (published): dome Cd 0.42 vs box Cd 1.05 = 60% less load
```

Less skin means less to lose heat through in winter and less to gain it through in summer, and it is true before a single layer goes on. Run the arithmetic a heat-loss engineer would, with the same wall in both buildings and a middling American climate – the climate and the power price are borrowed, and named:

```
wall build-up: R-24.4 -> U = 1/R = 0.0410
heat loss rate = skin x U:
dome: 583 sq ft x 0.0410 = 24 BTU/hr/F
box:  997 sq ft x 0.0410 = 41 BTU/hr/F
x 4200 heating degree days and 1200 cooling
dome: 756 kWh/yr = $129 at 17 cents
box:  1,292 kWh/yr = $220
```

The dome wins by exactly its missing skin, every year. One more multiplier is available on the outside face: the white radiative paints developed in the last few years reflect almost all of the sun and shed heat to the sky even in daylight. That is a published research result rather than something this model computes, so it is left out of the figures above.

The wall as a filter

A dome holds a lot of air behind a lot of wall, and that ratio does something useful. Push fresh air in through the shell itself, even hard, and it crosses the wall so slowly that nobody inside feels a draught:

```
dome radius 60 in: shell 14.6 m2, air 262 cu ft
     6 ACH ->    26 cfm, through the wall at 0.17 ft/min
    20 ACH ->    87 cfm, through the wall at 0.56 ft/min
    40 ACH ->   175 cfm, through the wall at 1.11 ft/min
```

A draught you can feel starts at many times those speeds. So a breathable shell could be a filter face for the whole building, with the flow spread over every square foot instead of rushing past one vent. Chapter 29 and the seam chapters are what the air does once it is inside the frame.

The shell so large the air barely crawls through it.

How much shell can carry solar?

The shell is also the dome's largest surface facing the sky. Three ways to use it, on the same dome:

```
SAME DOME: 20 ft across
Panel areas are measured from the model's faces.
Assumed panel density: 18.0 W/sq ft
one side: 292 sq ft -> 5.2 kW
top half, skirt below: 292 sq ft -> 5.2 kW
whole shell: 583 sq ft -> 10.5 kW
No panels = zero installed solar capacity.
Selected layouts: 5.2 to 10.5 kW nameplate.
Face area sets capacity, not guaranteed energy yield.
Sun angle, shade, climate and losses set actual kWh.
The useful limit is the available shell area.
```

Cells on one side give a rotating pad (Chapter 37) a clear purpose: turn the useful face to the sun. Cells on the top half with a metal skirt below make the skirt a rain catchment. Cells everywhere give the most capacity and the least reason to turn. In every case, face area sets the nameplate, not the year's energy.

One side, top half, whole shell.

Where the analogy stops

Two places, and both matter.

A person moves and a building does not. Your layers work partly because you are generating heat inside them and moving air through them. A building generates less heat per unit of surface than you do and does not move at all, so it relies on the vented gap of Chapter 28 to do what your movement does.

You take your suit off indoors. The dome does not. The cap is on it in July, in the sun, which is the ultraviolet problem at the end of the last chapter and the reason the cap is a consumable.

Neither of those breaks the model. They are the two places where a builder who reasons from the analogy alone will get it wrong, so they are written down.

PART 6

The Seam at Work

“Send air toward the wood only when it is drier than the water you want to remove -- unless it is going to a cold metal surface built to catch and drain it.”

What fits in the channel and what does not, the rule that decides which way its air moves, what each level of the dome does, where the desiccant belongs, and the metal that makes it work.

What Fits in the Channel

Water and wire, yes. A round duct, no -- and the channel does not need one, because it already is one

Chapter 29 fitted the seam out as gutter, vent and dehumidifier and priced it. It did not say how much room there is inside it, and every idea anybody has for the channel – a water line, a circuit, a duct, a desiccant – runs into that number first.

First, what the key is. It fills the V between the two members of a seam, and it is what the seam's bolts pull the two panels against. Make it solid and it is a spline. Make it hollow and it is a pipe running the length of every seam – 309 feet of it, touching every panel, reaching the crown and the ground. The room inside that pipe is the whole budget for everything below.

So here it is, from the solver's own seams, for the reference build.

The V between two members is 22.5 degrees at the tighter of the two seam types and 27.0 at the wider, and as deep as the log's radius. The key that fills it is printed with 3 millimetre walls, and it stops short of the ridge where the V is too narrow to be any use. What is inside those walls is the budget:

```
inside the key, tighter seam      5.3 sq in
inside the key, wider seam        6.5 sq in
largest round thing, tighter      1.69 in
largest round thing, wider        1.97 in
```

Everything in this chapter has to fit in that.

A water line, a circuit, the plates' pair and the drain, in the tighter of the two seams.

The sizes this chapter assumed

The pipe and tubing sizes are nominal, from their standards. The cable sizes are estimates, because every brand's sheath is different. The rule for how full a channel may be is borrowed from electrical conduit, which allows 40 percent of the area for a mixed bundle so the last thing can still be pulled through. The channel is not a listed raceway; the rule is used because it is a sensible allowance, not because it applies.

1/2 in PEX water line	nominal
3/4 in PEX water line	nominal
14/2 NM-B cable, 15 A circuit	estimate
12/2 NM-B cable, 20 A circuit	estimate
12 V pair to the Peltier plates, 14 AWG	estimate
condensate drain, 1/4 in ID vinyl	nominal
3 in round air duct	nominal
4 in round air duct	nominal
fill fraction	0.4 of area: rule borrowed from conduit fill (NEC Chapter 9, Table 1)
key wall in	0.11811 in: assumption
printer bed in	9.84252 in: assumption
filament g per cm3	1.27 g/cm3: nominal
filament usd per kg	20 USD/kg: assumption
print mm3 per s	15 mm3/s: assumption
quiet air fpm	400 ft/min: rule of thumb

What fits, item by item

item	size	22.5 deg seam	27.0 deg seam
1/2 in PEX water line	0.62 in round	fits	fits
3/4 in PEX water line	0.88 in round	fits	fits
14/2 NM-B cable, 15 A circuit	0.40 in x 0.20	fits	fits
12/2 NM-B cable, 20 A circuit	0.46 in x 0.22	fits	fits
12 V pair to the Peltier plates, 14 AWG	0.30 in round	fits	fits
condensate drain, 1/4 in ID vinyl	0.38 in round	fits	fits
3 in round air duct	3.00 in round	no	no
4 in round air duct	4.00 in round	no	no

Water and wire fit, with room to spare. A half-inch PEX line, a fifteen-amp circuit, the twelve-volt pair to the condensing plates and the condensate drain go into either seam together, filling 11 percent of the tighter one and 9 percent of the wider one, against the 40 allowed.

Round ducts do not. The smallest ordinary duct is three inches round. The largest round thing either seam passes is 1.97 inches. If the plan was to run ducts through the seams, the plan does not work as drawn, and a bigger log does not rescue it.

The smallest ordinary duct, against the V it would have to go in.

The channel is the duct

It does not need a duct in it, because it already is one.

With the water, wire and drain inside, the free space in one tighter seam still moves 13 cubic feet of air a minute at a quiet 400 feet a minute. The whole dome needs 18 for fresh air. So **2 seams carry the building's ventilation**, and every other seam is free for drying the frame, which is the job of Chapter 32.

If you want a duct anyway: the key becomes a spacer

There are reasons to want a real duct: a heat-recovery ventilator with round spigots, a stove exchanger, a fan you already own. Then the key stops being a filler and becomes a spacer.

Push every panel outward along its own face and every seam opens. The two seam types have different folds, so one push opens them by different amounts, and the push is set by whichever needs it most:

```
to fit a 3 in duct                panels out 7.0 in, dome +6%
a 4 in duct beside the bundle     panels out 16.4 in, dome +14%
seams open at the ridge           6.4 / 5.1 in
```

A spacer is not free. Moving every panel out makes the whole dome bigger: more wood, more skin, a bigger pad. And every opened ridge needs a cap over it. All of that, to carry air the plain channel was already carrying.

As built, and with the seams opened for a four-inch duct.

Or start with a bigger log

The other lever is the log. The room in the channel grows with the square of the log's radius:

```
10 in log    3.5 sq in inside, 1.37 in round
12 in log    5.3 sq in inside, 1.69 in round
15 in log    8.8 sq in inside, 2.17 in round
18 in log    13.1 sq in inside, 2.65 in round
```

Even the biggest of those still falls short of passing a three-inch duct. A bigger log buys more room for water and wire; it does not buy a round duct.

Printing the key in halves

The key is printed, and it is printed in halves. Split it down the middle of the seam and each member carries its own half, screwed to its sawn face before the panel is built. When two panels meet, the two halves close into one channel. Nothing has to be threaded into a finished seam.

1. **Two profiles.** The dome has two seam types, so it has two key profiles: 2 files, not 55.
2. **Print after the frame has dried** (Chapter 17), to the V the dried wood actually has.
3. **Print in pieces** no longer than the printer's bed, assumed at 250 millimetres: 8 pieces to a member, butted end to end.
4. **Bolts cross near the ridge**, where the V is narrowest. Route services through the wide belly below them.
5. **Print only the seams that carry something.** Every seam in the dome is 110 half-keys, 820 pieces, 121 kilograms of PETG, about $2,425 and 1,768 printer hours. One seam is 2.2 kilograms, $44 and 32 hours. Cut solid keys from the offcuts for the rest.

Each member carries its own half of the key.

Where the channels meet

The frame has no hubs, but the services need them. Channels meet at 6 five-way and 10 six-way vertices, and at 10 more on the rim, where they drop to the pad.

As built, the Vs close to a point at each vertex, so each junction is a printed **rosette** fixed to the inside of the vertex: a small box where a pipe turns and a wire is joined, somewhere a hand can reach. With the four-inch spacer the vertex opens into a hole about 11 inches across, and the rosette becomes a node the channels plug into.

A printed rosette inside a five-way and a six-way vertex.

Two rules go with every rosette. **No pressure fitting inside a seam**: run the PEX in one continuous length and make every joint in a rosette you can open, because a hidden leak inside a wooden seam is the one failure this whole system exists to prevent. And **whether a cable may run in this chase is your inspector's decision**, not this book's.

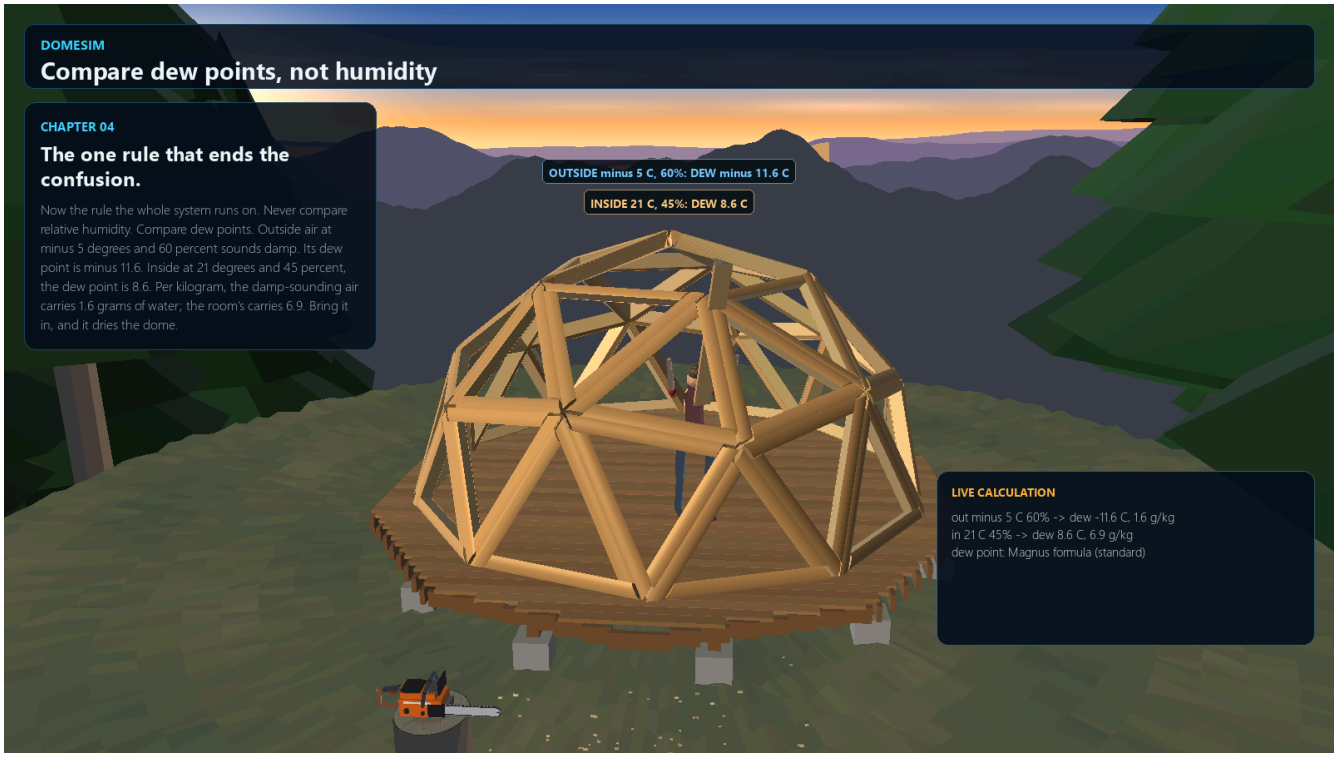
CHAPTER 32 · EXPLAIN

Compare Dew Points, Not Humidity

Three questions decide every move the seam's air makes, and none of them is about relative humidity

The seam channel can move air in almost any direction you like. Outside air in, inside air out, round in a loop, through a desiccant or past it, over a cold plate or a warm one. That is a lot of choices, and the way a system like this goes wrong is by making them on a feeling: it is a damp day, so shut everything.

This chapter makes them on one number.



Two airs, and the only number that says which one is drier.

The one rule

Never compare relative humidity. Compare dew points.

Relative humidity says how full the air is compared to how much it could hold **at its own temperature**. Cold air holds very little, so cold air reads as damp while carrying almost nothing. The dew point is the temperature at which the water in the air starts coming out as liquid, and it depends only on how much water there actually is. The lower the dew point, the drier the air, whatever the thermometer says.

Take a winter day:

```
outside   minus 5 C at 60%    dew point minus 11.6 C    1.6 g/kg
inside    21 C at 45%      dew point 8.6 C     6.9 g/kg
```

The outside air sounds damp. It carries 1.6 grams of water in every kilogram; the room’s air carries 6.9. Bring it in, warm it up, and it dries the dome. A rule that looked at humidity would have kept it out.

The dew point comes from temperature and relative humidity by the Magnus formula, which is what every weather station uses. A cheap sensor gives both inputs.

The three questions

The controller asks three questions, in this order.

1. Which air is actually drier? If the outside dew point is below the inside one, outside air has drying power. If it is above, outside air is a source of water, however cool it feels.

2. Will this air condense on that surface? Water comes out of air on any surface colder than the air's dew point. So a condensing plate must be colder than the dew point, and every piece of wood must be warmer than it, with a margin. This book keeps wood 3 degrees above the dew point of any air that touches it.

3. Which way should the air move? The answer to the first two decides it: toward the structure only with air drier than the water you are trying to remove – unless you are deliberately sending wetter air to a cold metal surface built to collect and drain it (Chapter 35).

The modes

That gives the one channel 7 ways to move air:

```
CLOSED        rain: exposed intakes shut
INTAKE        outside -> seam -> inside
EXHAUST       inside -> seam -> outside
CONDENSE      inside -> cold seam -> drain -> outside
FRAME_LOOP    seam -> desiccant -> seam
PURGE         outside -> seam -> outside
DRY_SUPPLY    dehumidified air in, slight positive pressure
```

Two of them never touch the living space at all. They are the ones that make this more than a ventilation system.

Eleven weathers, one controller

To test it, here is the table the owner of this design drew up – the weathers that matter and what the seam should do in each – plus one more: muggy outside with a wet frame. The temperatures and humidities are this book's estimates of typical conditions, not measurements from a site. Wet weathers carry a wood moisture reading over 19 percent.

The controller sees only the numbers, and works only from dew points:

weather (estimated)	out	in	dew out/in	mode
Cold + dry outside	-5 C 60%	21 C 45%	-12 / 9	INTAKE
Cold outside + humid inside	-5 C 80%	21 C 68%	-8 / 15	EXHAUST
Cold outside, harvesting water	-2 C 80%	21 C 68%	-5 / 15	CONDENSE
Mild + dry outside	16 C 40%	21 C 45%	2 / 9	INTAKE
Hot + dry outside	33 C 18%	21 C 50%	6 / 10	INTAKE
Hot + humid outside	32 C 70%	21 C 50%	26 / 10	DRY_SUPPLY
Hot + humid, cooling on	35 C 60%	24 C 50%	26 / 13	DRY_SUPPLY
Rain	14 C 97%	21 C 45%	14 / 9	CLOSED
After rain, dry outside	18 C 45%	21 C 45%	6 / 9	PURGE
Sun on a wet skin	24 C 55%	21 C 45%	14 / 9	PURGE
Muggy outside, wet frame	36 C 60%	24 C 55%	27 / 14	FRAME_LOOP

Every one of the 11 gets the answer the owner's table gives, and the book's own checks fail if one ever does not.

The loop and the purge

The two modes worth the extra plumbing are these.

The frame loop takes the seam's own air, dries it in the desiccant and sends it back round the seam. Nothing conditioned is thrown outside and nothing muggy is pulled in. On a hot, humid day with a wet frame – the one weather where neither the inside nor the outside air can help – it is the only thing in the design that dries the wood.

The purge takes dry outside air in at one port, through a wet seam, and straight back out at another. The wet air never enters the room. It is how a frame recovers after rain, and on a sunny day it vents the vapour the sun drives out of a wet skin before that vapour gets driven inward.

Both need something the plain barrier fan does not: two outside ports per circuit, and a gate that can route a seam to outside, to the room or to the drawer. That is the price of separating **drying the structure** from **ventilating the house**. It is worth paying. They are different jobs, and a building that does both with one airflow does neither well.

The controller, as code

The whole decision fits on one page. This is the function the book's tables come from, trimmed of its comments:

```
def decide(s):
    dp_in, dp_out = dew_point(s.t_in, s.rh_in), dew_point(s.t_out, s.rh_out)
    dp_seam = dew_point(s.t_seam, s.rh_seam)
    margin = C["dp_margin_k"]
    wet_frame = s.wood_mc >= C["mc_wet"] or s.rh_seam >= C["seam_rh_wet"]
    if s.rain:
        return "FRAME_LOOP" if wet_frame else "CLOSED"
    if wet_frame:
        return "PURGE" if dp_out < dp_seam - margin else "FRAME_LOOP"
    if s.rh_in > C["rh_high"]:
        if dp_out < dp_in - margin:
            if s.harvest and s.t_out <= dp_in - C["plate_below_dp_k"]:
                return "CONDENSE"
            return "EXHAUST"
        return "DRY_SUPPLY"
    if dp_out < dp_in - margin:
        return "INTAKE"
    if dp_out > dp_in + margin:
        return "DRY_SUPPLY"
    return "IDLE"
```

The sensors it needs are ordinary ones: temperature and humidity inside, outside and in the seam; a wood moisture pin; a rain switch. The margin stops the controller flapping between two modes when the two airs are nearly the same. Every value it compares against is declared, with its kind:

```
magnus_a              17.62 -          standard
magnus_b_c            243.12 C         standard
air_density           1.2 kg/m3        standard
air_viscosity         1.81e-05 Pa s    standard
gravity               9.81 m/s2        standard
sieve_uptake          0.2 kg/kg        nominal
sieve_bulk_density    650 kg/m3        nominal
bead_diameter_mm      2.5 mm           nominal
bed_voidage           0.37 -           standard
sieve_regen_c         200 C            nominal
silica_regen_c        120 C            nominal
sieve_heat_kj_per_kg  3700 kJ/kg       estimate
petg_soft_c           80 C             nominal
seam_air_max_c        60 C             assumption
pine_dry_density      420 kg/m3        estimate
mc_wet                0.19 kg/kg       standard
mc_target             0.12 kg/kg       estimate
dp_margin_k           1 K              assumption
```

wood_margin_k	3 K	assumption
plate_below_dp_k	6 K	assumption
plate_approach_k	2 K	assumption
frost_floor_c	1 C	assumption
rh_high	0.6 -	assumption
seam_rh_wet	0.85 -	assumption
fan_static_pa	250 Pa	estimate
drawer_face_m	0.3 m	assumption
drawer_depth_m	0.05 m	assumption
stove_exchanger_kw	2 kW	estimate
indoor_t_c	21 C	assumption

CHAPTER 33 · EXPLAIN

Four Levels, and the Ring That Cannot Drain

The solver's dome stands on five heights, its seams fall into five bands, and the slope of each one decides its job

The owner of this design describes the dome in levels. The first ring is the floor. Above it is the band of triangles that stands on the floor. Above that, the band that meets the pentagon round the crown. At the top, the cap. And the question that goes with it: should each level of seam do a different job?

It should, and the solver says which.

The seams, coloured by the band they run in.

The levels, counted from the solve

The dome's 26 corners stand at five heights, not four: 10 on the rim; a belt of 5 and 5 corners that zigzags between two slightly different heights; 5 round the crown; and the apex. The zigzag is what makes it five rather than four, and it matters, as you will see.

The 55 seams run between those heights, and they fall into five bands. What decides each band's job is its slope:

band	seams	feet	slope	
lower band, rim to belt	20	113	57 deg	drains
the zigzag belt	10	53	8 deg	drains
upper band, belt to pentagon	15	87	37 deg	drains
the pentagon ring	5	30	0 deg	dead level
the cap, to the apex	5	27	16 deg	drains

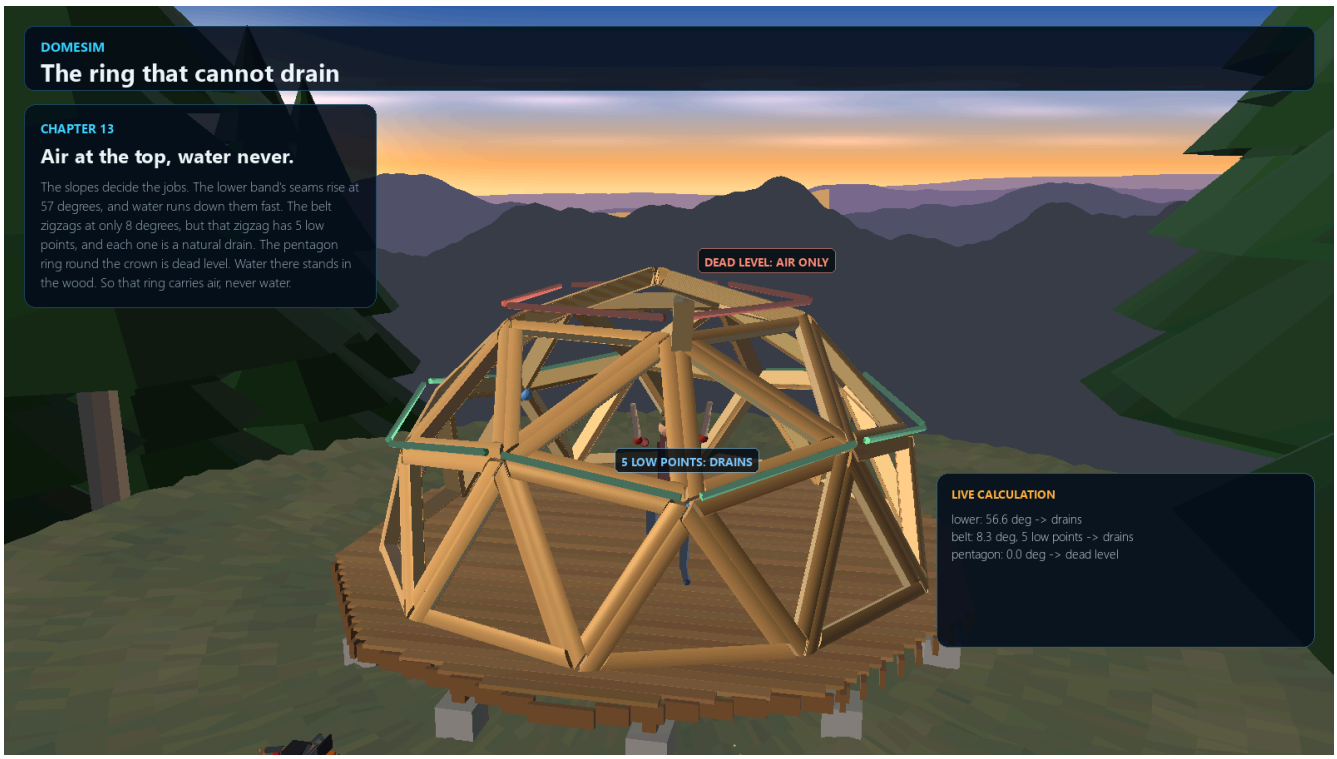
Read down the last column. Four bands slope, so water in them runs downhill to the rim. One does not.

The ring that cannot drain

The 5 seams round the crown – the pentagon ring – are dead level. Every one of them runs between two corners at exactly the same height. Water that gets into them does not go anywhere. It stands in a wooden seam at the top of the building until it evaporates, and it evaporates into the wood.

So the pentagon ring carries air, never water. Which suits it, because air wants to be there: warm, humid air rises to the crown. The pentagon ring is the natural **exhaust manifold** – a closed loop of channel at the top of the room, touching every seam above the belt – and the cap's 5 seams carry that air on up to the apex vent.

The belt is the opposite surprise. Its 10 seams rise at only 8 degrees, which looks like another level ring. But because the belt zigzags, it has 5 low corners, and every one of its seams runs downhill to one of them. **Each low corner of the belt is a drain point**, and the rosette there (Chapter 31) is where the belt's water leaves.



The pentagon ring is dead level; the belt's five low corners drain it.

Condense low, drain at the rim

The owner's first instinct was to put the condensing plates on the first ring, where the frame meets the floor. That instinct is right, and the reason is gravity.

Condensate falls. A plate at the rim drops its water straight into a drain at one of the 10 rim corners, with no seam for it to travel through on the way. A plate at the crown sends its water down the whole height of the dome inside the frame, which is the one place you least want water travelling.

The lower band's 20 seams, at 57 degrees, are the steepest in the building. They are the drains, and they are the path dry air takes up from the rim ports into the frame.

Rain does not come in through the ridge

Chapter 29 described **a slot along the outer cap of every seam** taking rain in at the ridge, with a gutter under it. That was wrong, and this book corrects it here rather than quietly changing the earlier chapter.

Three things are wrong with it. The slot is an opening into the structure along the entire length of every seam, on the face of the building the weather hits. Water taken in at the crown has to travel the whole height of the frame to get out, and on the pentagon ring it cannot get out at all. And every foot of seam that carries liquid water is a foot of wood with a leak waiting next to it.

Rain belongs outside. Let it run off the cap – the one watertight layer of Chapter 28 – to a gutter round the rim, outside the frame, and from there to the tank. The roof still delivers 185 gallons an inch; it just never enters a seam to do it. The only liquid water inside the channels is condensate, and it is made at the rim, where it has no distance to go.

And the rim gutter is a water plant. The cap's brim throws rain off a ring wider than the dome, and collecting what falls on that ring is the whole design:

```
hat rim radius = 8.9 ft; add an 18 in brim -> 10.4 ft
catchment = pi x 10.4^2 = 343 sq ft of plan area
x 40 in of rain a year x 0.6233 gal per sq ft inch
x 0.85 runoff
= 7,259 gallons a year, 20 a day, into a tank by the door
no pump, one downpipe -- the roof shape does the collecting
```

No pump and one downpipe: the roof shape does the collecting. (Those figures use the films' declared rainfall and cap brim, not a measurement of any one site.)

Warm air rises, and the fan still does the work

Warm air rising is free, so use its direction: take air in low at the rim ports and let it out high at the pentagon ring and the apex.

But do not count on its strength. With the inside ten degrees warmer than the outside, the height of this dome produces 1.2 pascals of pull between the rim and the apex. The barrier fan of Chapter 29 holds 4. The chimney effect helps the fan along; it does not replace it, and on a summer afternoon it runs the wrong way.

A tube around the bottom

A dome has an unhelpful habit: warm air, and anything it carries, rises to the apex and stays there. The answer that came out of a badly ventilated workshop dome is a tube right round the base, on a blower, so the whole perimeter is one duct instead of the building having one extract point on one wall.

Run it either way. Push air in at the ring and it leaves through the upper shell and the pentagon ring, carrying fumes out through the whole surface. Reverse it and outside air is drawn in through the shell, warming against the structure on its way: a breathing wall, or dynamic insulation. Same hardware, one switch.

The flows are computed (Chapter 30); the wall is not. Direction decides it, because pushing warm, wet air outward through a cold wall condenses it inside the wall – exactly the failure Chapter 32 exists to prevent – and a sealed skin cannot breathe at all, so any permeable band has to be designed in.

The layout

Put together, every level has one job:

- **the rim** – the ports in and out, the condensing plates, the drains, the two desiccant drawers of Chapter 34;
- **the lower band** – air up, water down;
- **the belt** – its low corners drain it;
- **the upper band** – air up toward the crown;
- **the pentagon ring** – the exhaust manifold, air only;
- **the cap** – the apex vent.

Sensors inside, outside and in the seam read temperature and humidity; a pin in the wood reads its moisture; a switch on the cap says whether it is raining. Chapter 32 is what they feed.

The Desiccant Belongs in a Drawer

Molecular sieve holds a fifth of its weight in water -- and a seam packed with it passes almost no air and can never be dried in place

A desiccant is anything that pulls water vapour out of air and holds on to it. Two kinds matter here.

Silica gel is the beads in the packet in a new pair of shoes: porous glass that holds water in its pores. **Molecular sieve** is a zeolite – a mineral with a crystal lattice full of holes the exact size of a water molecule. The common grade, 13X, holds about a fifth of its own weight in water, keeps pulling vapour out of air long after silica gel has given up, and is what people mean when they talk about a solar-regenerated fridge or a dryer that works in the tropics.

Both do the same thing in the seam: they are the frame loop's dehumidifier (Chapter 32), the one way to dry the wood when neither the inside air nor the outside air is dry enough. The question is where to put them. The candidates are the obvious ones: packed along a seam, packed round a ring, arched over the top of the dome from one side of the floor to the other like a rainbow, or in one place.

The answer is one place, and two drawers. Here is why.

Why not in the seams

A seam packed with beads is a very long, very thin filter, and air does not like going through filters.

The resistance of a packed bed is well understood – it is the Ergun equation, which chemical engineers use for exactly this – and it rises steeply with both the speed of the air and the length of the bed. Take one seam's share of the dome's air, 13 cubic feet a minute, and push it through one seam packed with beads:

```
resistance of one packed seam        65,848 Pa
what a small in-line fan pushes      250 Pa
what that fan gets through           0.29 cu ft a minute
```

A packed seam does not slow the air down. It stops it. The rainbow, rim to apex to rim along the seams, is 32 feet of it, and worse again.

One seam packed with beads, and the pressure it would take to push the air through.

It would also hold a lot of beads to no purpose – packing every seam takes 233 kilograms of them – because a bed the air cannot get through is a bed whose water nobody can reach.

And why it cannot be dried where it sits

A desiccant fills up. To use it again you have to drive the water back out, and that takes heat:

```
13X molecular sieve gives its water back at    200 C
silica gel                                     120 C
the printed key goes soft at                    80 C
the hottest air this book sends through a seam 60 C
```

Beads in a seam could never be regenerated where they sit. You would have to bake the frame to do it, and the key would have melted long before the beads gave anything up.

Two drawers and a gate

So the desiccant lives in **drawers, at the rim**, where the channels come down to the pad and a person can reach them.

One drawer 30 centimetres square and 5 deep holds 2.9 kilograms of beads, which hold 0.58 kilograms of water between them, and it passes the whole dome's air at 20 pascals: a fan barely notices it. A short, wide bed is everything a long, thin one is not.

1. **Use two drawers.** One works while the other is out being dried. This is the arrangement every industrial compressed-air dryer uses, and for the same reason: the air never waits for the desiccant.
2. **Regenerate them out of the wall.** A drawer of 13X wants 0.60 kilowatt hours to dry out: about 18 minutes of a stove exchanger's 2 kilowatts, or an oven, or a black box in the sun on a hot day.
3. **Switch between drawer and bypass with a gate that fails open.** The gate is a damper with two positions: through the drawer, or round it. Build it so that when the power goes, a spring puts it on the bypass. Air that stops moving in a wooden seam is the failure; air that skips the desiccant for an afternoon is not.
4. **Put the gate where the channels meet**, in the rim rosettes, not halfway along a seam where you cannot reach it.

A drawer of beads and a bypass, switched by a gate that falls back open.

What the desiccant will not do

And the number that does not help.

A frame that has got properly wet – over 19 percent moisture – has 54 kilograms of water to lose to get back to 12. That is 93 drawer fills, and 56 kilowatt hours of heat to regenerate them.

The desiccant does not dry a soaked frame. It is for the weather when no air is dry enough and the frame needs to stay where it is, not for getting it back after a leak. Drying a wet frame is the purge's job, with dry outside air, on the next dry day. The desiccant keeps it from getting wet in the meantime.

The stove's heat, never its air

A wood stove is a large, free source of heat in winter, and the owner's thought was to have it drive the seam's air: pulling it in or pushing it out depending on which way it draws.

Use its heat, never its air.

- **A sealed exchanger on the flue** – smoke on one side of a steel wall, clean air on the other – warms air that can dry the frame or regenerate a drawer. The smoke side never meets the seam's air.
- **Keep that air under 60 degrees** before it reaches a seam.
- **Give the stove its own outside air** for combustion, through its own duct. Building-science guidance says the same for any tight building: a stove that draws its air through the gaps in the envelope competes with every fan in the house.
- **Never connect the stove's draw or its flue to the seams.** A stove that can pull on the seam network is a stove that can back-draught smoke into the walls the first time a fan pulls harder than the chimney.

The flue sealed away; clean air warmed across a wall of steel.

CHAPTER 35 · EXPLAIN

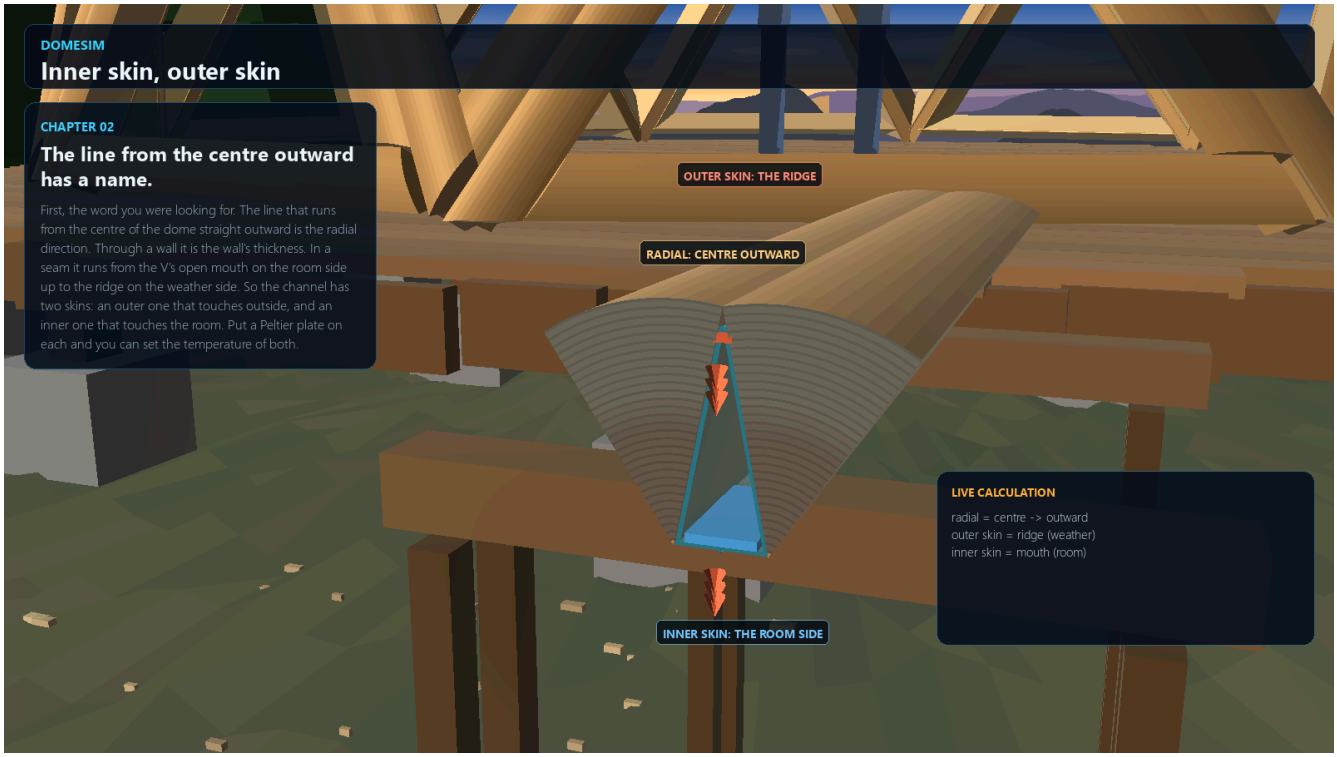
Metal in the Seam

A Peltier plate is a heat pump that runs both ways, and aluminium, copper and stainless make a battery the moment the seam is wet

The line that runs from the centre of the dome straight outward has a name: it is the **radial** direction. Through an ordinary wall it is the wall's thickness. Through a seam it runs from the V's open mouth on the room side up to the ridge on the weather side.

So the seam has two skins, the way a living cell has an inner and an outer membrane. The **outer skin** is the ridge, under the cap, touching the weather. The **inner skin** is the mouth of the V, touching the room. Put a thermoelectric plate on each and you can set the temperature of both independently.

That arrangement is the owner's, and it is a good one. This chapter is about what it takes to make it work: the plates, the metal channel that carries their cold, and the metals that must not touch.



The outer skin at the ridge, the inner skin at the room side, and a plate on each.

The plate runs both ways

A Peltier plate is not a cooler. It is a small solid-state heat pump: current through it moves heat from one face to the other, and reversing the current reverses the direction.

So the rule for both plates is simple: **throw the heat to the side you want warmer.**

- **In winter**, the cold face is in the channel and the heat goes into the room. The outer skin is already cold – often cold enough to condense on for free, with no power at all.
- **In summer**, the cold face is still in the channel, but the heat goes to the outer skin and out under the cap. Throwing it into the room would mean paying twice: once to make the cold, and again to the air conditioner to remove the heat you just put in the room.

Chapter 29 drew the plate throwing its heat into the room. That is right in winter and wrong in summer.

Winter: heat to the room. Summer: heat outside.

Fifteen to twenty degrees colder

The owner's question was this: if the framing wood is at the dew point, but the aluminium or copper channel is fifteen to twenty degrees below that, will the water go to the metal and leave the wood alone?

Take a room at 21 degrees and 55 percent. Its dew point is 11.6. The middle of the owner's range, 17.5 degrees under that, puts the plate at minus 5.9.

The plate freezes. Below zero, water comes out of the air as frost rather than liquid. Frost does not run to the drain; it builds up on the fins, blocks the air, and insulates the plate from the very air it is meant to dry.

So hold the plate 6 degrees under the air's dew point and never below 1: here, 5.6 degrees. That is cold enough to condense steadily and warm enough to drain.

A plate driven that far under the dew point freezes.

Plate first, wood second

And the real answer to the question: **a colder surface somewhere else does not protect wood that is already at the dew point.** Water comes out on every surface colder than the air touching it. Wood at the dew point, washed by raw room air, gets wet, however cold the plate across the channel is.

What protects the wood is **order**. Send the air over the plate first. It leaves carrying a dew point of 7.6 – about the plate's own temperature plus a couple of degrees – and then it reaches the wood. Wood at 11.6 is now comfortably above it; anything above 10.6 is safe with the margin this book keeps.

So in every mode that sends damp air along the frame, the plate is upstream of the wood. In the rim layout of Chapter 33 that happens by itself: the air comes in at the rim, over the plates, and then up the lower band.

Air over the plate first leaves the wood with room to spare.

The cold metal must never touch the wood. Aluminium carries heat 1,025 times better than the printed key's plastic. A metal channel laid against a sawn face chills that face toward its own temperature, and then the wood is the cold surface. Keep the metal inside the key, or behind a strip of closed-cell foam, never bare against the pine.

Which metals may touch

Put two different metals in contact, add water, and you have made a battery. The less noble metal dissolves into the more noble one. How fast depends on how far apart they are on the galvanic scale, and whether they stay wet. In a seam that condenses water on purpose, they stay wet.

The engineering rule, from military and aerospace practice, is that metals kept wet may touch only if their anodic index differs by 0.15 volts or less:

metal	against	difference	wet service
copper	stainless (passive 304/316)	0.15 V	yes
copper	aluminium	0.55 V	no
copper	galvanised steel	0.85 V	no
stainless (passive 304/316)	aluminium	0.40 V	no
stainless (passive 304/316)	galvanised steel	0.70 V	no
aluminium	galvanised steel	0.30 V	no

Three rules come out of that table.

1. **Copper and aluminium never touch**, and water never runs from copper onto aluminium: at 0.55 volts apart, dissolved copper plates out on the aluminium and pits straight through it. If the plates are copper and the liner is aluminium, put a plastic break between them and keep the copper downstream.
2. **Stainless bolts through aluminium need sleeves.** At 0.40 volts apart, a stainless bolt in a wet aluminium liner eats the liner round the hole. Nylon washers and sleeves isolate it.
3. **Copper and stainless may touch** (0.15 volts). If you are choosing one metal for the condensing channel and the bolts are stainless, that is the argument for copper. It also carries heat 1.9 times better than aluminium, which is the argument for it as a plate; it costs more and weighs more, which is the argument against.

Keep galvanised steel out of the seam entirely.

A liner grows against the wood

Metal moves with temperature much more than wood does along its grain. Over the longest seam, 72 inches, and a 50-degree swing between a winter night and a summer afternoon under the cap:

aluminium liner	2.1 mm
copper liner	1.6 mm
the pine beside it	0.4 mm

So fix a liner at one end only, and let the other end slide in a slot. A liner screwed down at both ends either buckles or pulls its screws, and the screw holes are where the water gets in.

The figures behind this chapter:

k aluminium	205 W/m K	standard
k copper	385 W/m K	standard
k stainless	16 W/m K	standard
k petg	0.2 W/m K	nominal
k pine	0.12 W/m K	standard
alpha aluminium	2.3e-05 1/K	standard
alpha copper	1.7e-05 1/K	standard
alpha pine	4e-06 1/K	standard
seam swing k	50 K	assumption
wet limit v	0.15 V	standard

PART 7

The Floor and the Ground

“Everything you own sits on the floor, and the ground is the one part of the building you cannot take with you.”

The second thing that has to be right, the five ways to build it, and the argument for standing on somebody else's ground.

CHAPTER 36 · HOWTO

The Floor Is the Other Thing to Get Right

It carries everything you own, you touch it all day, and it is where the ground sends its water

You will spend more hours in contact with the floor than with any other part of this building, and it is the only part in permanent contact with the ground.

Both of those are reasons to spend money here rather than on the frame.

What “right” means

Dry. The ground is wet. It is wet under a slab, under a deck, and under gravel, and it will move moisture upward into anything that touches it unless something stops it. A capillary break — gravel, a membrane, an air gap — is not optional and it is cheap at the time and impossible afterwards.

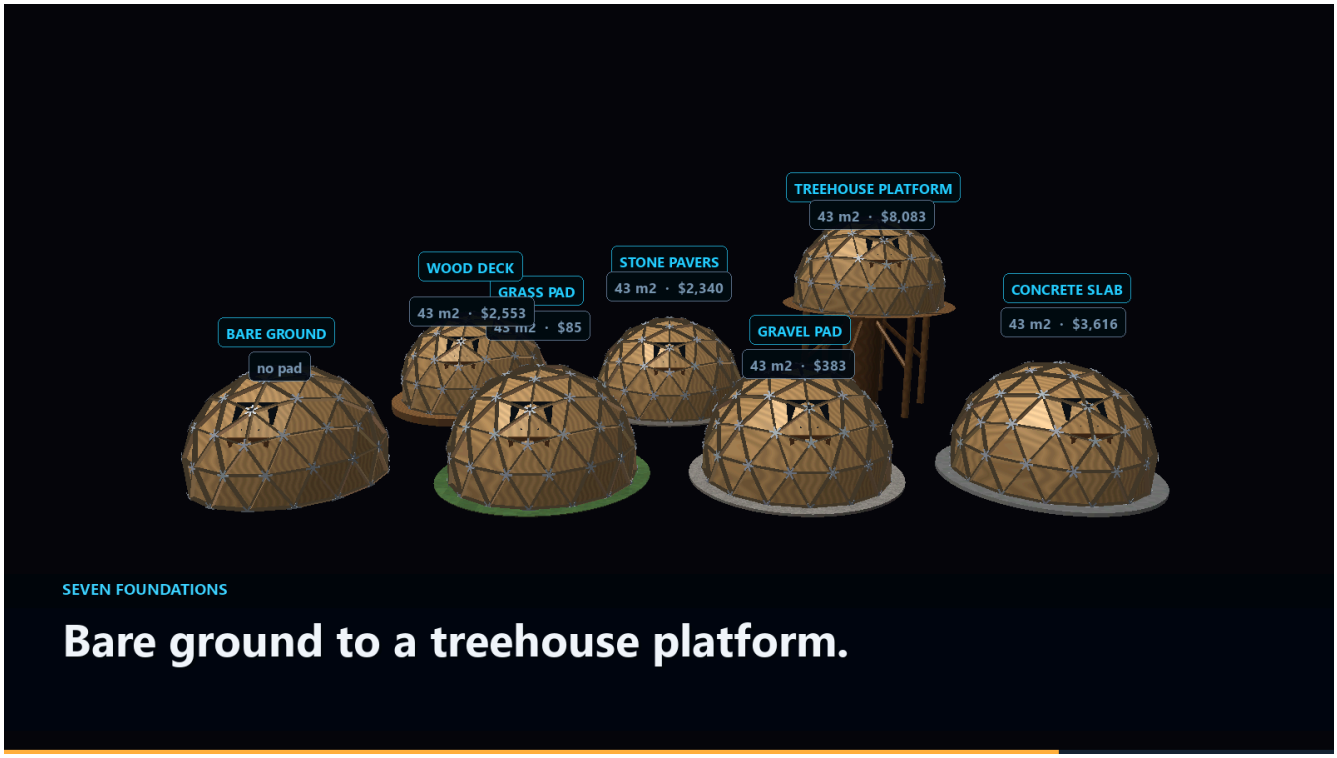
Warm underfoot. A cold floor makes a warm room feel cold, because you are touching it and you are not touching the air. This is worth more comfort per dollar than the same money spent on the walls.

Flat and stiff. You notice a floor that flexes every single day.

Level, to a few millimetres. Not because the dome needs it — the dome would stand on a slope — but because everything you put in the dome needs it.

Five ways to do it

Priced on this footprint, taken off board by board rather than rated per square foot:



Bare ground to a treehouse platform.

```
compacted gravel base                $443
full concrete slab                   $2,468
framed deck on piers, sealed         $2,916
concrete ring with a timber middle   -- see the table
framed deck, ply and epoxy over it   -- see the table
```

The staple is the framed deck on piers at $2,916: it is dry by construction, it is warm because there is air under it, it is repairable, and it does not require a concrete truck to reach your site.

The gravel base at $443 is not a floor. It is a prepared ground that a floor can go on. It is in the list because it is what a lot of people actually start with, and it is honest to say that it is a starting point rather than a cheap answer.

The thing that surprises people

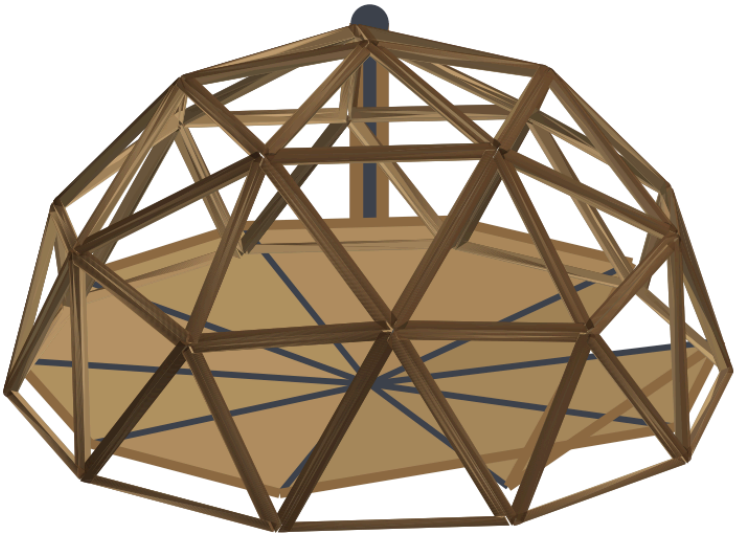
Half a pour is not cheaper than a whole one.

The concrete ring with a timber middle costs *more* than the full slab. The saving on concrete is smaller than the cost of the timber deck that replaces it, and the ring's formwork is fiddlier per cubic yard than a slab's.

Meeting the ground: piers, a ring beam, or a slab.

This is in the book because the intuition is so strong and so wrong, and because the model prints the number that does not flatter the argument.

The floor that clamps



The mast through the column, and the floor that clamps to it, sitting on the base ring where a floor goes.

There are two ways to build a floor in a dome.

Framed into the base ring. Joists between the ring segments, decking over. Simple, strong, and permanent: it is now part of the structure and removing it is a demolition.

A hub, spokes and decking. A steel hub in the middle, radial spokes out to the base ring, timber over the top. It is a component. It comes out, it goes back, it can be replaced with a better one in ten years, and it can be left behind when the dome moves.

Chapter 3 argues for the second on general principles — the floor is one of the two layers that matters, so keep it changeable. It also turns out to be what makes Chapter 55 possible, which was not the reason for doing it.

The hub wants something in the middle to clamp to. In this design that is the utility column, which is already there, already centred, and already the one thing running floor to apex.

The part of a house you never take with you, priced across the catalogue.

Under-floor is storage

A deck on piers has a void under it. That void is dry, it is out of the way, and it is the cheapest cubic footage in the building.

Design for access to it from inside — a hatch, not a crawl from outside — and it is the difference between a small building that feels cramped and one that does not.

What this chapter will not tell you

Frost depth, soil bearing, seismic anchorage.

Those are local, they are the reason foundations are the part of a building most tightly bound by code, and this book does not know where you are. The five constructions above are priced take-offs of a design; what depth your piers go to is a question for somebody who has stood on your ground.

CHAPTER 37 · EXPLAIN

A Pad, Not a Plot

The ground is the one thing you cannot take with you, so stop buying it

The ground is the one part of this building you cannot take with you.

Everything else — 120 members, forty panels, a column, a cap, a floor — comes apart and moves. The ground does not, and it is usually the most expensive thing in the whole arrangement.

So stop buying it.

The landowner builds the pad. It stays when the dome leaves.

The split

The host owns the ground and everything permanently attached to it. The pad, the service port coming up through it, the water and power to that port, the access track, the under-floor storage if there is any.

The tenant owns the building. The frame, the panels, the shell, the column, the floor, the cap.

That line is not arbitrary. It falls exactly where “can this be carried away” falls, which means it never has to be argued about. Nothing straddles it.

Two people, and what each of them owns.

Why it is a clean line

The usual version of this arrangement — renting land and putting something on it — is messy because the something is half-attached. Whose slab is it? Whose septic? Who pays when the deck rots?

Here the pad is a single object with a price. In this model it is $6,240, built once, by the host, to a specification the tenant can check before arriving. The dome is another object with a price, $12,036, and it arrives on a trailer.

When the arrangement ends, the tenant takes a building and the host keeps a platform. Both of those are worth something to their owner and neither is worth arguing about.

The line that never moves.

What the host is really selling

Not land. A **prepared, serviced, level place to put a building**, which is a much more specific thing and much more valuable than an acre of field.

Preparing that place is the expensive, skilled, permitting-heavy part of building anywhere. It involves earth, concrete, a trench, and somebody who knows where the frost line is. Doing it once and having it used for decades is a far better use of that effort than doing it once per owner.

And what the tenant is buying

A foundation, without a mortgage.

The pad is cheaper than the foundation they would otherwise pour, because it is amortised over more than one occupancy. It is available now rather than after a land purchase. And — the part nobody expects to value until they have it — it is *leavable*.

What a pad actually is

Graded ground. A deck – gravel, concrete or timber, whatever the host wants to pay for. A rim of wood round the edge that the dome latches to, which is the part that makes it a pad rather than a patio. Then power, water and drain coming up through the middle.

A deck, a rim, and the services up the middle.

The rim can be a fixed ring. Or it can be an **iris**: leaves on a track that open and close like a camera aperture, so one pad takes more than one size of dome without being rebuilt. That mechanism is a development concept, priced, and it belongs in the upgrade column, not the starter one.

One pad, more than one dome size.

Pad sizes are not invented. Every design the Dome Creator ships has a foundation diameter the tool computes from its own geometry; round each up to the next four feet and you have the pad catalogue:

```
 the Dome Creator ships 12 designs, and each one has a
foundation diameter the tool computes from its own geometry.
round each up to the next 4 ft and you have the pad sizes:
      36 ft    1,018 sq ft   takes  1 of 12
      40 ft    1,257 sq ft   takes  4 of 12
      44 ft    1,521 sq ft   takes  7 of 12
      48 ft    1,810 sq ft   takes 10 of 12
      64 ft    3,217 sq ft   takes 11 of 12
      80 ft    5,027 sq ft   takes 12 of 12

but one pad does not have to be one size. the three domes
this system is for, by floor area and longest member:
   small personal    314.00 sq ft  ·  20.0 ft across  ·  6 ft member
   medium            822.49 sq ft  ·  32.4 ft across  ·  10 ft member
   large           1,184.39 sq ft  ·  38.8 ft across  ·  12 ft member
an iris rim -- leaves on a track, opening and closing like
a camera aperture -- spans 19 to 40 ft and takes all three.
the mechanism is $12,400 on top of the deck.

the treehouse canopy dome needs 35 ft of pad;
the whole trunk lodge - 20 ft needs 76 ft.
4 ft steps because decking and ring stock come in even lengths,
and the second pad a host builds wants the first one's cut list.
```

The pad is the floor

This is the move the whole idea turns on. The dome that arrives does not bring a foundation, because it does not need one. The deck it lands on is the deck it lives on, the services come up through the middle of it, and when the owner leaves, the pad is exactly as it was. No slab poured, no footing dug, no permit pulled for a structure that will be gone. Bring your carpet if you want one.

Dome rentals already exist, and most of them are glorified tents rented a weekend at a time. Own the glorified tent and rent the place to put it, and both sides have a reason to make it good.

Start with the cheapest working version

Before any iris or turning base, price the starting point: a small framed deck on blocks, a fixed rim, and four pads sharing one power panel and water manifold:

```
 STARTER SCENARIO / declared unit prices
20 ft pad = 314 sq ft
deck on blocks, 314 sq ft: $4,398
share of one hub panel, 1 of 4: $600
spur from the hub: $450
permits: $1,200
site access + trunk service allowance: $6,500
TOTAL: $13,148
Author's all-in target: $10,000
Fixed rim; shared services; no iris or rotation.
Site allowance must cover the actual local works.
Start with a working floor and services; add features later.
```

That is above the target, and it is printed anyway. It shows exactly where the design work has to go – the site allowance, mostly – rather than pretending the difference is gone.

The pad turns

A pad on a rotating base can aim whatever stands on it. The panels are on the dome; the pad turns them. Worked for the films' medium, two-person dome:

```
every figure here is for a 10 ft radius dome -- the medium,
two-person size. a number a month means nothing without that.
the panels are on the dome, not the pad. the pad turns.

   one side               20 of 40 panels  ·    292 sq ft  ·    5.2 kW
   top half, skirt below  20 of 40 panels  ·    292 sq ft  ·    5.2 kW
   whole shell            40 of 40 panels  ·    583 sq ft  ·   10.5 kW

   one side               fixed   734  ·  tracking   918 kWh a month
   top half, skirt below  fixed   734  ·  tracking   918 kWh a month
   whole shell            fixed 1,468  ·  tracking 1,835 kWh a month

the skirt below the cells is the catchment -- the same metal that
keeps rain off the bottom ring collects it.
a tenant uses about 420 kWh a month.
and the one that does not flatter the tracking ring: cover the
whole shell and it is always facing the sun with something, so
turning the pad buys much less than it does on a half-clad dome.
the whole shell also makes 4.4 times what a tenant uses. that is surplus,
and the model refuses to credit surplus at retail: it is worth $0.045 sent back,
not the $0.14 it would have saved.
```

The host bought the ring. The tenant bought the panels.

The number that decides it

The pad is $6,240. The floor a dome brings with it is a few thousand more. Against a piece of land, which in most places is a five-figure number before anything is built on it and a thirty-year commitment after.

That ratio is the entire argument for the next chapter, which works it through from both sides and says what would have to be true for it to hold.

PART 8

What It Costs and What It Is Worth

“The tree is the same on every rung. What moves it up the ladder is how much of it you keep, and how many of the steps between the tree and the wall you are willing to take yourself.”

What a tree is worth on every rung, every hand between a stump and a stud, how much shelter an hour of your life buys, where a house's money goes, and the invoice line by line.

A Twenty-Dollar Pine Is Not a Twenty-Dollar Pine

One tree priced on every rung: on the stump, burned, sawn, split, built in, and carried on a mortgage -- the wood never changes, the use does

What is a pine worth? Not one price. A ladder of them, one for every use the same tree can be put to – and the tree does not change from rung to rung. Only what happens to it does.

This chapter climbs the ladder for one representative pine: 60 feet of usable stem, 15 inches at the butt and 5.5 at the top – the same worked tree as Chapter 12.

What the ladder rests on

Measured, published and the author's own figures, kept apart, because the ladder is only as honest as its inputs:

```
THE AUTHOR'S  a representative pine: 60 ft of stem, 15 in to 5.5 in
THE AUTHOR'S  60% kept by a mill; 88% by splitting; 90 cu ft of wood a cord
THE AUTHOR'S  sawing $0.60/bd ft; a cord $275 picked up, $300 delivered
THE AUTHOR'S  $4,800 of framing for the shell; it uses 1.5 trees, 2 felled
PUBLISHED  pine $1.50/bd ft; stumpage $23.23/ton; a cord 128 cu ft stacked
CHECK  this log in the simulator: sawn to 2x4s 45%, split 88.4% before trimming
So the mill gets the benefit of the doubt, not the wedge
```

The stem, in board feet

The stem is 36.8 cubic feet of solid wood: 442 board feet, which is the fixed budget every method below starts from. Sold standing, at the published stumpage price, the whole tree brings about **$24**. That is the bottom rung, and it is roughly what the chapter title calls it.

Burned, it is heat

The first rung up is fire. A cord is 128 cubic feet stacked, but air fills the gaps and only about 90 of them are wood. So the stem makes 0.41 of a cord: **$113** picked up, about $123 delivered. Every rung above this one has to beat the value of the tree's heat.

Sawn, or split

Next, lumber. A mill keeps about 60 percent of the stem as boards, 265 board feet. Splitting keeps about 88 percent, 389 – the only wood a split loses is the width of the saw.

```
split against sawn          46.7% more usable wood
left behind, sawn          177 bd ft
left behind, split          53 bd ft (70% less)
trees sawn to match 2 split   2.93
```

Priced as wood at a sawmill's $1.50 a board foot, the sawn lumber is **$398** and the split wood **$584**: splitting is worth $186 more from the same tree. The mill gets the benefit of the doubt throughout – the simulator's own sawn recovery for this log is 45 percent, well under the 60 used here.

Built, it replaces framing

The next rung is not a price for wood at all. In the dome, these trees become a shell, and the framing that shell replaces is worth $4,800 at commercial rates. The frame uses 1.5 trees' worth of trunk, but 2 are felled, so each tree's share is **$2,400**.

Most framing is not paid for in cash; it rides on a mortgage. On 30 years at 6.5 percent, that framing becomes $10,922 of payments, and this tree's half is **$5,461**. Read that rung carefully: it is nominal dollars, paid monthly for decades, not money in your hand today. But it is money a household never sends

to a lender.

The ladder

on the stump	$24
burned	$113
sawn, as lumber	$398
split, as lumber	$584
built into a shell	$2,400
that framing, financed	$5,461

As structure it is worth 21 times its firewood, 6 times its sawn lumber and 4.1 times its split stock. The wood did not get better on the way up. It got used.

One tree, every rung.

What the tree is worth to its owner

The honest figure is not the top rung. It is what the tree replaces, less what it costs the owner to make it do so: about 40 hours of the owner's own work for this tree, worth $1,000 at the author's wage, and $400 of cash. Net, **$1,000** – or $1,152 with the fortnight's actual receipts.

What the ladder does not say: the structure rung assumes you needed the house; the finance rung is spread over thirty years; and the author's own figures, though marked, are an author's. The generous numbers and the ones against them are both printed above.

The household as the whole chain

This is vertical integration at the scale of one household. A commercial house pays every hand between the stump and the wall: the logger, the hauler, the mill, the kiln, the yard, the framer, and very often the lender. Each takes a margin, and every margin is inside the price. This method does not make those businesses unnecessary for everyone. It lets one owner do the few steps that matter for one building – and keep the margins. The next chapter counts the hands.

Fifteen Sets of Hands

Every machine and every middleman between a tree and a stud, counted -- and the same frame bought at the rack, with everything nobody counts

Almost none of a stud's shelf price is the tree.

There are two ways to get from a standing tree to a wall. The long chain: earn an income, pay tax on it, drive to a store, buy lumber somebody else has sawn, dried, graded and shipped, pay a contractor to put it up, and very often borrow to do all of that. The short chain: tree, saw, wall. This chapter counts what the long chain is made of, so the short one can be priced honestly against it.

Counting the machines out

Every conversion step between a log and a stud is a building full of machinery, with its own setup, labour and margin. Station by station, which of them the wedge path still needs:

```
what has to happen to a log before it can hold a roof up:
  BOTH  fell and buck            chainsaw
        identical in both, so it cancels
  SAWN  primary breakdown        SAWMILL
        wedge path splits radially with the same chainsaw
  SAWN  edging                   EDGER
        there is no waney edge to remove from a split face
  SAWN  drying and grading       kiln / grader
        wedges are used green; the shell is not a graded-lumber assembly
  SAWN  surfacing to dimension   PLANING MILL
        the sawn faces come off the split already flat and already exact
  BOTH  crosscut to length       crosscut saw
        one square cut, both paths
  BOTH  end joinery              compound saw / jig
        sawn path measures two compound ends; wedge path cuts one butt and flush-cuts the head in place
machines the wedge path never buys, feeds or houses: 3
  SAWMILL, EDGER, PLANING MILL
what is left on the wedge side is one chainsaw and a flat board
```

The wedge path drops the sawmill, the edger, the kiln and grader, and the planing mill. It keeps the chainsaw, the crosscut and the joinery.

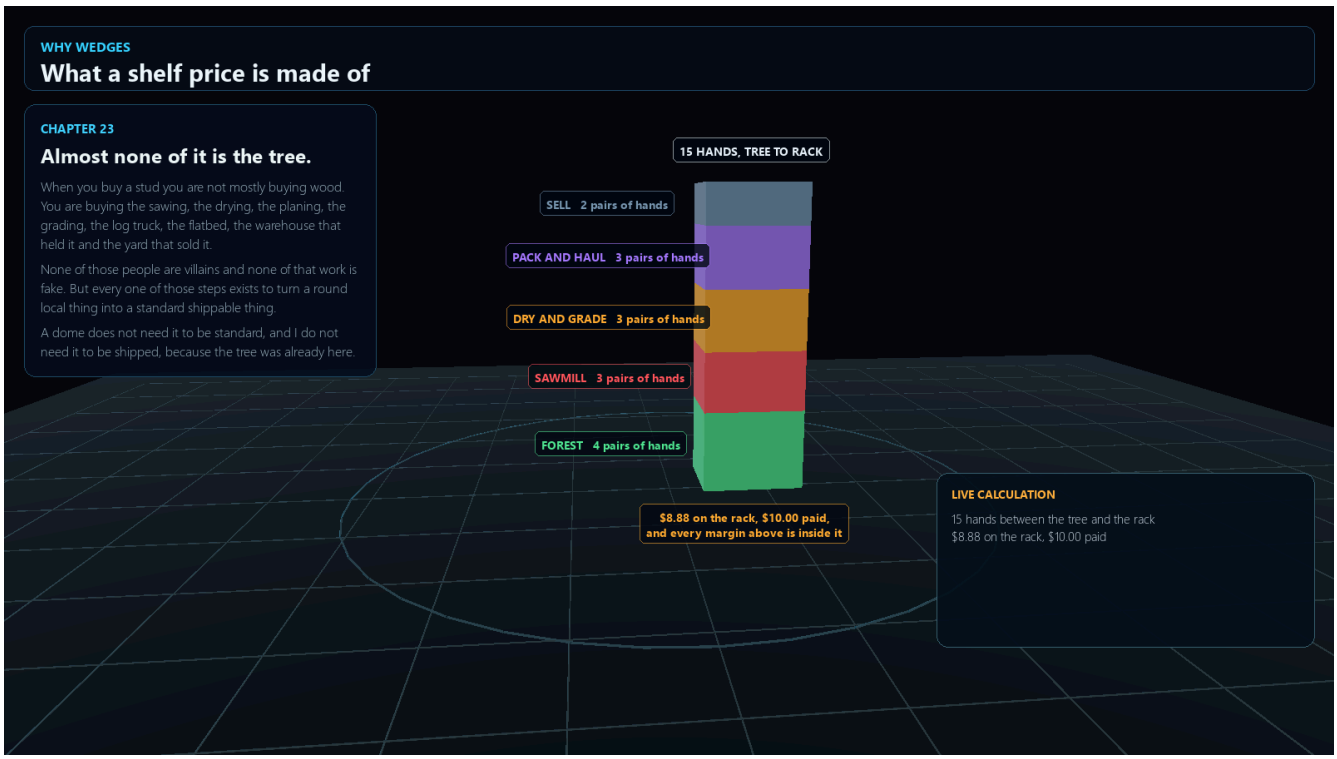
What a stud has been through

And the hands. A stud on a rack has been sold and handled this many times on its way there, listed rather than rounded, because a round number is easy to say and hard to check:

```
a stud on a rack has been sold and handled this many times:
   1. timber broker        sells the standing timber
   2. logging contractor   fells and skids it
   3. log haulier          trucks it to the mill yard
   4. log scaler           measures and grades the log
   5. headrig sawyer       breaks the log down
   6. edger operator       squares the waney sides
   7. trimmer operator     squares the ends
   8. kiln operator        dries it to a moisture spec
   9. lumber grader        stamps a grade on it
  10. planing mill         surfaces four sides to dimension
```

```
  11. bundler              packs and wraps a unit
  12. mill sales           sells the unit
  13. freight broker       books the flatbed
  14. distributor          holds stock and breaks bulk
  15. retail yard          sells you one stick
then sales tax: $8.88 on the rack becomes $10.00 paid
every one of the 15 took a margin, and every margin is in that price
the wedge path is: fell the tree, rip the trunk, use the wood
```

Each of them is a real job done well. Each of them also takes a margin, and all of those margins are in the price on the rack.



Every hand between the stump and the rack.

What one wedge is worth

So what is a wedge worth, priced against the bought sticks it would take to replace it? Two ways, and how far apart they land:

```
a 2x4x16 untreated pine is $8.88 on the rack, $10.00 with tax
  = $5.00 per eight-foot section
one 1/8 wedge of a 12 in trunk has 14.14 in2 of section
a dressed 2x4 has 5.25 in2
  computed multiplier = 14.14 / 5.25 = 2.69 x
  the estimate on site was 2.5 x
  the multiplier actually used for planning was 2.0 x -- lower than both
RULE OF THUMB:
  2.0 x $5.00 = $10.00 per wedge
STRICT, counting section AND the length difference:
  2.69 x 0.75 x $5.00 = $10.10 per wedge
  the two disagree by 1.0 %
at 5 wedges an hour that is $50.00/hour gross, $45.00 after fuel
one 6-hour afternoon: 30 wedges, $300 of wood, $30 of fuel
```

The rule of thumb and the strict calculation agree to about a percent, and the planning figure is set below both.

The same frame, bought

Now buy the frame instead: enough studs to match the section the wedges provide, plus everything a shopping list never shows – the boards bought and rejected as bowed, twisted or wet, the ones lost on the way home, the truck, and the hours of driving, picking and queueing. The overheads are estimates, and they are all on the store-bought side, which is the side it flatters least to guess about:

```
to match the section 120 wedges provide you would need 2.69 dressed studs alongside each one
  120 x 6 ft x 2.69 = 1939 stud-feet
  / 16 ft = 121 sticks of actual wood needed
then the part nobody counts:
  10 % bought and rejected -- bowed, twisted, wet, split
  3 % lost to handling and weather on the way
  so you buy 139 sticks, 18 of which become firewood ($176)
lumber = 139 x $10.00 = $1388
4 truck loads x 40 mi = 160 mi at $0.55/mi = $88
  and 6.0 h of driving, picking, queuing, loading and unloading = $150 of time
BOUGHT: $1476 cash, $1626 with the time
CUT: 24 h of ripping, $120 of fuel and oil, no trips at all
```

The whole argument on one page

Wood, machines, hands, rate, value, cost and strength, each from the chapters that worked it out:

```
WOOD     88 % of the tree kept split, 45 % sawn -- 1.95x
MACHINE  3 fewer: SAWMILL, EDGER, PLANING MILL
HANDS    15 middle men removed between tree and wall
RATE     6 wedges/hour measured, 5 used for planning
VALUE    $10.00 of wood per wedge, $45/hour net of fuel
COST     $120 to cut this dome's frame against $1476 to buy it
STICK    226 % of a dressed stud in bending at 12 in trunk; under 9.1 in it would lose
EDGE     every interior edge carries two, 539 % of a stud's section
ANGLE    4.43 deg of dihedral variation lives in the key, not the timber
CUTS     one batched butt per member, one flush cut in place, 4 fixtures for 40 panels
```

None of this says split wood is better wood. It says the long chain has fifteen sets of hands in it, and the short one has yours.

How Much Shelter Does an Hour Buy

Not how cheap the house is: how much house an hour of your life buys, how debt multiplies it, and the wage above which you should just buy

The usual question is how cheap a house is. The better one is how much shelter an hour of your life buys – because an hour is the thing you actually have a fixed number of, and money is only stored hours.

The figures here are a mix, and they are labelled: the author's own round numbers for one cycle of the shell, beside published ones.

```
THE AUTHOR'S  $100 a sq ft; framing 16% of it; 80 hours of shell work
THE AUTHOR'S  6.5% for 30 years; $25 an hour take-home; $800 cash
PUBLISHED  new manufactured home: $86.62 a sq ft (Census)
PUBLISHED  new site-built house: $216.94 a sq ft without land (NAHB)
PUBLISHED  framing 16.6% of building cost; median take-home $19.69 an hour (BLS)
So the author's figures sit on the cautious side of the published ones
```

What an hour produces

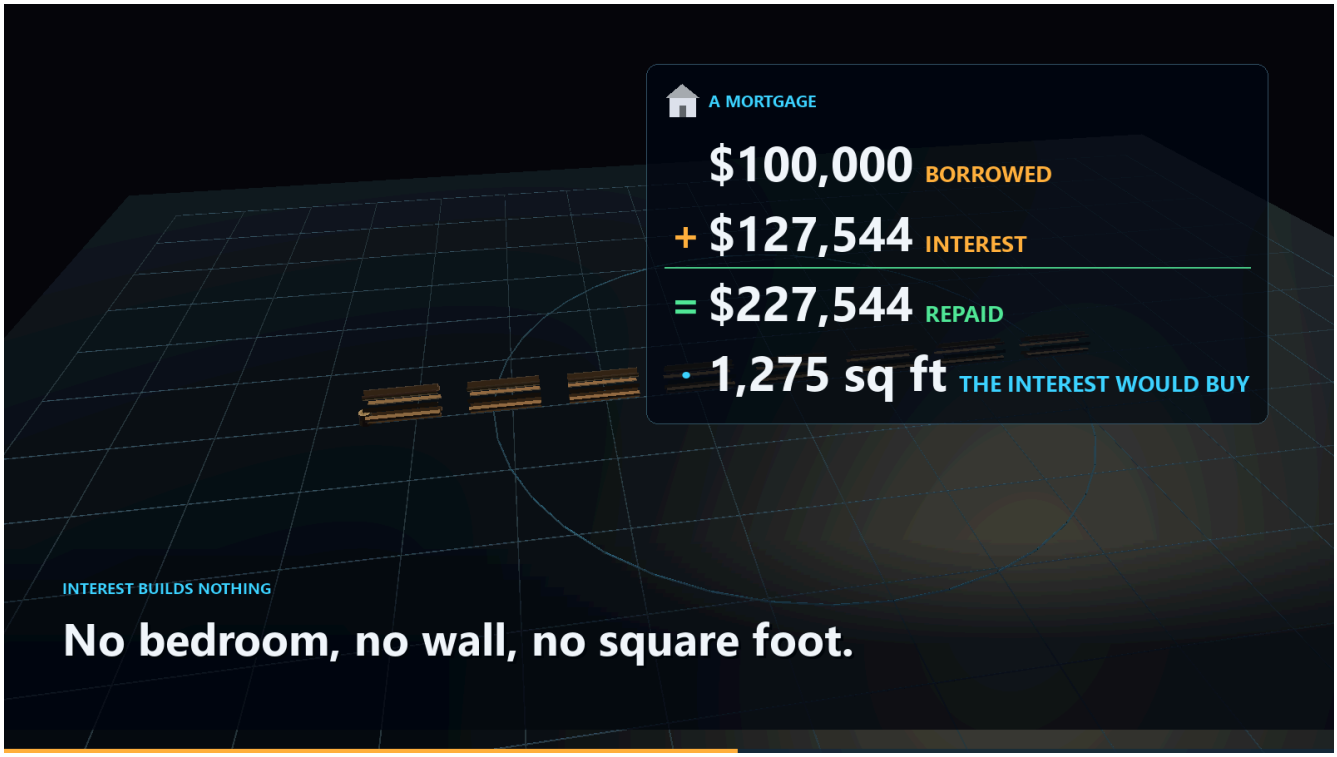
One cycle builds 300 square feet of shell in 80 hours: 3.75 square feet an hour. At $100 a square foot, that floor is worth $30,000 finished, but framing did not create all of it; plumbing, wiring, windows and finishes are still to come. So take only the framing, 16 percent: **$4,800**. Divided by the hours, that is **$60 an hour**.

Nobody hands that over as a wage. It is value you no longer have to buy. And it is more than the labour in the framing would earn: labour is 40 percent of that framing, $1,920, or $24 an hour. The dome way produces 2.5 times as much, because you supply the labour *and* replace the wood. Two other things help: a wedge arrives with its angle already in it (Chapter 15), and the jig solves the geometry once instead of for every panel (Chapter 24).

Debt makes it larger

Now suppose that framing would otherwise ride on a 30-year mortgage at 6.5 percent. $4,800 does not stay $4,800; it becomes $10,922 of payments. Over the same hours that is $137 an hour. That is not a wage and it is not today's money: it is payments that never have to be made, spread across decades. Today's value is still $60 an hour.

Take a whole $100,000 mortgage at the same rate. Over its term you pay **$127,544 of interest**, $227,544 in all. At $100 a square foot, the interest alone would buy 1,275 square feet. Interest builds no bedroom, no workshop, no wall. It is the price of having the house before you have the money. Building directly does the other thing: effort becomes equity, one hour at a time.



Interest builds no bedroom.

Count it in hours of your life

Money is stored labour, so price the frame in hours. At an example take-home of $25 an hour, buying that framing costs 192 hours of pay. Building it costs 80 hours of work plus 32 hours of pay for the $800 of cash it still needs: 112. **80 hours of life kept**, about 2 working weeks.

Out of the commodity chain

A conventional house also runs a long critical path – about 33 weeks from start to finish, by the Census count – carrying financing, weather, scheduling and overhead the whole way. And it rides on commodity lumber, whose record price in one recent spring added tens of thousands of dollars to an average new home (Chapter 41). A dome still needs fasteners, glazing and wiring from the same supply chain, but every member that comes from a nearby tree is one less thing exposed to that chain's shocks.

When to build, and when to buy

This is a rule, not an ideology. What building earns per hour is the money it avoids, less the cash it still costs, over the hours it takes: $4,000 over 80 hours, **$50 an hour**. If your take-home is under that, those hours are worth more building than working extra to buy the same frame. If it is well over, buying may be the better use of your time, and the model says so.

What the model does not promise: it is a model. The author's figures are round, the mortgage is illustrative, and the payments avoided are nominal dollars over decades. It is a way to see a decision clearly, not a guarantee.

The score

One cycle, on one sheet:

```
300 sq ft of shell in 80 hours: 3.75 sq ft an hour
$4,800 of commercial framing displaced
$60 an hour production value; 2.5x the framing labor alone
$10,922 of 30-year payments avoided: $136.53 an hour, not a wage
80 hours of life kept, against earning $25 an hour to buy it
Worth building while your take-home is under $50 an hour
```

```
One shell a year for 30 years: $144,000 of framing, 9,000 sq ft
One hour: $60 of shelter today, $137 of payments never made
```

Built once a year for 30 years, that is $144,000 of framing and 9,000 square feet of shell. More shelter from each hour of a finite life is the whole of the why.

CHAPTER 41 · REFERENCE

Where a House's Money Goes

The builders' own survey of a new house, the factory's split of a trailer, and where the dome's saving actually comes from

Before claiming a saving, look at what a house actually costs and who gets it. This chapter uses the builders' own survey of new houses, the Census, and the factory figures for manufactured homes, each one labelled with where it came from:

```
PUBLISHED  NAHB 2024 cost survey: who gets a house's price, and 37 line items of building it (41 builders; not representative)
PUBLISHED  Census: 7.6 months, start to finish; BLS wages, May 2025
PUBLISHED  Lumber peak: $1,500 per 1,000 bd ft; +$35,872 on an average home
PUBLISHED  Manufactured homes: $86.62 a sq ft; factory split from 2002
THE AUTHOR'S  $200,000 house, $100,000 trailer, 40% labor, 80 hours of shell work
DECIDED    the 12% federal bracket; state tax left out
ESTIMATED  dealer 25%, labor by item, run share 35%, curve +25%
So every estimate below is a guess that can be changed, and says so
```

Where a house's money goes

Of a $200,000 new house, in the shares the builders' survey found:

the land	$27,373
the building itself	$128,729
the builder's overhead and profit	$33,434
selling it: commission, marketing, loan	$10,464

The land, the building, and the people who sell it.

Where its time goes

A new house takes about seven and a half months from the first shovel to finished, by the Census count, and much longer when its owner builds it. One builder's week-by-week schedule shows where that time goes, and most of it is inside: the ground and foundation are a small share, framing a smaller one, and insulation, drywall, floors and trim the largest. The frame – the part this book is about – is one of the shortest stages of building a house.

Labor is not most of it

So is labour most of what you pay for? Not in a new house. Take the building's labour as the survey's share and it is $51,491: **25.7 percent of the price**. Lumber at its record high moves the number further toward materials, not away.

A manufactured home is built the other way. Of a $100,000 one, the dealer's share is $20,000, the factory's materials $45,969, its payroll $16,817 and its overhead and profit $17,214. Factory labour is only **16.8 percent** of the price. Building indoors on a line is what makes it cheap – the same lesson as Chapter 23.

Every part of the house

The same sum, stage by stage: what each part costs bought, against the dome way if every hour of the work is yours, and how many hours of take-home pay the bought version is:

```
Site work: buy $4,973 · dome $3,708 · 64 h of pay
Foundations: buy $6,801 · dome $1,816 · 253 h of pay
Framing: buy $10,788 · dome $845 · 505 h of pay
Exterior: buy $8,741 · dome $3,788 · 251 h of pay
Pipes, wires, HVAC: buy $12,511 · dome $3,759 · 444 h of pay
Interior: buy $15,714 · dome $7,420 · 421 h of pay
Yard, driveway: buy $4,212 · dome $1,771 · 124 h of pay
Other: buy $1,343 · dome $631 · 36 h of pay
All: buy $65,082, dome way $23,738, if every hour of the work is yours
```

The frame line is the one the harvest of Chapter 12 builds: two weeks of one person against buying it.

Where the saving comes from

Stack it up and the saving on the floor has four sources, and the two largest have nothing to do with the dome:

```
bought whole                              $65,082
the builder's and seller's share you keep $16,550
labour you do yourself                    $17,291
lumber from your own trees                $3,183
the shape itself                          $4,319
```

Most of the saving is building your own, which anyone who builds their own house keeps, whatever its shape. The dome's own contribution is real – less framing, less skin, a shorter foundation, shorter runs from the column – and it is a small line beside those two.

What this does not show

And the half of the sum that does not flatter the dome. **37 percent of what a house costs to build does not care about its shape at all**: fees, windows, cabinets, appliances, floors, the driveway. Only the frame's hours were measured; the rest of the labour is weeks more of somebody's time, and wiring, plumbing and gas usually need a licensed trade and an inspection. And a triangle is harder to finish inside than a flat wall.

CHAPTER 42 · REFERENCE

What It Costs, Line by Line

The invoice, the price guide, the pad, the stay and the factory -- every figure typed in declared before anything is claimed

What was typed in, before anything is claimed

Every price in this chapter is one of two kinds of number, and they are kept apart. Quantities are measured off the dome's solved geometry and cannot be argued with. Prices and rates are typed in by a person, each with a unit and a reason, and they are assumptions, not market research:

```
this film has two kinds of number in it.

MEASURED -- off the dome's own solved geometry:
   120 members, 664 ft of stock
   40 panels, 549.8 sq ft of surface
   277 sq ft of floor, 19.42 ft across
   none of these can be argued with.

DECLARED -- prices and rates, typed in by a person:
   140 inputs, each with a unit and a reason
   31 of them borrowed from suppliers' own lists
   these are assumptions. they are not market research.

change one and every figure after it moves,
```

Change one and every figure after it moves. That is the point of declaring them.

What the wood was never

Start with the frame, priced the way anybody can check it:

```
start with a board anybody can go and price.

   one 2x6x12 at Lowes  ·  $14
   rips and cuts into 4 struts of 2x3x6  ·  $4 each
   120 of those  ·  $420

that is the whole frame of this dome in bought sticks.

a wedge off a 12 inch trunk is bigger than a 2x3:
   one wedge  ·  1,018 cu in  ·  3.77 x a 2x3x6
   so 120 of them, bought as timber  ·  $2,465
   felled on site instead  ·  $692

and here is what splitting does to a tree:
   split into wedges  ·  88.4% of the trunk  ·  391 bf
   milled into 2x4s  ·  45.2% of it  ·  200 bf
   so a wedge takes  ·  1.95 x as much  ·  +191 bf a tree

squaring a round log throws away the round part.
we do not square it.
```

```
the pinwheel costs more stock, though:
   shared struts  ·  369 ft
```

The wood was never the expensive part of a house. That is the whole argument of Chapter 38, in one shopping list.

The invoice

What one costs, all of it, with the maker's margin shown rather than hidden in a round number:

```
what it costs, all of it.

   Wedge-cut frame  ·  $692
   Shower-cap shell  ·  $3,265
   Utility column and seal cap  ·  $1,405
   Light, fan and cooling  ·  $485
   Utility panels  ·  $491
   Build labour  ·  $1,884
   ------------------------------------------------
   materials and labour  ·  $8,221
   shop overhead  ·  $1,480
   warranty reserve  ·  $329
   COST TO BUILD  ·  $10,030

   our profit, 20% marked up on cost  ·  $2,006
   WHAT YOU PAY  ·  $12,036

   per square foot of floor  ·  $43.45

and the ground is quoted beside it, not inside it:
   a platform to stand on  ·  $6,240
```

That is about $43.45 a square foot of floor, and it does not include the ground. The ground is a different bill, paid by a different person.

Ten thousand to build. Two thousand is ours.

Four ways to buy it

The same dome bought new at the till, from the cheapest honest version to a pristine one, for the films' 20-foot dome:

```
one dome, 314 sq ft of floor (20 ft across), priced four ways,
every part bought new at the till:
CHEAPEST: $5,116  ($16/sq ft)
BUDGET: $6,583  ($21/sq ft)
PRISTINE, HAT: $6,943  ($22/sq ft)
PRISTINE, FULL: $9,496  ($30/sq ft)
```

The ground's bill

The landowner builds the pad once and keeps it (Chapter 37). Itemised:

```
two people are paying here, and they are not the same person.

THE LANDOWNER builds the pad, once, and keeps it:
   framed deck on piers, sealed, 369 sq  ·  $2,916
   moisture barrier  ·  $152
   service port up through the deck  ·  $340
   water tank, 120 gal  ·  $162
   under-floor storage  ·  $420
   share of one hub panel, 1 of 4  ·  $600
   spur from the hub  ·  $450
   permits  ·  $1,200
   THE PAD  ·  $6,240

THE DOME BUYER pays none of that.
```

```
the deck stays when the dome is lifted off.
the next dome lands on the same port.
```

A bigger, gravelled pad for a park, built and run as a business – including the lines people leave out, the management and the tax:

```
a 48 ft gravel pad, built:
   compacted gravel base, 1810 sq ft     $2,031
   electrical pedestal                   $1,400
   water and drain                       $1,100
   site infrastructure, per pad          $6,500
   permits                               $1,200
   to build                             $12,231

lease            $667 a month, leased 80% of the year
utility margin    $10 a month -- paperwork money, not rent
management, tax     -$920 a year (insurance too)
net            $5,578 a year

and the cheap version, for a host starting with nothing:
a framed deck on precast blocks, and one power panel and water
manifold in the middle of 4 domes instead of a pedestal and a trench each.
   deck on blocks, 1257 sq ft           $17,593
   share of one hub panel, 1 of 4          $600
   spur from the hub                       $450
   permits                               $1,200
   built                                $19,843
services: $2,250 against $10,060 -- nothing dug, nothing poured.
the deck is still the cost of a pad. it always is.
```

And the deluxe one, with rotation, a column, solar and the iris, which is an optional future build rather than anything a first site needs:

```
DELUXE SCENARIO / declared unit prices
Same model's 48 ft concrete pad:
full concrete slab, 1810 sq ft: $11,315
electrical pedestal: $1,400
water and drain: $1,100
site infrastructure, per pad: $6,500
permits: $1,200
rotating base: $9,120
utility column: $2,600
solar, 15,339 W: $17,640
Add adjustable iris: $14,880
TOTAL WITH IRIS: $65,754
A larger pad with rotation, column and solar.
Optional future build; outside the first-site campaign scope.
```

The declared inputs behind the park figures:

```
60 numbers in this film were typed by a person.
they live in one table, with a unit and a reason each:
   a serviced pad rents for $450 a month, plus $0.12 per square foot
   power costs the host $0.14 a kWh; the host adds $0.02
   a hotel room is $140 a night, a short let $95
   a one-bedroom is $1,450 a month
   the sun delivers 4.6 peak hours a day here
   pads are leased 80% of the year
everything else on screen is measured off the Dome Creator:
which domes fit which pad, how much deck that is, how much
```

```
shell can carry panels, what the frame and the skin cost.
none of the typed numbers are market research. they are the
owner's working assumptions, printed where you can replace them.
```

What a stay costs

A year in one place, four ways, with power and water on every line so the compare is not rigged:

```
 a year in one place, four ways. every line carries power
and water, because two of them include it in the nightly rate
and dropping it from the other two would rig the compare.

   hotel room            $4,256 / month    $51,072 for the year
   short let             $2,888 / month    $34,656 for the year
   apartment lease       $1,641 / month    $19,696 for the year
   own dome on a pad     $1,277 / month    $15,325 for the year

the dome line is the split-log homestead at $16,601 on a pad,
plus $1,800 to move it, and it carries the two
costs an ownership pitch usually hides: $600 a year of upkeep
and 6% on the money tied up in it.
at the end of the year it is worth $13,640, after a
15% secondhand haircut taken on day one.
```

The smallest version

At the other end of the scale: the 62-inch dome from one sheet of Chapter 28, cut by a laser in an afternoon. You do not stand up in it and you barely sit up in it. It is a hole you can afford, above ground. With a riser wall under it, the same sheet becomes somewhere to sit; with a taller one, somewhere to stand.

The factory’s view

The software that wrote this book also prices domes the way a factory would: a fifteen-station line in trailer-plant order, every element with a material cost and an install time. Its consumer price separates three things that a sticker merges – the bill of materials, the labour at a burdened wage with its overhead, and the margin a comparable product carries. A cheaper shell does not erase labour, and a fast build does not make glass cheap. And what that line costs the people on it is Chapter 23, in one sentence:

```
 simulate every part placement, all six motions, crew of 2:
time on task = 191 hours each = 23.9 eight-hour shifts
food energy = 54,870 kcal per worker
mechanical work actually raising things = 87 kcal (0.16% of the fuel)
fastening alone = 49,366 kcal (90% of the fuel) -- and it lifts nothing
so a factory should not chase the lifting;
it should chase the screw gun
```

PART 9

The Network

“Own the building. Rent the ground. Move when you want to, and leave a platform behind rather than a house.”

One chapter, the economics of a pad network from both sides of it: what the dome owner saves, what the landowner earns, and what would have to be true for either.

CHAPTER 43 · EXPLAIN

Why a Network Beats a Park

An RV park rents you a slot for a week. This rents you a foundation for a decade, and both sides come out ahead

An RV park rents you a slot for a week. This rents you a foundation for a decade.

That is the whole difference, and everything below follows from it. The arrangement only works if it is better for the landowner *and* for the person who arrives with a building, so this chapter works it out twice.

First, why it is not a caravan site

An RV park sells **nights**. Its costs are turnover: booking, cleaning, hookup, disconnect, a pitch standing empty out of season, and a tenant who is leaving from the moment they arrive and behaves accordingly.

A pad sells **years**. It is built once, connected once, metered once, and occupied by somebody who owns the building standing on it. The revenue per square foot is lower. The cost per occupied month is very much lower. The difference is turnover, and turnover is most of what makes short-stay accommodation a job rather than an income.

The tenant's own building is also the deposit. Nobody abandons a dome.

An RV park, for houses.

The tenant's side

Four ways to be housed, on this book's dome and on rates declared in the park model with the assumption written beside each one:

	per month	at the door	over 10 yr	owned after
hotel room	$4,200	$0	$504,000	$0
short let	$2,850	$0	$342,000	$0
apartment lease	$1,450	$3,625	$175,450	$0
own the dome, rent the pad	$528	$12,036	$75,365	$12,036

Two things in that table are worth slowing down for.

Four ways to have a roof, for a year.

The monthly figure includes the dome wearing out. $528 a month is the pad's rent plus the dome's own depreciation over a 30-year service life. It is not rent-and-ignore-the-asset; the building is being used up and the table says so.

The entry cost is the real objection. $12,036 at the door against an apartment's $3,625. That is the catch and it does not go away. What the table shows is how quickly it is repaid: owning overtakes renting after **11 months**, and after ten years the difference is about $100,085 — with a building still standing at the end of it that you own.

How long you have to stay.

What the tenant actually gets

No land purchase and no mortgage. The most expensive item in almost every housing decision is removed, not reduced.

A building that moves. Not "portable" in the sense of a caravan — this is a real timber shell — but demountable. It comes apart into panels, a column, a floor and a cap, and it goes onto a trailer. The thing you cannot take is the pad, and the pad is the cheap part.

Flexibility in how long. A season or a decade, decided as you go. There is no twelve-month lease to break because the thing tying you to the place is a rent on a platform, not a claim on a home.

Flexibility in where. This is the part that has no equivalent. If the work moves, or the climate turns, or the neighbours do — you take the house. In every other arrangement on that table, moving means losing either a deposit or a building.

A floor that is better than the pitch. An RV park gives you a slab and a hookup. A pad in this arrangement is a prepared, levelled, serviced platform sized for the building that is going on it, and the dome's own floor clamps to its own column.

The host's side

Three things to do with a piece of ground:

	upfront	a year, net	payback
bare ground, unused	$0	$-380	never
one serviced pad	$14,280	$4,868	2.9 yr
furnish a unit and let it short	$26,316	$26,238	1.0 yr

And here is the number that does not flatter the argument, printed rather than buried: **furnishing a unit and letting it short earns the host more.** More than five times the annual net, and it pays back in about a year against a pad's 2.9 years.

So why would anybody build pads?

The host's side, against the obvious alternative.

Because those two lines are not the same kind of thing

The short-let line is given every night of the year full, which nobody achieves. Halve the occupancy and it halves.

It also requires the host to own a *building* — to build it, insure it, furnish it, clean it three times a month forever, hold a reserve against damage, renovate it every year, and answer the phone when the heating fails at eleven at night. That is not an investment. It is a small business, and the income in the table is the pay for running it.

The pad is an annuity. $14,280 once, $4,868 a year net, and the work is reading a meter and sending an invoice. The tenant owns the building, so the tenant fixes the building.

The honest framing is: **a short let earns more per pound of effort at the start and a pad earns more per hour of attention forever.** Which of those you want is a question about your life, not about your land, and this book is not going to pretend the money answers it.

What the host is really selling

Not land. A prepared, serviced, level place to put a building — which is the expensive, skilled, permit-heavy part of building anywhere.

Doing that once and having it used for thirty years by a succession of owners is a far better use of the effort than doing it once per owner. The pad is also worth something while it is empty, which bare ground is not: it is an improvement with a rent attached to it.

Why a network and not a park

One host with one pad is a transaction. The thing that makes it a network is that the pads are **the same**.

A dome built to this book's geometry fits any pad built to this book's specification, which means a tenant can move between them without modifying anything, and a host can take any tenant without knowing in advance which dome is coming. Chapter 37's clean split — the host owns what is attached to the ground, the tenant owns what can be carried — is what makes that interchangeability possible, because nothing straddles the line and so nothing has to be negotiated.

That is the same argument as Chapter 22, applied to land instead of labour: a short, fixed interface is what lets a thing scale.

Why this has to be a network.

What would have to be true

This chapter is a design possibility, not a tested arrangement. Four things would have to hold, and none of them is proven:

Zoning would have to permit it. A demountable dwelling on a leased pad is not a category most jurisdictions have. Some will treat it as an RV, some as a permanent dwelling, some as neither. This is the single largest risk and it is entirely local.

The rates would have to survive contact. $494 a month for a serviced pad is a declared assumption against local RV and storage rates, not a price anybody has charged.

Enough domes would have to exist. A network of pads with nothing to stand on them is a field of concrete. The first hosts carry that risk, which is why the model's test build is three pads and not thirty.

Somebody would have to underwrite the first year. A host who builds a pad on the strength of a book deserves better than hope, and the park model carries a twelve-month manufacturer guarantee for exactly that reason.

State those four plainly and the argument is still a good one. Leave them out and it is a brochure.

PART 10

Other Shapes

“A zome is swept, a hex dome is tiled, and a geodesic dome is subdivided. Three answers to one question about curvature.”

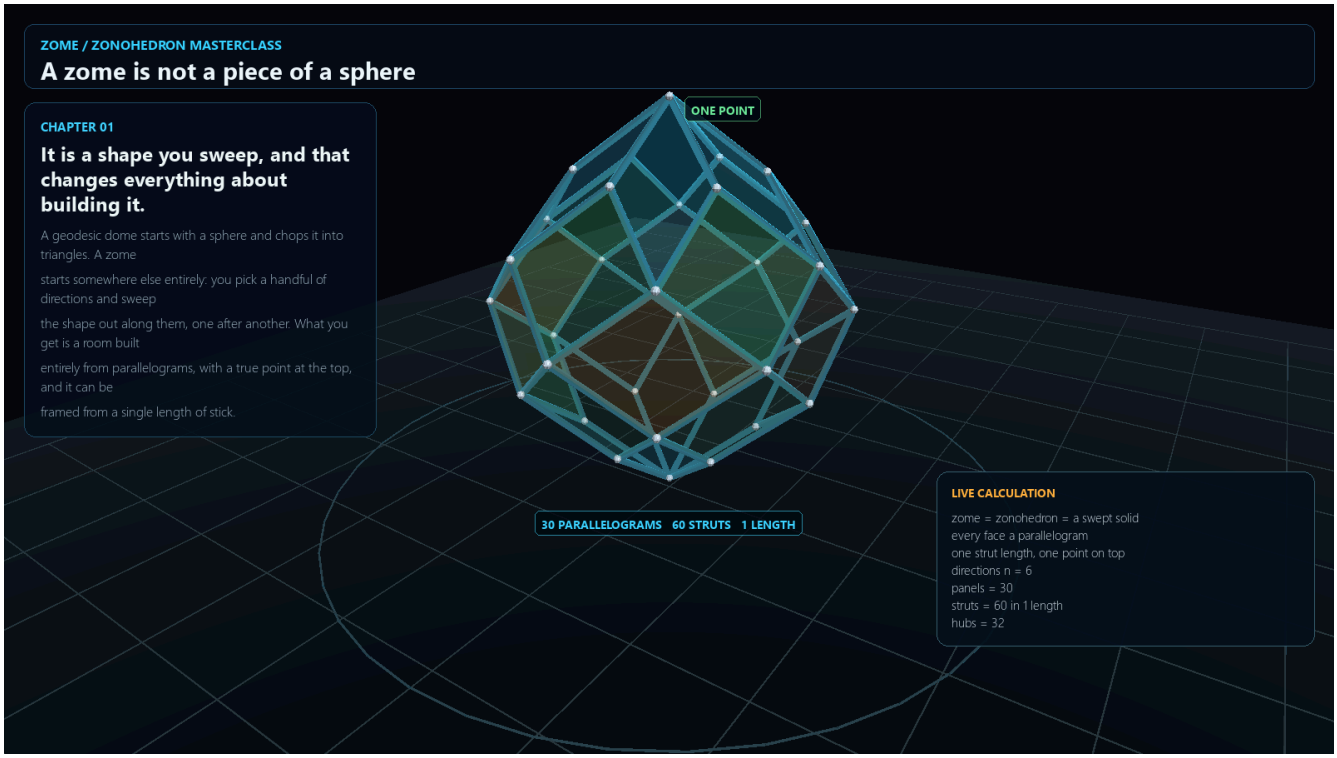
The two other round buildings this project solves, what each one is good at and what it costs against the dome in the rest of this book -- and the framed-panel method most bought dome plans use, solved again from this book's own geometry.

CHAPTER 44 · EXPLAIN

A Zome Is Not a Piece of a Sphere

Swept from a star of directions instead of subdivided from a solid -- which is why every panel comes out flat, guaranteed

Everything so far has been subdivision: take a solid, cut its faces smaller, push the new points onto a sphere. A zome is made the other way and the difference decides everything about it.



A zome is not a piece of a sphere.

The sweep

Start with a **star of directions** — a set of vectors radiating from a point, evenly spread around a vertical axis.

Now take a stick. Point it along the first direction and walk to the end. Point it along the second and walk to the end of that. Keep going through all the directions and you come back to where you started, having traced the outline of a solid.

Every face of that solid is a **parallelogram**, because every face is made by two of the directions and nothing else. Two directions define a plane. A parallelogram made of two directions is flat, necessarily, always, without anybody checking.

That is a zome.

Every panel flat, guaranteed

This is the property worth the whole chapter.

In a geodesic dome, a panel is flat because a triangle is always flat. Fine — but it means every panel must *be* a triangle, and triangles tile a sphere with a lot of members.

In a zome, panels are four-sided and still guaranteed flat, because of how they were constructed rather than because of how many sides they have. You get big flat quadrilateral panels, which is a genuinely better thing to sheathe and to glaze than a triangle is.

And the star means **one strut length**. Make the directions equal in length and every stick in the building is the same stick. Not two lengths. 3 is not the number of anything. One.

What it costs

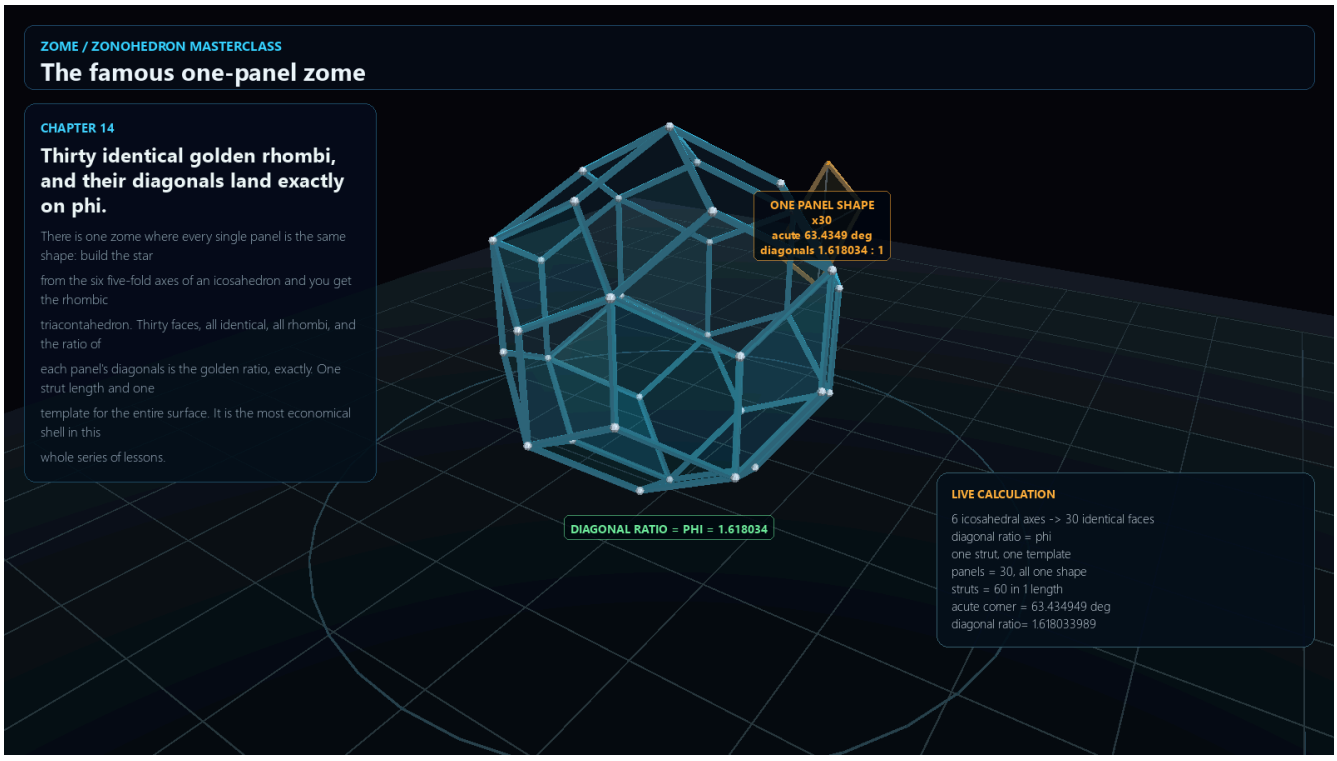
Three things, and they are real.

The hubs get particular – in a frame of single sticks. In a geodesic dome, the hub problem is what the pinwheel of Chapter 20 exists to remove. Build a zome as a frame of single sticks and the problem comes back: the sticks arrive at a hub in a pattern set by the star, at angles that are not all the same, and the joint has to hold that pattern. An earlier draft of this chapter stopped there and said zome hubs are where zome builders spend their time. That was wrong for the way most zomes are actually built: frame each panel on its own and the hub disappears, exactly as it does in the wedge dome. Chapter 45 shows how.

The top comes to a point. The sweep converges on the axis, so a zome finishes in a spike rather than a cap. That is either a feature or a detailing problem depending on your rainfall.

The shape is not a sphere. It is taller for its footprint, it has more surface per square foot of floor, and if you wanted a hemisphere you did not want a zome.

The famous one-panel zome



The famous one-panel zome.

Here is where the golden ratio comes back, and this time it really is the design.

Choose the star's directions using φ and the pitch to match, and every single panel in the zome becomes the **same rhombus**. Not two panel shapes, not five. One.

One strut length and one panel template, for a whole building. You can cut the entire skin from one jig and the entire frame from one saw setting.

That is the golden zome, and it is the single most elegant result in any of the geometries this project solves.

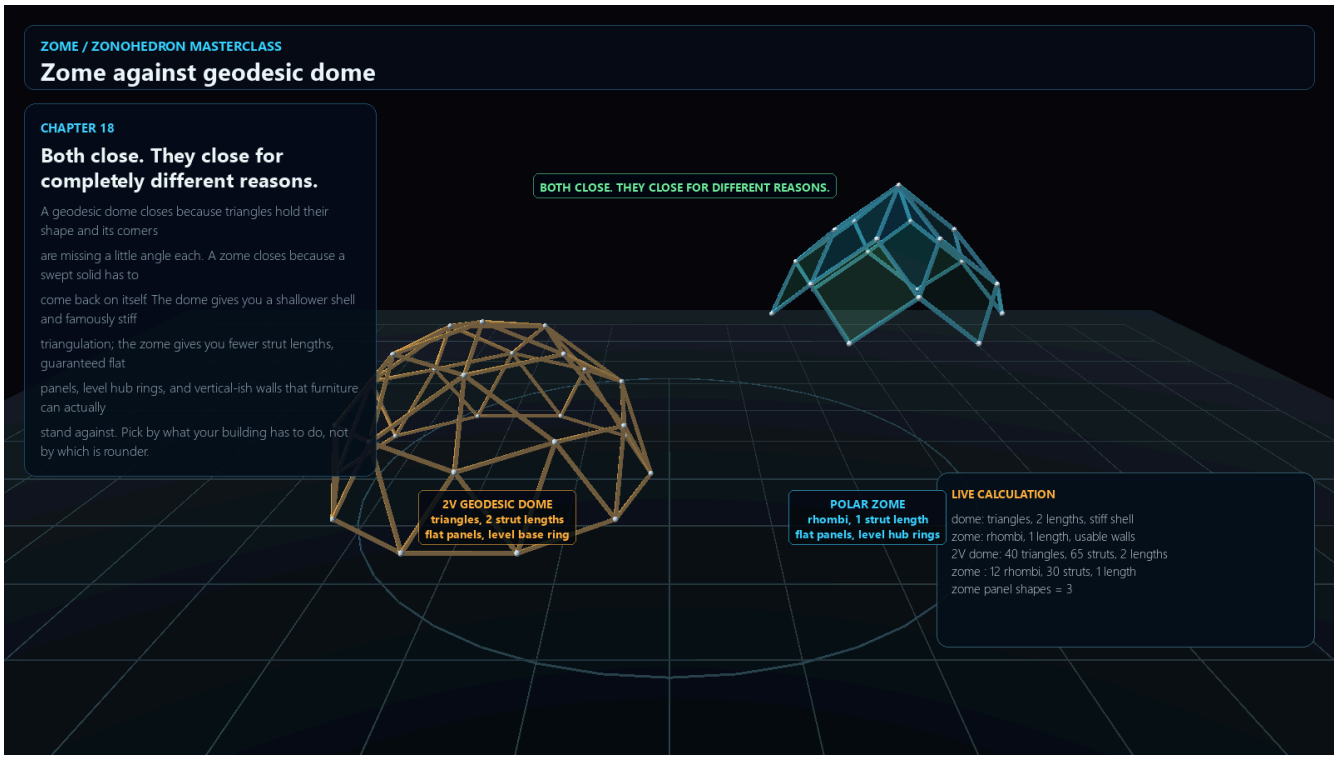
What the golden zome charges you

It charges you the freedom to choose your proportions.

The reason every panel comes out identical is that the star and the pitch are both pinned by φ. Change the height-to-width ratio you want and you lose the property — you are back to several panel shapes, and the one thing that made it worth doing is gone.

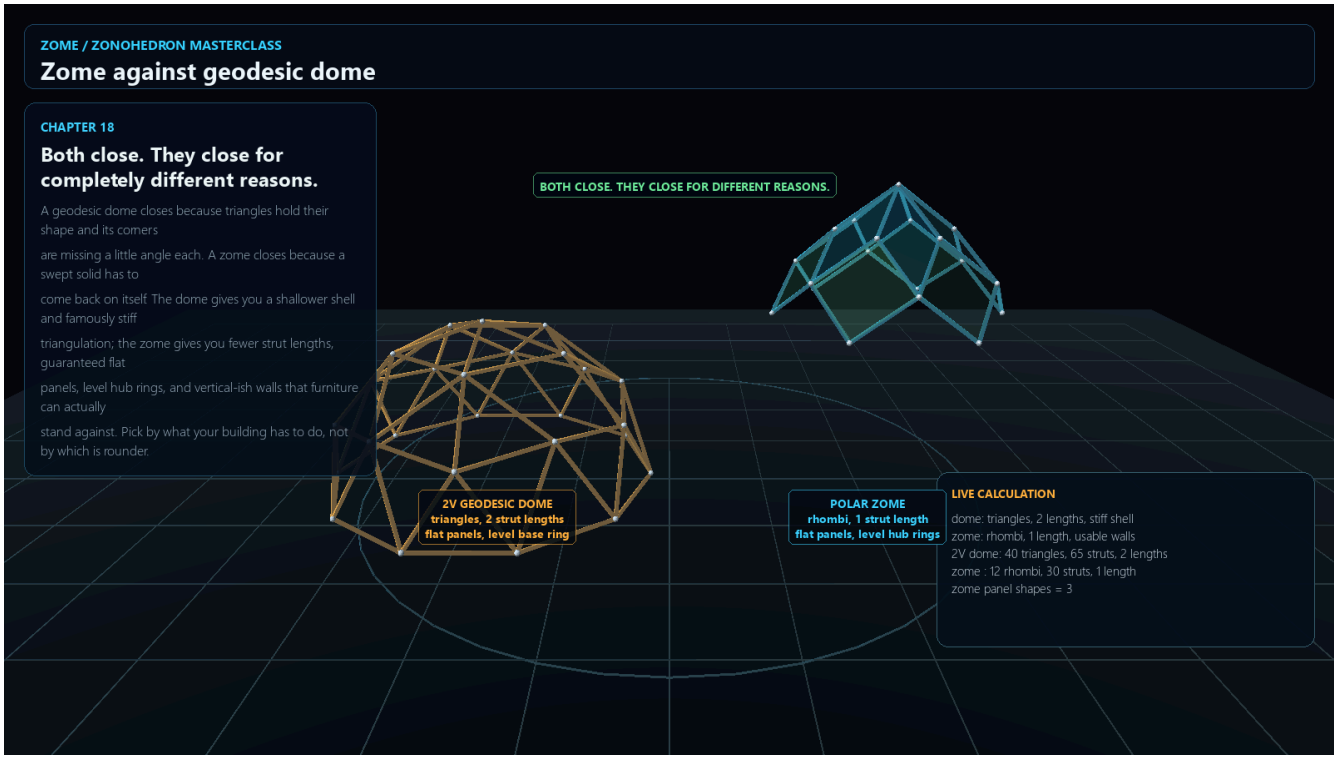
So the golden zome is beautiful and it is exactly one building. If its proportions suit your site, it is very hard to beat. If they do not, you are building an ordinary zome and paying the hub cost without the one-panel prize.

That trade is worth stating, because the property is seductive enough that people design around it and then discover they have designed around a constraint rather than a feature.



A zome, drawn by the tool that solves it.

Zome against geodesic dome



Zome against geodesic dome.

strut lengths	zome 1	geodesic 2
panel shape	parallelogram	triangle
panel flatness	by construction	by being a triangle
hubs	particular	removed by the pinwheel
top	a point	a cap
surface per floor	more	less

Neither wins. They are answers to different questions, and the question a zome answers well is *"I want big flat panels and one stick length and I do not mind a hub."*

This book builds a geodesic dome because its question was *"I want to use a split log and avoid a hub entirely"*, and the wedge's flat sawn faces make the pinwheel possible in a way they would not in a zome's hub.

But if you have a bandsaw and a box of brackets and no chainsaw, the zome may be your building, and this chapter exists so you can find that out here rather than after the tree is down.

A Zome Without a Bevel

Four identical sticks to a panel, a miter at each end and no bevel anywhere -- and the gap on the outside that the method allows on purpose

Chapter 44 said a zome's hubs are where zome builders spend their time, because the sticks arrive at a hub in a pattern the joint has to hold. That is true of a zome built as a frame of single sticks. It stops being true the moment each panel is framed on its own, the way Chapter 48 frames a dome's triangles. Then the panels meet each other along their edges, and there is no hub at all.

It is called the *non-beveled* method, and builders who have framed zomes both ways overwhelmingly choose it. The figures are this book's, solved for its own zome:

`base_ft`	16	ft	the reference 3V dome, base corner to base corner: a round size, and the largest that stands beside th
`stock_w_in`	1.5	in	a 2x3's actual thickness: the stick lies wide side down, so this is its width on the jig
`stock_d_in`	2.5	in	a 2x3's actual width, ripped in half along a bevel to make two sticks
`kerf_in`	0.125	in	a common table-saw blade
`base_board_in`	5.5	in	a 2x6 on edge under the lowest base corner: lifts the first panels off the deck's wet
`zome_n`	9	directions	a nine-sided zome
`zome_pitch_deg`	54	deg	the pitch the zome chapter uses, from the zome module
`zome_side_m`	0.9	m	panel side: puts the rim about four metres across, a garden-studio size
`zome_w_in`	1.5	in	a 2x4's thickness, flat on the bench
`zome_d_in`	3.5	in	a 2x4's width, on edge through the wall

The zome is nine-sided – 36 rhombus panels in 4 rows of 9 – about 13.8 feet across its rim and 8.7 feet tall above it, every panel side 35.4 inches.

One stick, one miter, per row.

Four sticks in a pinwheel

Each panel is four sticks in a pinwheel, the arrangement of Chapter 20 once more: each stick runs to one corner and butts the next at the other. A rhombus's corners alternate sharp and blunt, and the sine of an angle is the sine of its supplement, so the shortening at every tail is the same – **all four sticks of a panel are one stick**. Its two ends are cut at the same angle and parallel, long point to short point.

Nothing is bevelled. The sticks stay square through their whole depth, so there is no compound cut, no tilted blade, and no jig with a bevel in it. Cut one end, butt two sticks together on the bench, mark the panel length across the joint, scribe the miter and cut to the mark; set a stop block and cut the rest of that row's sticks from full-length boards. Screw pairs into halves and halves into panels.

Both ends cut parallel; nothing bevelled.

Row by row

A zome's rows differ, so each row has its own stick:

row	panels	sharp corner	miter both ends	fold	outside gap, in	block bevel
1	9	32.1	57.9	24.2	1.50	12.1
2	9	62.7	27.3	19.3	1.19	9.6
3	9	89.0	1.0	27.8	1.73	13.9
4	9	74.4	15.6	27.8	1.73	13.9

One stick and one miter for every panel in a row; that is the whole cut list.

The gap, and what fills it

Square-edged panels meet only along their *inside* edges. Where two panels fold away from each other, that leaves a V-shaped gap on the outside whose width is twice the stick's depth times the tangent of half the fold. In a two-by-four on edge it opens from **1.19 to 1.73 inches** on this zome.

The gap is allowed on purpose. On a small zome, screw the inside edges together and let the roof bridge it – a zome's panels each slope one way, so roofing laps down the structure like shingles. On a large one, rip a **wedge block** to half the fold on each face and fix it between the panels: it fills the V and gives the joint something to bear on. The block's bevels are in the table.

Assemble with two screws per stick near the inside edge, not tight, so the panels can hinge. As the inside edges line up, the fold sets itself. Brace the top rows – they sag under their own weight until the ring closes – then go back and add the rest of the fasteners.

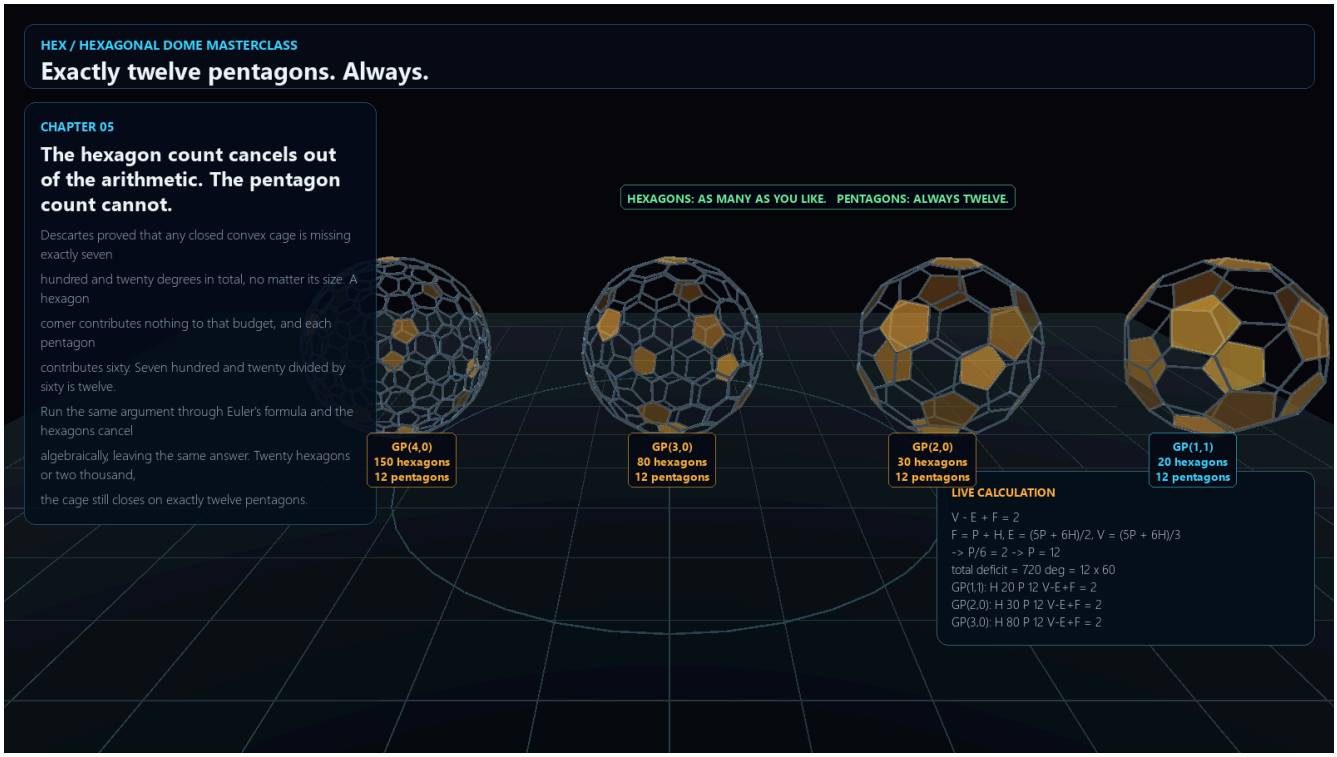
CHAPTER 46 · EXPLAIN

Exactly Twelve Pentagons, Always

A sheet of hexagons will not curve. Curvature is bought with missing angle, and the price is always twelve pentagons

Everybody reaches for hexagons. Honeycomb, graphite, the compound eye — the hexagon is nature's answer to tiling a plane, and it is the shape people sketch when they imagine a dome that is not made of triangles.

There is a catch, and it is absolute.



Exactly twelve pentagons. Always.

A sheet of hexagons will not curve

Six regular hexagons meet at a point with their angles summing to exactly 360 degrees. A full turn. No angle left over.

That is what makes hexagons perfect for a **flat** surface, and it is exactly what stops them curving. Curvature is bought with *missing* angle: to make a surface close up into a ball you need the angles around each point to sum to less than a full turn, so that the surface has to pucker.

Hexagons have nothing to give. A sheet of them stays flat forever, however large you make it.

So where does the angle come from

A pentagon.

Put five hexagons and one pentagon around a point and the angles no longer reach 360. The surface puckers. Do it enough times and it closes into a ball.

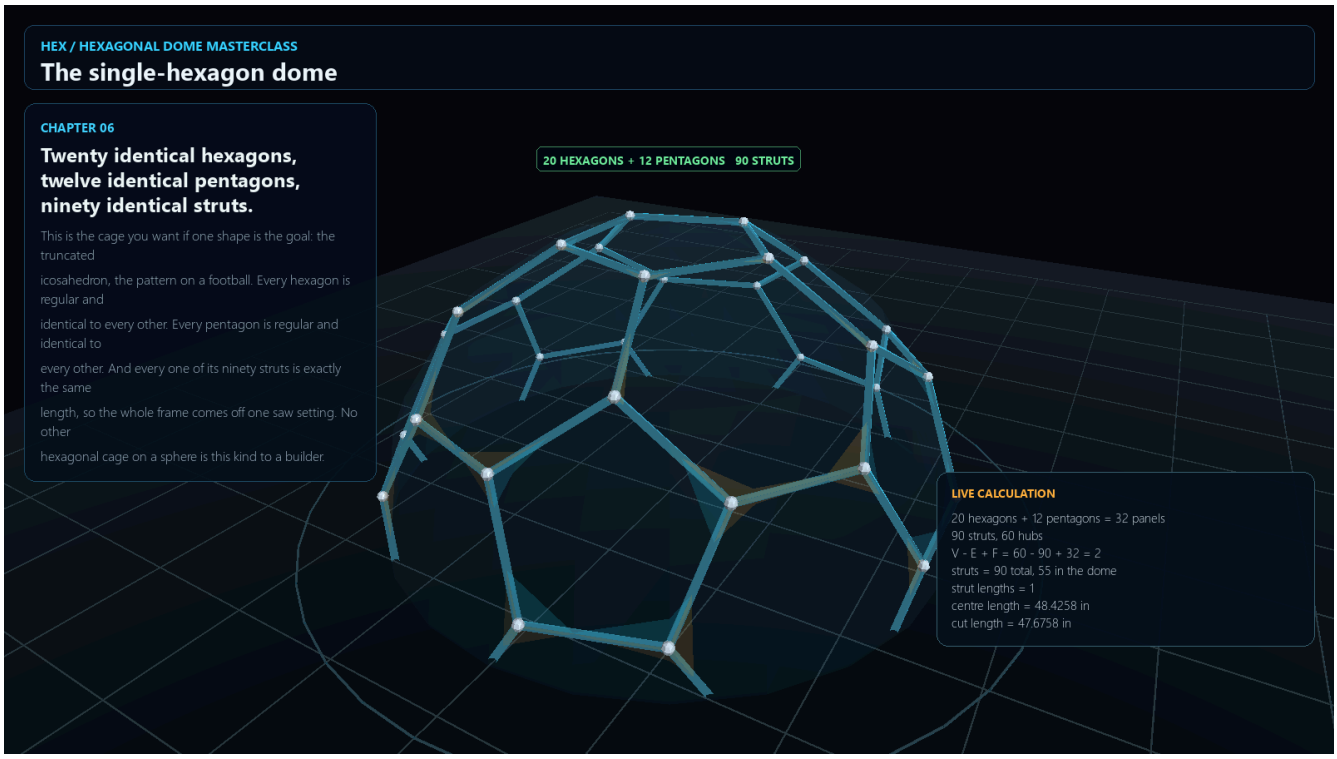
And the number of times is not a design choice. Euler's formula fixes it: a closed surface tiled with hexagons and pentagons, three faces meeting at every corner, needs **exactly twelve pentagons**. Twelve. Not eleven, not thirteen, and no amount of hexagons changes it.

A football has twelve. A fullerene has twelve. A hexagonal dome has twelve, or it is not closed.

The single-hexagon dome

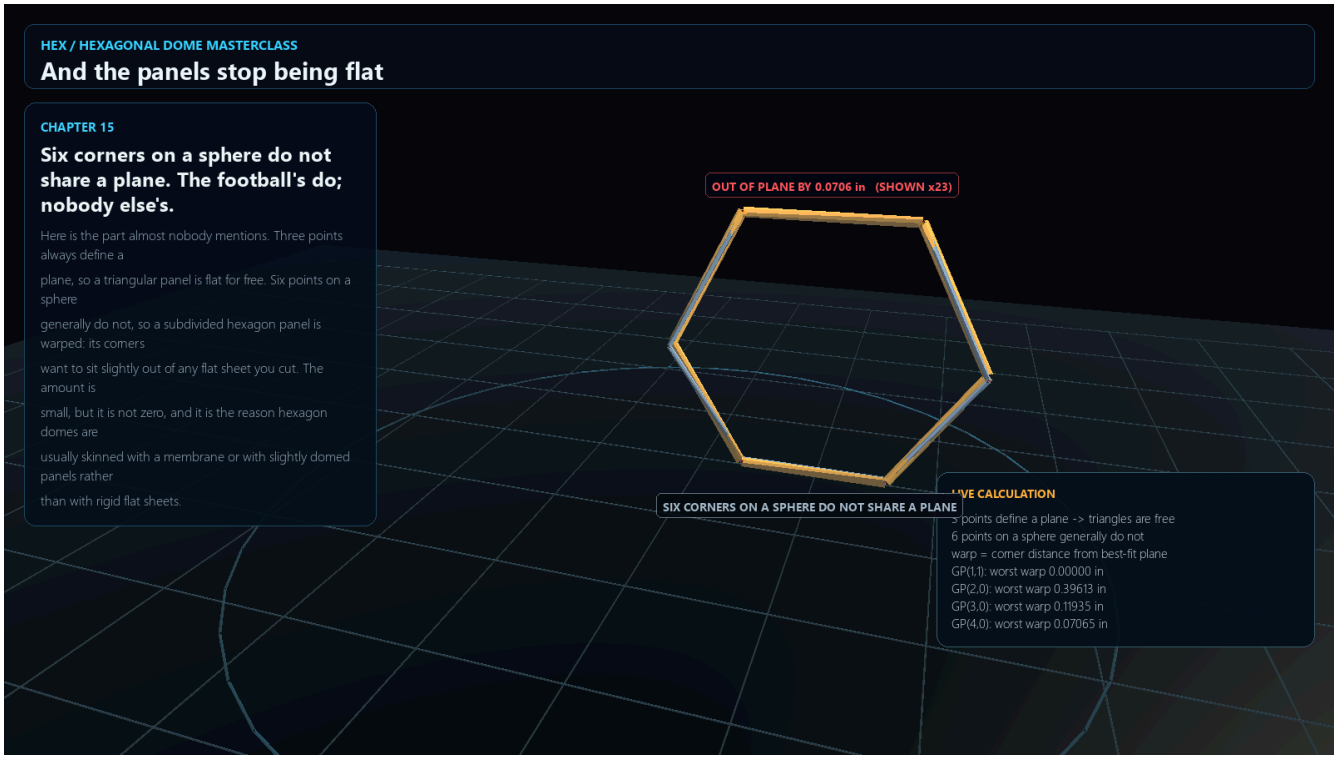
The smallest useful case is charming: one hexagon at the crown, a ring of pentagons, and you are done.

It has **one strut length**, cut ninety times, and **two flat panel templates** that cut the entire skin. For a small structure — a shelter, a shade, a play house — that is an extremely cheap parts list.



The single-hexagon dome.

Then you want it bigger



The panels stop being flat.

Here is where it goes wrong, and it goes wrong in a specific way worth knowing before you commit.

Scale a hexagonal dome up by subdividing and two things happen at once.

The hexagons stop being regular. Away from the pentagons the tiling stretches, so the hexagons become slightly different from each other — and each different one is another panel template. Every step up multiplies the shapes.

And the panels stop being flat. A hexagon has six corners. Three points define a plane; six do not, and on a curved surface the sixth corner will not lie in the plane of the other five. The panel warps.

A triangle cannot warp. That is the property Chapter 6 opened with, and this is where the bill for not having it arrives.

What the warp costs

You have three ways out and all of them cost something.

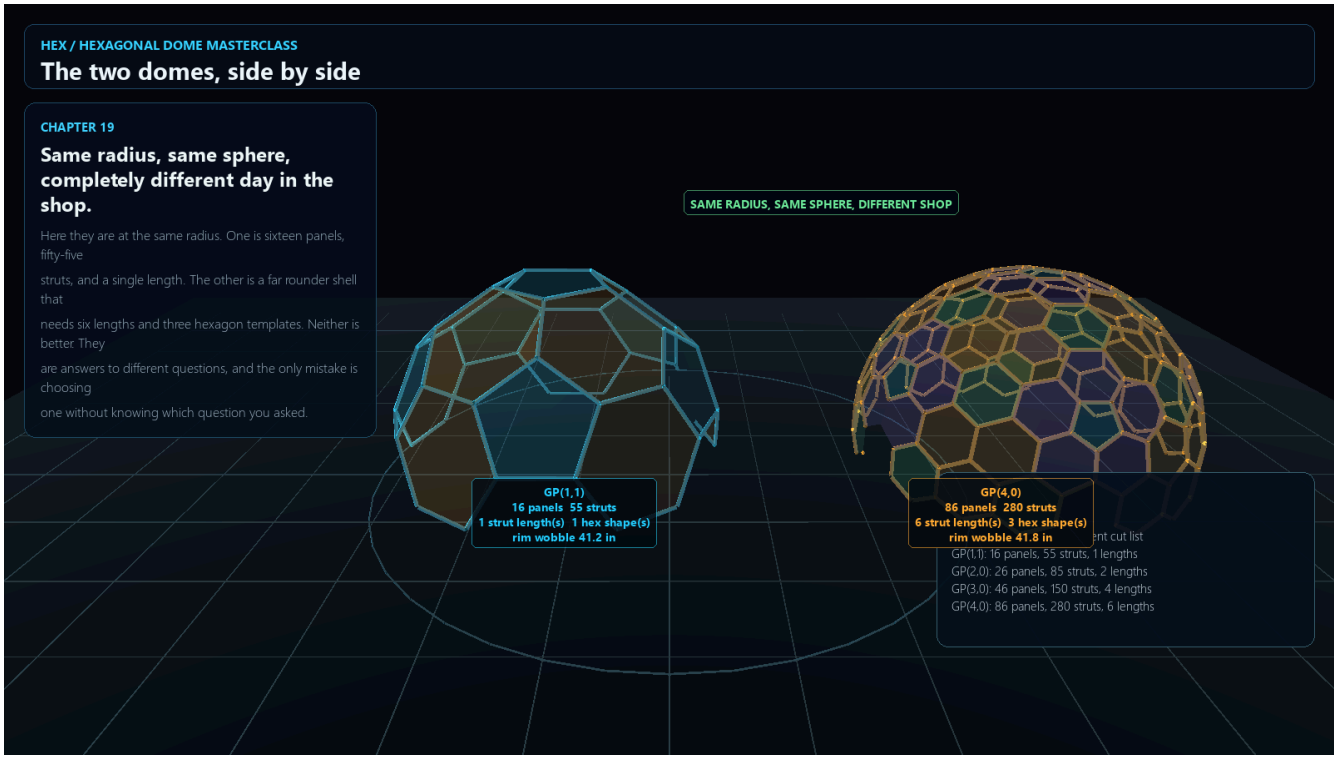
Accept the warp and force the panel flat when you fix it. It goes in stressed, it wants to spring back, and every fastener is holding that.

Fold the panel along a diagonal, which is admitting it is two triangles and building it as one.

Make the panels smaller until the warp is below what the material tolerates, which means more of them.

None of those is fatal. All of them are work that the triangulated dome simply does not have.

The two domes, side by side



The two domes, side by side.

The honest summary:

A hexagonal dome is better when it is small, when you want few strut lengths, and when big flat panels matter more than anything else.

A geodesic dome is better when it is large, because the triangle's flatness is guaranteed at any size and the hexagon's is not.

The crossover is not far up. Which is why this book builds triangles — and why it is worth understanding the hexagon properly rather than dismissing it, because for a chicken coop the hexagon wins and this method would be over-engineering.

The thing to carry away

Curvature costs angle, and something has to pay.

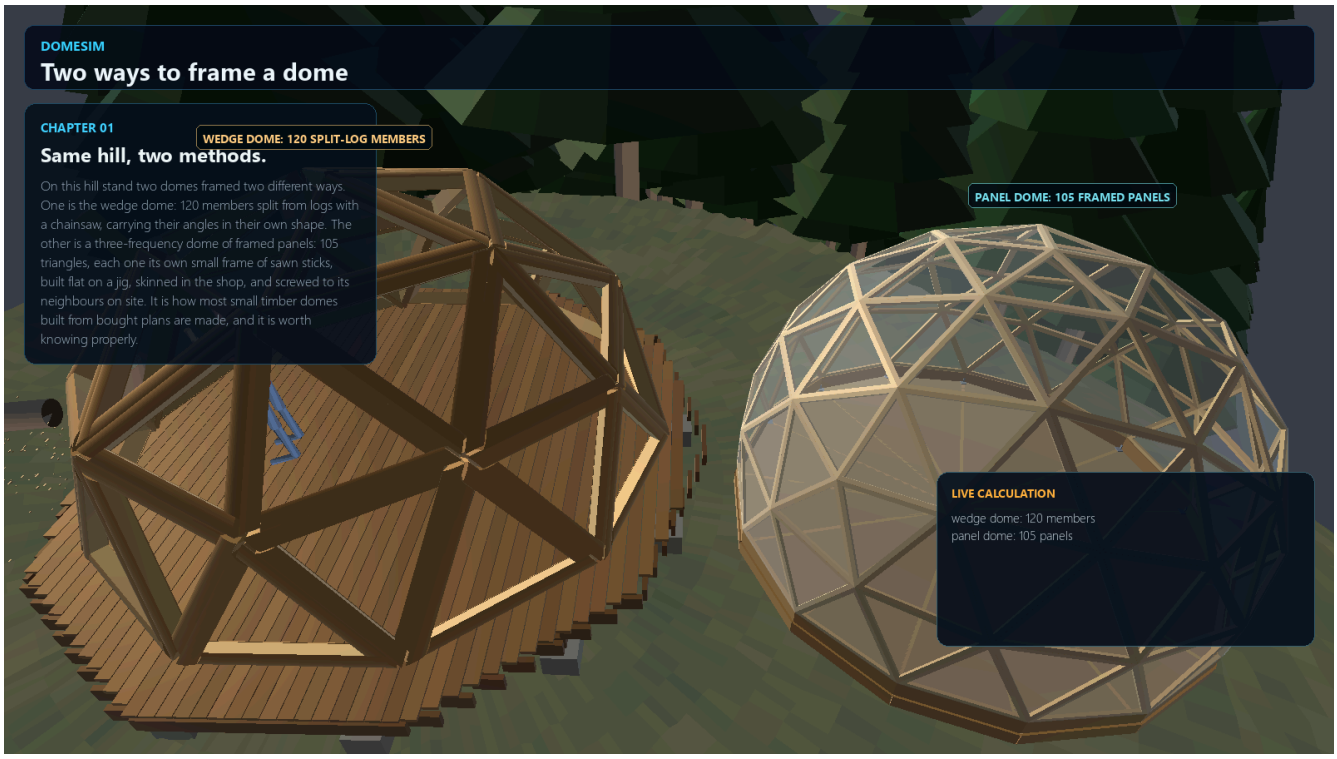
In a hex dome, twelve pentagons pay it. In a geodesic dome, the subdivision pays it — every one of those twenty faces becomes triangles that are slightly different from each other, which is precisely why there are two chord lengths instead of one.

There is no tiling that curves for free. Every round building in this book is a different answer to the same bill.

Three Frequency, Five Eighths

The dome most bought plans build: a three-frequency sphere cut below its middle, a base that zig-zags, and two panel shapes

Most small timber domes built from bought plans are not split from logs and are not hub-and-strut frames. They are **framed panels**: every triangle of the dome is its own small frame of sawn sticks, built flat on a plywood jig, skinned in the shop, and screwed to its neighbours on site. It is a good method with a great many domes behind it, and anyone who builds the dome in this book should understand it, because it is the road most people take.



Two domes, framed two ways.

Why it is here. The three-frequency five-eighths dome of framed panels is a popular build, tried and true: a great many are standing. That makes it the fairest comparison for the wedge dome, and this chapter and the next set it beside it. Every length, angle and count is solved by this project's own geometry (`two_v_demo/panel_method.py`), for a dome of this book's own choosing, and drawn by this project's own software.

The reference dome, and the stock its sticks come from, are choices rather than measurements, and they are declared before anything is claimed:

`base_ft`	16	ft	the reference 3V dome, base corner to base corner: a round size, and the largest that stands beside th
`stock_w_in`	1.5	in	a 2x3's actual thickness: the stick lies wide side down, so this is its width on the jig
`stock_d_in`	2.5	in	a 2x3's actual width, ripped in half along a bevel to make two sticks
`kerf_in`	0.125	in	a common table-saw blade
`base_board_in`	5.5	in	a 2x6 on edge under the lowest base corner: lifts the first panels off the deck's wet
`zome_n`	9	directions	a nine-sided zome
`zome_pitch_deg`	54	deg	the pitch the zome chapter uses, from the zome module
`zome_side_m`	0.9	m	panel side: puts the rim about four metres across, a garden-studio size
`zome_w_in`	1.5	in	a 2x4's thickness, flat on the bench
`zome_d_in`	3.5	in	a 2x4's width, on edge through the wall

Three lengths, two kinds of point

Chapter 8 made this book's dome by cutting each face of the icosahedron into four and pushing the new points out. Cut each face into *nine* instead – three to an edge – and push every point out to the sphere, and you have a **three-frequency** dome. It is rounder, its triangles are smaller, and it has only three strut lengths. At 16 feet across the base:

```
A    34.00 in   (864 mm)
B    39.36 in   (1,000 mm)
C    40.23 in   (1,022 mm)
```

Twelve points have five neighbours – the icosahedron's own corners – and every other point has six. Chapter 46 explained why it must be exactly twelve. The triangles round a five-way point make **pentagons**; all the others make **hexagons**. That is the whole vocabulary of the dome: three lengths, two kinds of point.

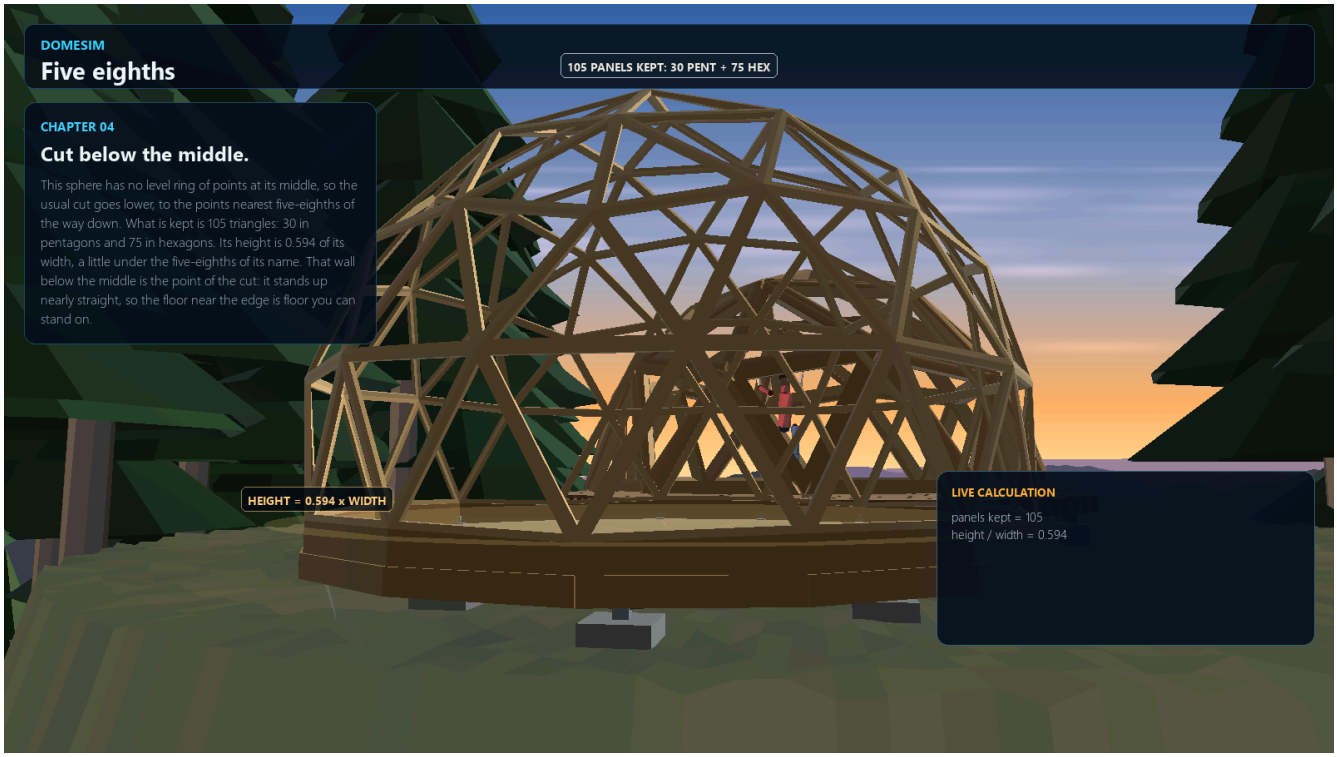
Three chords: A, B and C.

Why five eighths

A two-frequency sphere has a level ring of points at its middle, which is why this book's dome is a clean hemisphere. A three-frequency sphere does not. Its points sit at heights that never line up across the middle, so the cut goes lower, to the ring of points nearest five-eighths of the way down.

That cut keeps **105 triangles**: 30 that belong to pentagons and 75 that belong to hexagons. Its height is **0.594 of its width** – a little under the five-eighths in its name – which for the reference dome is 9.65 feet.

The extra wall below the middle is the point of the cut. Near the base it stands almost upright, so the floor near the edge is floor you can stand on. A hemisphere's wall leans in from the floor; this book's dome answers that with a riser wall under the shell (Chapter 27), and the five-eighths cut answers it with more of the sphere.



Cut below the middle.

A base that zig-zags

The price of the five-eighths cut is its base. The bottom edge is 15 struts long and its corners do not lie on one level: they alternate between two heights **40 mm apart** – about 1.56 inches. So the base sections, the boards the bottom panels sit on, are not all alike. **10 must be tapered**, low at one end and high at the other, and **5 sit level**.

Build that ring first. Set the base sections out on the deck, fasten them to each other, pull the diagonals until they agree, and fasten them down. Every panel above lands on this ring, and every error in it is passed up the dome.

Ten tapered, five level.

Two panels, over and over

Every panel in the dome is one of two triangles, and the jig is drawn for each of them full size:

	pentagon	hexagon
panels	30	75
sides (letters)	B-A-A	B-C-C
base, in	39.36	39.36
left side, in	34.00	40.23
right side, in	34.00	40.23
apex angle, deg	70.73	58.58
base angles, deg	54.63	60.71
height on jig, in	27.73	35.08

The outlines are the *outside faces* of the panels – the face the weather sees. Nothing may stick out past them and nothing may fall short of them, which is what the jig of the next chapter enforces.

A pentagon panel and a hexagon panel.

A floor the shape of the base

A dome on a deck wants the deck to copy its base. A **radial deck** does: one joist from the centre to every base corner – 15 of them – and a rim board along every base edge, 39.9 inches long point to long point, each end cut **12 degrees** off square. That angle is not looked up anywhere: it is half of the 360 degrees round the deck divided by the number of sides, 180 over 15.

Under the joists, the girders follow the dome's own symmetry. A three-frequency dome has five-fold symmetry about its top, so the girders make a ring of 5, each end cut 180 over 5, 36 degrees, with a post under every joint where two girders meet. Mark the centre on the ground, lay the girders out on it to place the posts, set the post tops to a level line, then roll in the joists. The deck is 195 square feet. Chapter 36 is about why the floor is the other thing to get right, and Chapter 37 about where it stands.

Joists to the corners, girders on the five-fold.

A Panel on a Jig

Rip, draw, rail, pinwheel, skin, raise: the framed-panel method start to finish, and the bevel no single saw setting gets right

This is the framed-panel method from first cut to anchored shell, for the three-frequency dome of Chapter 47: a popular, tried-and-true way to build, shown here for comparison. The figures are this book's, solved for its own reference dome; the pictures are this project's own renderings.

The idea is to move the work indoors. Each triangle is built flat, on a bench, at a comfortable height, out of the weather. It can be finished and skinned before it ever leaves the shop. On site, the dome is mostly a matter of screwing finished frames together.

Finish, then rip in half

The sticks are sawn lumber: a 2x3, ripped in half down its length, gives two sticks 1.50 inches wide and 1.19 inches deep. A 2x6 ripped into two 2x3s first is often straighter than any 2x3 on the rack, and straightness matters more here than species or grade.

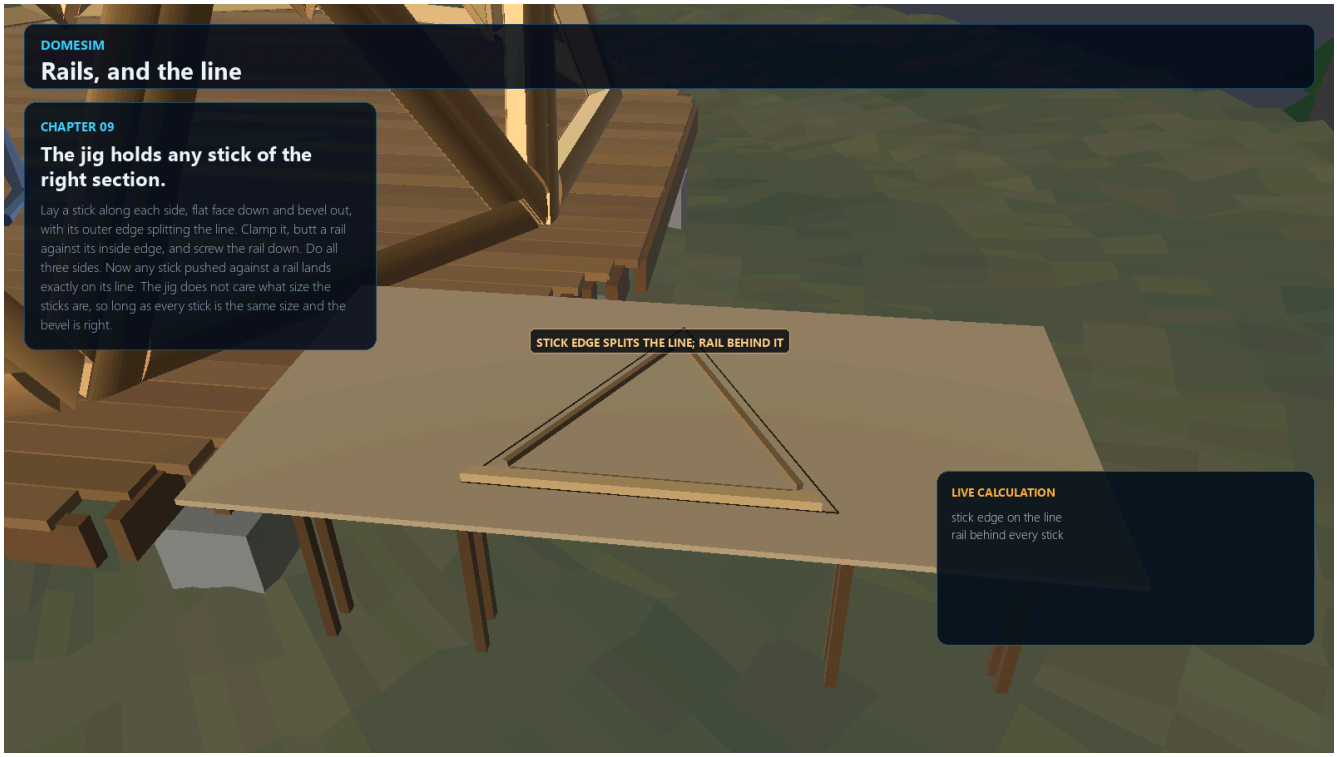
Finish the boards before you rip them. The face the saw makes is hidden in the joint, so finishing whole boards is half the work of finishing sticks.

Then rip on the bevel, through the centre. Tilt the table-saw blade to the bevel and set the fence so the cut runs through the middle of the board's end. A cut through the centre of a rectangle leaves two pieces that are copies of each other, turned end for end – the same symmetry that makes a split log give identical wedges in Chapter 13. Turn the offcut over and send it back through for a hair more, and the two halves come out exactly alike. A stick a little smaller than planned is fine; sticks that differ from each other are not. The bevel itself must be right, so set it with a digital angle gauge, not the saw's scale.

Draw the jig with two arcs

The jig is a sheet of 3/4-inch plywood – thick enough not to warp – with the panel drawn on it full size. Snap a straight line for the base and mark the base length on it. Make a **trammel**: a thin stick with a nail through each end. Set it to one side length and swing an arc from one end of the base; set it to the other side length and swing an arc from the other end. Where the arcs cross is the apex. Measure all three sides again before joining the points. For the hexagon panel the apex lands 35.08 inches above the base line; for the pentagon panel, 27.73. Cut the sheet down to an inch or two outside the lines.

It is the same jig, in spirit, as Chapter 24: geometry solved once on a flat board and then copied by every panel.



Where the arcs cross is the apex.

Rails, and the line

Lay a stick along one side, flat face down and bevel out, with its outer edge *splitting* the pencil line. Clamp it. Butt a straight rail against its inside edge and screw the rail down hard. Unclamp, and do the other two sides. Now any stick pushed against a rail lands exactly on its line. The jig does not care what section the sticks are, so long as every stick is the same.

Sticks that turn like a pinwheel

The three sticks do not meet at a point. Each runs all the way to one corner and butts into the inside face of the next, so seen from above the corners turn like a pinwheel. It is the pinwheel of Chapter 20, met again in a different dome: nothing meets at the corner, so no corner needs a hub.

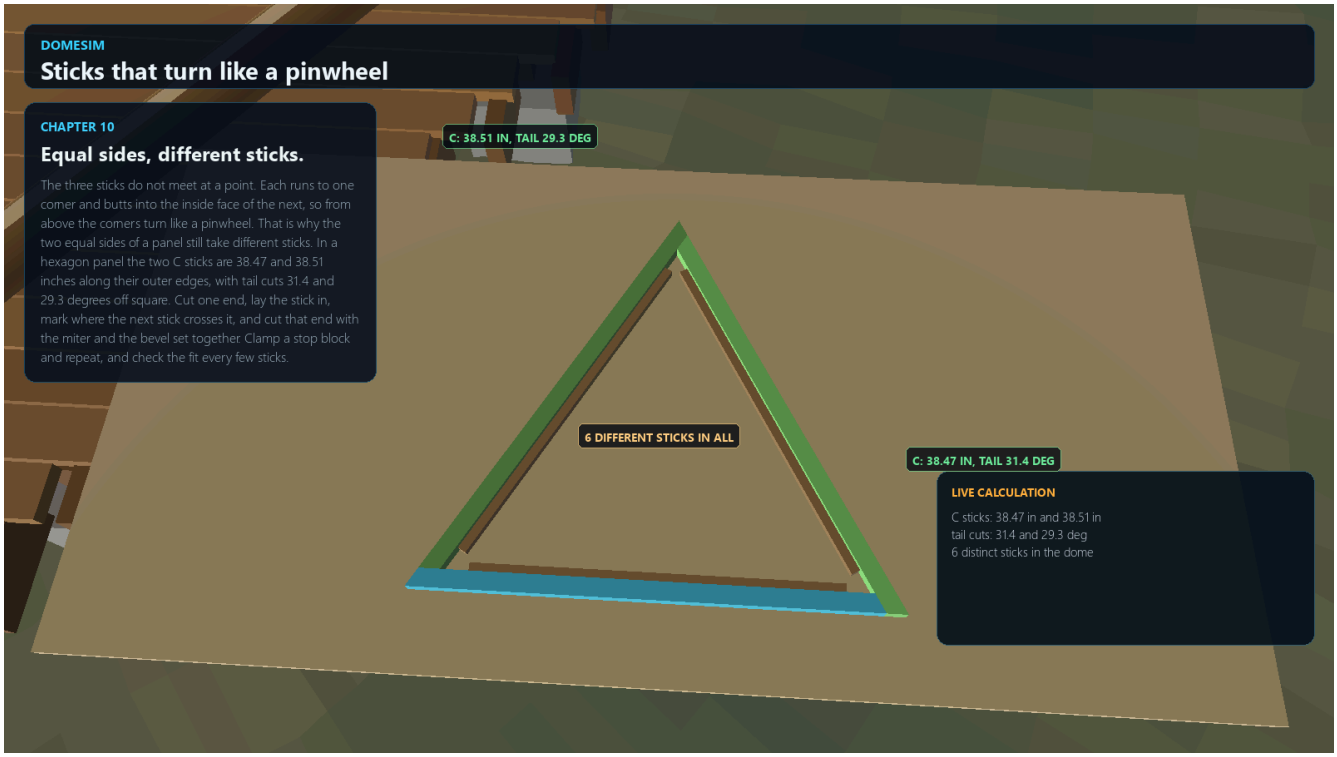
A pinwheel has a consequence that surprises people: the two equal sides of a panel do **not** take the same stick, because one butts and the other is butted at the apex. A stick of width w that butts at a corner of angle *phi* stops w / sin *phi* short of the corner and is cut 90 - *phi* off square. Every stick in the dome:

panel	side	each panel	outer edge, in	tail miter	bevel
pent-BAA	B	30	37.52	35.4	6.5
pent-BAA	A	30	32.42	19.3	7.2
pent-BAA	A	30	32.17	35.4	7.2
hexa-BCC	B	75	37.64	29.3	6.5
hexa-BCC	C	75	38.47	31.4	6.8
hexa-BCC	C	75	38.51	29.3	6.8

That is **6 different sticks**, 315 in all, about 972 feet of them.

Cut each stick's first end, lay it in the jig against its rail, and mark where the next stick's inside face crosses it. That second end is a compound cut – the tail miter and the bevel together. A miter saw that bevels both ways makes it in one pass; a handsaw, cutting flush along the next stick after the panel is screwed, works too and is pleasantly slow. Clamp a stop block for the repeats, and check a stick in the jig every few cuts in case the block has crept.

Half panels – at a door, or where the base needs one – are drawn on the same jig, but their middle stick runs across the triangle rather than along an edge, so it takes a different bevel or none at all. Check it before you cut it.

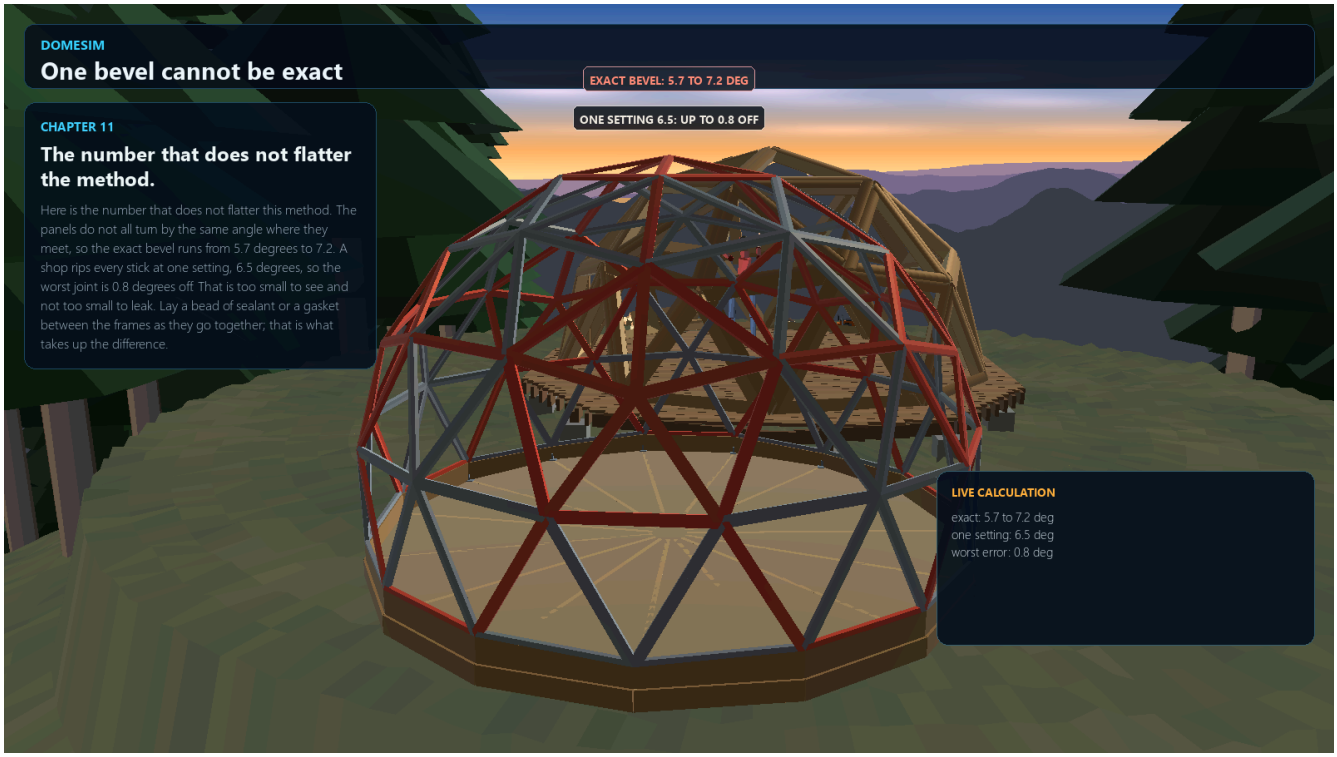


Three sticks, six kinds in the dome.

One bevel cannot be exact

Here is the number that does not flatter this method. The panels do not all turn by the same angle where they meet. Across a B edge alone the fold runs from 11.4 to 14.5 degrees, and over the whole dome the exact stick bevel runs from **5.7 to 7.2 degrees**. A shop rips every stick at one setting – 6.5 degrees, halfway – so the worst joint is **0.8 degrees** off.

That is too small to see and not too small to leak. Lay a bead of sealant or a strip of gasket between the frames as they go together; that is what takes up the difference.

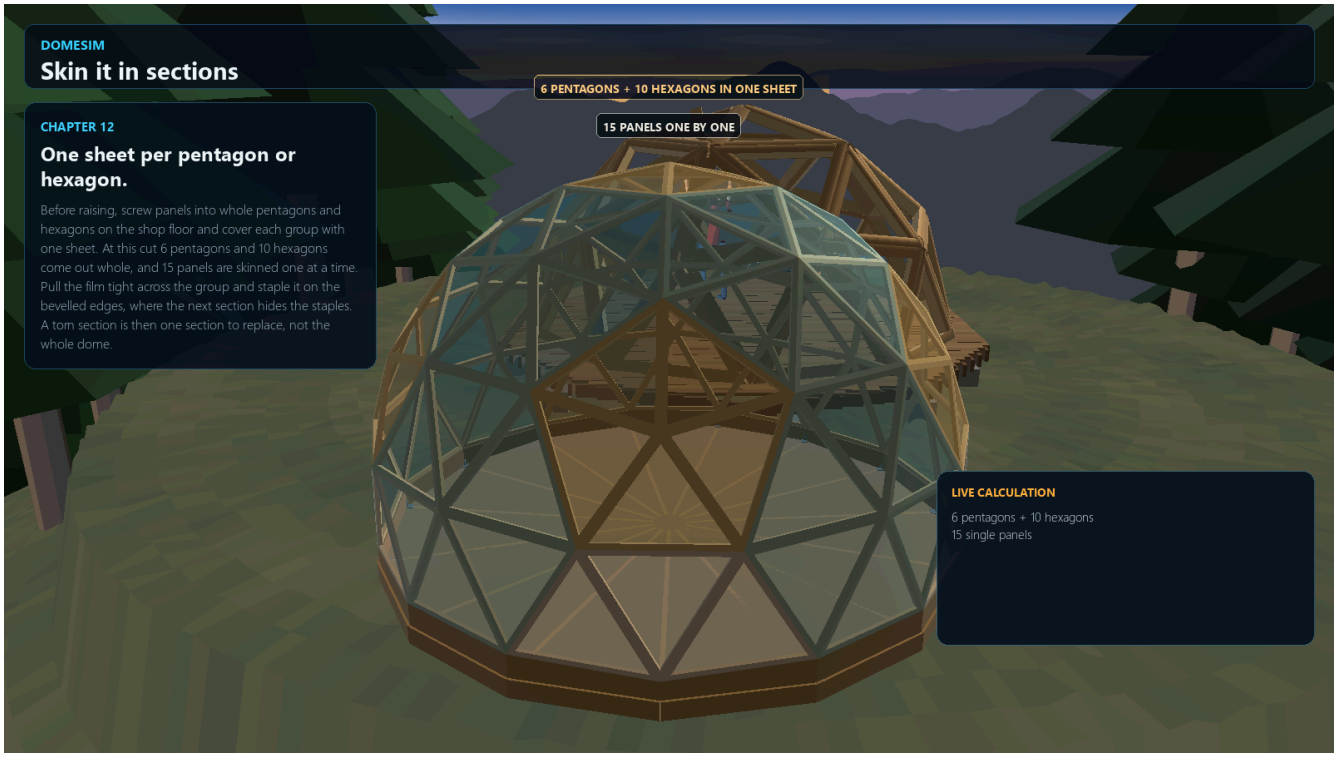


Red where one setting is furthest off.

Skin it in sections, raise it loose

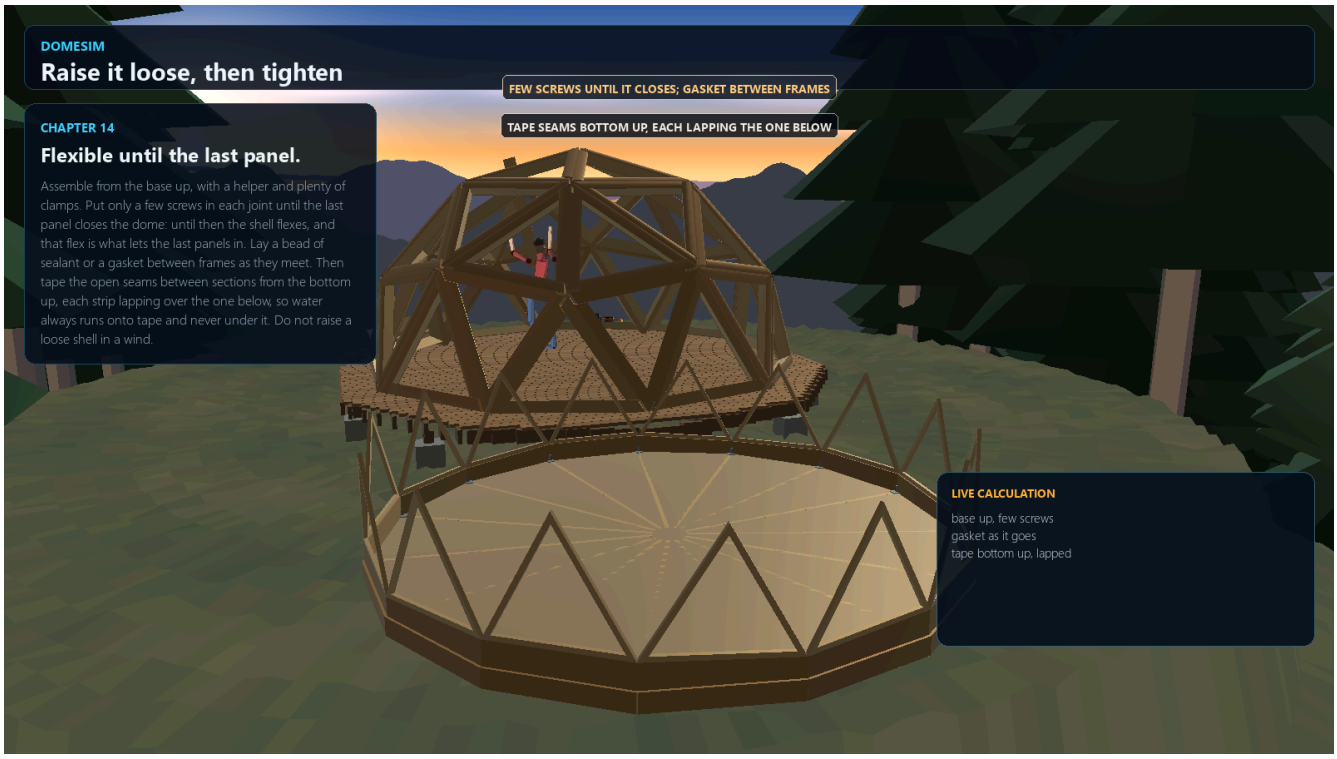
Sections. Before raising, screw panels into whole pentagons and hexagons on the shop floor and cover each group with one sheet. At this cut 6 pentagons and 10 hexagons come out whole, and 15 panels are skinned one at a time. Pull the film tight and staple it on the bevelled edges, where the next section hides the staples. A woven poly film stretches less than a thin greenhouse film and forgives a cold day; a thin film tears where a woven one does not.

This is a different trade from Chapter 28, which argues for one continuous watertight layer over the whole frame. Sections are repairable – a torn one is one section to replace – but every seam between sections is a joint to tape and to keep.



One sheet per pentagon or hexagon.

Windows. Build one extra frame for each window. Leave the skin off that panel in the dome, cover the spare frame, and hinge it along its top edge. A small gas strut of the kind kitchen cabinets use holds it open. Weather-strip the face it closes on, so rain running down the dome cannot creep under it, and tape the hinge seam once the dome is up.



A spare frame on two hinges.

Raising. Assemble from the base ring up, with a helper and plenty of clamps. Put only a few screws in each joint until the last panel closes the dome: until then the shell flexes, and that flex is what lets the last panels in. Lay the sealant or gasket between frames as they meet. Do not raise a loose shell in wind.

Taping. Tape the open seams between sections from the bottom up, each strip lapping over the one below, so water always runs onto tape and never under it. The surfaces must be clean and dry or the tape is decoration.

Anchoring. A dome is light for its size, and wind lifts. On a deck, angle brackets tie every base section to the frame below. On gravel or earth, steel stakes driven at a slant and strapped to the base sections do the same job. Anchor before the skin is finished, not after the first storm.

A doorway that sticks out

A five-eighths dome's own wall gives room for a door, but a door set into triangles is a door with a slanted head. The better answer is a short projecting entry – an arch doorway – built from the same kind of panels: a row of roof panels and a row of wall panels standing out from the dome, with two posts and a header inside them carrying an ordinary rectangular door. The same piece built twice, back to back, is a covered passage joining two domes into one building, which is the network of Chapter 43 at the scale of a single site.

Its panels are not triangles of the sphere, so they are built on **templates** rather than a trammel jig: each panel's outline is cut full size from a sheet, the sticks are fitted inside it, and the template is then turned over to build the panel's mirror twin for the other side. One stick on each roof panel – the one that lands on the dome – is the only thing allowed past the template's line, because its bevel is what seats the entry against the existing frame. That bevel is sharp; it is easier to rip with the board standing on edge, the saw set to the complement of the angle. The wall panels are drawn the same way, then shortened by the depth of the roof's edge stick so they fit snugly under it. Put the roof panels up first, then the walls, then the posts and header.

This book does not solve the entry's geometry – its angles depend on where it meets the dome and how far it projects – so it gives no figures for it.

Against the wedge dome

	2V wedge dome	3V panel dome
across the base, ft	19.42	16.00
height, ft	9.71	9.65
floor, sq ft	277	195
triangles	40	105
sticks	120	315
distinct sticks	2	6
base sections that taper	0	10

The panel dome is the better-trodden road: many people have built one from bought plans, and the method forgives a beginner. It asks for a table saw, a miter saw, bought boards, and a lot of identical cuts. The wedge dome asks for trees and a chainsaw, and gives back fewer, bigger pieces that carry their angles in their own shape. Neither is wrong. Choose by what you have.

PART 11

The Catalogue

“Twelve finished buildings, drawn by the tool that builds them, with the price of every choice in each one.”

The whole substitution space -- frame, section, panel, cladding, foundation, finish and fit-out -- and the two arguments that decide most of it.

Every Other Way of Doing It

Eight frame materials, eight sections, sixteen panels, nine claddings and twelve foundations -- priced on this dome, not in the abstract

This book recommends one way of doing each thing. You are owed the others.

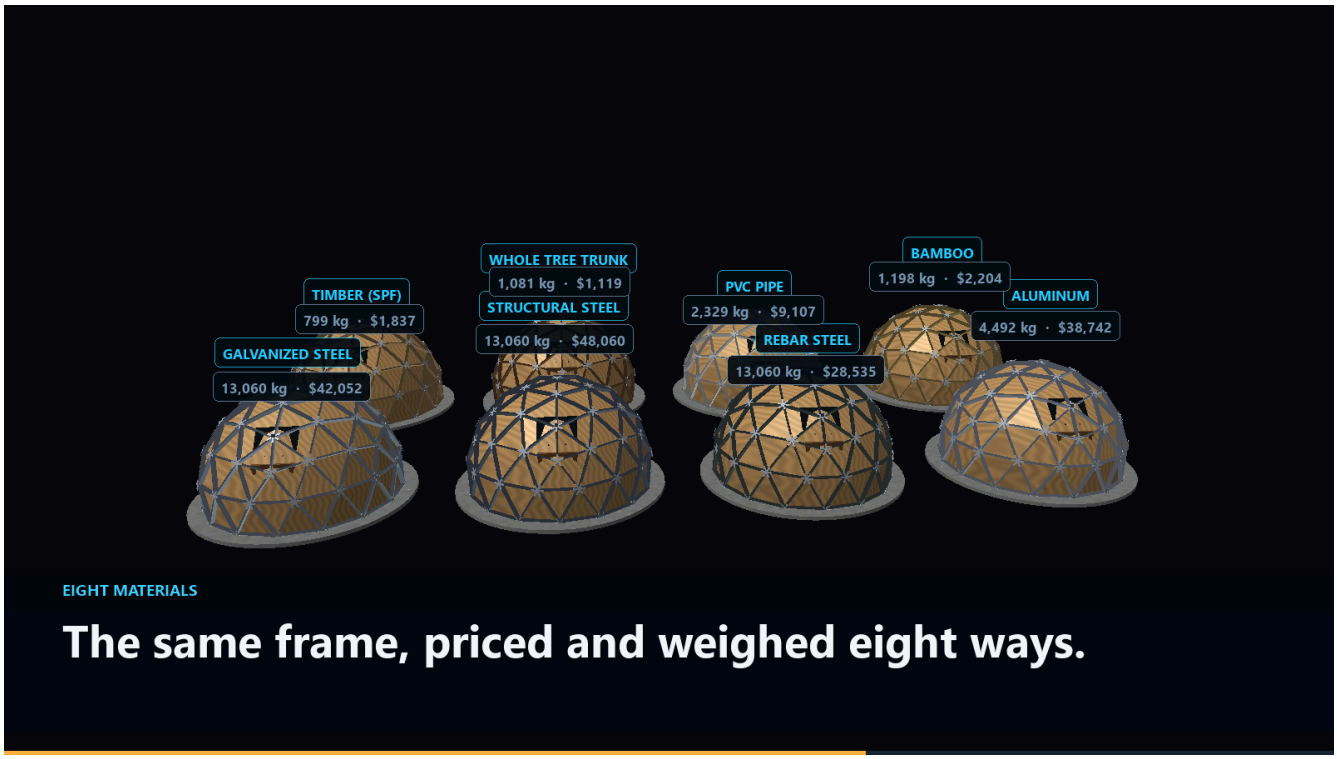
Not a paragraph saying alternatives exist — the alternatives, named, with what each one costs and weighs **on this building**, and a mark against the one the reference build uses.

How to read these tables

A list of materials with prices per square metre is not a decision. A decision is *"on a 19.42-foot dome with 550 square feet of shell, cedar shakes are this much and this heavy, and asphalt is that"* — and that is arithmetic this project can do, because it costs domes for a living.

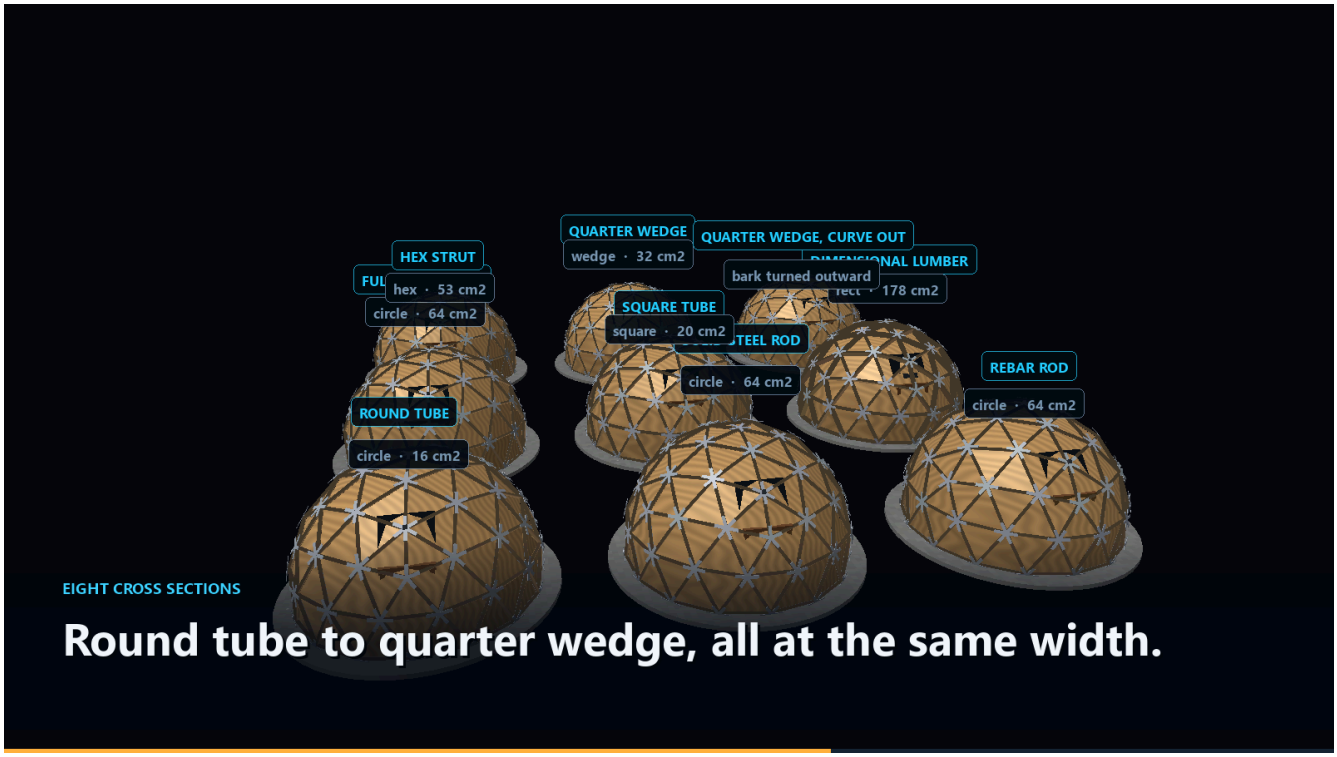
Every option comes from a tool that already models it. Nothing is invented for the book. The full tables are in the reference section at the back; these pages are the shape of the choice.

The frame



The same frame, priced and weighed eight ways.

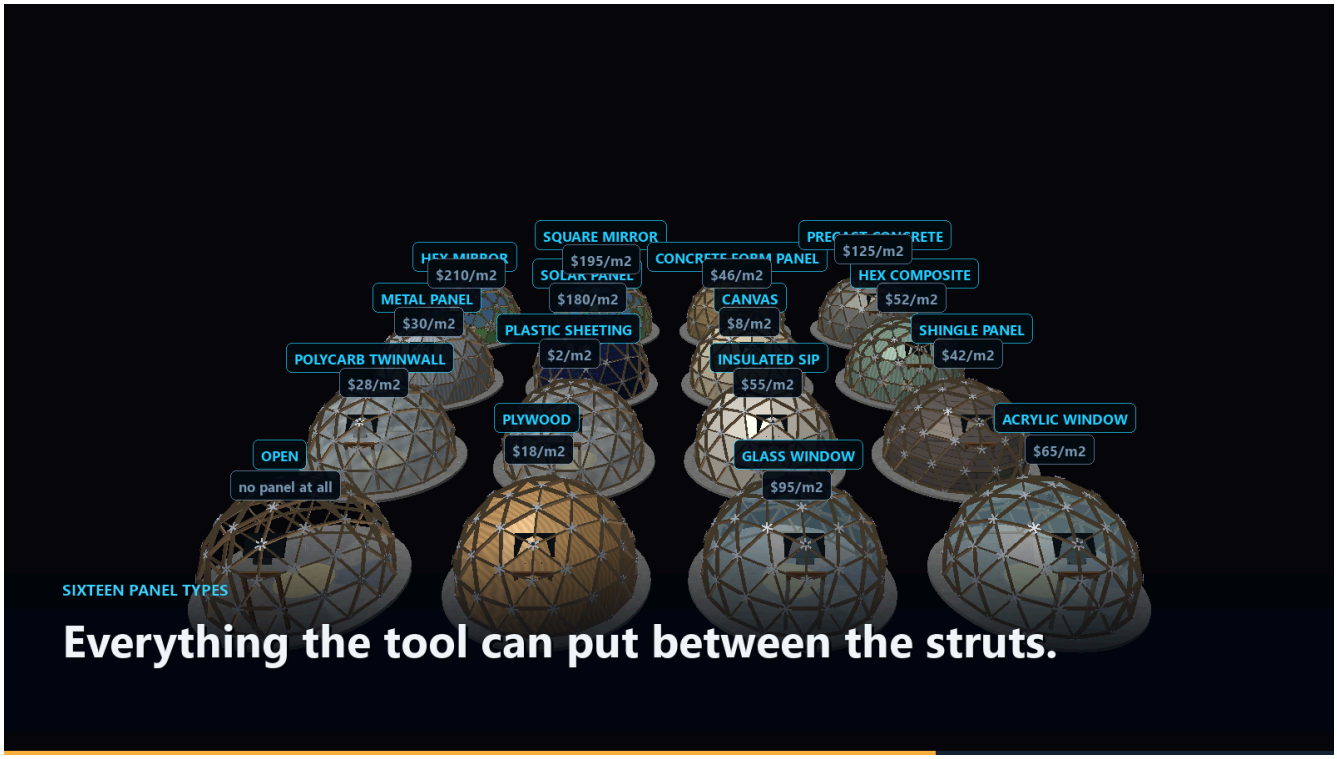
Eight materials, the same 120 members. The reference build is the whole tree trunk, and it is the cheapest by a long way — which is the entire economic argument of this method in one row.



Round tube to quarter wedge, all at the same width.

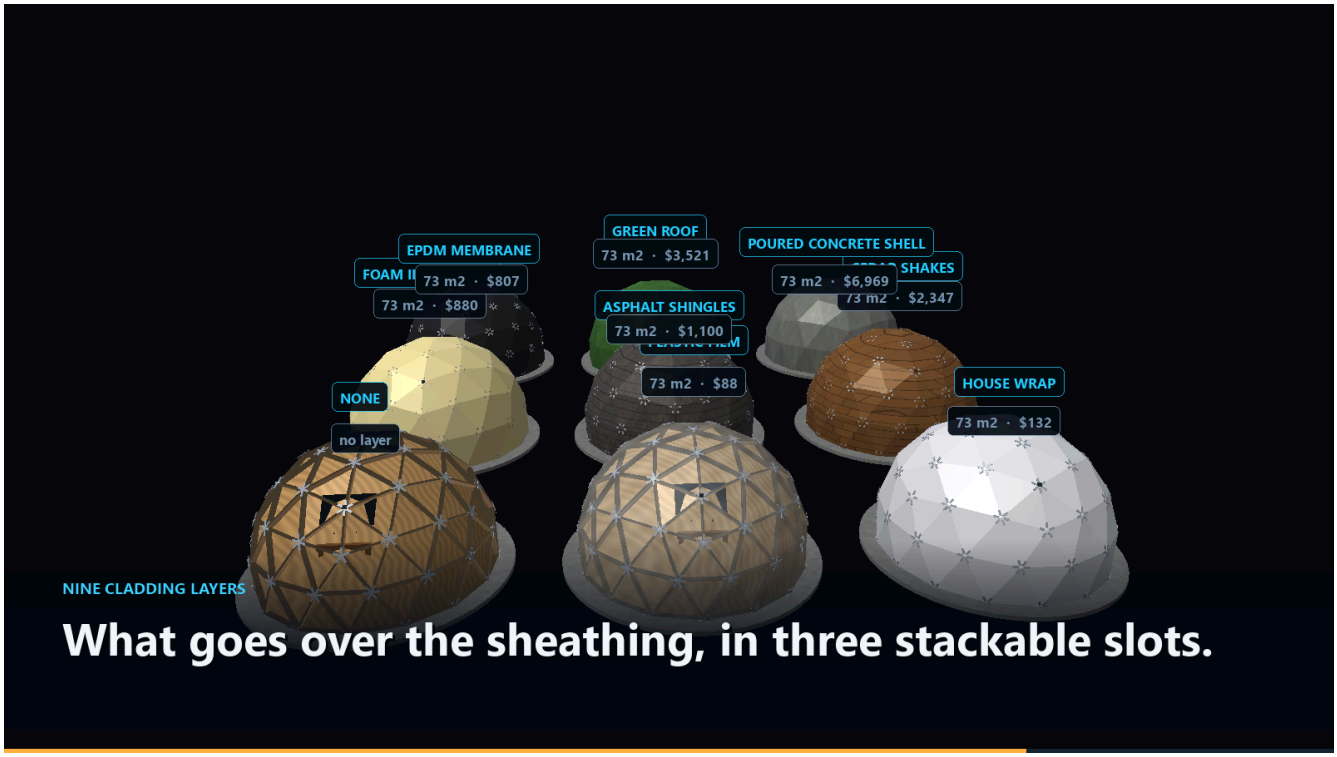
Eight cross sections at the same nominal width. Read the wedge row carefully: the Creator's quarter wedge and this book's member are both split log, sized differently — one is a quarter of a log scaled to the strut's width, the other an eighth of a 12.0-inch one. Same family, different stick.

The envelope



Everything the tool can put between the struts.

Sixteen panel types, from canvas to solar, over 550 square feet. The span from cheapest to dearest is more than a hundred to one, which is why this is the decision that actually sets what your dome costs.



What goes over the sheathing.

Nine claddings. A green roof is twenty-four times the weight of house wrap, and the frame has to carry whichever one you pick.

What is deliberately missing

There is no straw-bale row and no earthbag row.

Not because they are bad — because this project cannot cost one, and a table with a plausible number in it that came from nowhere is worse than a table with a gap.

The same applies to anything else you find missing. If it is not here, it is because nothing in this repository models it, and the honest thing is to say so rather than to estimate.

The four you cannot change later

Everything above can be decided after the dome is standing. Four things cannot, and they are worth a page of their own in the back:

- which way the wedge's point faces
- how many sectors the log splits into
- how big the tree was
- which way the triangle pinwheels

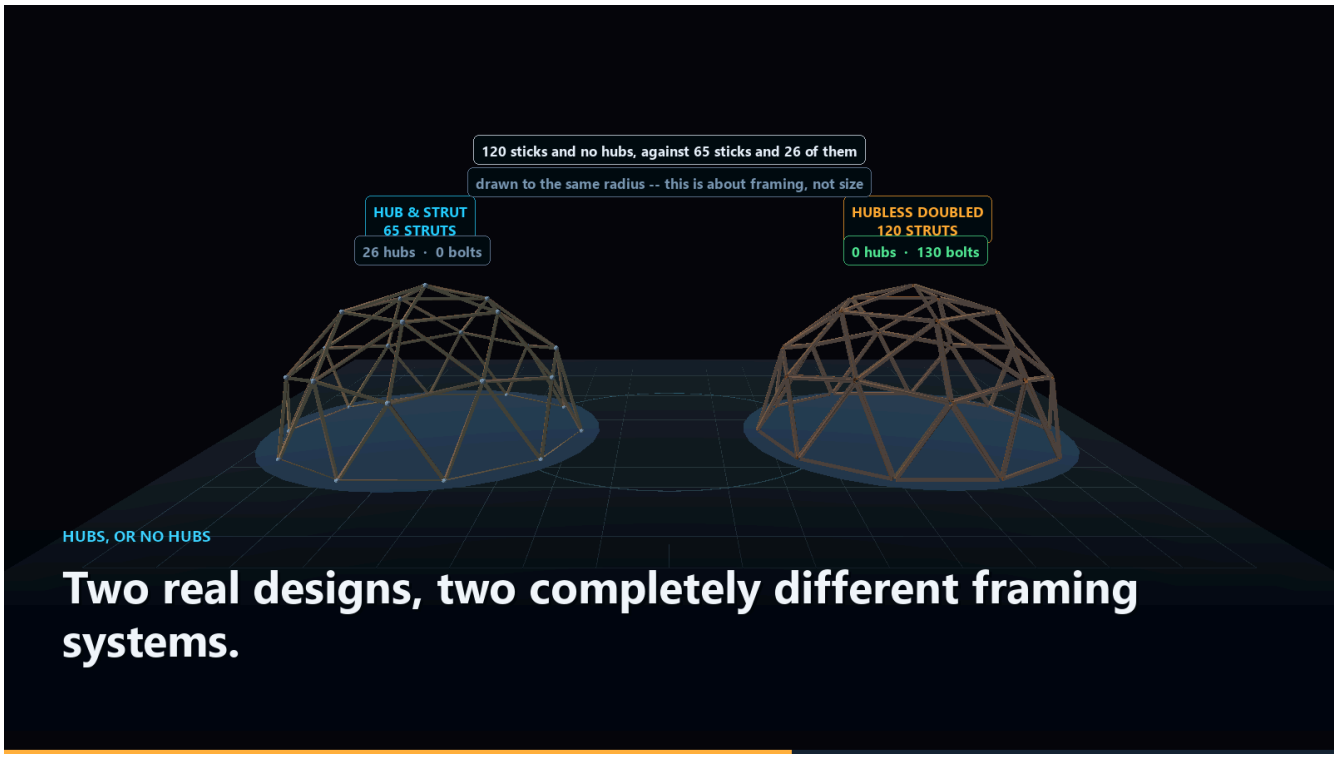
Each of those changes the frame itself, and each has a figure in this book showing what it does to the seam. Chapter 15 is where they are argued; the tables at the back are where they are priced.

CHAPTER 50 · EXPLAIN

Hubs, or No Hubs

The trade this whole method is an answer to, counted across twelve designs rather than argued

This is the decision every dome builder makes, usually without noticing they have made it, and this entire method is one answer to it.



Hubs, or no hubs.

The two ways

Hubbed. Every edge of the geodesic mesh is one strut. Where five or six edges meet, their ends all arrive at the same point, and a connector holds them there.

- Fewest members: 65 struts rather than 120.
- Least timber, by a long way.
- But the hub is now the whole engineering problem. It has to hold five or six sticks at five or six different angles, in three dimensions, and it has to do it 26 times.

Hubless. Every panel is a closed triangle of its own three members, and panels meet panels rather than sticks meeting sticks.

- Nearly twice the timber — 120 members for 65 edges.
- But there is no connector to design, buy, machine or get wrong.
- And a panel is a *thing*: it can be built flat, checked flat, sheathed and lifted as a finished unit.

Why the hub is worse than it looks

The strut count makes hubbed framing look obviously cheaper, and on a materials list it is.

The hub's cost is not on the materials list. It is:

- **Design.** Angles that are specific to the frequency, the vertex type and the strut section.
- **Fabrication.** Either bought, which is expensive per unit, or made, which needs a machine most people do not have.
- **Tolerance.** The one place in a triangulated frame where error *does* accumulate — five sticks whose ends all want the same cubic inch, and if they are each a little long the hub cannot close.
- **Time at height.** A hubbed dome goes up stick by stick, up a ladder, with somebody holding things.

Chapter 1 argued that a dome forgives error. That is true of the shell and untrue of the hub, and the hub is the place a hubbed dome's precision requirement lives.

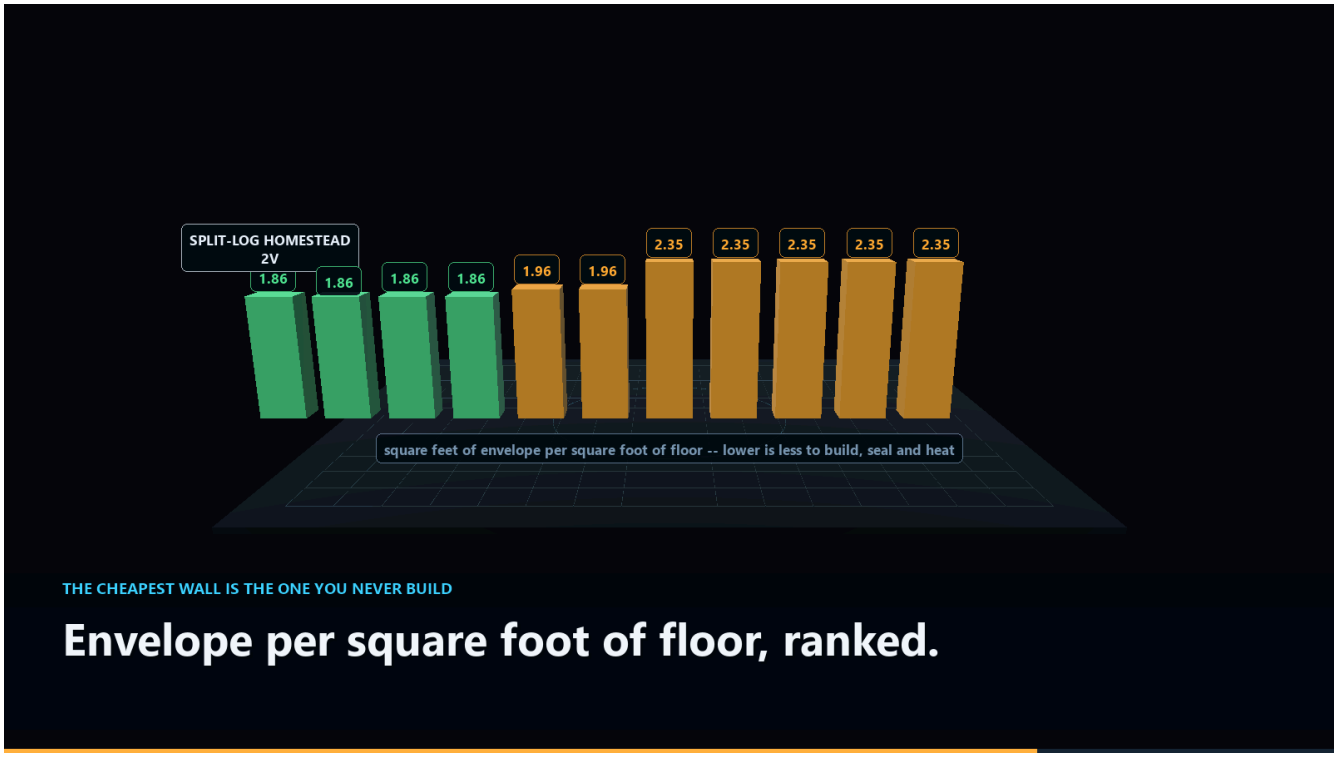
What the duplicated timber buys

Everything in Chapter 24.

The extra members are what make a panel a closed object. A closed object can be made on a bench at waist height, in a shed, in February, by one person, and checked before it goes anywhere. Then the site work is lifting 40 finished things into place instead of assembling 65 sticks in the air.

Timber is the cheapest thing in this method — Chapter 51 prices it against seven alternatives and it wins by a factor of several. Buying more of the cheapest input to remove the most expensive process is a good trade, and it is more or less the only trade this book makes.

The cheapest wall is the one you never build



Envelope per floor, measured.

One more argument from the same family, and it is about the shape rather than the joint.

A dome encloses a given floor area with less envelope than a box does. Less envelope is less material, less cladding, less insulation, less surface losing heat, and less area for water to find a way through.

That saving is permanent and it compounds with everything else: a smaller envelope means the skin of Chapter 2 — which is the part this book says matters most and costs most — is smaller too.

It is measured rather than asserted, across the whole catalogue, in the figure above.

When to take the hub anyway

Be fair to the other side.

If you are working in steel or aluminium, the hub is a bracket and a bolt and none of the above applies — the machine that makes it is a drill press. If you are buying a kit, somebody else has solved the hub and you are paying for it once. If the dome is very large, the timber duplication stops being cheap.

The hubless argument is strongest exactly where this book is: **timber, split rather than sawn, one person, no shop.** Outside that, it is a real trade with two defensible sides.

CHAPTER 51 · REFERENCE

Twelve Buildings, One Tool

Every finish, floor division and fit-out the Dome Creator will put on a shell, priced against each other

The previous chapter priced the frame and the envelope. This one is everything else the Dome Creator will put on a shell — and the point of it is that all twelve of these are the same program, so the comparison between them is real rather than rhetorical.

Every design in the catalogue, priced.

What the catalogue contains

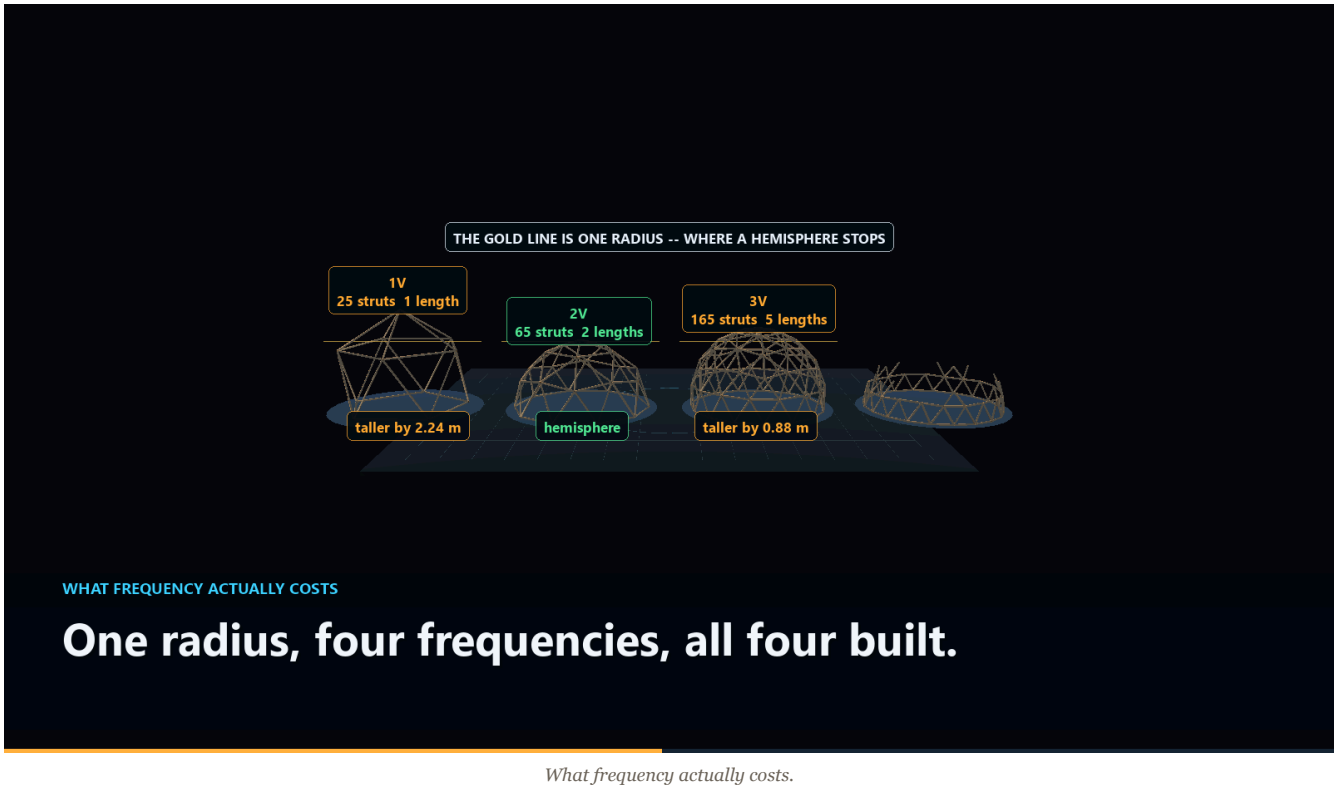
Twelve finished buildings, from a timber workshop to a concrete monocoque, a glass studio loft, a rebar garden dome, a treehouse canopy and a steel arc hangar.

They are not illustrations of an idea. Each one is a configuration of the same four dials — frequency, frame material, strut section, panel type — plus cladding, foundation and finish, and each one comes with its own bill of materials.

Set the dials differently and you get a thirteenth.

Frequency, and where the dome gets cut.

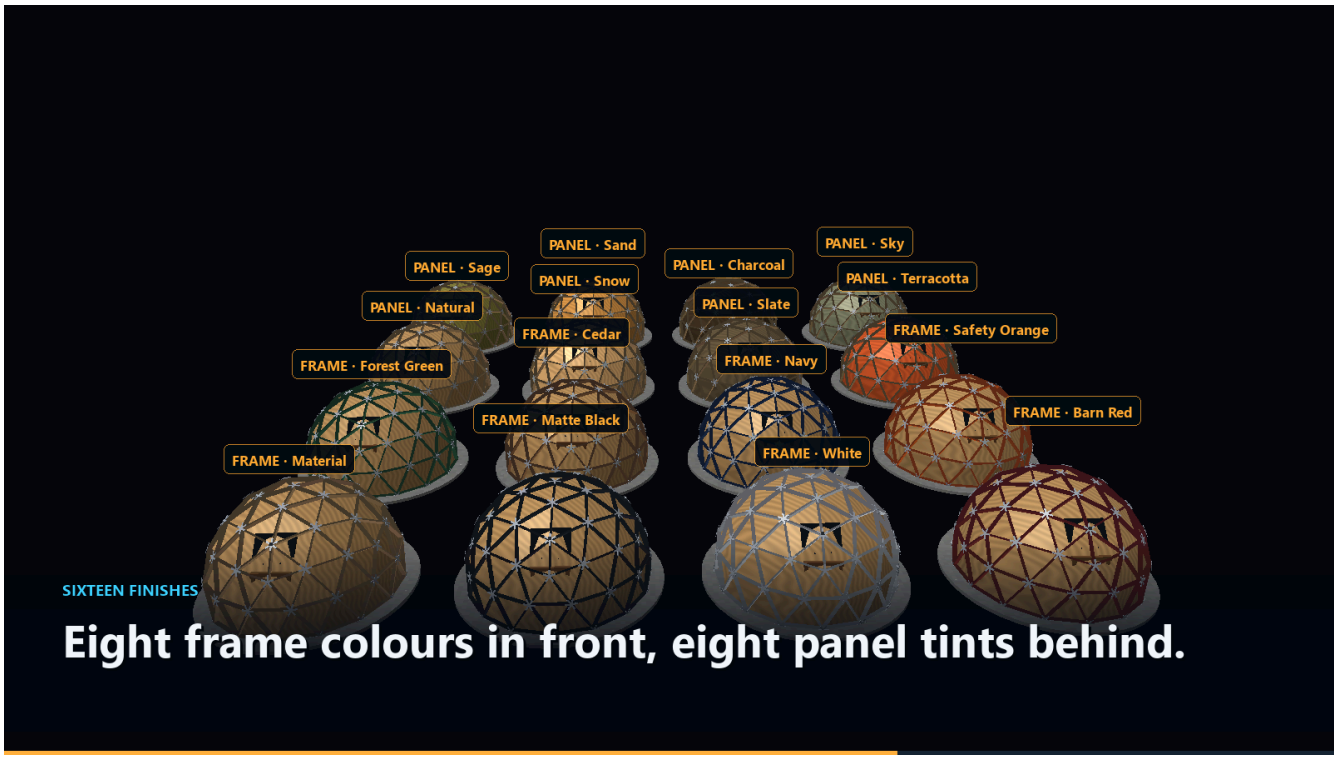
The catalogue is a set of saved positions, not a product range.



What frequency actually costs.

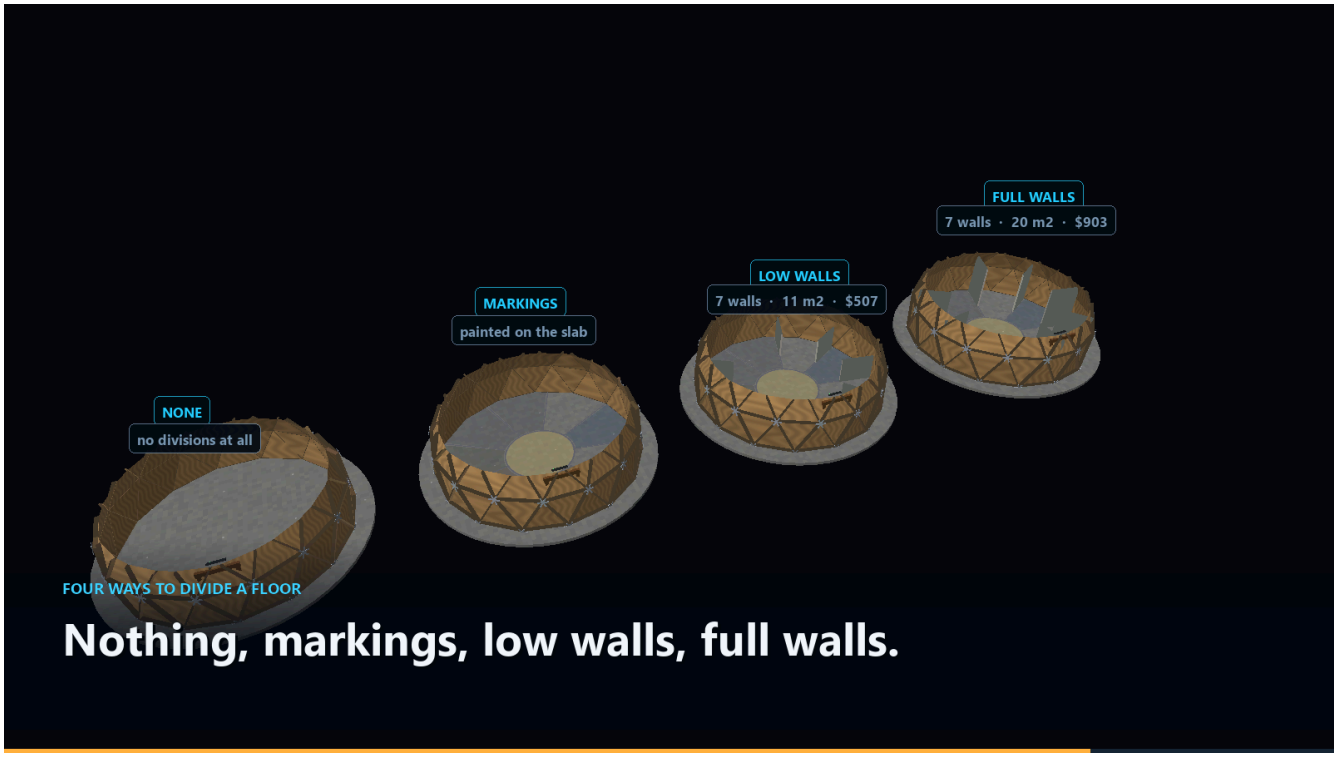
The decisions nobody prices

Three of them, and they change how a dome *lives* rather than how it stands.



Sixteen finishes.

Finish. Sixteen of them over the same shell. Cheap to change, and the only decision in this entire book that is purely aesthetic — which is worth saying, because almost every other choice here has a number attached and this one honestly does not.

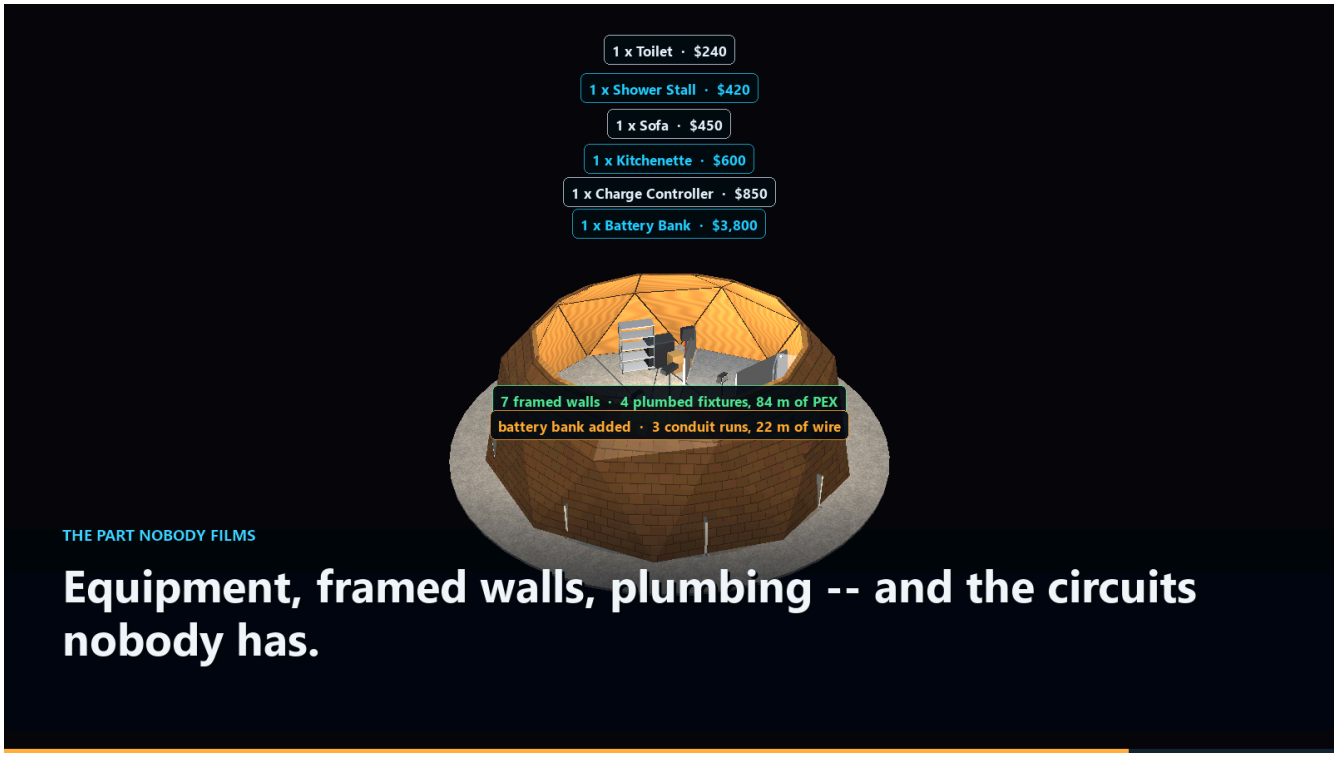


Four ways to divide a round floor.

Floor division. This is the question everybody asks second, right after “does it leak”, and it is the one round buildings are genuinely worse at.

A round floor does not take rectangular rooms. The four divisions the tool offers — radial, a central core, a chord wall, and open — are four answers to that, and the honest summary is that a dome wants **one big room and a periphery**, and fighting that produces awkward rooms with curved outside walls.

The periphery is also where headroom runs out. Chapter 11’s 277 square feet of floor is not 277 square feet of usable room, and a knee wall is the cheapest fix there is.



The part nobody films.

Fit-out. Benches, storage, services, the things that make it a workshop rather than a shell. It is most of the cost of a finished building and none of the glamour, which is why it is in the tool and not in the photographs.

How to use this chapter

Do not read it. Open the tool.

The tables at the back have every figure in them, but a table of sixteen panel types is a worse instrument than sixteen domes side by side with the prices on them. Chapter 57 says how to start it, and the one thing worth doing first is building the dome you were going to build before you read this book, pricing it, and then building this one.

That comparison is yours rather than mine, and it is the only one that will change your mind.

PART 12

The Stem Cell

“Buy the plumbing once. The shell is the thing that grows, and the core is the thing that moves.”

The product line: one undifferentiated body, the utility core that is thirty per cent of it and outlives it, and the fifteen buildings it becomes.

CHAPTER 52 · EXPLAIN

Why We Call It a Stem Cell

One body, undifferentiated, that becomes fifteen different buildings depending on what you put in it

A stem cell is a cell that has not decided what to be yet.

It has the full machinery — a nucleus, a membrane, everything a working cell needs — and no specialisation. Put it in one place and it becomes liver. Put it somewhere else and it becomes bone. The body is the same body; the environment does the differentiating.

That is the product line in this book, and the word is not decoration.

One body, many things it can become.

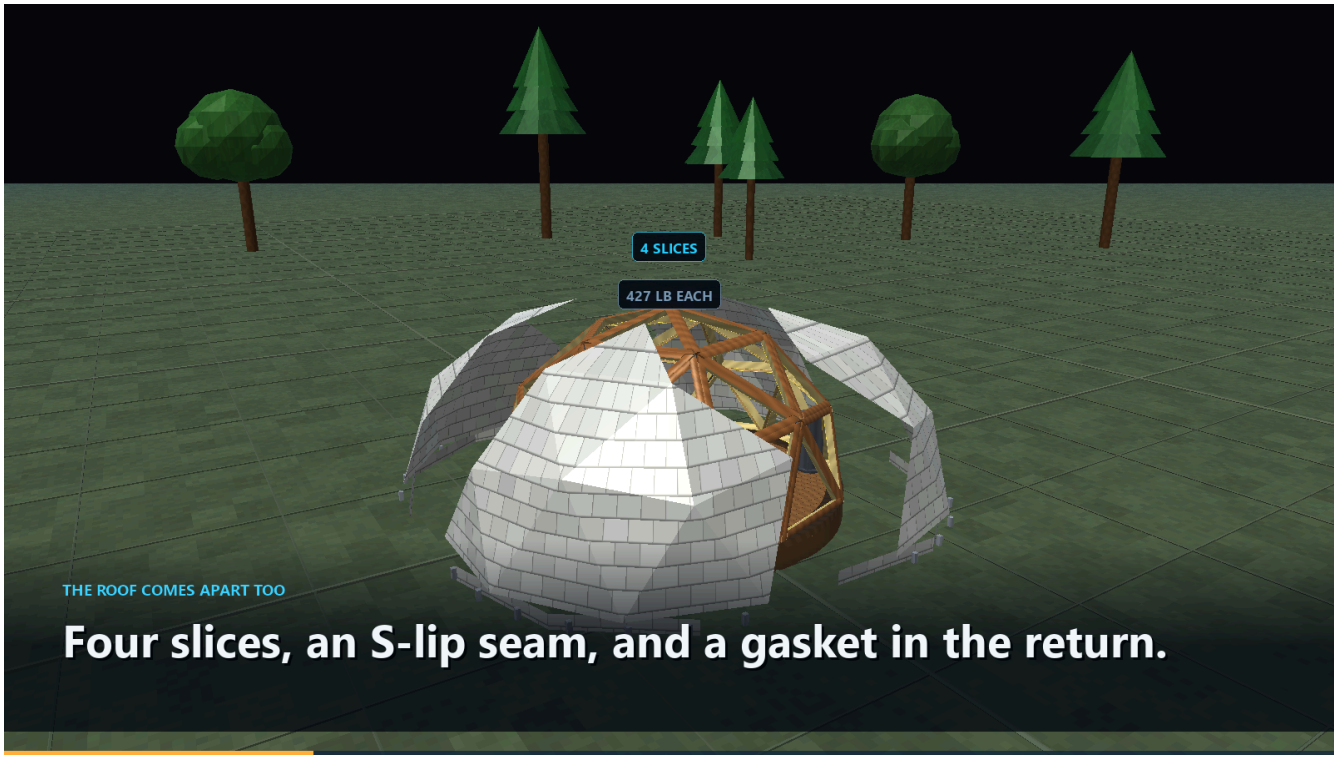
One body, undifferentiated

The standard article is a dome: 120 members, 40 panels, a shell, a floor, and a utility column running from the floor port to the apex. It lists at $12,036.

It is not a house. It is not a workshop. It is not a studio, a gym, a sauna or a guest room. It is the **body** of all of those, and what it becomes depends on what you put in it and what you plug into the column.

There are 15 of them in the catalogue — Chapter 54 — and every one is this same body. The frame line of the quote does not change between them. Not by a dollar.

The socket at the top



The roof comes apart too.

What makes differentiation possible is that there is exactly **one** place where things connect.

The utility column is the only object in the building that touches the ground, the wall and the sky. It carries 5 services in 14 documented parts, and it makes exactly one penetration in the weather surface — the apex sleeve, with a gasketed seal cap over it that is meant to stay shut for years and come off in ten minutes.

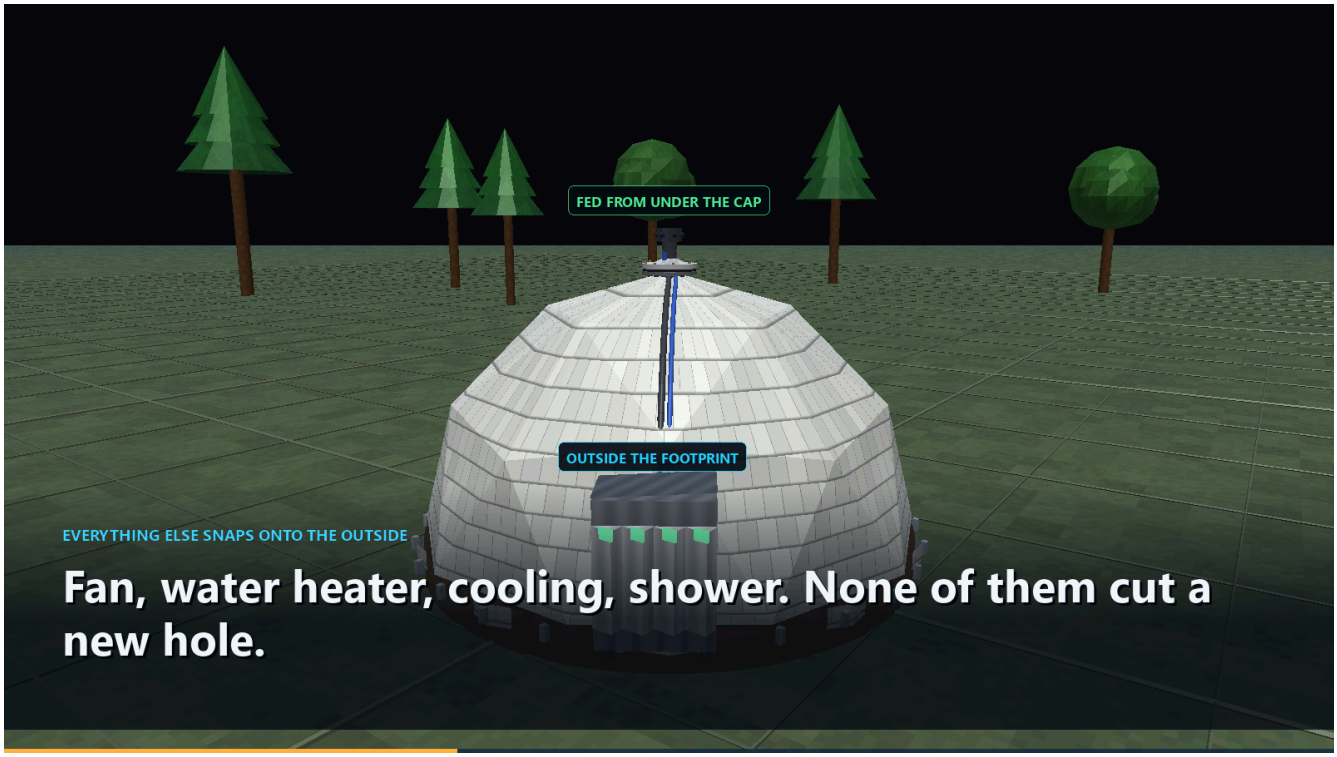
Everything a fit-out needs comes off that column. Power at a sub-panel inside it. Water at a four-port manifold. Drain down a stack inside the same chase. Air, eventually, through the seam channel of Chapter 20.

Nothing is plumbed into a wall. The wall is a shell that resolves loads and keeps weather out, and it has no services in it to disturb. That is what makes a fit-out a thing you do rather than a thing you commission.

And a skin that snaps on

The outside works the same way. Services that want to be outside the footprint — a water heater, a cooling unit, a shower — hang off the shell on threaded inserts rather than cutting a new hole. The design calls them polyps and the name is the same joke as the stem cell: an organism growing an appendage rather than a builder cutting an opening.

Every one of them is reversible. Take it off and there is a panel with four inserts in it.



Everything else snaps onto the outside.

Why this is a different kind of product

A normal building is specialised on the day it is designed. A workshop has a workshop's floor, a workshop's power, a workshop's door. Turning it into a guest room is a renovation.

This one is specialised on the day it is **fitted**, and un-specialised by taking the fit-out out again. The most expensive parts — the frame, the shell, the column — never learn what the building is for.

That is the argument, and the next chapter is the economic consequence of it, which is the part that actually decides whether any of this is worth doing.

CHAPTER 53 · EXPLAIN

Sink the Money Into the Part That Moves

The utility core is thirty per cent of the dome and the only part that transfers -- so it is the part worth overbuilding

Here is the theory, in one sentence.

Put the money in the hardware, because the hardware transfers and the shell does not.

What buying it once is worth.

The two halves of a dome

Every building in this book is two things wearing one roof.

The shell — 120 members, 40 panels, the cap, the floor. It is shaped by *this* dome: its members are cut to this diameter, its panels are this geometry, its cap is this size. Move to a different dome and none of it comes with you, because none of it fits.

The core — the utility column and everything in it. A chase, a sub-panel, a manifold, a stack, a seal cap. 14 parts carrying 5 services. It is shaped by *what a small building needs*,

One hardware set, any size of house.

which is the same in every small building.

The shell is specific. The core is generic. And the core is **30 per cent of what the dome costs to build** — $1,890 of it.

What generic means for money

A shell is consumed by the building it is part of. When that building's life ends, so does the shell's.

A core is not. It was never shaped by this dome, so it fits the next one. Its declared service life is 25 years — not because a pipe wears out in 25 years but because at some point it is outgrown rather than worn out — and moving it to the next dome costs $260.

Two hundred and sixty dollars to move 30 per cent of a building.

What does not move when the house does.

That is the whole theory. The expensive, fiddly, skilled, permit-adjacent third of the project gets bought **once** and then amortised across every dome you ever put it in.

Why this changes what you should overbuild

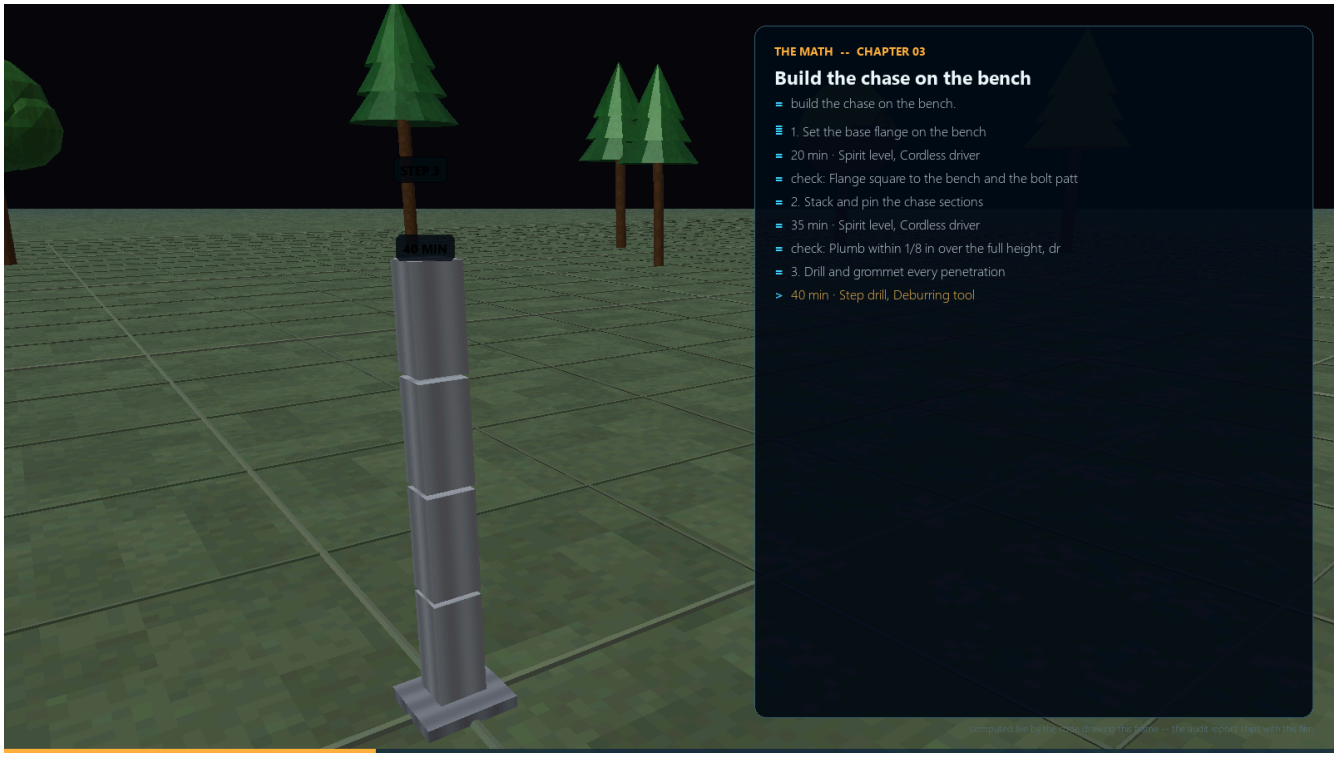
Ordinary building advice says spend where it shows and economise where it does not.

This says something different: **spend where it transfers.**

Every dollar in the shell is spent on this dome. Every dollar in the core is spent on this dome *and the next one and the one after*. So the core is the place where overbuilding is not overbuilding — it is prepayment.

Concretely, this is why the design specifies:

- a sub-panel with a **spare breaker**, so the first snap-in module does not need a panel change;
- a four-port manifold with **capped stubs**, in a dome that ships with no plumbing at all, because the alternative is opening the chase later;
- a feeder that **unplugs** — a cord and a recessed inlet rather than hard wiring, which is what makes the dome moveable without an electrician;
- a floor-port tie-in that the **host** makes once and nobody disturbs when a dome is swapped.



Build the chase on the bench.

Every one of those is a small extra cost now against a large avoided cost later, and each one is only worth it because the thing carrying it survives the building it was bought for.

Close it, prove the cap, stand it up.

The same argument, one level up

This only works as a system.

The core is to a dome what the pad of Chapter 37 is to a site and what the jig of Chapter 24 is to a panel.

In all three cases, something is built once, carefully, and then used many times without being rebuilt:

```
the jig     built once, makes 40 panels
the core    bought once, serves several domes
the pad     built once, hosts a succession of buildings
```

And in all three cases the thing being amortised is not material — it is **precision and setup**. The jig holds the angles so you do not have to. The core holds the services so an electrician does not have to come back. The pad holds the level and the connection so a foundation does not have to be poured again.

That is what “this only works as a system” means. Any one of them alone is merely a nice idea. Together they move almost all of the difficulty out of the building and into three objects that are built once.

What a site can share, so the dome does not have to.

The honest objections

It raises the entry price. A dome with a proper core costs more than a dome with a light socket and a hose through the wall. The transfer argument only pays if there is a second dome, and for somebody who will only ever build one, this chapter is wrong and they should spend the money on the skin.

Transfer is not free. $260 is a declared assumption, not an invoice. It covers disconnection, handling and reconnection on a pad that already has the port. It does not cover a drive across a country.

The core moves to the next dome.

Generic is a claim, not a fact. The core is generic across the 15 structures in this catalogue, which were all designed around it. Whether it is generic across somebody else's building is unproven, and the honest answer is that it is generic across this system and that is all anybody has tested.

The sentence to keep

If you are building one dome: put your money in the skin and the floor, as Chapter 2 says, and keep the core simple.

If you are building more than one: **put it in the core.** It is the only third of the project you will not have to buy again.

CHAPTER 54 · REFERENCE

Fifteen Buildings From One Body

The catalogue a homestead wants second, priced from the same frame and the same core

The catalogue a homestead wants second, and third, and fifth.

The catalogue, priced.

The list

Every one of these is the same frame, the same shell and the same utility core. What differs is the fit-out — and the fit-out is the only line in the quote that moves.

Stem cell	$ 12,036
Sauna seed	$ 15,726
Jacuzzi seed	$ 17,383
Treehouse seed	$ 17,907
Bunker seed	$ 18,009
Cold storage seed	$ 18,068
Nursery seed	$ 18,588
Gym seed	$ 20,022
Studio seed	$ 20,132
Guest house seed	$ 21,914
Garage seed	$ 22,621
Home seed	$ 22,844
Advertiser seed	$ 23,181

Workshop seed	$	24,279
Food seed	$	25,511

15 structures. The bare body is the Stem cell at $12,036; the most specialised is the Food seed at $25,511.

That spread — about two to one across the whole range — is the differentiation cost. Everything below it is the body, and the body is the same body.

What the spread tells you

Water is what makes a building expensive. Sort that list and the ones with plumbing in them are at the top. A sauna, a jacuzzi and a food seed are all wet buildings, and the ones that are merely rooms are cheap.

Power is not. The core already carries a sub-panel and a spare breaker, so a building that wants more sockets wants almost nothing.

Structure is never the variable. No building on that list changes the frame. Chapter 53's argument, made visible: the expensive parts never learn what the building is for.

The ladder

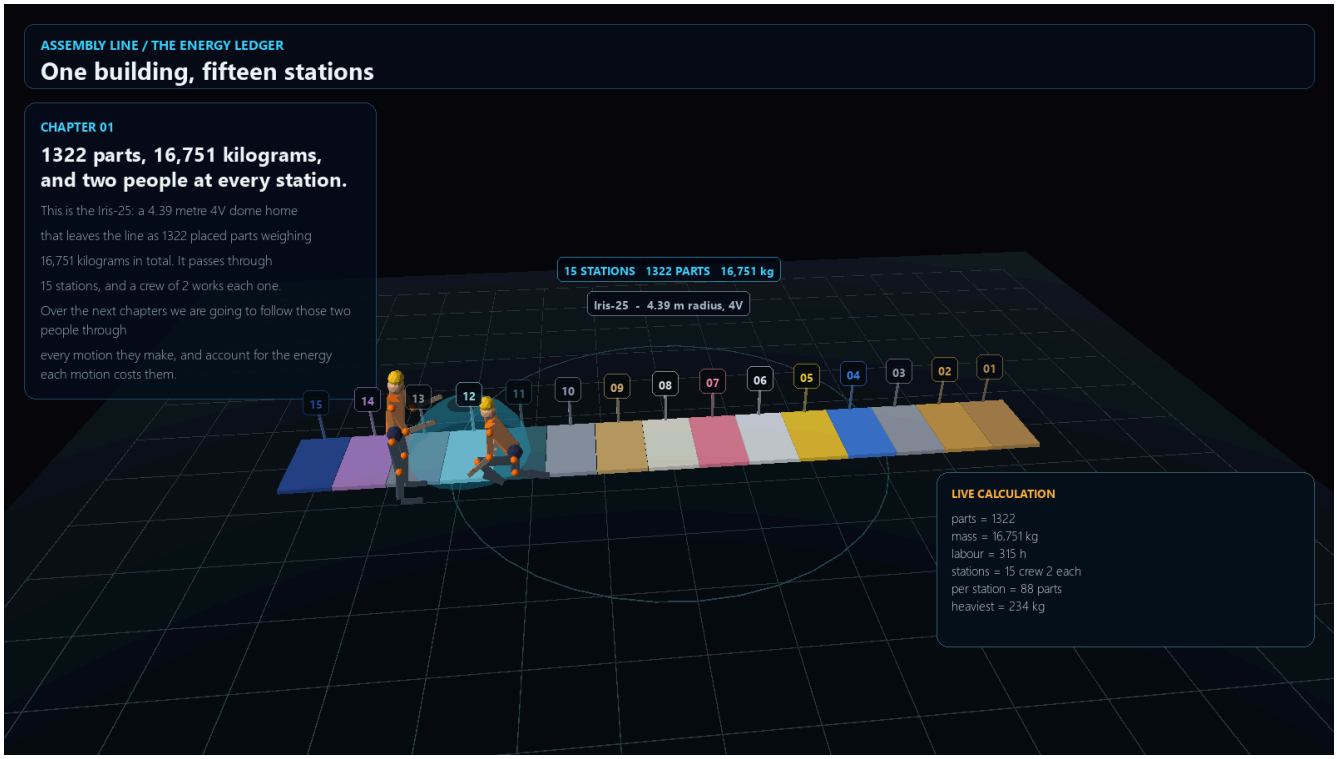
How far down the ladder goes.

Every one of these can be bought in stages, and the ladder matters more than the list does.

The rungs are, roughly: the bare body; the body with a floor; with a cap and one quilted layer; with the fit-out; with the polyps. You can stop at any rung and live there, and each one is a real building rather than a half-built one.

What makes that possible is Chapter 3 — the four features that keep the skin and the floor swappable. Without them the ladder is just a half-finished building with a tarpaulin on it.

One building, fifteen stations



One building, fifteen stations.

There is a manufacturing view of this, and it is worth a page because it is where the argument stops being about one person with a chainsaw.

The nine processes of Chapter 22 do not change between these 15 structures. Neither does the jig, the cut list, or the core. What changes is which station a particular dome stops at on its way through.

That is a **production line**, and the thing that makes it one is not machinery — it is that the parts list and the verb list are both short and neither grows with the catalogue.

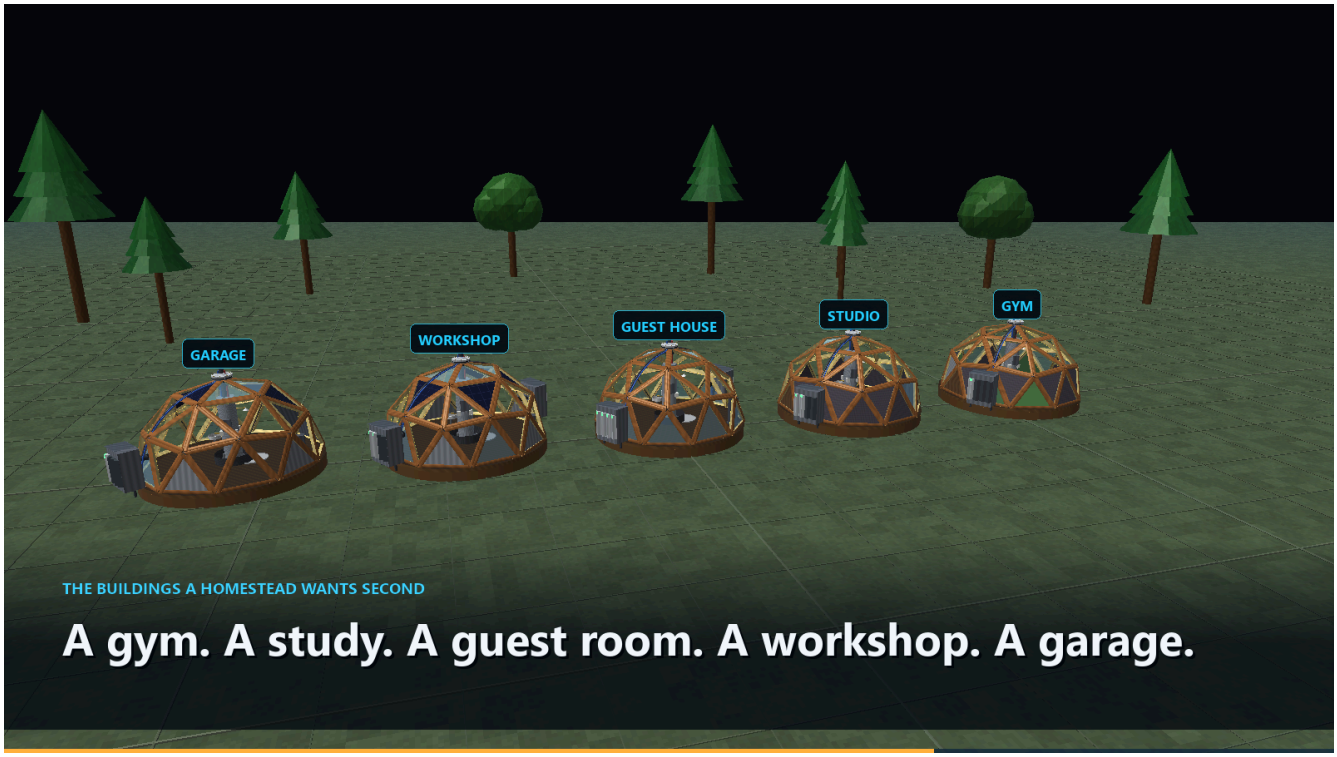
A workshop that builds one dome a month is building the same nine operations whether this month's is a sauna or a garage. That is the difference between a product line and a range of products.

What this catalogue is not

It is not a range anybody has built. The Stem cell is the one this project has solved end to end; the rest are configurations of the same model with their fit-outs costed, and they are **priced rather than proven**.

The difference matters. A food seed's price is real arithmetic over real line items. Whether a food seed is a good building — whether the ventilation is right, whether the floor drains where it should — is not something a cost model can tell you, and this book will not pretend otherwise.

Treat the list as what it is



The buildings a homestead wants second.

: evidence that the body generalises, and a set of starting points rather than a set of finished designs.

PART 13

Variations

“The same frame can wear more than one roof, stand on more than one ground, and hang from nothing at all.”

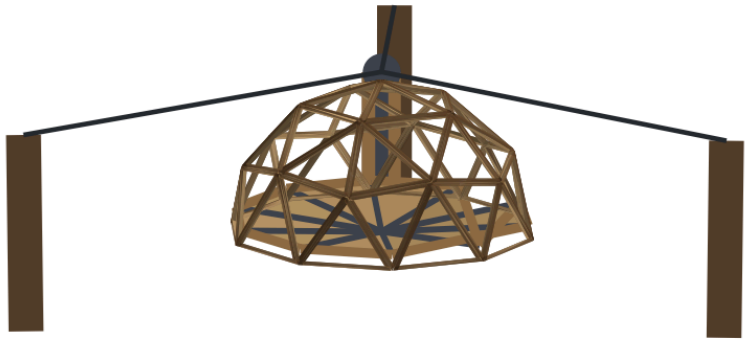
The one variation that is a design possibility rather than a tested build, priced and explicitly not rated.

Hang It Between Two Trees

The most speculative corner of the method: a mast, three cables, and no ground at all -- priced, and explicitly not rated

This is the most speculative corner of the method, and it exists because the floor of Chapter 36 clamps to a mast rather than being framed into the base ring.

Once the floor hangs off something in the middle, the ground stops being structural. And once the ground stops being structural, you can take it away.



The same mast and floor, hung on three cables to saddles on two trees.

What changes

Everything the earlier chapters assumed about the ground quietly reverses.

The base ring stops being what holds the building up; the mast is. The floor stops being something the dome stands on and becomes something the dome **carries** — clamped to the mast, hanging under the shell, level because the mast is level.

The utility column keeps its job. Services come up a flexible drop instead of a rigid rise, which is a fitting rather than a redesign.

The parts

A steel core through the utility column, timber-clad, floor to apex, with a lifting ring under the seal cap. Three galvanised cables from that ring to three saddles on two trees. A brake winch and a block.

The mast through the column, and the floor that clamps to it.

The rig is $1,490. With the mast and the dome's own floor the whole upgrade is $4,011, against a frame weighing 2,478 pounds in green pine.

Saddles, not holes. Nothing is drilled into the trees. A saddle spreads the load around the trunk and comes off without leaving a wound, which matters because a tree with a bolt in it is a tree with a problem in ten years.

Why you would

Flood, mostly. Getting a floor above water is the whole reason in some places and no amount of foundation solves it.

After that: slope without earthworks, airflow underneath that keeps timber dry, and the space below, which becomes storage or shade.

Hang it between two trees.

The sentence that has to go with it

That is a design possibility, not an engineered structure.

The loads on the trees, the cables, the saddles and the mast need an engineer before anybody stands under it. The model prices the parts; it does not rate them, and the difference between those two things is the whole of somebody's safety.

Trees are also not static. They move in wind, they grow, they shed limbs, and two of them move independently of each other. A rig that is correct on a still day in June is a rig that has not yet been tested.

Nothing in this chapter has been built. It is here because the floor design makes it possible and because a book that only prints the parts of a method that are finished is not documenting a method, it is advertising one.

PART 14

The Tools

“Every number in this book came out of a program about a second before it reached the page. Here are the programs.”

What ships with this book: the solver, the worlds, the cost model and the calculators, with what each is for and how to drive it.

The Software That Wrote This Book

Eight three-dimensional worlds, a cost model and a solver -- what each one is for, what it can do, and how to open it

Not one figure in this book was typed into a sentence.

Every dimension, count, angle, percentage and price is a token in the manuscript that gets filled in, at the moment the book is made, by the same software that draws the three-dimensional models. When this book says the dome is 19.42 feet across, that number came out of a solver about a second before it reached the page.

That software comes with the book. 11 programs — 5 worlds you can fly around in, 4 that are arithmetic with no picture, and the desks where the book and the films are made.

Why a builder would open any of it

The short answer: **because your tree is not my tree.**

Every number in these pages is for one specific dome, cut from one specific size of log, at one split count, with one orientation. Change any of those and every downstream figure changes — the chords, the cuts, the seam key, the timber, the price. The tables at the back are a snapshot of one solve.

The tools are how you take your own.

Three kinds of thing

Worlds are three-dimensional environments with a window, a camera and keys. This is where you go to *look* at something — to turn one decision and watch what it does to the joint.

Models are arithmetic with no picture. They answer a question and print a table: what does this cost, how long is that stick, what does the platform take off to.

Desks are where you make something: this book, the films, the calculators.

What each one is for

```
WORLDS -- things you fly around in

  The raw-wedge solver
      You want to see the dome this book is about, and turn
      one decision at a time to watch what it does to the
      joint.
      start it:
        py -3.12 geodesic_raw_wedge_dome_dihedral.py
      - solves the whole 2V frame from one number: the long
        member
      - four wedge orientations, and the seam changes under
        each
      - two seam keys, three things to put in the channel
      - explodes the panels apart without moving the geometry,
        so you can see into a seam
      - switches off the wireframe, the node markers, the
        debug lines and the sacrificial head stock -- which is
        what turns a screenshot into an illustration
```

```
    - exports a full-size panel drawing as SVG, to print and
      lay a stick on
    - carries the cost console, so the price is over the
      geometry rather than in another program
    keys: W A S D walk   mouse look   E explode the panels
          apart   I the mathematical wireframe on and off
          H the head-end stock on and off   J the
          fabrication jig   C the cost console

The fabrication jig
    You are about to build the fixture and you want to see
    it assembled, one step at a time, before you cut the
    board.
    start it:
      py -3.12 geodesic_raw_wedge_dome_dihedral.py   (then J)
    - twelve steps from a bare bench to a finished panel
    - each step says what it is enforcing, not just what it
      is
    - the red and green fences that make a rotated stick not
      fit
    - the two cut planes: the flush-cut fence and the
      receiving face
    - the sacrificial tail drawn separately from the
      finished member
    - camera stations at each corner, because the corner is
      where the method either makes sense or does not
    keys: J show the jig   [ ] step through the twelve   V
          walk the camera round the corners   P which panel
          the jig is set for

The Dome Creator
    You want a dome that is not this one: a different
    frequency, a different material, glass instead of ply, a
    different foundation -- and the price for each.
    start it:
      py -3.12 dome_creator.py
    - frequency 1V to 4V, and the truncation
    - eight frame materials, from bamboo to structural steel
    - eight strut cross sections including the quarter wedge
    - sixteen panel types: ply, glass, acrylic, twinwall,
      SIP, shingle, metal, solar, canvas, mirror, precast
    - nine cladding layers over the panels
    - seven foundations
    - a bill of materials for whatever you just built
    keys: F frequency   M frame material   P cycle the panel
          under the cursor   L cladding layers   G
          foundation   B print the bill of materials

The seed world
    You want to see the product version: the utility column,
    the shower cap, the quilted layers and the mast, on the
    dome this book prices.
    start it:
      py -3.12 -m two_v_demo.app --lesson seed_pitch
    - the utility column and the gasketed cap at the apex
    - the shower cap, and the hats that stack under it
```

```
      - the mast through the column and the floor clamped to
        it
      - the floating rig: three cables to saddles on two trees
      - the composite member, drawn at an exaggerated section
        and labelled as exaggerated
      keys: 1-9 the chapters of the campaign film   mouse
            orbit   wheel zoom

  The pad and the park
      You are thinking about the ground: what a pad is, what
      it costs, and what a network of them looks like.
      start it:
        py -3.12 -m two_v_demo.app --lesson dome_park
      - a pad built one course at a time
      - amber for what the host owns, cyan for what the tenant
        owns
      - several domes on several pads, and the shared services
        between them
      - the rotating pad, which is a design possibility and is
        marked as one
      keys: mouse orbit   1-9 the chapters of the park film

MODELS -- arithmetic, no picture

  The geometry
      You want the numbers: chords, cuts, angles, areas,
      counts -- for your dome, not for this one.
      start it:
        py -3.12 -m wedge_book.numbers
      - two chords and two cut lengths at any diameter
      - the bite the pinwheel takes, which does not scale
      - face areas, seam runs, member volumes, timber totals
      - the whole builder's reference, regenerated

  The cost model
      You want to know what one costs, line by line, with
      every assumption named and changeable.
      start it:
        py -3.12 -m seed_model
      - every line carries its own source constant
      - materials, labour, overhead and warranty kept apart
      - the maker's markup stated as markup on cost, not
        margin on price, because those are different numbers
      - the pad priced separately, because it is the host's
        bill
      - fit-outs, shells and resins swapped without editing
        anything

  The platform take-off
      You are pricing the ground and you want boards and piers
      counted rather than a rate per square foot.
      start it:
        py -3.12 -m pad_deck
      - five constructions: gravel, slab, ring, deck, deck
        with ply
      - counted line by line, not rated
```

```
      - sized to the dome plus a walking margin

  The shell model
      You are deciding what to put over the frame, and you
      want the R-value and the price of each layer.
      start it:
        py -3.12 -m soft_shell
      - one watertight cap, and the quilted layers under it
      - each hat a size up from the one beneath
      - against a laminated hull, priced four ways
      - the concerns it will not hide, printed with the answer

DESKS -- where you make something

  The book studio
      You want to change this book: its outline, its prose,
      its figures, or the dome it is written about.
      start it:
        py -3.12 -m wedge_book.build
      - the outline is code, so reordering is an edit and not
        a rewrite
      - every number is a token resolved at build time
      - an audit that fails the build if a figure in the prose
        is one the code cannot produce
      - the pictures render from the solver and from the films

  The film exporter
      Thirty-seven narrated films are built from the same
      geometry, and you can re-render any of them with your
      own numbers in it.
      start it:
        py -3.12 -m two_v_demo.app --lesson wedge
      - one lesson is one Python module: chapters, narration,
        scenes
      - landscape and phone cuts from one render
      - captions, chapter thumbnails and description copy as a
        set
      - a still from any second of any film
```

The one that matters most

If you open exactly one thing, open the solver, and press the key that explodes the panels apart.

Everything difficult about this method is in the gap between two panels — the key, the angle, why the point faces inward, where the error goes. It is invisible in an assembled dome and obvious the moment you can see into a seam, and no drawing in this book, including the section in Chapter 20, teaches it as quickly as turning it in your hands for a minute.

How the numbers get from there to here

Worth stating once, because it is the unusual part.

The manuscript is Markdown with holes in it. A sentence that needs a figure names the number it wants – `dome.diameter_ft` in double braces – rather than printing one. At build time a resolver fills every one of them from the solver, the geometry and the cost model, and an **unknown token is a hard error** — the book refuses to build rather than printing a blank.

The reason is not showing off. A number typed into a paragraph is correct on the day it is typed and silently wrong from the first time anybody changes anything. Nothing fails. The book just quietly becomes untrue in places nobody remembers.

So the arithmetic in this book is a test suite. It runs, it checks itself against the geometry, and it fails loudly.

It caught four things

This is not theoretical. An audit reads the finished prose and checks every figure in it — every number carrying three or more decimals, and every dollar figure — against what the code can actually produce. It found:

- a set of reference designs that **scaled the pinwheel's bite with the dome**, which is the exact error Chapter 16 spends a page warning about. Every stick of the smallest dome was an inch too long;
- a triangle altitude printed as 52.520 where the solve says 52.516 — in the sentence that tells a reader whether plywood fits;
- four prices from before the shell lost a waterproof layer, still being quoted as current;
- a chapter quoting the cost of a whole upgrade as the cost of two of its three parts, double-counting the third.

Every one of those was written by somebody being careful. That is the point: care is not a method for keeping numbers true across time

What one costs, line by line.

, and a program that re-checks them on every build is.

Change the tree and rebuild

The whole of it — geometry, prices, figures, this book — regenerates from one set of inputs.

Change the trunk diameter to what you actually felled. Change the split count to what your saw and your log will give you. Rebuild. Every chapter that mentions a figure now says your figure, without anybody going looking, and the pictures are redrawn from your solve.

That is what "the software comes with the book" means here. It is not a companion download. It is the thing the book is a printout of.

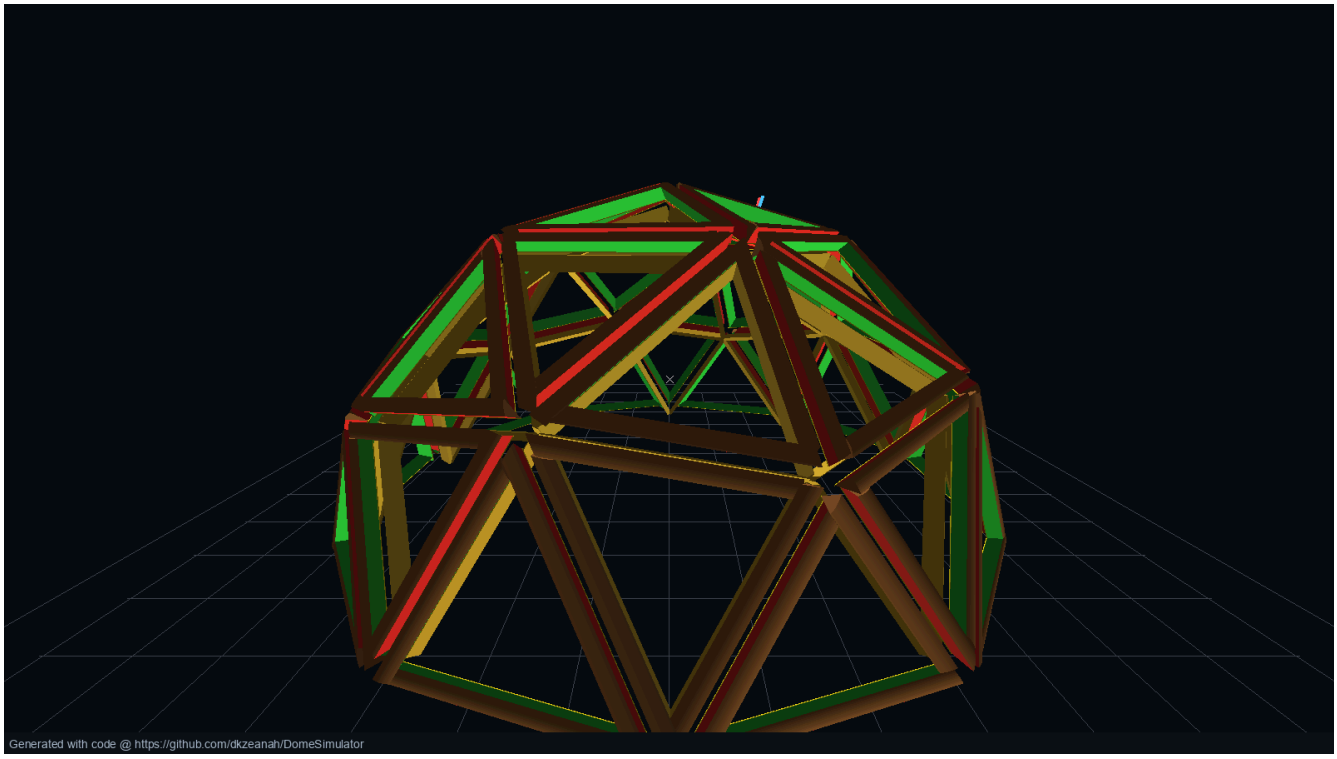
Three things this says against itself.

What Each World Shows You

One picture from every three-dimensional environment in the project, with what it is for and what to do in it

One picture from every three-dimensional environment that ships with this book, what it is for, and what to do once you are in it.

The raw-wedge solver



The reference build, opened eight inches.

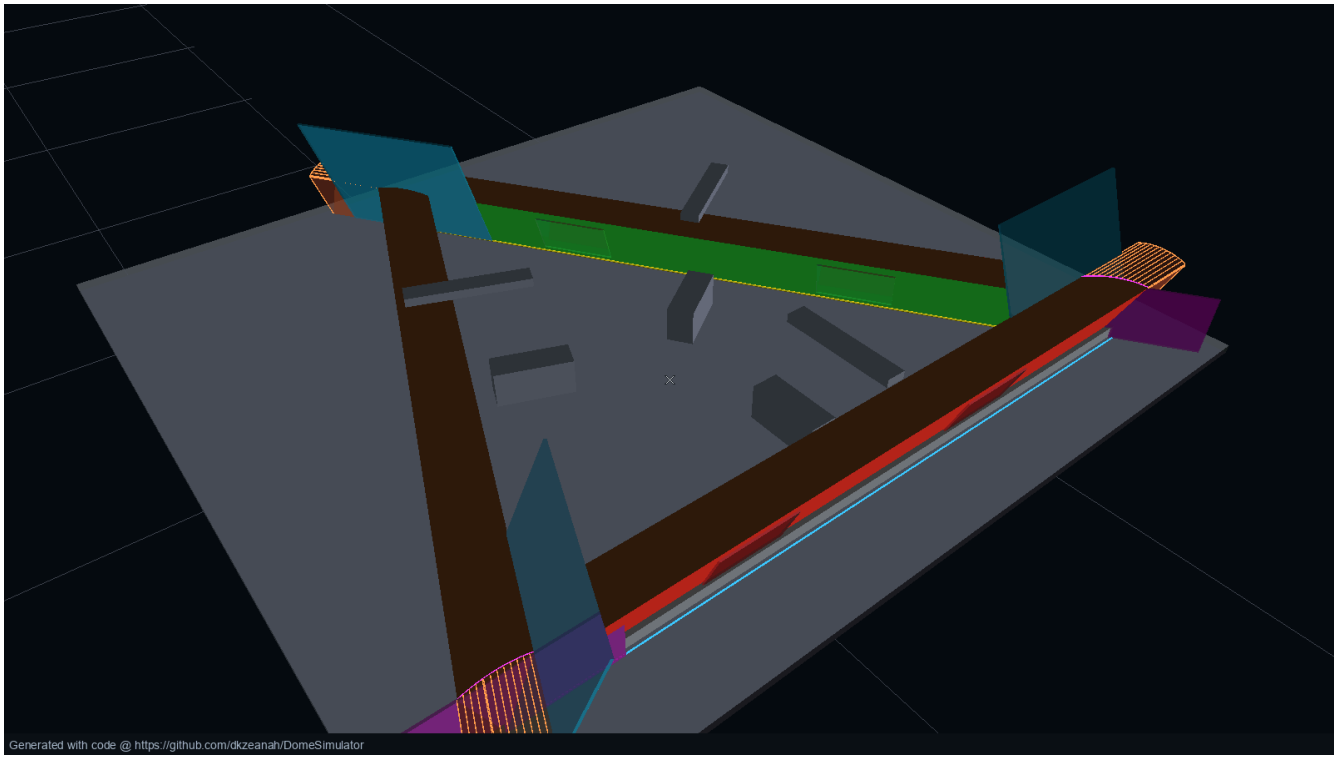
What it is. The dome this whole book is about, solved from one number: the long member. 120 members, 40 panels, 55 seams, every angle derived.

What to do in it. Press the explode key. The panels slide apart along their own face normals — nothing in the geometry moves, it is the same solve drawn with a gap — and for the first time you can see into a seam and watch the key sitting in it.

Then turn one decision at a time. Change the wedge orientation and watch the key's size change. Change the split count from 8 to twelve and watch the seam channel shrink to something that will not take a hose.

What it is not. It has no doorway, no foundation and no utility column, and it builds 2V hemispheres only. Those live in the other worlds.

The fabrication jig



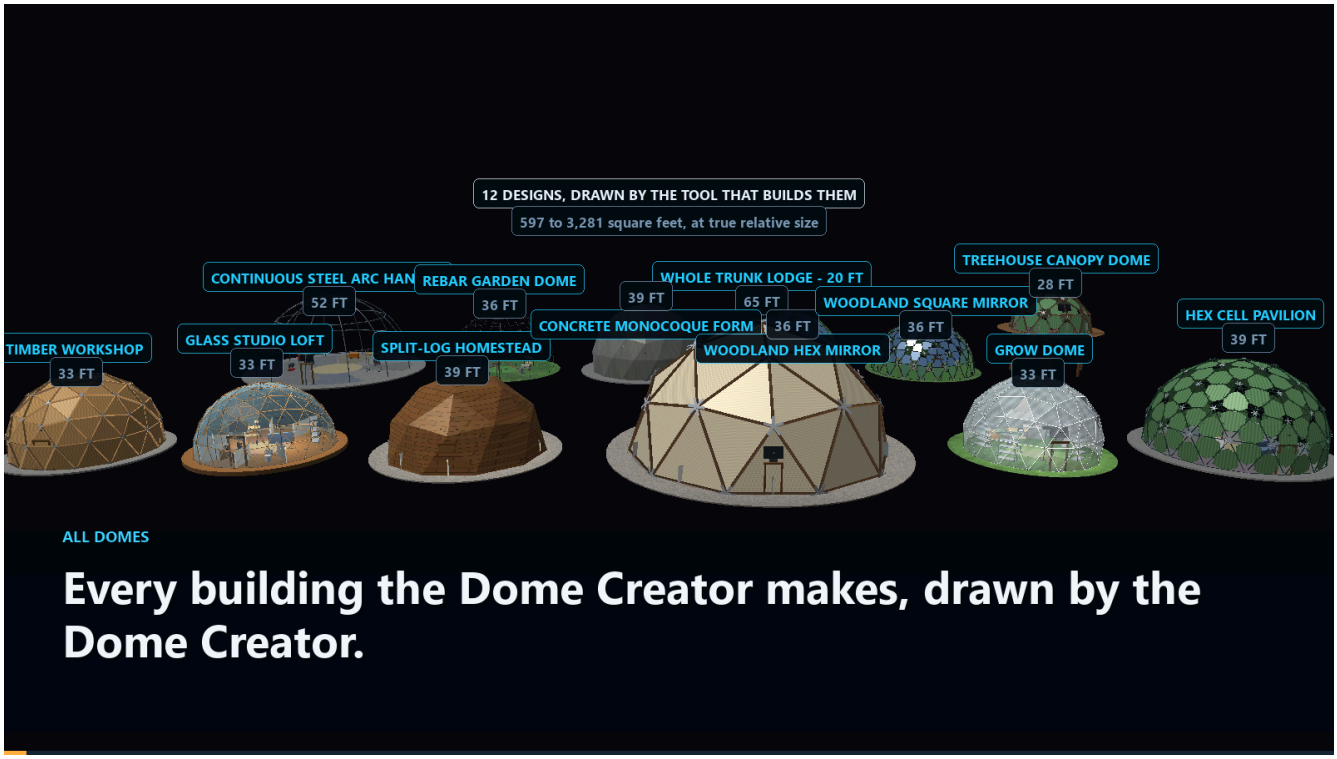
A panel closed on the jig.

What it is. The fixture of Chapter 24, in twelve steps, each one saying what it *enforces* rather than just what it is.

What to do in it. Step through all twelve before you cut a board. The corner stations are the ones worth flying to — the corner is where this method either makes sense or does not, and seeing the butt land on a radial face rather than on an end is the moment it clicks.

Then set it for the other panel family and step through again. The fixture does not change. That is the claim, and you can check it.

The Dome Creator



Twelve designs, drawn by the tool that builds them.

What it is. Every dome this project can describe that is *not* this one. Frequencies 1V to 4V, eight frame materials, eight strut sections, sixteen panel types, nine claddings, seven foundations — and a bill of materials for whatever you just assembled.

What to do in it. Build the dome you were going to build before you read this book, price it, then build this one and compare. That comparison is more persuasive than any argument in these pages, and it is yours rather than mine.

The seed world

Power and water up the middle and out through the apex.

What it is. The product version: the utility column, the gasketed cap at the apex, the shower cap, the quilted layers, the mast and the floor.

What to do in it. Look at the column. It is the one object that touches the ground, the wall and the sky, and almost every decision in Part 5 and Part 7 of this book is downstream of it existing.

Then add layers to the shell and watch the cap grow. Chapter 30 is arithmetic on a page; here it is a thing changing size.

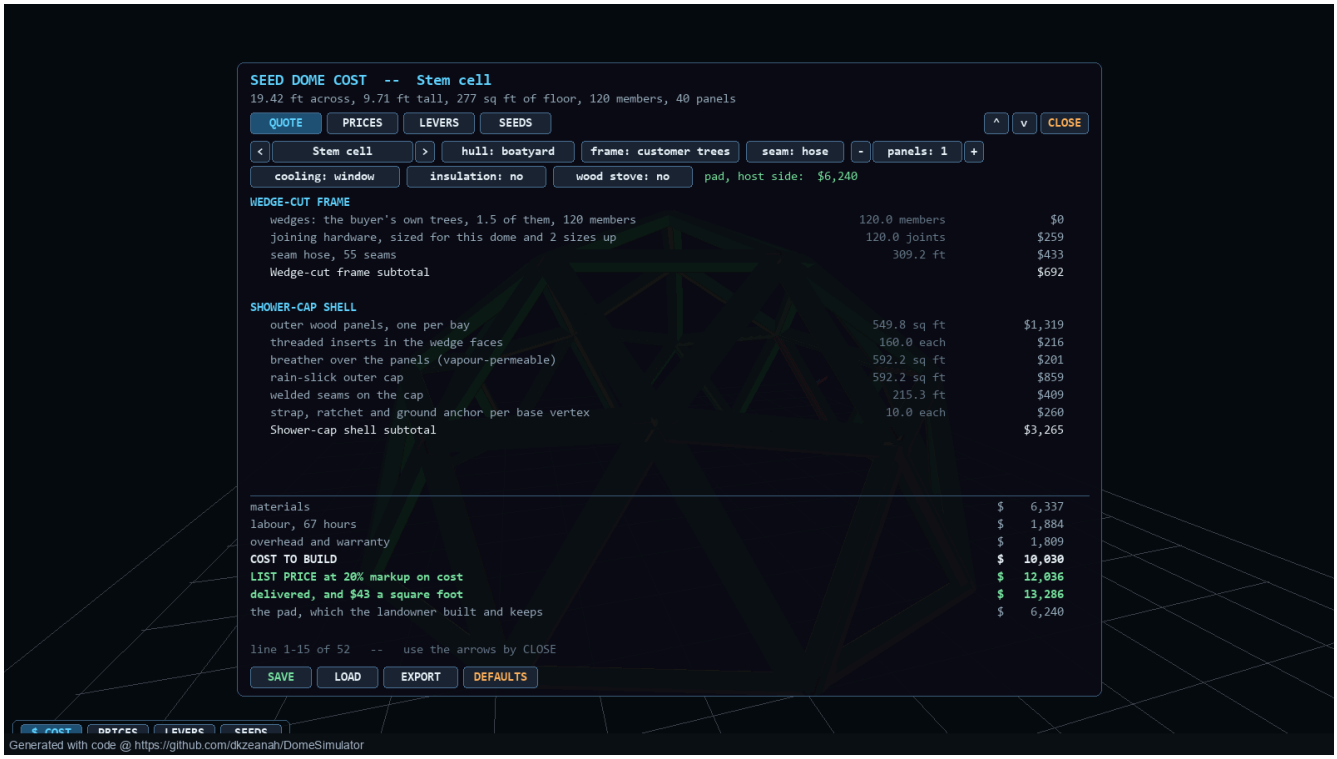
The pad and the park

The platform, built one course at a time.

What it is. The ground of Chapter 37: a pad built course by course, and a network of them with the shared services between.

What to do in it. Watch the colours. Amber is what the host owns, cyan is what the tenant owns, and the clean split of Chapter 43 is something you can see rather than something you have to be told.

The cost console



The geometry and the price in one program.

What it is. The cost model, open over the solved dome. Every line carries its own source constant, and changing an assumption moves every figure downstream of it immediately.

What to do in it. Find a price you disagree with — there will be one — and change it. The whole quote re-derives. That is the right way to argue with this book: not by doubting the total but by editing the input you think is wrong and seeing what it was worth.

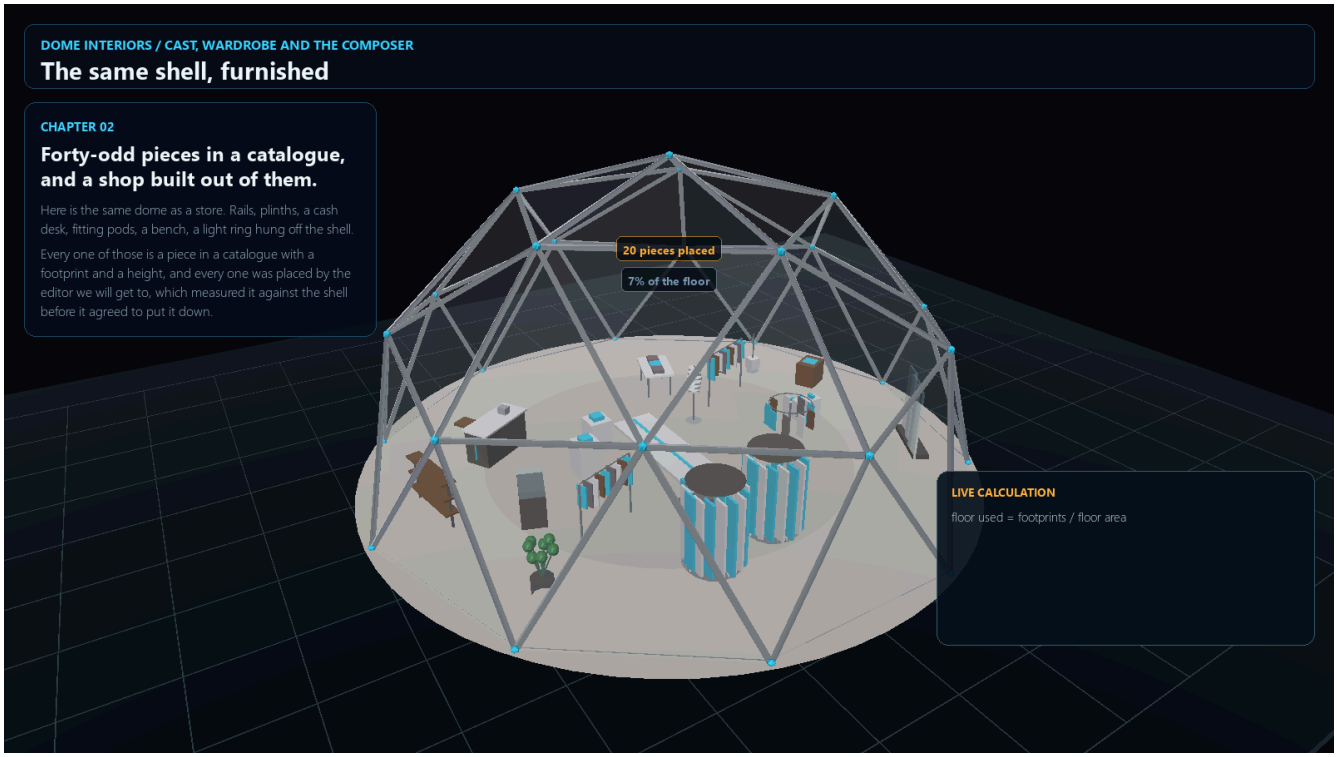
The number to look at first. $12,036 for the standard article, built to $10,030, at $43.45 a square foot of floor. The maker's markup is stated as a markup on cost rather than a margin on price, because those are different numbers and the one a reader can check with a calculator is the first.

Where to start

If you have an afternoon: the solver, exploded, then the jig's twelve steps.

If you have a tree: the cost console, with your own trunk diameter typed into it.

If you have ground: the park.



The same shell, furnished.

CHAPTER 58 · REFERENCE

Every Part, Priced

The column, the pad, the hardware, the power bench and the water bench -- the whole bill, with a source on every line

Six bills. Every line carries the constant it came from, so any of them can be argued with by editing one number rather than by doubting the total.

What is separate from what, and why

The dome is $12,036, and it is priced in Chapter 53 and the chapters around it. Nothing on this page changes it.

The pad is the host's bill, not the buyer's — Chapter 37. It is here because somebody has to build it and because a book that prices only half of an arrangement is not pricing the arrangement.

Everything else on this page is an upgrade. A dome works without the seam module, without the solar bench and without the water bench; it sits on a pad's pedestal and drinks from the pad's stub. These are what you fit when you want it to work with nothing plugged into it.

Added up, the systems on this page come to $20,409.

The utility column, complete

Including the steel, which is the part most bills leave out.

column housing, insulated chase	12.0 ft	$6.00	$72
steel core, service port to apex	12.0 ft	$18.00	$215
base flange onto the service port	1.0 each	$95.00	$95
apex sleeve and gasketed seal cap	1.0 each	$240.00	$240
apex lifting ring	1.0 each	$260.00	$260
floor hub that clamps the mast	1.0 each	$310.00	$310
radial steel floor spokes	10.0 each	$34.00	$340
sub-panel, breakers and outlets	1.0 each	$310.00	$310
water manifold and rise	1.0 each	$240.00	$240
drain stack and floor-port tie	1.0 each	$165.00	$165

Utility column, complete			$2,247

Two lines there are doing more than they look.

The steel core runs the full height, floor port to apex ring, and it is what the floor clamps to and what the lifting rig hangs from — Chapter 55. It is a pipe. It is also the single component that makes a floating dome possible, and it costs $622 in a $12,036 building.

The radial floor spokes are the 10 steel arms from the hub out to the base ring. They are what make the floor a component rather than framing, which is Chapter 3's third feature.

The power bench

solar array	800.0 W	$0.95	$760
MPPT charge controller	1.0 each	$195.00	$195
battery bank, LiFePO4	10.0 kWh	$410.00	$4,100
DC energy meter and shunt	1.0 each	$62.00	$62

Power bench			$5,117

A 8-way split log frame does not need electricity, but the condensing plates and the barrier fan of Chapter 29 do, and so does everybody who lives anywhere.

The array is sized at 800 W and the bank at 10 kWh, which is the owner's stated arrangement rather than a calculation from a load list. The metering is in the bill deliberately: a system you cannot read is a system you cannot size next time.

The water bench

under-floor tank	120.0 gal	$1.35	$162
first-flush diverter	1.0 each	$68.00	$68
three-stage filter	1.0 each	$210.00	$210
inline UV lamp	1.0 each	$165.00	$165
on-demand pump and accumulator	1.0 each	$145.00	$145
pulse-output water meter	1.0 each	$78.00	$78

Water bench			$828

This is what turns 309 feet of seam gutter into something you can drink.

The first-flush diverter is the cheapest item and the one that matters most. It throws away the first gallons off a dry roof — the ones carrying whatever has landed on the cap since the last rain — and it is the difference between roof water and drinking water.

The UV lamp is the other one. A filter makes water clear; a lamp makes it safe. Neither does the other's job.

Joining hardware, stainless

A2 stainless bolt, nut, two washers	440.0 sets	$0.92	$405
threaded insert in a sawn face	880.0 each	$0.74	$651
flat V bracket, four holes a flap	120.0 each	$5.80	$696

Joining hardware, stainless			$1,752

Edge to edge, into threaded inserts rather than into end grain — Chapter 3.

Stainless throughout is not a luxury here. Chapter 36 notes that wet timber and plain steel together are a battery; on a frame whose whole argument is that it lasts decades and gets taken apart, a corroded fastener is the one failure that turns a demountable building into a permanent one.

The V bracket

The flat V bracket is the connector this method settled on: two flaps meeting at the panel's angle, **4 holes in each flap**, folded from stainless on a bench brake.

The flaps run out to as much as **half the member's length** — 35.1 inches on a long member — which puts about 8.8 inches between holes. That spacing is the point of the thing. Screws clustered at the end of a stick are all working the same short grain; spread along it, each one is into fresh wood, and the joint stops depending on any single fixing.

At $5.80 each, 120 of them is not the expensive part of anything.

The pad's own materials

compacted gravel base over fabric	369.2 sq ft	$1.20	$443
deck boards, 2x6 laid flat	902.2 ln ft of 2x6	$1.17	$1,053
joists at 16 in centres	310.1 ln ft of 2x6	$1.17	$362
beams, doubled 2x6	179.3 ln ft of 2x6	$1.17	$209
precast piers	20.0 each	$12.00	$240
hangers, structural and deck screws	369.2 sq ft	$0.55	$203
two coats of penetrating sealer	369.2 sq ft	$1.10	$406

```
                                                                  ---------
  Pad materials (framed deck on piers, sealed)                    $2,916
```

The host's bill, counted board by board rather than rated per square foot — which is why it can be checked against a merchant's quote.

What this page does not include

Labour, which is Chapter 53's territory and is 67 hours on the standard article.

Tools. A chainsaw, a circular saw, a drill and a bench brake, none of which are consumed by one dome.

And freight, permits, and the ground itself, all of which are local and none of which this book can know.